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Experimental investigation of Russian

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**Fine-tuning (in)definiteness in the absence of articles:
Experimental investigation of Russian**

A dissertation presented

by

Daria Bikina

to

The Department of Linguistics

in partial fulfillment of the requirements

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Fine-tuning (in)definiteness in the absence of articles: Experimental investigation of Russian

Abstract

Languages vary in the ways they encode (in)definiteness. In languages with articles, determiners like *the* and *a/an* in English explicitly mark whether a nominal is definite or indefinite. By contrast, in languages without articles, nouns can appear *bare*, with no special (in)definiteness marking. This raises a fundamental question: in the absence of articles, do bare nouns have a uniform semantic profile, and if so, is it definite-like, indefinite-like, or something else? Is this semantic profile the same in systems where bare nouns co-exist with overtly (in)definite nouns, like English, vs. in systems that have no articles whatsoever?

This dissertation aims to answer these questions, focusing primarily on Russian, and using English as a comparison. Through a series of controlled experimental studies, I investigate (in)definiteness across the following dimensions: uniqueness, maximality and anaphoricity. The results bear on competing approaches to nominal semantics: the neo-Carlsonian approach (Carlson 1989; Chierchia 1998; Dayal 2004, 2013), which derives bare noun meanings from kind reference and iota; existential-based approaches, which treat bare nouns as fundamentally indefinite; and property-based approaches, which allow free alternation between definite and indefinite construals.

Across the board, the findings rule out a fully existential-based analysis of bare nouns. Russian bare singulars display systematic sensitivity to uniqueness, and apparent violations reflect contextual domain restriction rather than existential semantics. Both Russian and

English bare plurals tend to be interpreted maximally, which is again inconsistent with a purely existential analysis. Russian bare plurals show robust anaphoric potential that English bare plurals lack, consistent with the prediction that covert *iota* is available in Russian but blocked in English by the overt definite article. However, this anaphoric potential is strongly modulated by discourse context, suggesting that *iota* is accessible rather than obligatory. More broadly, the cross-linguistic contrast supports a gradient rather than categorical version of the Blocking Principle (Chierchia 1998).

Taken together, this dissertation provides controlled experimental evidence bearing on a long-standing debate in formal semantics that has largely relied on introspective judgments and corpus data, with only a few experimental investigations to date. I argue that Russian bare nouns should not be analyzed as standard existential quantifiers, as they consistently display properties associated with definiteness. However, they are also not simply covert definites. Like overt definites in languages with articles, they require uniqueness/maximality; but unlike overt definites, they are exempt from the Novelty-Familiarity condition (Heim 1982), in line with the position outlined in Dayal (2026). This exemption follows from the absence of overt (in)definiteness morphology rather than from an ambiguity in the underlying semantic operation. The findings point towards an analysis in which bare nouns in articleless languages are realized through an *iota*-like operator or kind formation, but not through a standard existential operator or a choice function.

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I was a weird kid. At the age of 6, I scared my mom’s friends because I asked her in public if Russian converbs (*deepričastija*) inflect for case¹. So in a way, no one from my family or school friends was really surprised that my interests didn’t change much since 1999, and I ended up entering a PhD program in Linguistics. What is surprising, though, is that I made it till the defense. This only happened because of those who supported me along the way.

First and foremost, I want to thank my committee. There could be no better team to do experimental work on bare nouns with. I am especially indebted to my dissertation chair, Gennaro Chierchia. Our first meaningful interaction was during Semantics II. I volunteered to present the first week of class; the task was to show that every single logical operation can be defined from just lambda and identity function. I immediately regretted volunteering; I had no idea how to even approach it; and I remember how gracefully Gennaro explained it to me, and it was the most elegant thing I have ever seen. Till this day, when I need to show someone the beauty of semantics, I tell them about this experience. I am very grateful that Gennaro was so patient with me, my slowness, my extreme tendency to drown in unnecessary details, and believed in me during the hardest parts of this journey. I think our meetings in Boylston were probably the most insightful hours in my PhD experience, if not in my life.

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¹When I entered Moscow State University and started working with Uralic, I learned that in some languages converbs do, in fact, inflect for case. This revelation brought me a lot of joy.

²It was actually Chierchia (1998).

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*To the memory of Valentina Ivanovna Bakhur (1933–2012)
and Lyubov Viktorovna Aldakova (1950–2020)*

*There are several opinions about the relationship between the complexity of human language
and its linguistic description:*

- 1. Language is complex, therefore its model must be complex; otherwise the model is inadequate (I.A. Mel'čuk).*
- 2. Language is simple, therefore its model must be simple; otherwise the model is inadequate (A. E. Kibrik).*

*A third point of view is closer to me, which is relatively unpopular in Russian linguistics:
language (and the world in general) is extremely complex, therefore its model should be
simple. Only in this case – if, of course, we do not confuse the model with the language –
can we hope that we might have understood something.*

Yakov Testeleets

Glossing Abbreviations

1, 2, 3	person	M	masculine
ACC	accusative	NEG	negation
COMP	complementizer	N	neuter
DAT	dative	NOM	nominative
DEF	definite	PFV	perfective
F	feminine	PL	plural
GEN	genitive	PREP	prepositional
IMP	imperative	PRS	present
INDEF	indefinite	PST	past
INF	infinitive	PTCL	particle
INSTR	instrumental	REFL	reflexive
IPFV	imperfective	SG	singular

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1 The puzzle

Russian, like many other languages of the world, lacks articles. In such languages, argument nouns can appear bare, that is, without any determiner signaling (in)definiteness. At first glance, this might suggest that Russian nouns are underspecified for (in)definiteness, freely allowing for both definite and indefinite interpretations. Consider the following examples. In the first sentence in (1), the bare singular *devočka* ‘girl’ introduces a new individual into the discourse. Its continuation features the same noun referring back to that individual¹².

- (1) V komnat-u zaš-l-a **devočka_i**. **Devočka_i** se-l-a na stul.
in room-ACC enter-PST-F girl girl sit-PST-F on chair
‘A girl entered the room. The girl sat on the chair.’

A similar pattern holds for Russian bare plurals:

- (2) V komnat-u zaš-l-i **devočk-i_j**. **Devočk-i_j** se-l-i na stul’-ja.
in room-ACC enter-PST-PL girl-PL girl-PL sit-PST-PL on chair-PL
‘Some girls/The girls entered the room. The girls sat on the chair.’

So, on the surface level, Russian bare nouns appear ambiguous between definite and indefinite interpretations. This observation has motivated the hypothesis that in languages without articles, bare nouns may freely shift between indefinite and definite construals, as there is

¹Examples (1) and (2) are inspired by the tests on (in)definiteness suggested in Dayal (2026c), see p. 51–52. These patterns were reported specifically for Russian in Geist (2010) (for bare singulars) and Soloveva et al. (2026) for both singulars and plurals.

²Unless a citation is provided, all examples are elicited data collected by the author. While various authors citing Russian used different romanization systems, I standardized them using the scholarly romanization system described in Timberlake (2004), and used my own glossing for consistency.

no overt marking of (in)definiteness.

However, as we will see throughout this dissertation, the empirical picture is more nuanced. Let us look at some cases that raise some doubt about a simple ambiguity analysis.

1.1 Russian bare arguments: definite or indefinite?

We will start with some descriptive observations about Russian in parallel with English. If bare nouns in languages without articles can readily switch between definite and indefinite interpretations, we would expect all occurrences of English definites and all occurrences of English indefinites to be appropriately translatable using the Russian bare noun. Without diving into theoretical details yet, let me present some cases where this doesn't seem to hold true. The critical contexts will concern anaphoricity (section 1.1.1), scope flexibility (sections 1.1.2-1.1.3), maximality (section 1.1.4), and number contrasts (section 1.1.5).

The tests discussed in this section largely rely on the (in)definiteness questionnaire proposed in Dayal (2026c), as well as the data from the application of this questionnaire to Russian, presented in Soloveva et al. (2026). Some examples were modified to be more explanatory, in which case they were elicited; I refer to the original examples from Soloveva et al. (2026) that served as a starting point nevertheless.

1.1.1 Bare nouns are bound to a constant referent

From (1), we saw that the bare singular both introduces a new referent and refers back to an already introduced individual. However, it seems like a Russian bare noun has to refer back to an already introduced referent whenever possible. Example (1) cannot have a continuation like (5-b), which suggests that these were two different girls:

- (3) a. V komnat-u zaš-l-a **devočka**. **Devočka**_{*i*/**j*} se-l-a na stul.
in room-ACC enter-PST-F girl girl sit-PST-F on chair
'A girl entered the room. The girl sat on the chair.'

-
- b. #Tu, čto zaš-l-a, zov-ut Maša, a tu, čto
 that.ACC.F COMP enter-PST-F call-PRS.3PL Masha and that.ACC.F COMP
 sela, Sveta.
 sat-PST-F Sveta
 ‘The one who entered is Masha, and the one who sat down on a chair is Sveta.’

That would not be problematic for an overt indefinite in English:

- (4) A girl_i entered the room. **A girl**_j sat down. The one who came in is Mary, and the
 one who sat down is Claire.

In order to refer to some other girl, the Russian bare singular has to be modified by a word like ‘another’; without it, the bare noun has to be anaphoric.

- (5) a. V komnat-u zaš-l-a **devočka**_i. ***(Drug-aja) devočka**_j se-l-a na
 in room-ACC enter-PST-F girl another girl sit-PST-F on
 stul.
 chair
 ‘A girl entered the room. Another girl sat on the chair.’
 b. Tu, čto zaš-l-a, zov-ut Maša, a tu, čto
 that.ACC.F COMP enter-PST-F call-PRS.3PL Masha and that.ACC.F COMP
 sela, Sveta.
 sat-PST-F Sveta
 ‘The one who entered is Masha, and the one who sat down on a chair is Sveta.’

If Russian bare singulars were fully ambiguous between definite and indefinite, there would be no problem using one in a sentence where the reference is switched. Yet it seems to be impossible for the bare noun to not be anaphoric in this situation.

There exists another diagnostic related to the constancy of referents that distinguishes definites from indefinites. It concerns nominal behavior in conjunctions. Löbner (1985); Löbner (2011) observes that definite descriptions cannot occur in conjunctions that attribute incompatible properties; Coppock and Beaver (2015) interpret this test as a diagnostic for uniqueness in definites, and Dayal (2026c) extends it as a test for (in)definiteness in bare

nouns. In (6-a)³, the repeated definite phrases have ostensibly the same referent, which results in a contradiction. By contrast, in (6-b), each indefinite may introduce a distinct individual, and the sentence is coherent.

- (6) a. #The rabbit is in the cage and the rabbit is outside the cage.
 b. A rabbit is in the cage and a rabbit is outside the cage.

Russian bare nouns pattern with English definites in this respect: repeating a bare singular in a conjunction with incompatible predicates results in a contradiction (see Soloveva et al. (2026), p. 229). The bare singular behaves as if it refers to a single referent across the two conjuncts.

- (7) #Krolik sid-it v kletke i krolik sidit rjedom s kletkoj.
 rabbit sit-PRS.3SG in cage and rabbit sit-PRS.3SG next with cage
 ‘#The rabbit is sitting in the cage and the rabbit is sitting next to the cage.’

If Russian bare singulars freely alternated between definite and indefinite interpretations, we would expect them to be able to “switch” to the indefinite option and allow reference to distinct individuals across conjuncts. The systematic unavailability of such readings suggests that bare singulars behave as referentially constant expressions in these environments.

Bare plurals exhibit similar effects. Plural indefinites in languages with articles typically introduce independent existential domains. For example, English plurals marked by the indefinite determiner *some* are not bound to any constant referent, and therefore can be used in sentences with two contradictory predicates. Definite plurals resist such uses:

- (8) Vogeleer and Tasmowski (2006, p. 2)
 a. Some children are in the garden and some are not.
 b. #The children are in the garden and the children are not.

³Examples (6-a) and (6-b) are inspired by examples involving English possessives from Coppock and Beaver (2015, p. 417–418).

Bare plurals pattern with definite plurals in this diagnostic. The following sentence is degraded on the intended reading where two distinct groups of children are introduced:

- (9) #**Deti** v sad-u i **deti** ne v sad-u.
 children in garden-PREP and children NEG in garden-PREP
 #‘Some children are in the garden and some children are not.’

This mirrors the behavior of English bare plurals, which likewise are degraded in such contexts:

- (10) #Children are in the garden and children are not.

These anaphoricity-related diagnostics indicate that both singular and plural bare nouns in Russian do not behave like canonical indefinites. In our examples, they are unable to introduce new referents independently in each conjunct. Instead, they behave as if anchored to a stable referent or referential domain. This behavior is difficult to account for with a view on which bare nouns simply shift between definite and indefinite without restriction.

1.1.2 Lack of wide-scope existential readings

Formally, overt indefinites are often analyzed as introduced by an existential quantifier:

- (11) a. Existential quantification (Partee, 1986)
 $\exists: \lambda P.\lambda Q.\exists x[P(x) \wedge Q(x)]$
 b. $[[\text{A girl dances}]] = \exists x[\text{girl}(x) \wedge \text{dances}(x)]$

Quantifiers are typically flexible in scope: when combined with operators like negation or modals, they can be interpreted either below or above that operator (Fodor and Sag, 1982; Karttunen, 1969; Kratzer, 1998, among others). This flexibility is a core property of indefinites in languages with overt articles, like English:

(12) **A cat** is not sitting on the chair.

✓ Narrow scope: ‘There is no cat such that it is sitting on the chair.’

$\neg\exists x.\text{cat}(x) \wedge \text{sitting-on-the-chair}(x)$

✓ Wide scope: ‘There is a cat such that it is not sitting on the chair.’

$\exists x.\text{cat}(x) \wedge \neg\text{sitting-on-the-chair}(x)$

(13) John wants to marry **a French woman**.

✓ Narrow: ‘He doesn’t know any yet.’

✓ Wide: ‘He got to know her last summer.’ (Geist, 2010, p. 211)

If genuine existential quantification was freely available to Russian bare nouns, we would expect similar scope flexibility. However, in the Russian example parallel to (12), the bare noun can get a definite reading (a unique, familiar cat is not on the chair) or a narrow-scope indefinite reading (there are no cats on chairs at all), but not a wide-scope reading (one of the cats is not on the chair):

(14) **Kot** ne sidit na stul-e.

cat NEG sit-PRS.3SG on chair-PREP

✓ Definite reading: ‘The cat is not sitting on the stool.’

✓ Narrow reading: ‘There isn’t any cat such that it is sitting on the stool.’

× Wide reading: ‘There is a cat such that it is not sitting on the stool.’

(Dayal, 2004, p. 405)

(15) Džon xoč-et žen-it’-sja na frantsuženk-e.

John want-PRS.3SG marry-INF-REFL on French.woman-PREP

✓ Narrow: ‘He doesn’t know any yet.’

× Wide: ‘He got to know her last summer.’ (Geist, 2010, p. 211)

The unavailability of wide scope for bare singulars has been reported in many other languages, e.g. Hindi (Dayal, 2004, 2015), Mandarin (Yang, 2001; Dayal, 2004), Brazilian Portuguese (Schmitt and Munn, 2002; Pires de Oliveira and Rothstein, 2011), Kannada

(Srinivas, 2021; Srinivas and Rawlins, 2021), Turkish (Sağ, 2019), Laz (Sağ et al., 2025a,b), Wolof (Fong, 2023).⁴ If bare singulars were simply existentially quantified expressions, their behavior should parallel that of overt indefinites.

The pattern repeats in bare plurals, which also resist a wide-scope reading.

- (16) **Kot-y** ne sidj-at na stul’j-ax.
 cat-PL NEG sit-PRS.3SG on chair-PREP.ACC
 ✓ Definite reading: ‘The cats are not sitting on the chairs.’
 ✓ Narrow reading: ‘No cats are sitting on the chairs.’
 × Wide: ‘Some cats are not sitting on the chairs.’

The systematic absence of wide scope therefore challenges an unrestricted ambiguity analysis⁵.

1.1.3 Quantifier interactions: absence of intermediate scope

In sentences with two quantifiers, English indefinites can interact with both of them, giving rise not only to wide and narrow, but also to intermediate scope. This has been demonstrated for English *certain* and *some*-phrases, although some acceptability has been reported for *a*-indefinites as well (Karttunen, 1969; Farkas, 1981; Fodor and Sag, 1982; Kratzer, 1998; Abusch, 1993).

- (17) Martí and Ionin (2019, p. 1)
 Every student read every article that **a (certain) professor** recommended.
 ✓ Wide: ‘There is a professor x such that for every student y , y read every article that x recommended.’

$$\exists x.\text{professor}(x) \wedge \forall y[\text{student}(y) \rightarrow \forall z[[\text{article}(z) \wedge \text{recommended}(z, x)] \rightarrow \text{read}(y, z)]]$$

⁴Reported counterexamples, bare singulars with variable scope, include Malagasy (Paul, 2016), Kaingang (Navarro, 2025), and Xhosa Carstens et al. (2026).

⁵It is worth noting that the generalization about obligatory narrow scope of Russian bare nouns has been challenged in the literature. Specifically, Bronnikov (2006) provides examples (based on introspection), which are derived from those in Dayal (2004) and where both bare singulars and bare plurals can have a wide scope reading. Similarly, Golosov (2026) claims that examples like (15) have a wide scope reading.

✓ Intermediate: ‘For every student y , there is a (potentially different) professor x , such that y read every article that x recommended.’

$\forall y[\text{student}(y) \rightarrow \exists x[\text{professor}(x) \wedge \forall z[[\text{article}(z) \wedge \text{recommended}(z, x)] \rightarrow \text{read}(y, z)]]]$

✓ Narrow: ‘For every student y , y read every article that was professor-recommended.’

$\forall y.\text{student}(y) \rightarrow \forall z[[\text{article}(z) \wedge \exists x[\text{professor}(x) \wedge \text{recommended}(z, x)]] \rightarrow \text{read}(y, z)]$

In contexts like that, Russian bare singulars are far more restricted: the only reading they confidently allow is the definite one. The intermediate scope is unavailable; the narrow scope is marginal at best⁶.

- (18) Každ-yj student pročita-l každ-uju stat’j-u, kotor-uju **professor**
 every-M student read-PST.M every-ACC.F article-ACC which-ACC.F professor
 porekomendova-l.
 recommend-PST.M

✓ Definite: ‘Every student read every book that the professor recommended.’

× Wide: ‘There is some professor x such that for every student y , y read every article that x recommended.’

× Intermediate: ‘For every student y , there is a (potentially different) professor x , such that y read every article that x recommended.’

? Narrow: ‘For every student y , y read every article that was professor-recommended.’

The absence of wide and intermediate scope suggests that Russian bare singulars are not freely interpreted as existential quantifiers. Their behavior aligns more closely with definite descriptions, which do not give rise to such scope flexibility.

⁶Soloveva et al. (2026) report the narrow scope existential reading to be available for Russian bare singulars, as long as they denote unique individuals, see p. 239–240. However, their examples use postverbal bare singulars, and the position might have influenced interpretation (see section 2.2.1 for an overview). For the sake of parallelism, I provide a sentence with a preverbal subject.

1.1.4 Bare plurals: the contrast between English and Russian

In 1.1.1, we saw that Russian bare plurals behave similar to English bare plurals in sentences with incompatible predicates. However, there exist contexts, where Russian bare plurals are licensed but English bare plurals aren't. For example, English bare plurals cannot normally appear in bridging contexts:

- (19) I bought a car. **The wheels/*Wheels** need to be replaced. Dayal (2013)

In Russian, by contrast, the bare plural is fully acceptable:

- (20) Lena kupi-l-a mašin-u. U neë nado pomenj-at' **kolës-a**.
Lena buy-PST-F car-ACC by her needed change-INF wheel-PL
'Lena bought a car. The wheels on it need to be replaced.' (Soloveva et al., 2026,
p. 230)

English bare plurals also do not exhibit maximality in contexts like (21):

- (21) **The wheels/*Wheels/#Some wheels** of my car need to be replaced.
modified from Dayal (2013)

In Russian, the bare plural is again acceptable:

- (22) **Kolës-a** moej mašin-y nuždaj-ut-sja v zamen-e.
wheel-PL my-GEN.F car-GEN need-3PL-REFL in replacement-PREP
'The wheels of my car need to be replaced.'

These facts suggest that Russian bare plurals cannot be reduced either to plural indefinites or to English-like bare plurals. Like definite or bare plurals, they are not felicitous in contradiction contexts; yet, they also display distributional properties not shared by English bare plurals. I am not explaining these facts here; but one can notice that the resulting profile is not one of free oscillation between definite and indefinite construals, not in singulars, and

not in plurals. So far it looks like Russian bare plurals might be just definite. In the next section, I will show that this is also not quite empirically correct.

1.1.5 Number contrasts

What we have seen so far should already cast doubt on a simple ambiguity analysis. A more fundamental challenge arises once we see that bare singulars and bare plurals do not behave alike. Consider the following contrast, first pointed out by Dayal (2004) in a number of article-less languages, including Russian:

(23) Dayal (2004)

- a. #**Sobaka** by-l-a vezde.
dog be-PST-F everywhere
#‘The dog/A particular dog was everywhere.’
- b. **Sobak-i** by-l-i vezde.
dog-PL be-PST-PL everywhere
‘Dogs were everywhere.’

The sentence with a plural allows a reading where different dogs are present in different places. The singular, however, only permits the implausible reading that a single individual is simultaneously everywhere. If bare nominals were freely ambiguous between definite and indefinite construals, number morphology alone should not determine whether differentiated scope readings are available. Yet the contrast is observed in many languages, and in the languages it is present, it is systematic.

In this section, I presented initial evidence that a simple ambiguity account, where bare nouns freely alternate between existential and definite construals, leads to wrong predictions. In fact, bare nouns lack the scope flexibility of genuine indefinites. Russian bare plurals align with English bare plurals in some contexts, but not the others. We also see that (in)definiteness interpretation is sometimes number-sensitive.

At the same time, the ambiguity hypothesis remains a natural starting point, and, in

fact, is defended in literature. In the absence of overt articles, it is tempting to assume that bare nouns simply have access to both indefinite and definite interpretations, and the context determines which construal is realized in a given utterance. The theories that employ this view vary in how exactly they implement ambiguity and what restricts it, if anything. In the next section, I will review some of such approaches relevant for the discussion of Russian.

1.2 The ambiguity approach

The works presented in this section all share the following point: bare nouns in languages without articles can be definite or indefinite depending on the context. Formally, these two interpretations are derived via distinct type-shifting mechanisms needed in order for a noun to serve as an argument. Since lexical nouns are properties of the type $\langle e, t \rangle$, they need to be transformed into an argument type (e or $\langle \langle e, t \rangle, t \rangle$) in order to participate in subsequent derivation. The two type-shifting mechanisms resulting in different (in)definiteness interpretations are *iota* and existential quantification. We have already introduced existential quantification in (11-b); it builds a generalized quantifier and asserts the existence of at least one entity satisfying the predicate of the noun. In this way, existential quantification yields an indefinite-like reading; the scopal properties of English indefinites follow from them being existential quantifiers.⁷

In contrast, the *iota* operation returns the unique maximal entity satisfying a predicate, if such an entity exists, and is considered responsible for overt definite nouns such that English definites with *the*. Unlike existential quantification, *iota* presupposes uniqueness: it is defined only when there is exactly one relevant individual in the context. The infelicity of English definites in sentences with incompatible predicates follows exactly from that uniqueness/maximality property attributed to *iota*.

⁷In this chapter, I am staying neutral on the distinction between choice functional analyses and generalized quantifier approaches, as long as they make the same predictions about scopal flexibility.

-
- (24) a. *Iota* operation
 $\iota: \lambda P.\iota P_s$, if there exists a unique maximal entity in P , undefined otherwise.
b. $[[\text{The girl dances}]] = \text{dances}(\iota x.\text{girl}(x))$

In this way, what ambiguity accounts assume is that in the absence of special morphological marking, both existential quantification and *iota* are available.

In fact, there are two kinds of approaches that I will treat as ambiguity approaches. One of them is ambiguity in the narrow sense: analyses on which bare nouns in languages with articles can occur both via *iota* and via existential quantification, based on certain contextual or structural factors. Another family of approaches treats bare nouns as semantically equivalent to indefinites, with definite-like uses being derived through some pragmatic mechanism. I will start with the latter one.

1.2.1 The origins: Heim (2011)

The work that is usually cited as a central source of indefinite-based approaches to bare nouns is Heim (2011). In this overview chapter, Heim suggests treating definiteness and indefiniteness as abstract features, thereby loosening the correspondence between (in)definiteness and article morphology. The aspect of Heim (2011) that has most influenced work on bare nouns in articleless languages is her idea of competition between definites and indefinites. Indefinites are often characterized by two properties. The first one is non-uniqueness: indefinites are typically infelicitous in a context where the referent is expected to be unique. The second property is novelty: indefinites introduce a new discourse referent. Heim (2011) claims that at least for English singular indefinites, neither non-uniqueness nor novelty are part of the meaning of the indefinite itself. Instead, she suggests that these properties occur as a result of pragmatic strengthening due to competition with English definites. For instance:

- (25) The bicycle was fine after Heim (2011, p. 1005)

-
- a. **the seat** was replaced.
 - b. #**a seat** was replaced.

Rather than concluding that the indefinite article brings in a non-uniqueness requirement, which results in the infelicity of (25-b), Heim suggests to treat definites and indefinites as competing alternatives on a scale. The definite comes with presuppositions of uniqueness and existence, while the indefinite is logically weaker. If a speaker uses an indefinite in a context where uniqueness is presupposed, like (25), it signals that the presuppositions of the definite NP are not met. This triggers an anti-uniqueness implicature:

(26) $|SEAT| \neq 1$ adopted from Šimík and Demian (2020)

Heim further demonstrates it in application to domain restriction:

- (27) There were four chairs. John sat down on a chair. Then Mary (Heim, 2011, p. 1006)
- a. knocked **the chair** over.
 - b. knocked **a chair** over.

The only available interpretation of (27-a) is that John's chair was knocked over. This is because the definite can be construed with a covert restrictor. Out of all options of this restrictor's scope, only the narrowest one allows the singular definite to escape the failure of the uniqueness presupposition. In contrast, the indefinite in (27-b) cannot be interpreted anaphorically. Again, for each of these potential options of the covert restrictor's scope, we must consider the corresponding definite competitor and compute a non-uniqueness implicature. In the case of the narrowest restrictor, this implicature is the one in (28); as it is odd, the speaker disambiguates in favor of the wider covert restrictor.

(28) $|CHAIRS-THAT-JOHN-SITS-AT| \neq 1$

Importantly, anti-uniqueness implicatures do not arise in all cases where the indefinites are used. However, Heim (2011) does not go into much detail as to when they do and when they don't.

Heim further hypothesizes that if a language has no articles, competition is absent, and bare nouns are therefore “semantically equivalent to English indefinites” (Heim, 2011, p. 1006). If this is correct, bare nouns should have a broader range of felicitous uses in such systems, since they are not subject to strengthening via competition with definites. Heim does not directly say that the use of bare nouns is completely unrestricted in languages without articles.

Several works on Slavic bare nouns (Šimík and Demian, 2020; Seres and Borik, 2021; Seres et al., 2021) rely on this hypothesis from Heim, interpreting it to mean that bare nominals in articleless languages lack the presuppositions of uniqueness and familiarity associated with definiteness. However, as noted in Dayal (2026a), it is worth emphasizing that Heim's remark on articleless languages is brief and incidental; it is not the central focus of either her dissertation (Heim, 1982) or her later work on (in)definiteness (Heim, 1991, 2011).

1.2.2 Geist (2010): ambiguity constrained by information structure

A more articulated version of the ambiguity approach is proposed by Geist (2010). She focuses on Russian bare singulars and argues that they are in principle compatible with both definite and indefinite construals. However, the availability of the indefinite interpretation is structurally constrained. In this way, Geist's account is not a fully flexible ambiguity account as she does not treat *iota* and existential construal to be freely interchangeable. Instead, she proposes that existential type-shifting is restricted by information structure.

As a starting point, Geist assumes that nouns denote predicates of type $\langle e, t \rangle$ and require type-shifting in order to serve as arguments, and accepts *iota* and existential closure as available options. She further demonstrates that the indefinite interpretation of bare singulars is more restricted, and argues that these restrictions are best captured in terms of

topic-comment structure. Specifically, she formulates the following generalization:

(29) **Geist’s (2010) generalization**

A bare NP can receive an indefinite interpretation only if it belongs to the comment.

By the comment position, Geist understands any part of the utterance that is not topical. Empirically, this proposal builds on the interpretational asymmetries in Russian word order. Preverbal NPs are more likely to be topical material, and Geist shows that they resist indefinite readings in neutral contexts, while postverbal NPs can be either definite or indefinite. Geist argues that this contrast cannot be reduced to linear position per se, but reflects the interaction between information structure and (in)definiteness. Formally, we can summarize her points as follows:

(30) **Two conclusions of Geist (2010)**

- a. Topical bare singulars are interpreted via *iota*.
- b. Existential closure is only licensed in the comment position.

This restriction follows from felicity conditions on topics. Geist claims that in languages with overt indefinite articles, indefinite NPs can be aboutness topics (in the interpretation of Reinhart (1981)), especially when supported by rising prosody – see an example from German:

- (31) /EINEN Linguisten, den kennt \JEDER.
 one linguist res.pron knows everybody
 ‘One linguist everybody knows.’ (Geist, 2010, p. 205)

However, in a parallel example from Russian, the topical NP has to be marked by *odin* ‘one’, a quasi-determiner that can introduce novel referents. Otherwise, it is not interpreted as indefinite:

-
- (32) *(ODNOGO) lingvist-a, ego znaet každyj.
 one linguist res.pron knows everybody.
 ‘One linguist everybody knows.’ (Geist, 2010, p. 209)

According to Geist, the reason bare singulars cannot occur as topics is that aboutness topics have to be specific. That is, they must introduce a discourse referent that is identifiable from the speaker’s perspective. However, bare singulars (unlike overt indefinites) are weak and non-specific when existential, as evident from their obligatory narrow scope. They introduce new discourse referents but lack the anchoring properties of specific indefinites. Therefore, they fail to satisfy the requirement that indefinite topic be specific, and therefore, the existential construal is excluded from topical environments.

Importantly, this system is asymmetric in the sense that it only restricts topical bare singulars, which must be interpreted via *iota*. Non-topical bare singulars are compatible with either type shift. Nothing in Geist’s analysis forces them to be existential; rather, existential closure is only available in the absence of a topic, and the notion of topic is understood broadly.

In this way, Geist (2010) provides a middle ground between fully existential analyses (inspired by (Heim, 2011)), which deny any definite-like semantics for bare singulars at all, and fully flexible ambiguity analyses, which leave the choice between *iota* and existential construal unconstrained.

1.2.3 Property-based accounts

Finally, there are approaches that allow full ambiguity to bare nouns. On this view, all nouns begin as predicates and may undergo *iota* or existential closure, depending on the context. There are only two restrictions: (i) type shifting is a last resort, and it occurs as locally as possible; (ii) if a language has an overt determiner responsible for *iota* or existential shift, that shift cannot happen covertly. If a language has no articles, *iota* and existential shift are equally available, and there is no structural preference of one over the other. No intrinsic

asymmetry between bare singulars and bare plurals is predicted either.

This account, often referred in the literature as property-based account, was first articulated in Krifka (2003) for bare plurals. Liu et al. (2023a,b) provide *prima facie* support for this view and extend this account to singulars. In a parallel corpus study, they aligned noun phrases in 6 languages, including Russian, with their uses in English, using the overt marking of (in)definiteness in English as a comparison standard. Their study demonstrated that not only English definites, but 80% of English indefinite singulars were translated into Russian using a bare singular. Based on this, they conclude that Russian bare nouns must shift between the definite and indefinite construal freely.

1.2.4 Predictions for Russian, summarized

Let us summarize what the approaches we have seen so far predict for Russian.

On approaches inspired by Heim (2011), bare nouns in languages with no articles are semantically equivalent to indefinites, and definite readings arise pragmatically. For Russian, this means that the existential reading should be always available, as it is the default option. Bare singulars should freely introduce new discourse referents whenever we want. In conjunction diagnostics, mentioned in 1.1.1, bare singulars are predicted to introduce distinct individuals across conjuncts. Bare singulars are also predicted to pattern like overt indefinites in languages with articles in terms of scope flexibility. This should include wide-scope existential readings under negation or modals, and intermediate scope readings. Finally, since both singulars and plurals are existentials in this system, number morphology alone should not block differentiated scope readings. In short, Russian should behave like a language with covert indefinites across the board.

The account of Geist (2010) maintains the availability of *iota* and existential shift, but restricts the existential construal to non-topical positions. For Russian, this predicts positional asymmetry: since preverbal bare singulars are typically topical, they should strongly favor definite readings; postverbal bare singulars should allow existential readings,

but are not expected to block definite ones. In this way, linear position should partially track (in)definiteness interpretation. Since Geist treats bare singulars with indefinite readings as weak existentials, the unavailability of wide and intermediate scope readings is what her account expects.

Finally, the most radical ambiguity accounts remove structural constraints altogether and claim both *iota* and existential closure to be freely available. For Russian, this essentially predicts free alternation: bare singulars should be equally compatible with definite and indefinite interpretations across positions. Obligatory narrow scope may be accounted for via the requirement for type shifts to apply locally, although requires more explicit machinery. Singular and plural bare nouns should not differ fundamentally in the availability of existential interpretations.

1.3 The neo-Carlsonian approach

Despite their differences, all ambiguity approaches share one central commitment: Russian bare nouns have access to existential type shifting. They vary in how freely this option is available and what restricts it, but none of them predict a systematic ban on existential interpretations for bare nouns. Yet, even the preliminary data presented in section 1.1 suggest that existential readings of bare nouns are limited. The abundance of data like this should motivate us to seek for an explanation in which existential interpretation is not freely available.

The neo-Carlsonian approach, developed in Chierchia (1998) and Dayal (2004), provides such a framework. Building on Carlson (1977b,a, 1989), it adds to the list of available type shifts the kind formation operation, and assumes that bare nouns can denote kinds rather than properties of atomic individuals or existential quantifiers. Under this approach, the three argument-forming operations (*iota*, existential quantification, and kind formation) are not equally available but ranked. Higher-ranked operations are available whenever possible,

but lower-ranked operations are only used when other shifts are unavailable. Existential closure is ranked lower than kind formation and *iota*.

When applied to languages without articles, this ranking makes clear predictions. In the absence of the overt determiner for existential quantification, an articleless language simply never has access to existential quantification. This predicts that bare nouns have access to kind or definite interpretations, but never to a genuine existential interpretation.

In this section, I introduce the formal architecture of the neo-Carlsonian approach, including the argument forming operations and restricting principles. I then discuss how kind-denoting expressions combine with object-level predicates via Derived Kind Predication. Finally, following Dayal (2004), I show how this approach can be applied to Russian and how it accounts for the behavior of bare plurals, bare singulars, and the asymmetries between the two mentioned earlier.

1.3.1 The architecture

Chierchia (1998) proposes the Nominal Mapping Parameter, which divides languages based on whether nominals can appear as predicates, as arguments, or as either. He further assumes that in languages where nominals may serve as predicates and as arguments⁸, nouns denote properties of type $\langle e, t \rangle$ and must undergo type shifting in order to function as arguments. In addition to the *iota* (24-a) and existential closure (11-b), there is a kind formation operation, nominalization (*nom*) (33). This operation maps a property to a kind: it takes an index and returns the maximal entity that is in the extension of the predicate at that index. In case there is no corresponding kind for the property P , $\cap P$ is undefined.

(33) Nominalization (Kind Formation)

For any property P and world/situation s ,

$$\cap P = \lambda s. \iota P_s, \text{ if } \lambda s. \iota P_s \text{ is in } K, \text{ undefined otherwise,}$$

⁸Meaning languages that classify as [+pred, +arg] with respect to the Nominal Mapping Parameter; this includes both Slavic and Germanic languages.

where P_s is the extension of P in s and K is the set of kinds (Chierchia, 1998)

Chierchia further proposes three constraints into the system of argument formation. The first constraint can be summarized as “don’t do covertly what you can do overtly” (Chierchia, 1998, p. 360). Known as the Blocking Principle, it asserts that if a language has a determiner that overtly performs a specific type-shifting operation, the covert application of that operation should be blocked:

(34) **Blocking Principle** (Chierchia, 1998, p. 360)

For any type shifting operation τ and any X :

$*\tau(X)$,

if there is a determiner D such that for any set X in its domain,

$D(X) = \tau(X)$

The second constraint, Avoid Structure, reflects a broader economy principle in grammar. It stipulates that type-shifting operations should apply at the earliest possible level of derivation. For instance, if a type shift can be performed at the NP level before the projection of D , it should be applied at that point to minimize syntactic complexity:

(35) **Avoid Structure** (Chierchia, 1998, p. 393)

Apply SHIFT at the earliest possible level.

Finally, the neo-Carlsonian approach establishes a hierarchy among type-shifting operations. The original Ranking was proposed in Chierchia (1998) and placed kind formation (*nom*) above *iota* and $\exists$. This means that, as long as the Blocking Principle holds, kind formation is the preferred type shift whenever it is defined. If kind formation cannot apply, *iota* and $\exists$ become equally viable alternatives:

(36) **Ranking**

$$nom > \{iota, \exists\}$$

If a language has no overt determiners, like Russian, the Blocking Principle does not apply. Thus, the version of the Ranking in (36) yields the following prediction: whenever possible, apply *nom*; if *nom* is unavailable, both *iota* and existential quantification should in principle be freely available. However, Chierchia further claims that kind readings are available for bare plurals in Russian, but not for bare singulars:

(37) Adapted from Chierchia (1998, p. 361)

- a. **Dinozavr-y** vymer-l-i.
dinosaur-PL die.out-PST-PL
'Dinosaurs are extinct.'
- b. ***Dinozavr** vymer.
dinosaur die.out.[PST.SG]
Expected: 'The dinosaur is extinct.'

As a result, bare plurals in languages like Russian are expected to allow generic, definite, and indefinite interpretations depending on context. Russian bare singulars do not allow kind readings (according to Chierchia's data), and have to undergo argument formation via *iota* and existential quantification.

As we saw in 1.1, this prediction is too strong. Bare nominals systematically lack wide-scope existential interpretations, and exhibit uniqueness requirement similar to definites. Based on the data from Hindi, Russian, and Mandarin Chinese, Dayal (2004) suggests that existential quantification is not an equal alternative to *iota* once kind formation is excluded, and proposes a revised ranking, which makes the existential shift a true last resort:

(38) Revised Meaning Preservation (Ranking) (Dayal, 2004)
 $\{nom, iota\} > \exists$

1.3.2 Kind readings in episodic sentences and Derived Kind Predication

In languages with articles, *iota* and existential closure are linked to specific morphology. Kind formation typically does not have special morphological marking. When there is no specific morphology, additional mechanisms may be needed. Let us consider the English bare plural. If the predicate is kind-level, *nom* turns the bare plural into an argument that can be fed into the verb meaning directly:

- (39) a. **Dinosaurs** are extinct.
b. $\text{be-extinct}(\cap \text{dinosaurs})$

However, English bare plurals also occur in characterizing sentences with non-kind-level predicates (40), or even in episodic sentences (41):

- (40) **Candies** are extra sweet in the US.
(41) **Dogs** are barking.

In these cases, a simple application of *nom* would not be enough, as the predicate is not from the set of kind-level predicates. Chierchia (1998) further adjusts the system by adding two new operations. One of them, predicativization, is meant to take the extension of the kind at a certain index:

- (42) **Predicativization**

$$\cup k = \lambda x. x \leq k_s \text{ if } k_s \text{ is defined,}$$

$$\lambda x[\text{FALSE}] \text{ otherwise,}$$

where k_s is the plural individual that comprises all of the atomic members of the kind

The second operation is called Derived Kind Predication (DKP). It uses predicativization in

(42) as an intermediate step, converts its result into a predicate of objects, and existentially bounds it:

(43) **Derived Kind Predication**

If P applies to objects and k denotes a kind,
 then $P(k) = \exists x. [\cup k(x) \wedge P(x)]$

Now the system is fully equipped to deal with examples like (40) and (41). Let us go through the derivation of (40), repeated here as (44):

(44) **Candies** are extra sweet in the US.

Sentences like that can have different truth conditions based on what goes into the restrictor of the GEN operator⁹. If predicativization (42) shifts the type of the bare plural from kind to predicate and generically binds it, we get the following denotation in (45). Such a reading says of candies in general that, given that the contextual condition $C(s)$, they are extra sweet in the US:

$$(45) \quad \text{GEN } s \ x \ [\cup \text{candies}(s)(x) \wedge C(s)] [\text{be-extra-sweet}(s)(x) \wedge \text{in-the-US}(s)(x)]$$

If the GEN operator binds the locative adjunct first, a slightly different reading is yielded, namely, that of all situations involving the US, if the contextual condition $C(s)$ holds, it is generally true that there are candies that are extra sweet (46-a). Having the bare plural in the nuclear scope of the predicate also requires the application of DKP. Since *be extra sweet* is not in the set of kinds K , we need DKP in order to be able to operate on instantiations of kinds (46-b).

⁹The original examples similar to (44) come from Milsark (1974); the further discussion developed in Carlson (1989); Krifka et al. (1995); Dayal (2013), among others. The derivations in (45) and (46) follow the model outlined in Dayal (2013); following Dayal, the situation variable s is assumed to be indexical in episodic sentences, rather than existentially bound.

-
- (46) a. $\text{GEN } s \ x \ [\text{the-US}(s)(x) \wedge C(s)] \ [\text{be-extra-sweet-in-}x(s)(\cap \text{candies})]$
GEN-binding of the locative adjunct, DKP is needed
- b. $\text{GEN } s \ x \ [\text{the-US}(s)(x) \wedge C(s)] \ \exists y \ [\cup \cap \text{candies}(s)(y) \wedge \text{be-extra-sweet-in-}x(s)(y)]$
the result of DKP

In a similar manner, DKP assists with the derivation of bare plurals in episodic contexts, such as (41), repeated here as (47-a):

- (47) a. **Dogs** are barking.
an episodic sentence with a bare plural and an object-level predicate
- b. $\exists [\text{are-barking}(s)](\cap \text{dogs})$
 $\cap \text{dogs}$ can not be directly fed into an object-level predicate scope, so DKP is needed
- c. $\exists x. [\cup \cap \text{dogs}(s)(x) \wedge \text{are-barking}(s)(x)]$
the result of DKP

As a result, the sentence in (47-a) asserts the existence of instances of the dog-kind that satisfy the predicate. Existential force occurs locally, as part of the repair operation triggered by the type mismatch. This has two important consequences: first, existential quantification is not contributed by the nominal itself; second, it necessarily takes narrow scope with respect to higher operators like negation, modals, or quantification environments. This is what we see borne out: the sentence in (48) cannot mean that some dogs are barking while some are not.

- (48) Dogs are not barking.

While DKP has originally been proposed to account for the behavior of English bare plurals, it becomes particularly important for languages without articles. As we will see below, Dayal (2004) relies on DKP in arguing that existential interpretations of bare nouns in

such languages happen through DKP rather than existential type shifting. This allows the neo-Carlsonian approach to capture the obligatory narrow scope facts.

1.3.3 Bare plurals

Let us go back to bare plurals, and look at their distribution in Russian, a language without articles. We have already seen that they have a complex profile. They allow for differentiated scope readings in the presence of aspectual operators, and they systematically lack wide-scope existential interpretations. This would be difficult to explain if bare plurals had free access to existential quantification. However, these properties follow naturally if bare plurals denote kinds.

First, Russian bare plurals are compatible with predicates that select for kinds, similar to English bare plurals (Chierchia, 1998). Under the neo-Carlsonian approach, the plural undergoes *nom* and can directly saturate the argument position of a kind-selecting verb:

- (49) **Dinozavr-y** vymer-l-i.
 dinosaur-PL die.out-PST-PL
 ‘Dinosaurs are extinct.’ Chierchia (1998, p. 361)

And similar to English bare plurals, Russian bare plurals occur in episodic contexts, too:

- (50) Zakroj, požalujsta, okno! **Sobak-i** laj-ut.
 close.IMP please window dog-PL bark-PRS.3PL
 ‘Please close the window! Dogs are barking.’

On the neo-Carlsonian analysis, these readings are derived through DKP, exactly like in English. The bare plural shifts to a kind via *nom*, and in case it needs to combine with an object-level predicate, DKP intervenes. This mechanism introduced existential quantification over instances of the kind, and the resulting interpretation is the following:

- (51) $[[\textit{Sobaki lajut}]] = \exists x. [\text{dog}(x) \wedge \text{barking}(x)]$

Because the existential quantifier is introduced inside the predicate, the obligatory narrow scope is predicted. It follows directly from the assumption that existential quantification is part of the DKP mechanism rather than an independent type shift.

- (52) **Sobak-i** ne laj-ut.
 dog-PL NEG bark-PST.3PL
 ‘Dogs are not barking.’
 Available: $\neg\exists x.[\text{dog}(x) \wedge \text{barking}(x)]$
 Not available: $\exists x.[\text{dog}(x) \wedge \neg\text{barking}(x)]$

At the same time, bare plurals sometimes appear to allow definite-like interpretations:

- (53) Zakroj, požalujsta, okno! **Sobak-i** laj-ut.
 close.IMP please window dog-PL all time bark-PRS.3PL
 Context: My three dogs, Spotty, Rover, and Daisy, bark all the time when anyone
 passes by the house.
 ‘Please close the window! The dogs are barking.’
- (54) Ja podobra-l-a na ulic-e dv-ux sobak. Teper’ sobaki
 I pick.up-PST-F in street-PREP two-ACC.PL dog.ACC.PL now dog-PL
 lajut.
 bark-PRS.3SG
 ‘I took two dogs from the street. Now the dogs are barking.’

Within the neo-Carlsonian framework, an additional definite shift is actually not required to account for such cases. Instead, they can be analyzed as contextually restricted kind readings¹⁰. The relevant instantiations of the kind introduced by the bare plurals are limited to a salient subset. This yields an interpretation that resembles a definite plural.

In what follows, I will cover how Dayal (2004) accounts for the asymmetry between bare plurals and bare singulars, mentioned before.

¹⁰See Dayal (2011); Trinh (2011) for the discussion of possible ways for kind-based readings to yield definite-like interpretations.

1.3.4 Bare singulars

We now turn to bare singulars. As shown in 1.1, singular and plural bare nominals differ in their interpretive possibilities. While bare plurals can show differentiated scope readings under a scopal adverbial (56), bare singulars cannot (55):

- (55) **#Sobaka** by-l-a vezde.
dog be-PST-F everywhere
‘The dog/A particular dog (the same one) was everywhere.’
- (56) **Sobak-i** by-l-i vezde
dog-PL be-PST-F everywhere
‘There were dogs (different ones) everywhere.’

The contrast holds with object NPs as well:

- (57) Lena neodnokratno ubi-va-l-a **#krolik-a** / **krolik-ov**.
Lena repeatedly kill-IPFV-PST-F rabbit-ACC.SG rabbit-ACC.PL
Unavailable for singular (*krolik-a*): ‘Lena repeatedly killed a (possibly different) rabbit.’

Available for plural (*krolik-ov*): ‘Lena repeatedly killed rabbits.’

(Soloveva et al., 2026, p. 241)

Dayal (2004) proposes that the contrast follows from how singular and plural kinds relate to their instantiations. For plurals, DKP is what accounts for differentiated scope. The kind can be instantiated by multiple individuals in a situation, and existential quantification over these instances allows differentiated scope readings. Singulars do not allow such readings. Dayal argues that this is because singular kinds are grammatically atomic. Even though they are conceptually associated with a plurality of instances across situations, they do not permit free semantic access to multiple instantiations within a single situation. In this, singular kind terms resemble group terms (Link, 1983; Landman, 1989), like *team* or *committee*, as analyzed in Barker (1992); Schwarzschild (1996): the group term *team* and the definite plural

the team members refer to the same set of entities (e.g., players a , b , c and their pluralities $a \oplus b$, $a \oplus c$, $b \oplus c$, $a \oplus b \oplus c$), but only the definite plural can be used in distributive contexts.

Because singular kinds are grammatically atomic, the *nom* operation is effectively undefined for them: singular morphology requires an atomic individual, but kind formation returns a maximal plural entity. To resolve this conflict, Dayal (2004) suggests that instead of forming a kind over ordinary individuals, the singular noun denotes a taxonomic property, whose extension in the relevant domain is a singleton set. While Russian bare singulars are indeed restricted when used with some kind-level predicates (see the contrast in (37) which involves the kind-level predicate ‘be extinct’), this generalization does not always hold (Dayal, 2004, p. 401). The acceptability of singular kind readings for Russian bare singulars improves when the context provides cues for taxonomic kind relevance (58) (see a more detailed discussion of this effect specifically for Russian in Mueller-Reichau (2015)).

- (58) Šerstist-yj nosorog vyper.
woolly rhinoceros die.out[PST.3SG]
'The woolly rhinoceros died out.'
(Mueller-Reichau, 2015, p. 279)

In this way, the derivation of the singular kind involves *iota*, which picks out the unique element of the singleton taxonomic set. In Russian, *iota* is supposed to operate covertly.

Let us now come back and account for the infelicity of the bare singular in (55) and (57). Technically, the singular has two options. One is *iota*-shift; in this case, the singular would denote a unique individual, yielding the implausible reading that a single dog was everywhere. The other option would be the kind formation. In principle, DKP would be able to introduce existential quantification over multiple instantiations of a kind. But since the singular kind is grammatically atomic, its instantiation set in a given situation is effectively restricted. The result collapses into the same infelicitous reading with a single individual. This is why the differentiated scope reading is available to bare plurals but not to bare singulars.

In this way, one of Dayal’s contributions to the neo-Carlsonian approach is the explanation of the number asymmetry. Number morphology appears to be of special importance in

languages with no articles. The systematic contrast between singular and plural bare nominals therefore argues against an ambiguity analysis.

In this way, under the neo-Carlsonian approach, neither bare plurals nor bare singulars have access to full-fledged existential quantification. Bare plurals denote kinds or definite descriptions, and have access to DKP, which allows them to have narrow-scope existential-like readings. Bare singulars are ambiguous between kinds and definite descriptions, and are never truly existential either.

1.3.5 Choosing between the two approaches for Russian

We are now in a position to compare the predictions of the neo-Carlsonian approach with those of the ambiguity approaches reviewed in section 1.2. The most important difference concerns the availability of the existential shift. On ambiguity-based accounts, bare nominals have access to existential closure as an argument-forming operation. The variation of interpretation is attributed to higher-level factors, which decide whether an existential or a definite construal is used. But the neo-Carlsonian approach as developed by Chierchia (1998) and extended by Dayal (2004) claims that existential force is not freely available. Bare plurals denote kinds, bare singulars are kinds or definite descriptions, and the only way to achieve an existential-like interpretation is derivationally through DKP.

We see that the neo-Carlsonian approach has several advantages. First, the absence of wide-scope existential readings in plural follows directly from DKP, which introduces existential quantification inside the predicate. No additional stipulations about bare nominals are required. On ambiguity approaches, though, the lack of wide scope must be restricted independently. Second, the absence of intermediate scope demonstrated in 1.1.3 is also expected if bare nominals do not introduce quantificational force. The existential quantification induced by DKP is embedded within the predicate and cannot escape. Again, any of the ambiguity accounts would need additional constraints in order to capture this fact. Third, the constancy of the referent of bare singulars follows if they are definite or kind-based.

In case a bare singular is *iota*-based, its referent is unique; in case it is kind-based, its instantiations are constrained in a way that does not permit existential introduction across different parts of the sentence. Ambiguity accounts must prevent independent existential readings in such contexts through separate assumptions.

Finally, the neo-Carlsonian approach accounts for the number asymmetry in differentiated scope contexts. The availability of differentiated scope for plurals and its absence for singulars follows from a structural difference: singular kinds are grammatically atomic and do not allow access to multiple instantiations within a situation. None of the ambiguity approaches known to me include modeling this number asymmetry, and number morphology itself does not constrain existential scope.

At the same time, there remain still open questions, specifically, the treatment of bare singulars. In Dayal’s account, the absence of wide-scope existential readings in bare singulars is accounted for by banning existential shift altogether. This correctly predicts the absence of differentiated scope readings for singulars and their strong anaphoric potential, but does not explain how apparently narrow-scope existential readings of bare singulars are derived. Dayal tentatively suggests that such readings lack familiarity, and that the uniqueness presupposition associated with *iota* may fail to project under operators such as negation. This tentative hypothesis has been criticized (Bronnikov, 2006; Šimík and Demian, 2020), as it does not fully resolve how quasi-existential readings of bare singulars should be derived or in which way they differ from genuine existentials. In this way, while the neo-Carlsonian approach successfully covers many of the empirical facts about bare nominals in Russian, the exact analysis of bare singulars remains an area of ongoing debate.¹¹

Despite this, I am adopting the neo-Carlsonian framework as more promising for Russian than ambiguity-based accounts. In the next section, I turn to a more general question.

¹¹See the discussion of the limitations of the neo-Carlsonian approach in application to a number of languages with bare arguments in Dayal (2026b).

1.4 What is (in)definiteness?

So far we have discussed competing analyses of bare nouns with respect to argument-forming semantic mechanisms. However, there is a broader question: what semantic properties are at stake when we label a nominal “definite” or “indefinite”? The patterns that we touched upon in this chapter are related to the two main dimensions of (in)definiteness: uniqueness/maximality and familiarity. Uniqueness is a condition on the cardinality of the N-set: a singular definite presupposes that exactly one individual satisfies the predicate in the relevant context; a plural definite picks out the maximal plural individual in the extension of the predicate. Familiarity is often operationalized as anaphoricity, the ability of a nominal to retrieve an already established discourse referent. These two dimensions are sometimes treated as competing analyses of definiteness (see Srinivas (2021) for an overview; a more thorough discussion of uniqueness vs. familiarity analyses follows in Chapter 2). Indefinites, by contrast, are underspecified in the sense that they do not require uniqueness (although are compatible with it); they are typically non-anaphoric. A repeated indefinite has to refer to a different individual (32). In this way, indefiniteness cannot be reduced to just existential force, as existential quantification does not by itself encode novelty, nor does it guarantee non-uniqueness.

(59) A girl_i entered the room. A girl_{j/*i} sat down.

The Russian data reviewed in 1.1 show that bare nouns do not pattern uniformly with either canonical definites or canonical indefinites. They do not have the full range of existential readings and display effects suggestive of a uniqueness condition, yet they can introduce discourse-new referents. We have also seen that bare plurals allow differentiated scope readings unavailable to singulars, and in later chapters we will see that they may also display maximality effects. All of these patterns indicate that we are dealing with a complex phenomenon, and (in)definiteness cannot be reduced to a simple definite/indefinite

opposition.

The empirical program of this dissertation is designed to test the predictions of the neo-Carlsonian approach for Russian bare nominals using controlled experimentation. Russian provides a particularly revealing testing ground for theories of (in)definiteness. As an article-less language with productive bare singular and plural arguments, flexible word order, and rich prosodic marking, it allows the semantic components of definiteness to be examined independently of overt determiner morphology. I separate the following three dimensions of (in)definiteness: (i) uniqueness, (ii) maximality, and (iii) anaphoricity, and examine bare singulars and bare plurals separately. Chapter 2 presents a set of experiments that focus on bare singulars and test whether they exhibit a uniqueness requirement, and if that requirement interacts with familiarity and word order. Chapter 3 shows a set of studies targeting maximality effects in bare plurals in Russian vs. English. Chapter 4 investigates the anaphoric potential of bare plurals, comparing Russian to English. Chapter 5 summarizes the findings in the light of existing theories, identifies the questions that the data leave open, and lays out directions for future work.

In general, the studies presented in this dissertation probe which semantic components associated with (in)definiteness are encoded in the interpretation of singular and plural bare nominals. But more broadly, the results bear on the general theory of argument formation. Russian, as a language without any articles but with systematic restrictions on bare nominal interpretation, provides a great testing ground. If effects associated with (in)definiteness systematically emerge in Russian in the absence of the dedicated (in)definiteness morphology, this supports the view that uniqueness/maximality and anaphoricity are part of a universal semantic toolkit, variably encoded in the languages of the world. On the other hand, if bare nominals alternate between definite and indefinite without structural constraints, this would suggest that (in)definiteness is more tightly linked to article morphology and competition. By decomposing (in)definiteness into measurable parameters and experimentally probing their influence on nominal interpretation, this dissertation aims to shed light on a broader

theoretical question: which aspects of (in)definiteness are hard-coded in the noun itself, which occur from syntactic or information-structural configurations, and which are pragmatically inferred in the absence of specialized morphology.

2 Uniqueness and its mirrors

Uniqueness is the core component of the denotation of the *iota* operation, which is assumed to be responsible for the meaning of definite markers in languages that have them; therefore, uniqueness is often taken to be essential in licensing definiteness. As a result, the debate around potential definite-like semantics of bare singulars often centers around whether they display uniqueness. Different approaches, introduced in Chapter 1, will have different predictions in this regard. Since the Meaning Preservation prioritizes *iota* and kind formation over the existential type-shift (Dayal, 2004), the Neo-Carlsonian approach maintains that bare singulars are introduced via an *iota* or *iota*-like operation. Under this view, bare singulars are predicted to display uniqueness relative to the relevant domain. The approaches that either allow bare singulars to be occasionally existential, or claim that bare singulars are uniformly existential, do not predict that the referent of a bare singular must be unique in the relevant domain.

This chapter investigates whether Russian bare singulars impose a uniqueness requirement, and if so, whether it happens universally or under certain syntactic conditions. Building on the observation that uniqueness is interpreted relative to a contextual domain that can potentially be restricted, I argue that apparent violations of uniqueness do not constitute enough evidence in favor of existential semantics of bare singulars. Instead, they reflect cases where the domain of interpretation is narrowed, and the uniqueness requirement is satisfied trivially. To test this claim, I present a series of experimental studies that progressively control for factors known to interact with noun phrase interpretation in Russian, such as

familiarity and word order. The main empirical result is that Russian bare singulars exhibit systematic sensitivity to uniqueness.

This chapter proceeds as follows. Section 2.1 clarifies what it means to diagnose uniqueness in the absence of overt definiteness marking and discusses what factors must be controlled for. It also distinguishes uniqueness from familiarity as a separate factor requiring independent control, and sets out the predictions that different theories of bare nouns have with respect to uniqueness. Section 2.2 introduces word order as a methodological tool for controlling familiarity, supported by an independent baseline study (Experiment 1) that establishes a reliable correlation between the linear position of the subject NP and the familiarity of its referent under controlled prosody. Sections 2.3 and 2.4 report the main experimental results: Experiment 2A shows that preverbal familiar bare singulars are strongly sensitive to uniqueness, Experiment 2B shows that postverbal non-familiar bare singulars appear insensitive to uniqueness under presentational contexts that allow domain restriction, and Experiment 3 demonstrates that this apparent insensitivity disappears once domain restriction is blocked by incompatible predicates, revealing a uniqueness requirement across syntactic configurations. The chapter concludes with an interim summary that situates these findings within broader debates about definiteness and bare nominal interpretation, and identifies remaining open questions.

2.1 What is uniqueness?

2.1.1 Formalisms: *iota* and cardinality

A natural starting point for talking about uniqueness in formal semantics is the *iota* operator. I assume the classical definition of the *iota*, which takes a predicate and returns the unique individual satisfying that predicate when such an individual exists. Under this definition, $\iota x[P(x)]$ is defined only if there is exactly one x in the relevant domain D such that $P(x)$ holds:

$$(1) \quad \iota x[P(x)] \text{ is defined in domain } D \text{ iff } \exists x \in D[P(x) \wedge \forall y \in D, [P(y) \rightarrow y = x]]$$

How exactly this uniqueness requirement enters the meaning of definite descriptions is a long-standing question in formal semantics. One way of doing so would be through encoding it as a part of the at-issue content. Such a denotation would include both existence and uniqueness. In the denotation below, the semantics of *the* contains an existential quantifier and requires uniqueness of the referent to hold with respect to a contextually restricted domain. This suggests that the composition of Russellian sentences like *The king of France is bald* (Russell, 1905) returns falsity, as the existence condition is not met in the actual world. This understanding of the semantics of definiteness is typically called Russellian, as it follows the logic proposed by Russell (1905).

$$(2) \quad [[the]] = \lambda P. \exists x. P(x) \wedge \forall y [P(y) \rightarrow y = x]$$

Alternatively, the uniqueness component can be treated as a presupposition, following Frege (1892); Strawson (1950). Under this view, the definite description presupposes existence and uniqueness of the referent. Such an understanding of the definite article would leave Russellian sentences like *The king of France is bald* undefined rather than false: the existence of the king of France is considered a necessary precondition to evaluating the truth value of the sentence, not a precondition to it being true. Since it is not met in the current world, the truth value of the sentence cannot be evaluated. This understanding of the definiteness semantics is usually referred to as Fregean.

$$(3) \quad [[the]] = \lambda P : \underbrace{\exists! x. P(x)}_{\text{presupposition}} \underbrace{\iota x. P(x)}_{\text{assertion}}$$

But there is a further division even within presuppositional approaches. One can distinguish between Fregean presuppositional analyses, on which definites presuppose both existence and uniqueness (similar to (3); relevant approaches are presented in Elbourne (2013, 2016, 2018, 2021)), from quantificational presuppositional analyses, which claim that definite

descriptions presuppose uniqueness but assert existential force (various versions of this are presented in Karttunen and Peters (1979); Abbott (2008); Hawthorne and Manley (2012)).

Although I will be treating uniqueness as a definedness condition similar to (1), I will not attempt to definitively decide among these implementational options yet. What matters for our purposes is that descriptions introduced with *iota* impose a cardinality constraint relative to a certain domain¹. If bare singulars are introduced via *iota*, we must be able to catch the evidence of this requirement in their behavior.

2.1.2 Uniqueness vs. familiarity

Uniqueness is not the only parameter associated with definiteness; the other one is *familiarity*. Familiarity is often defined in terms of some sort of accessibility through either previous discourse or shared knowledge between the speaker and hearer. The exact distribution of work between uniqueness and familiarity in licensing definite descriptions varies depending on the theory.

The debate around the contribution of uniqueness vs. familiarity often starts around definite NPs that can be licensed when only one component is satisfied. This goes both ways: familiarity can override uniqueness, and uniqueness can override discourse givenness. The first set of data comes from Heim (1982), who provides examples where an English definite NP is felicitous even though its referent is not necessarily unique. On Heim's explanation, example (4) is compatible with a situation where two glasses broke, which makes the following definite description *the glass* non-unique. This hints that uniqueness and familiarity cannot be treated as the same phenomenon.

- (4) A glass broke last night. **The glass** had been very expensive. (Heim, 1982)

By contrast, indefinite NPs oppose familiarity: if we replaced the definite NP in (4) with an indefinite, the anaphoric link between the two instances of *glass* disappears:

¹In the case of singular common nouns, I take the domain of individuals to be restricted to atoms.

-
- (5) A glass broke last night. #**A glass** had been very expensive.

Based on this, Heim proposes that definite NPs are only licensed if their referent has been introduced in the previous discourse or can be accommodated. For Heim, an NP is familiar if it has a co-indexed discourse referent in the common ground. Definite NPs presuppose familiarity, while indefinite NPs presuppose novelty (the absence of a co-indexed referent):

- (6) The Extended Novelty-Familiarity Condition Heim (1982)

For a Logical Form ϕ to be felicitous w.r.t. a context C it is required for every NP $_i$ in ϕ that:

- (i) if NP $_i$ is [-definite], then $i \notin Dom(C)$,
- (ii) if NP $_i$ is [+definite], then
 - (a) $i \in Dom(C)$, and
 - (b) if NP $_i$ is a formula, C entails NP $_i$.

In contrast, there are cases where uniqueness seems to license a definite in the absence of familiarity, reported in Coppock and Beaver (2015). In (7), the existence of the queen of the world is not exactly implied (in fact, the queen of the world is implied not to exist). Yet, the sentence is felicitous. Based on this evidence, the authors argue that such definites cannot be analyzed through just a familiarity/novelty condition similar to the one in (6).

- (7) You're not **the queen of the world**. (Coppock and Beaver, 2015)

These contrasts motivate different theoretical responses to the uniqueness-familiarity debate. In this respect, semantic accounts of definite descriptions can be divided into three groups: uniqueness-based theories, familiarity-based theories, and mixed theories. Importantly, the disagreement between the theories is mostly about which component is primary, not whether the other exists. So, Heim's account can be considered familiarity-based, as the accessibility of an antecedent is the primary licensing condition for definites. Consequent

familiarity-based accounts, such as Roberts (2003), extend the licensing of definites to weak familiarity, under which a referent need only to be inferable in some way. This significantly broadens the predicted distribution of definites, but also, as argued in Coppock (2022), blurs the empirical distinction between uniqueness- and familiarity-based accounts: in most cases where weak familiarity is satisfied, situational uniqueness holds, too.

Uniqueness-based accounts place the cardinality restriction as the main requirement for licensing a definite. However, such accounts put additional weight on the importance of the contextually restricted domain, as they analyzed apparent violations of uniqueness, such as (4), through domain restriction. One influential example of uniqueness-based approaches is developed within situation semantics in Elbourne (2005) and subsequent work, where definite descriptions are taken to denote individuals of the type $\langle s, e \rangle$ and require uniqueness in order to be defined, but their uniqueness is evaluated relative to a contextually supplied situation s .

There are also accounts that explicitly encode both uniqueness and familiarity. Some of them include both uniqueness and familiarity in a single lexical entry with some mechanism of prioritization of one over another (Farkas, 2002), while others claim that any language with a definite article has two separate lexical entries for the definite marker, one responsible for uniqueness uses, and another one licensed by familiarity (Schwarz, 2009); in case of English, these lexical entries would be homonymous.

An important contribution to the uniqueness vs. familiarity debate comes from experimental studies. Through a series of comprehension and production studies that independently manipulate uniqueness and prior discourse mention, Srinivas (2021) shows that English definites can be successfully interpreted when only one of these factors is present. However, uniqueness was shown to be a more effective cue to definiteness than familiarity: non-familiar but unique referents are more reliably associated with definiteness marking than familiar but non-unique referents. Srinivas (2021) also argues against an ambiguity-based approach, along the lines of Schwarz (2009), as it does not model the observed asymmetry favoring

uniqueness over familiarity. Instead, she proposes a probabilistic account in which multiple candidate referential domains are considered in parallel and weighted by contextual factors such as uniqueness and familiarity, with successful reference corresponding to probabilistic convergence on a single referent across these domains. On this view, uniqueness promotes stronger cross-domain convergence than familiarity, explaining its greater effectiveness, while familiarity acts as a weaker bias rather than an independently sufficient condition for definiteness. Based on evidence from Kannada, Srinivas (2021) extends her approach to the definite readings of bare nominals.

The complex nature of the interaction between uniqueness and familiarity reported in the previous literature is what motivated the experimental paradigm in the present study. While uniqueness is our main object of inquiry, familiarity is methodologically important, and will therefore be treated as a control variable. Accordingly, the experiments in this chapter are designed to isolate uniqueness effects by controlling familiarity rather than explaining it. Our operational definitions of uniqueness and familiarity will be following those of Srinivas (2021):

- A referent is **unique** if there is only one discourse-relevant individual in the relevant domain that satisfies the predicate of the NP, regardless of whether it had been introduced in the prior discourse or not;
- A referent is **familiar** if it is part of the communicative context shared by both speaker and hearer, regardless of whether it is unique in a strong sense.

Armed by these preliminary definitions, let us proceed to the discussion of what predictions prior literature makes concerning uniqueness manifestation in bare singulars.

2.1.3 Uniqueness in bare nouns: what do we know?

Most of the literature about the role and nature of uniqueness in *iota*-introduced descriptions focuses on languages with overt definite articles. In the context of languages without articles, uniqueness turns into a diagnostic. If bare singulars are introduced via *iota*, they should

exhibit uniqueness effects comparable to overt definites in languages like English. If they are existential, they should not inherently impose uniqueness. Any observed effect of uniqueness in this case would need an alternative explanation.

The literature on bare nouns offers conflicting answers to whether bare nouns are expected to display a uniqueness effect or not. On the neo-Carlsonian approach (Chierchia, 1998; Dayal, 2004, and subsequent work) bare singulars are realized either through *iota* or through the taxonomic kind (which is also realized through the combination of *iota* with an atomic property of kinds; see section 1.3.4). This entails that a bare singular argument must satisfy the uniqueness requirement associated with *iota*, and this uniqueness requirement should be the same as that of definite descriptions in languages like English. This view is supported by several empirical diagnostics associated with uniqueness (see Chapter 1 for details), namely, the incompatible predicate test and the obligatorily unique interpretation of bare singulars in intermediate scope contexts (Dayal, 2026c). Dayal (2004) provides the data that shows that Hindi bare singulars do exhibit uniqueness according to these diagnostics. Similar data has been reported for other article-less languages, including Russian. In (8), the bare noun *sobaka* ‘dog’ is interpreted as unique in a context with incompatible predicates, rendering the sentence infelicitous. In (9), the bare noun having the intermediate scope is also interpreted as unique.

- (8) #**Sobaka** spa-l-a v uglu, v to vremja kak **sobaka** bega-l-a po
 dog sleep-PST-F in that time how dog run-PST-F over circle-DAT
 krug-u.

#‘The dog was sleeping in the corner while the dog was running around.’ (Soloveva et al., 2026, p. 229)

- (9) Každ-yj doktor po-prosi-l dv-e meditsinsk-ie kart-y u **medsestr-y**,
 every-M doctor PFV-ask-PST two-F medical-PL card-pl from nurse-GEN
 kotor-aja emu pomaga-l-a.
 which-F him help-PST-F
 ‘Every doctor requested two medical reports from the nurse/#a nurse assisting him’.

Under the neo-Carlsonian view, Russian bare singulars are predicted to exhibit a uniqueness effect: they should be felicitous in contexts where there is only one individual satisfying the predicate of the noun, and infelicitous otherwise. As uniqueness is treated as the part of the inherent semantics of the bare noun, it is predicted to manifest across syntactic configurations.

A competing line of analysis claims that Russian bare singulars are always existential, and therefore are not expected to exhibit uniqueness requirements. One of the works defending the existential analysis is Seres and Borik (2021). Their main argument comes from the evidence that the uniqueness presupposition of overt definites cannot be cancelled (10); Russian bare singulars, however, do not exhibit a parallel pattern (11).

- (10) **The director** of our school appeared in a public show. #**The other** / #**Another director** (of our school)...
- (Seres and Borik, 2021, p. 349)

- (11) a. **Direktor** našej školy pojavilsja v tok-šou.
 director our school.GEN appeared in talkshow
 ‘The director of our school appeared in a talkshow.’
 b. **Drugoj direktor** (našej školy) vystupil na radio.
 other director our school.GEN spoke on radio
 ‘The other director (of our school) spoke on the radio.’

From this, Seres and Borik (2021) conclude that for any bare noun in Russian, an indefinite interpretation is the only one derived semantically; non-anaphoric bare nouns are realized through classic existential quantification, and anaphoric bare nouns are analyzed as choice functions a-la Reinhart (1997). While the data in (11) may not be the most convincing argument that bare singulars are necessarily indefinite (specifically, one cannot omit the word *drugoj* ‘another’ in (11-b)), a conclusion like that entails that Russian bare singulars do not have uniqueness as an uncancellable requirement in their semantic profile.

(12) Denotations suggested in Geist (2010)

- $$[[\textit{devočka} \text{ ‘girl’}]] = \lambda x.\text{girl}(x)$$

- $$[[\textit{devočka} \text{ 'girl' }]] = \iota x. [\textit{girl}(x)]$$

- $$[[\textit{devočka} \text{ 'girl'}]] = \lambda Q. \exists x [\textit{girl}(x) \wedge Q(x)]$$

(13) Džon xoč-et žen-it'-sja na francuženk-e.
John want-PRS.3SG marry-INF-REFL on French.woman-PREP
'John wants to marry a French woman.'
(Geist, 2010, p. 211)

- 43

-
- (14) Maša čita-l-a každ-uju knig-u, ktor-uju rekomendova-l professor.
 Masha read-PST-F every-ACC.F book-ACC which-ACC.F recommend-PST professor

‘Masha read every book that was recommended by some professor or the other.’

(Geist, 2010, p. 212)

- a. #There exists a professor such that Masha read every book they recommended.

NP > $\forall$

- b. For each book Masha read, there was a professor that recommended it to her.

$\forall$ > NP

Geist takes this narrow-scope behavior to be diagnostic of non-specificity. As a contrast, she shows that Russian bare singulars modified by the numeral *odin* ‘one’ are wide-scope indefinites, as they show the opposite behavior in the same contexts (15)-(16); Geist analyzes them as existential quantifiers introducing referential anchoring to the speaker.

- (15) Džon xoč-et žen-it’-sja na odn-oj francuženk-e.
 John want-PRS.3SG marry-INF-REFL on one-PREP.F French.woman-PREP
 ‘John wants to marry a French woman.’ (Geist, 2010, p. 211)

- a. #But he doesn’t know any yet. want > $\exists$

- b. He got to know her last summer. $\exists$ > want

- (16) Maša čita-l-a každ-uju knig-u, ktor-uju rekomendova-l odin
 Masha read-PST-F every-ACC.F book-ACC which-ACC.F recommend-PST one
 professor.
 professor
 ‘Masha read every book that was recommended by some professor or the other.’
 (Geist, 2010, p. 212)

- a. There exists a professor such that Masha read every book they recommended.

NP > $\forall$

- b. #For each book Masha read, there was a professor that recommended it to her.

$\forall$ > NP

Based on these observations, Geist (2010) proposes that bare singulars with indefinite readings are derived via local existential type-shifting and are interpreted in situ. The requirement for the existential closure to apply locally is not encoded in the denotation itself (12-c). It is a part of the larger architecture Geist assumes, where type-shifting is a repair strategy for the type mismatch. As a repair strategy, it applies at the lowest possible level, and once applied, no further shift is triggered. Because of this, and because this architecture has no alternative mechanism that would allow bare singulars to outscope higher operators (such as quantifier raising), the existential remains narrow-scope in interpretation.

The central restriction in Geist’s analysis is related to information structure: bare singulars may shift through existential closure only if they belong to the comment (as opposed to the topic). This explains why the preverbal bare singular subject in (17-a) refuses an existential interpretation: preverbal noun phrases tend to be topical²

(17) (Geist, 2010, p.193)

- a. **Devočka** voš-l-a v dom.
 girl enter-PST-F in house
 ‘The girl/*A girl entered the house.’
- b. V dom voš-l-a **devočka**.
 in house enter-PST-F girl
 ‘A girl/The girl entered the house.’

What is important for our current purposes is that Geist’s account treats bare singulars with an “indefinite” interpretation as (i) purely existential, (ii) non-topical. This allows us to derive the following uniqueness-related prediction from her account: preverbal bare singulars are expected to be degraded if uniqueness is not satisfied. This follows from (18) and (19).

²Geist discusses two exceptions when preverbal bare nouns can receive an indefinite interpretation. First, a preverbal subject of athetic statement can be indefinite if stressed. A second case involves sentences where the preverbal noun is not part of the topic, as there is another topical element preceding it, such as a locative phrase.

-
- (18) According to Geist (2010):
- a. A bare singular can only have a non-anaphoric interpretation if non-topical.
 - b. An unstressed preverbal bare singular is typically topical.
∴ An unstressed preverbal bare singular cannot have a non-anaphoric interpretation.
- (19)
- a. An anaphoric bare singular must be introduced via *iota*; since *iota* presupposes uniqueness, any such bare singular must satisfy the uniqueness requirement.
 - b. An unstressed preverbal bare singular must be anaphoric.
∴ An unstressed preverbal familiar bare singular must display the uniqueness requirement.

Importantly, the condition Geist is proposing is one-directional. From her account, it only follows that if a bare noun is interpreted as non-anaphoric, it must be in the comment. It does not follow that if a bare singular is in the comment, it must be existential. Thus, the bare singular non-topical noun can occur either as an *iota* or an existential expression; Geist proposes no syntactic mechanism that would adjudicate between the two. For our investigation of uniqueness, it means that Geist does not predict postverbal bare singulars to require uniqueness, but she does not predict them to be non-unique either: either interpretation is possible, even when the referent of the noun phrase is non-familiar and therefore non-topical.

- (20) According to Geist (2010):
- a. Postverbal bare singulars are typically comment information.
 - b. Bare singulars in the comment information can be either *iota* expressions or existential quantifiers.
∴ Postverbal non-familiar bare singulars may or may not have a uniqueness interpretation, depending on the way they shift.

Second, while Geist formulates her restriction in terms of information structure, it has clear

implications for familiarity. Topic NPs are typically familiar, but the comment position itself does not require non-familiarity. Comment NPs may be either familiar or non-familiar. This means that the comment status or the syntactic position alone cannot serve as a diagnostic for an existential construal. Therefore, familiarity must be independently controlled.

Therefore, Geist’s predictions can be summarized as follows: topical bare singulars should display uniqueness; preverbal nouns are most often topical. Non-topical bare singulars are compatible with uniqueness, but not required to display it.

Finally, there are approaches that are even less restricted. Specifically, Liu et al. (2023a) argue in favor of an approach stemming from Krifka (2003)³, which they call the property-based approach, and against the neo-Carlsonian framework. The property-based approach takes nouns to denote properties that then shift through one of the available type-shifts. However, unlike in the neo-Carlsonian approach, under the property-based approach, type-shifts are unranked relative to each other:

- (21) a. Neo-Carlsonian approach (Dayal, 2004): $\{\iota, \cap\} > \exists$
 b. Property-based approach (Krifka, 2003; Liu et al., 2023a): $\{\iota, \cap, \exists\}$

The property-based approach still adopts the Blocking Principle (Chierchia, 1998), which claims that a covert type-shift cannot happen if the language has an overt determiner for that type-shift, and the local nature of existential closure if it occurs as a repair strategy. As a result, Russian bare singulars are predicted to be interpretable through either *iota* or existential closure in the absence of overt determiners.

Based on parallel corpus evidence, Liu et al. (2023a) argue that the distribution of Russian bare singulars aligns more closely with the predictions of the property-based approach, as they appear widely in contexts corresponding to English indefinites. On their interpretation, Russian bare singulars behave as flexible property-denoting expressions that may shift either

³It is important to mention that the original Krifka (2003) was proposed in the context of analyzing bare plurals; Liu et al. (2023a) extended it to bare singulars.

through *iota* or through existential closure.

However, the flexibility predicted by the property-based approach raises a further question. If both *iota* and existential closure are freely available due to the absence of overt articles, the theory must specify what determines the choice of the shift. Liu et al. (2023a) appeal to contextual factors, but do not provide an explicit mechanism deciding the shift selection. In particular, their study does not control for word order or information structure, factors known to affect nominal interpretation in Slavic (for Russian, see Slioussar (2007); Bailyn (2012); Titov (2017); see Šimík and Burianová (2020); Šimík and Wierzba (2017), among others, for other Slavic). When these factors are not controlled for, the flexibility of the interpretation of Russian bare singulars in the corpus may reflect variation in discourse configurations rather than the absence of semantic constraints.

What is important for our purposes is that neither the original approach of Krifka (2003) nor Liu et al. (2023a)’s interpretation of the Russian data predicts that bare singulars should display a uniqueness requirement systematically. Since bare singulars are taken to be highly flexible between existential and definite construals, uniqueness effects should be unstable across different configurations.

Let us summarize the predictions made for bare singulars by the theories overviewed in this section.

1. **Neo-Carlsonian approach** (Chierchia, 1998; Dayal, 2004, and following)

- Bare singulars shift via *iota* or denote kinds
- Narrow scope is acknowledged, but not explicitly accounted for
- Word order affects information structure, but uniqueness is tied to the inherent semantics of the bare singular
- **Preverbal and postverbal bare singulars are predicted to display uniqueness uniformly**

2. **Fully existential account** (Seres and Borik, 2021)

- Bare nouns are existential; anaphoric uses are in fact specific indefinites, arising

-
- via choice functions
 - Scope may vary via choice functions
 - Word order is not considered a crucial factor in nominal semantics
 - **No systematic uniqueness expected independent of the position of the bare singular**
3. **Ambiguity account constrained by information structure** (Geist, 2010)
- Bare singulars shift via *iota* or $\exists$, but $\exists$ is restricted to comment
 - Narrow scope is modeled through a local $\exists$ -shift
 - Existential construal is largely dispreferred in preverbal structures, as they tend to be topical
 - **Preverbal bare singulars are predicted to display uniqueness uniformly**
 - **Postverbal bare singulars are predicted not to display uniqueness systematically**
4. **Fully flexible property-based approach** (Krifka, 2003; Liu et al., 2023a)
- Bare singulars shift via *iota* or $\exists$, which are unranked
 - No built-in bias for either scope, and no formal encoding of word order in shift selection
 - **No systematic uniqueness expected independent of the position of the bare singular**

As we see, the approaches differ in their predictions regarding uniqueness. Despite these clear theoretical contrasts, empirical evidence bearing directly on uniqueness in Russian bare singulars remains limited. Most prior work has relied on theoretical argumentation or corpus observations, rather than controlled experimentation. The only relevant large-scale experimental investigation of specifically uniqueness in Russian to date known to me is Šimík and Demian (2021). In the next section, I review their findings and discuss how they might not have settled the debate about the (in)definiteness of bare nouns.

2.1.4 Previously found null effect: Šimík and Demian (2021)

The study by Šimík and Demian (2021) investigates whether bare nominals in Russian pattern with definite nominals in German in conveying uniqueness (and maximality). They test this using the covered box paradigm Huang et al. (2013). Participants were presented with pairs of pictures, one visible and one covered, and asked whether the visible picture describes a given sentence better than the covered alternative (that never gets revealed). Uniqueness was tested for Russian bare singulars, German indefinite singulars, and German definite singulars, and manipulated visually through pictures. Their results replicate the expected pattern for German definite singulars: participants reliably chose the covered option when the sentence contained a definite NP but the referent in the picture was explicitly not unique. However, in the task with Russian bare singulars, the presence of multiple potential referents did not lead participants to choose the covered alternative more often compared to the case where the referent was clearly unique. Šimík and Demian (2021) further manipulated word order and prosody, but their interaction with uniqueness manipulation did not appear to be significant. From these results, Šimík and Demian (2021) conclude that Russian bare singulars do not presuppose uniqueness and should be analyzed as existential.

However, this null result may not be decisive in the debate about bare singular semantics, as there is a methodological concern. As argued extensively by Alstott and Aravind (2025), covered-box paradigms are prone to false negative results when used for presupposition studies. Since such a task primarily tracks which interpretation makes a sentence a better description relative to salient alternatives, it does not block non-presuppositional strategies that participants may employ to not choose a covered alternative. This means that the absence of a presupposition-violation effect in a covered box study is challenging to interpret, as it may be the case that the presupposition was just not captured by the task behaviorally. Šimík and Demian (2021)’s experiment did capture a uniqueness effect in German definite singulars. However, the absence of a comparable effect in Russian bare singulars is still

compatible with an analysis on which both definite singulars and bare singulars encode uniqueness, but differ in how strongly the uniqueness constraint manifests in a picture selection task.

The experiments in this chapter investigate uniqueness in tightly controlled settings. A comprehensive investigation of uniqueness would manipulate familiarity, word order, and prosody, all of which have been argued to affect the interpretation of bare nominals (Srinivas, 2021; Šimík and Wierzba, 2017; Šimík and Demian, 2021, among others). However, it is not yet clear if any of these variables covary. I will approach this in the following way. The experiments in this study will hold prosody constant (neutral); this factor will be set aside. Before proceeding to the study of uniqueness, we will establish if word order (subject-initial vs. subject-final) reliably tracks familiarity under controlled prosody. This baseline will allow us to manipulate uniqueness while holding familiarity constant in the subsequent tasks.

2.2 Controlling familiarity: word order manipulation as a methodological tool

In this section, I demonstrate, through our first experimental study, that the position of the bare singular subject correlates with the familiarity of its referent. These results will allow us to use word order as a “booster” for familiarity in the subsequent experiments on uniqueness.

I begin with a brief overview of what we know about the role of word order in shaping nominal interpretation in Russian. I then introduce Experiment 1, and explain how its results provide the baseline needed for the main experiments (2A and 2B).

2.2.1 Word order and information structure in Russian

Slavic languages are often labeled as languages with a “free” word order. This usually means one can permute words in a sentence and still get something that is considered well-formed.

A popular example is that in a sentence with a subject, a transitive verb, and an object, all six logically possible permutations are grammatical sentences with roughly the same truth-conditional meaning (23)⁴. By contrast, a language with less flexible word order, like English, does not allow such permutations, at least without re-assigning grammatical roles (22).

- (22) a. Lenin cites Marx.
b. #Marx cites Lenin.
- (23) a. Lenin citiru-et Marks-a.
 Lenin cite-PRS.3SG Marx-ACC
SVO
- b. Lenin Marks-a citiru-et.
 Lenin Marx-ACC cite-PRS.3SG
SOV
- c. Marks-a Lenin citiru-et.
 Marx-ACC Lenin cite-PRS.3SG
OSV
- d. Marks-a citiru-et Lenin.
 Marx-ACC cite-PRS.3SG Lenin
OVS
- e. Citiru-et Marks-a Lenin.
 cite-PRS.3SG Marx-ACC Lenin
VOS
- f. Citiru-et Lenin Marks-a.
 cite-PRS.3SG Lenin Marx-ACC
VSO

However, Russian word order is not free in the sense that it is not arbitrary. Consider two parallel examples having the same lexical material, but in different order, from Slioussar (2007):

⁴The sentences in (23-a)-(23-f) do not have a translation gloss on purpose: the meanings slightly differ in terms of information structure, and some sentences are compatible with multiple scenarios. However, in all of these sentences, *Lenin* is still the Agent, and *Marx* is the Patient, which follows from the case marking.

-
- (24) a. Programmist kupi-l kofevark-u.
 software.developer buy-PST coffee.machine-ACC
- b. Kofevarku kupi-l programmist.
 coffee.machine-ACC buy-PST software.developer

Slioussar (2007) demonstrates that depending on the preceding context, (24-a) and (24-b) often are in complementary distribution. Specifically, what order is preferred depends on what information is given and what information is new in the discourse. For instance, if there is no mention of either software developers or coffee machines in the preceding discourse, the SVO order is preferred over the OVS one:

- (25) Slioussar (2007, p. 1)
- a. Naš novyj ofis bystro obustra-iva-l-sja.
 our new office quickly settle.in-IPFV-PST-REFL
 ‘Our new office was settled in very fast.’
- b. Programmist kupi-l kofevark-u.
 software.developer buy-PST coffee.machine-ACC
 ‘The software developer bought a coffee machine.’ SVO
- c. #Kofevarku kupi-l programmist.
 coffee.machine-ACC buy-PST software.developer OVS

By contrast, if a referent is explicitly mentioned in the previous discourse, the felicitous structure is one that places the given information before the new information:

- (26) Slioussar (2007, p. 1)
- a. Naš-i sisadmin i programmist reši-l-i obustroit novyj ofis.
 our-PL sysadmin and software.developer decide-PST-PL settle.in new office
 ‘Our sysadmin and software developer decided to settle in the new office.’
- b. Programmist kupi-l kofevark-u.
 software.developer buy-PST coffee.machine-ACC
 ‘The software developer bought a coffee machine.’ SVO

-
- c. #Kofevarku kupi-l programmist.
 coffee.machine-ACC buy-PST software.developer

OVS

(27) Slioussar (2007, p. 1)

- a. Vskore v novom ofise pojavil-s' čajnik, kofevarka i
 soon in new office appear-PST-PL-REFL kettle coffee.machine and
 toaster.
 toaster
 'Before long, there were a kettle, a coffee machine, and a toaster in the new
 office.'

- b. #Programmist kupi-l kofevar-k-u.
 software.developer buy-PST coffee.machine-ACC

SVO

- c. Kofevarku kupi-l programmist.
 coffee.machine-ACC buy-PST software.developer
 'The coffee machine was bought by the/a software developer.'

OVS

It is generally assumed that the SVO order is basic, or non-derived, as it has the least restricted distribution and the widest range of interpretations. The other word orders are typically considered derived by various kinds of movement. One of the most detailed accounts of word order syntax in Russian was proposed by Bailyn (2012). He introduced a special level of grammar, called Functional Form, which represents the distribution of topic and focus at the syntax-phonology interface. He further suggested that word order and intonation jointly determine Functional Form. This allows Bailyn to derive different surface orders corresponding to different Functional Forms even when truth-conditional content is held constant, much like in the examples above. Importantly, a non-neutral word order is only licensed if it expresses a Functional Form that the neutral word order cannot encode. A non-neutral order in a context where the neutral word order would in principle be available signals that this sentence encodes a different interpretation at the level of Functional Form.

It has long been observed that word order in Russian can affect the interpretation of bare nouns. For instance, Geist (2010) shows examples of preverbal bare singulars favoring

definite (anaphoric) interpretations, plausibly because this position is associated with topical material (28-a). By contrast, postverbal bare singulars allow both definite and indefinite interpretations, depending on the discourse context (28-b).

(28) (Geist, 2010)

- a. **Devočka** voš-l-a v dom.
girl enter-PST-F in house
‘The girl/*A girl entered the house.’
- b. V dom voš-l-a **devočka**.
in house enter-PST-F girl
‘The girl/A girl entered the house.’

Parallel effects are found in the ordering of objects. In a study of Russian ditransitive structures, Titov (2017) demonstrates that the relative order of direct and indirect objects can constrain their interpretation, even in presentational contexts. If a direct object precedes an indirect object, both arguments can be interpreted as either definite or indefinite (29-a); but if an indirect object precedes a sentence-final direct object, the former has to be definite and the latter must receive an indefinite interpretation (29-b).

(29) (Titov, 2017)

Context: “What happened?”

- a. Ivan pereda-l špion-a agent-u.
Ivan hand-PST[M.SG] spy-ACC agent-DAT
‘Ivan handed a/the spy to an/the agent.’ DO < IO
- b. Ivan pereda-l agent-u špiona.
Ivan hand-PST[M.SG] agent-DAT spy-ACC
‘Ivan handed a spy to the agent.’ IO < DO

Titov (2017) explains this asymmetry through a prominence principle applicable to non-canonical word orders: interpretively prominent material must precede non-prominent material. On her account, definite or specific noun phrases are considered [+referential], and

therefore prominent. Titov consider the DO $\prec$ IO order (29-a) to be canonical, and therefore compatible with multiple interpretations. When the word order is non-canonical, like in (29-b), the first object has to be [+referential] and the second object has to be [-referential]. This yields the non-ambiguous interpretation in (29-b). In this way, a non-canonical word order is only licensed when it encodes a prominence asymmetry between the two objects.

A similar but more nuanced prominence-based account, supported by experimental evidence, was proposed by Slioussar (2007). Specifically, Slioussar argues that Russian word order variation is sensitive not to information-structural notions such as topic or focus per se, but to relational notions of prominence. In her account, prominence is defined within two dimensions: (i) accessibility (whether and how a discourse referent is activated in the preceding discourse), (ii) salience (how much the discourse referent's activation increases as a result of the utterance in question). Since on this view, the word order is not defined by the topic/focus distinction⁵, prominence is expected to define word order even in presentational or all-focus contexts. The claim is supported by evidence from self-paced reading studies comparing neutral and non-neutral word orders. Participants read sentences presented in controlled contexts designed to manipulate the relative accessibility or salience of competing constituents. The results show that scrambled orders are processed more easily when they align with the predicted relational prominence pattern.

Slioussar further proposes a fully syntactic formalism for information structure within the Phase-Theoretic Minimalist framework (Chomsky, 2005). In her system, information structure is not represented as features, but derived through syntactic configurations. The general rule is as follows: if XP asymmetrically c-commands YP at the interface, XP is interpreted as more accessible and/or less salient than YP.

A recurring theme in the discussion of Slavic word order is that word order effects on interpretation are often gradient rather than categorical. This point is supported by a growing body of experimental and corpus-based research. For instance, Šimík and Burianová

⁵In this, Slioussar (2007)'s approach differs from a more recent account by Bailyn (2012): Bailyn's account relies on the distinction between topic and focus structures.

(2020) show on the basis of a large corpus study of Czech that the definiteness of bare nouns correlates strongly with absolute clausal position (clause-initial vs. clause-final), but not with their position relative to the verb (preverbal vs. postverbal). Their results showed that clause-initial bare nouns are overwhelmingly interpreted as definite, while clause-final elements tend to be indefinite; this effect was independent of syntactic function.⁶ These findings challenge accounts that tie definiteness directly to structural positions or to presuppositional domains defined relative to the verb (Diesing, 1992; Kučerová, 2007), and instead support approaches where word order encodes broader prominence relations.

The contribution of word order to noun interpretation must be disentangled from that of prosody. Through a series of acceptability judgment experiments on Czech, Polish, and Slovak, Šimík and Wierzba (2017) investigated what role word order and prosody play in formally expressing the givenness of subject and object NPs. They treated givenness in a very conservative way, considering an NP to be given only if it had an overt antecedent in the immediately preceding discourse and was not focused. Šimík and Wierzba (2017) were probing the following two approaches. The first approach claims that given information must linearly precede new information; this precedence constraint is what shapes the word order (Slioussar, 2007; Kučerová, 2007; Kučerová, 2012; Titov, 2017; Skopeteas and Fanselow, 2010). The second approach builds on two constraints, first being the Nuclear Stress Rule stating that the main sentence stress is typically sentence-final (originally proposed for English by Chomsky and Halle (1968)), second being the requirement for given information to not bear sentence stress. On the latter view, word order permutations are a secondary effect of the interaction of the prosodic constraints: since the stress must be sentence-final, and given information cannot be stressed, given information is scrambled to not be sentence-final. Some works that included similar logic in their approach to word order alternations include

⁶Šimík and Burianová (2020) relied on the following diagnostics of definiteness: whether the bare NP could be modified by a demonstrative or an indefinite determiner without the loss of the intended meaning; whether the referent was uniquely identifiable in context; whether the most natural English translation would require *the* or *a*. All three diagnostics were applied while manually annotating the corpus, and served as heuristics rather than mechanical tests. See the discussion of the limitations in Šimík and Burianová (2020).

(Szwedek, 1974a,b, 1976, 2011; Neeleman and Reinhart, 1998; Šimík and Wierzba, 2015).

The main result of Šimík and Wierzba (2017) was that givenness is expressed more robustly through prosodic means than through word order alone: given constituents resist being stressed, and violations of this requirement led to a substantial drop in acceptability. By contrast, the word order generalization often assumed in the literature (given must precede new) had a much weaker effect. While word order alternations improved acceptability in some configurations, they did not rescue violations of givenness-related constraints unless accompanied by appropriate prosody. Another important result of Šimík and Wierzba (2017) is that they revealed an asymmetry between the sensitivity to givenness in subject vs. object NPs. While the givenness of the object NP interacted strongly with both prosody and word order, givenness of the subject NP had a limited effect on acceptability in most conditions. Specifically, subject givenness often failed to produce the expected preference for subject-initial orders.

The findings from works overviewed in this subsection are important in two ways. First, they support a view in which word order effects are gradient and cannot be treated as categorical reflections of simple information structure manipulations. Second, the findings from Šimík and Wierzba (2017) suggest that the position of subject NPs may be driven by factors other than givenness narrowly construed. The first experiment in this chapter builds specifically on this result. In particular, I am looking at subject NPs, isolating familiarity as an independent variable, and testing whether it influences speakers' preferences for SV vs. VS order, independently of prosodic considerations.

2.2.2 Experiment 1: familiarity vs. word order

The goal of Experiment 1 was to establish whether the linear position of a bare singular subject reliably encodes familiarity, if the prosody is held constant (neutral). Building on the observations from previous literature, we looked at whether speakers of Russian are more likely to choose an SV word order over the VS word order if the subject NP was familiar,

and vice versa. In this way, Experiment 1 crossed the following conditions:

1. **Familiarity:** +familiar vs.- familiar
2. **Choice:** SV vs. VS.

2.2.2.1 Materials and methods

Experiment 1 was designed as a forced-choice task with one manipulated factor, familiarity, and a binary dependent variable, word order preference. Participants were choosing a preferred word order (SV vs. VS) given a scenario that introduced either a familiar or a non-familiar individual. Participants read short context scenarios describing a conversation between two people. In each scenario, one of the interlocutors was witnessing a particular situation, while the other was either familiar with the object under discussion (the **+familiar** condition) or unfamiliar with it (the **-familiar** condition). Familiarity was established through prior mention, and **+familiar** contexts were such that for both interlocutors, the referent was easily identifiable. The non-witness character would ask, ‘What is going on?’, and participants were then asked to choose between two possible responses from the witness character. The choices differed in word order. Below are two examples illustrating the **+familiar** and **-familiar** conditions, with contexts translated into English:

(30) **Non-familiar condition**

Lena and Tom don’t have pets. Lena calls Tom and hears barking in the background.

Tom looks outside and sees that a group of street dogs ran into the backyard.

Lena: What is going on?

Which of Tom’s answers would sound best?

- a. **Sobaka** laet vo dvore.
dog barks in backyard
- b. Vo dvore laet **sobaka**.
in backyard barks dog

(31) **Familiar condition**

Lena and Tom have a dog named Max, who barks whenever someone walks by. Lena calls and hears barking. It's Max, who's out in the yard.

Lena: What is going on?

Which of Tom's answers would sound best?

- a. **Sobaka** laet vo dvore.
dog barks in backyard
- b. Vo dvore laet **sobaka**.
in backyard barks dog

The experiment used a mixed-modality design. The context scenarios were still presented as text, while the response options were delivered in audio format. A screenshot of an experimental trial screen is provided in Figure 2.1.⁷

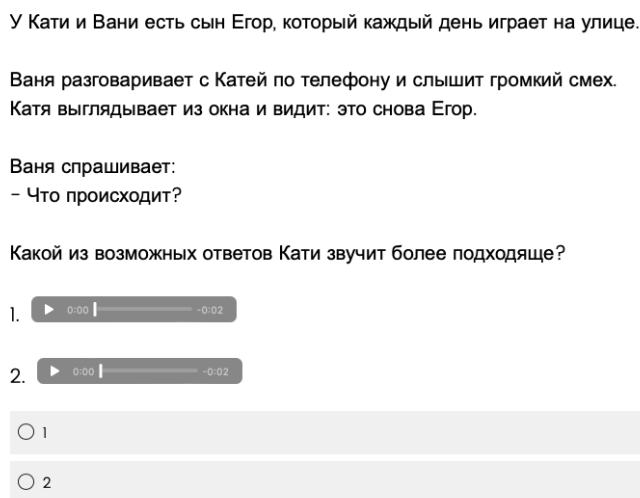


Figure 2.1: A screenshot of an experimental item in the baseline task.

⁷Translation of the text in Figure 2.1: Katya and Vanya have a son Egor, who plays every day in the street. Vanya talks to Katya on the phone and hears loud laughter. Katya looks out of the window and sees: this is, as always, Egor. Vanya asks: “What is going on?”. Which of the possible Katya’s answers sounds more appropriate?

2.2.2.2 Materials and participants

To minimize variability and control for prosodic effects, all Russian auditory stimuli were recorded using a consistent intonational contour, specifically, the neutral narrative intonation known in the studies of Russian prosody as Intonational Contour 1 (IK-1), following Bryzgunova (1980). It naturally appears in presentational sentences like (32), is characterized by a lowering of pitch on the stressed syllable of the final content word from the speaker’s average pitch to their individual average minimum. Prior to this fall, the contour includes a relatively high pitch level, with stressed syllables of meaningful words often raised above the speaker’s average. The final pitch drop is typically accompanied by a decrease in intensity and a lengthening of the final stressed syllable. The intonation center usually aligns with the last content word in the phrase. An example of an IK-1 pitch contour is given in 2.2

- (32) Marina sluša-l-a muzyk-u.
Marina listen-PST-F music-ACC
(What was happening?) ‘Marina was listening to music.’ (Jasinskaja, 2014)

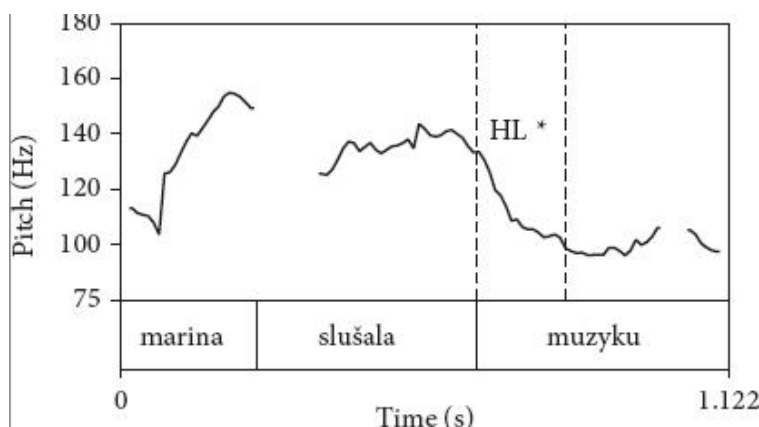


Figure 2.2: The IK-1 pitch contour for (32) (Jasinskaja, 2014, p. 712).

The auditory stimuli were recorded by a 31-year-old female native speaker of Russian from Moscow. To ensure accurate modeling of the narrative intonation, I followed its acoustic descriptions from Svetozarova (1982) and recorded her sample sentences. These recordings were used to establish the speaker’s average pitch, average maximum, average minimum,

and pitch values for stressed syllables. Using these reference points, I then recorded the experimental stimuli, aiming to reproduce the IK-1 contour as closely as possible. Where necessary, recordings were edited using Praat (Boersma and Weenink, 2025). Specifically, the stressed syllable of the first word (the subject NP in canonical order or the locative phrase in inverted order) was adjusted to match the speaker’s average pitch (331 Hz), and the final content word was modified to reach the speaker’s average minimum pitch (141 Hz). The intermediate elements of the sentence optionally bore postnuclear (phrasal) accents.

16 pairs of audio responses were recorded (one SV and one VS sentence per item). Each pair of recordings was used in both the **+familiar** and **-familiar** versions of the corresponding scenario, yielding 32 experimental trials in total. In addition, 20 filler tasks were created to assess participant attention and data quality. Participants were randomly assigned to one of four experimental lists, each comprising 8 experimental items and 10 filler tasks. The lists were counterbalanced such that no participant saw both the **+familiar** and the **-familiar** version of the same experimental item.

128 self-identified native speakers of Russian were recruited using Prolific⁸ and word of mouth. Filler tasks were used to filter for inadequate answers: the data from 6 participants who failed more than 3 filler tasks were excluded from analysis. The median survey completion time was 8 minutes.

2.2.2.3 Results

In the **+familiar** condition, participants preferred the direct word order (SV) in 58.9% of trials, 95% CI[0.54, 0.63], while they chose the inverted word order (VS) in the remaining 41.1% of trials, 95% CI[0.37, 0.46]. In the **-familiar** condition, the pattern reversed: participants chose the inverted word order in 58.21% of trials, 95% CI[0.53, 0.63], and the direct word order in 41.79% of trials, 95% CI[0.37, 0.47]. These patterns are shown in Figure 2.3 below.

⁸In this and the following studies, participants recruited through Prolific were located outside of Russia; participants recruited through word of mouth were located in Russia.

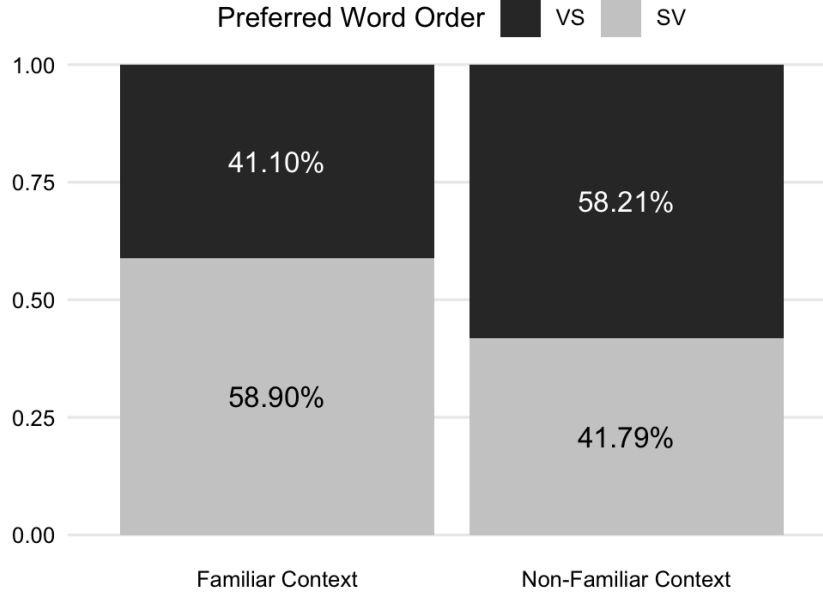


Figure 2.3: Proportions of participants' word order choices in Experiment 1.

The results were analyzed using mixed-model logistic regression with the following formula: $\text{choice} \sim \text{context} + (1|\text{ResponseId}) + (1|\text{item})$. The model predicted participants' choice of word order (direct vs. inverted) based on context type (familiar vs. non-familiar), with random intercepts for participant and experimental item. Following post-model residual diagnostics using the DHARMA package (Hartig, 2024), one experimental item pair exhibiting notably different residual variance was removed, and the model was re-fitted on remaining items.

The results confirmed that context had an effect on word order choice. In other words, participants were significantly more likely to choose the direct word order in familiar contexts compared to non-familiar ones ($\beta = -0.85$, $\text{SE} = 0.16$, $p < 0.001$). The intercept of the model was estimated as 0.43 ($\text{SE} = 0.16$, $p < 0.001$). The negative β coefficient reflects that the log odds of selecting the direct word order were significantly lower in non-familiar contexts than in familiar ones. The odds of choosing direct word order in a familiar context are approximately 2.34 times higher than in an unfamiliar context. This effect size can be considered moderate-to-high.

The effect of context explained approximately 4.2% of the variance in word order choice.

Random effects accounted for an additional 19.8% of the variance, which indicates meaningful variation across individuals and experimental items. There were no clear patterns in the distribution of either random effect. After removing the outlier experimental item pair, there was no heteroscedasticity or auto-correlation in the residuals of the model. In this way, the overall fit of the model can be evaluated as good, explaining approximately 24% of variance.

Due to technical limitations, response options were not randomized but counterbalanced across experimental lists. This resulted in a small but significant positional bias: participants were slightly more likely to choose the first-listed audio option. Specifically, there was a moderate-sized increase in the likelihood of choosing the direct word order in familiar contexts when it appeared first, and similarly, a slight preference for the inverted order in unfamiliar contexts when listed first. However, this effect was consistent across conditions and did not interact with the main effect of context.

In this way, the word order preferences of participants clearly depended on the familiarity of the NP in the preceding context. When the NP was familiar, participants tended to prefer the direct (subject–verb–location) order; when it was unfamiliar, they switched to the inverted order. This pattern was statistically robust and moderately strong. Although participants showed a small bias toward the first audio option, this did not affect the main result.

2.2.3 Discussion and methodological conclusions

Our results demonstrate that in a sentence with a bare singular subject, word order preference strongly correlates with the familiarity of the referent of the subject NP. When the referent is familiar, participants preferred sentences where the corresponding subject NP was preverbal, and when the referent was explicitly non-familiar, participants preferred the postverbal position of the subject NP. Importantly, these preferences are probabilistic rather than categorical: neither SV nor VS were exclusive to any context. Instead, manipulating the

familiarity of the referent of the subject NP reliably biased the choice of the participants in one direction or the other.

In order to interpret the results more clearly, let us clarify the way in which familiarity was manipulated in the experimental materials. In the **-familiar** condition, the context introduced a situation in which multiple potential referents were available and no individual entity was singled out as discourse-salient. Some examples of **-familiar** scenarios are below:

(33) Andrey calls Masha on the phone and hears honking. Masha looks outside and sees **several cars in the intersection**.

(34) Katya and Vanya are talking on the phone. Vanya hears children laughing. Katya looks outside and sees **some boys playing in the street**.

In these cases, the bare singular NP in the target sentence (*Mašina signalit na ulitse* (car honk-PRS.3SG in street) ‘A/The car is honking outside’ and its VS version for (33); *Rebėnok smeėtsja na ulitse* (child laugh-PRS.3SG in street) ‘A/The child is laughing outside’ and its VS version for (34)) did not correspond to any previously activated individual. The target NP has no anaphoric antecedent and its referent is not identifiable; theories that use the notion of relative discourse accessibility would consider such a noun relatively low in discourse accessibility.

By contrast, in the **+familiar** condition, the context explicitly established a salient individual referent shared by the interlocutors, often through prior mention or shared background knowledge; the **+familiar** counterpart of (33) described Masha seeing their own car, and that of (34) mentioned Katya and Vanya’s child playing outside. In all **+familiar** scenarios, the referent was already present in the discourse and highly accessible, as identifiability was guaranteed by the context.

One can notice that the stimuli did not manipulate the absolute ends of the accessibility scale. A minimally accessible referent would be a newly introduced one. In the **-familiar**

stimuli, the preceding contexts may have made the non-familiar referent D-linked to a degree. But even though the experimental design targeted contrasts in relative accessibility, not absolute accessibility, we still observe a strong word order preference. This aligns with the mapping rules proposed by Slioussar (2007): the difference in relative accessibility is sufficient to license different word orders even in the absence of clear topic-focus contrasts.

Another important thing about these results is that the observed behavior demonstrated a preference for a certain word order, rather than a sharp contrast between them. This is also exactly what accounts relying on relative prominence like Slioussar (2007) or Titov (2017) predict. If discourse accessibility is a continuum and the grammar only encodes relative ordering relations, word order effects should manifest as biases in preferences, not as categorical choices. Therefore, the fact that SV order was still chosen in over 40% of **-familiar** trials, and VS order in over 40% of **+familiar** trials, is therefore not noise or failure of the manipulation, but a reflection of the probabilistic mapping between discourse accessibility and syntactic linearization. A probabilistic interpretation is further supported by the moderate effect size and the substantial contribution of speaker and item variability.

The moderate effect size is also in line with the findings of Šimík and Wierzba (2017), who showed that the givenness manipulation had weaker effects on acceptability in subject NP vs. object NPs, especially when prosody is controlled. At first glance, the fact that Experiment 1 still revealed a robust effect of familiarity on word order might seem conflicting with Šimík and Wierzba (2017)’s findings. However, they manipulated givenness, defined as the presence of an explicit overt antecedent, and the dependent variable was the acceptability under controlled prosodic conditions. Experiment 1 manipulated familiarity (with a much looser understanding of accessibility than the presence/absence of an overt antecedent), and the dependent variable was word order choice, not acceptability. Perhaps givenness, understood narrowly, may be insufficient to trigger strong acceptability contrasts, but relative accessibility can show up as a robust effect in a forced-choice task.

To conclude, Experiment 1 provides direct experimental evidence that familiarity of the

referent of a subject NP biases word order choice in Russian. The way this bias shows up is predicted by relational prominence-based accounts of information structure. The observed preferences are gradient, but robust across speakers and items. The results fit well with the literature that Slavic word order is a probabilistic cue to discourse accessibility rather than a categorical marker of information structure.

Experiment 1 treated word order as a dependent variable. The subsequent experiments 2A and 2B reverse this: since we identified a systematic association between familiarity and word order, they will be treating word order as a controlled cue to familiarity. In either task, familiarity will be kept constant, and the target sentences will be using the word order that best fits the familiarity parameter. That is, Experiment 2A will only test sentences where the referent of the subject NP is familiar and the word order is SV. Experiment 2B will test sentences where the referent of the subject NP is not familiar and the word order is VS. These tasks will investigate how familiarity and word order constrain the interpretation of the subject NP in terms of uniqueness.

2.3 Main experiments: uniqueness under controlled familiarity

2.3.1 Predictions

Now that we have confirmed that the linear position of the subject NP correlates with the familiarity of its referent, we can proceed to the main question: do bare singulars presuppose uniqueness? If yes, does it happen independently of familiarity or does familiarity obscure the effect of uniqueness?

The experiments reported below manipulate uniqueness while holding familiarity constant within each task. As discussed in section 2.1.3, competing theories differ in their predictions. I will spell out three logically distinct hypotheses about the semantics of bare singulars.

The most radical version of the uniqueness effect in bare singulars is predicted by the neo-Carlsonian approach (Chierchia, 1998; Dayal, 2004, 2026b, among others). We can formulate it as follows:

(35) **Strong Uniqueness Hypothesis**

A bare singular is licensed if and only if its referent is unique (there exists only one individual satisfying the predicate in the discourse situation), independently of its familiarity.

Prediction: Bare singulars should be degraded in non-unique contexts regardless of whether they are familiar (and preverbal) or non-familiar (and postverbal).

Another theory we will probe is the account in line with Geist (2010). Such theory claims that bare singulars may be introduced either via *iota* or existential closure, with the existential construal restricted to non-topical content. Since existential closure is limited in terms of information structure rather than familiarity, controlling for familiarity is especially important in testing this theory. The hypothesis in line with Geist (2010) and its predictions would be the following:

(36) **Flexible Uniqueness Hypothesis**

A familiar bare singular is licensed if and only if its referent is unique. A non-familiar bare singular does not have restrictions on the uniqueness of its referents.

Predictions: Bare singulars are expected to be degraded in non-unique contexts when familiar. The manipulation of uniqueness in non-familiar bare singulars should not influence their acceptability.

Finally, if all bare singulars are existential, as suggested by works like Seres and Borik (2021); Šimík and Demian (2021), we are unlikely to see any difference in the acceptability of bare singulars in any syntactic position or discourse configuration:

(37) **No Uniqueness Hypothesis**

Bare singulars do not impose requirements on the uniqueness of their referents, no matter their syntactic position or information structure status.

Predictions: The manipulation of uniqueness should not influence the acceptability of bare singulars independently of their familiarity.

To test these predictions, we constructed two parallel experiments. **Experiment 2A** tested the ability of bare singulars to be used in unique and non-unique settings when they refer to a familiar individual. **Experiment 2B** examined the same contrast when the referent was explicitly novel (non-familiar). Experiment 1, presented in section 2.2, showed that word order and familiarity correlate: preverbal subjects tend to be interpreted as familiar, while postverbal subjects are more natural when non-familiar. Therefore, each experiment used the word order that best maintained the intended familiarity condition. Experiment 2A tested uniqueness in familiar contexts and used sentences with preverbal bare singular subjects. Experiment 2B tested uniqueness in non-familiar contexts and used sentences with postverbal bare singular subjects, making non-familiarity more accessible. The predictions of the three hypotheses we suggested in (35), (36), and (37) are summarized in Table 2.1.

Experiment	Familiarity	Position	Uniqueness	Uniqueness Hypothesis Version		
				Strong	Flexible	None
2A	Familiar	Preverbal	unique	✓	✓	✓
			non-unique	✗	✗	✓
2B	Non-Familiar	Postverbal	unique	✓	✓	✓
			non-unique	✗	✓	✓

Table 2.1: Expected results for the acceptability of bare nouns in Experiments 2A and 2B under different versions of the Uniqueness Hypothesis.

Previous experimental investigations of Russian (Šimík and Demian, 2021) showed no uniqueness effects in either pre- or postverbal bare singulars. The authors took this result as surprising, admitting that this could have reflected limitations of the covered box experimental design. This suggests that the potential effect of uniqueness may be subtle. This motivated

us to include a control condition in the experimental design. Specifically, the experiments included a comparison with singular NPs modified by the numeral *odin* ‘one’. Such NPs can either express cardinality or, following Geist (2010), introduce an existential quantifier together with referential anchoring to the speaker, yielding a specific indefinite. Thus, ‘one’-phrases do not impose uniqueness, but may encode speaker-oriented specificity. Example (38) illustrates that such phrases can function both as answers to questions addressing cardinality (e.g., it can be interpreted as an answer to *How many friends who live on a boat do you have?*) (38-a), or as a narrative opener that do not imply uniqueness (38-b).

- (38) U menja est’ **odin drug**, kotoryj živ-ët na lodk-e.
 by me there.is one friend which live-PRS.3SG on boat-PREP
 a. ‘I have (exactly) one friend who lives on a boat.’
 b. ‘I have a friend who lives on a boat.’

Importantly, the expectations for the behavior of ‘one’-marked nouns in Experiments 2A and 2B are independent of the bare noun theory. If the referent is familiar and unique, *odin*-phrases are expected to be pragmatically weird but not strictly ungrammatical. By contrast, if the referent is familiar but non-unique, the ‘one’-phrase should be felicitous, as its cardinality reading remains available. In Experiment 2B, which manipulates non-familiar NPs, *odin*-phrases are predicted to be fully acceptable regardless of uniqueness manipulation. No strong difference between unique and non-unique contexts is expected for them. These predictions are summarized in Table 2.2.

Experiment	Familiarity	Position	Uniqueness	‘One’-marked noun licensed?
2A	Familiar	Preverbal	unique	✗
			non-unique	✓
2B	Non-Familiar	Postverbal	unique	✓
			non-unique	✓

Table 2.2: Expected results for the acceptability of ‘one’-marked nouns in Experiments 2A and 2B.

2.3.2 Experiments 2A and 2B: Design and materials

Both of our main experiments employed a fully within-subjects 2×2 design manipulating two factors: **Uniqueness** (unique vs. non-unique) and **Noun Type** (bare singular vs. *odin*-singular). Each experiment was implemented as an acceptability judgment task with double presentation: in each trial, participants evaluated both levels of the **Noun Type** variable (bare singular vs. ‘one’-marked singular).

2.3.2.1 Experiment 2A: uniqueness when familiarity is supported

Participants read short context scenarios describing a conversation between two interlocutors. In each scenario, one person reported an event involving an individual familiar to both them and the addressee (e.g., their pet). Familiarity was kept constant across conditions by ensuring that the referent was part of the shared conversational background.

Two types of contexts were constructed:

1. Unique contexts, where only one individual satisfied the description of the NP,
2. Non-unique contexts, where two equally salient individuals were introduced.

An example of a unique context from Experiment 2A is shown below:

(39) **Unique condition**

Masha and Andrey have a pet hamster named Pyatachok. He lives in a cage in the living room. One day, Masha and Andrey were having dinner. Masha went to the room to get napkins and saw that Pyatachok was gone, and the cage door was open. Masha screamed.

Andrey asked from the kitchen: “What happened?”

How appropriate is each of these possible replies from Masha in this situation?

- a. **Xomjak** sbežal iz kletki.

hamster ran.away from cage

‘The hamster ran away from the cage.’⁹

- b. **Odin xomjak** sbežal iz kletki.

one hamster ran.away from cage

‘One hamster ran away from the cage.’

In the non-unique condition, the effect of non-uniqueness was achieved through introducing two equally salient individuals satisfying the description of the NP. A non-unique context parallel to (39) would be the following:

(40) **Non-unique condition**

Masha and Andrey have two pet hamsters, Busya and Pyatachok. They live in a cage in the living room. One day, Masha and Andrey were having dinner. Masha went to the room to get napkins and saw that Pyatachok was sitting in the cage alone. Busya was gone, and the cage door was open. Masha screamed.

Andrey asked from the kitchen: “What happened?”

How appropriate is each of these possible replies from Masha in this situation?

- a. **Xomjak** sbežal iz kletki.

hamster ran.away from cage

‘A hamster ran away from the cage.’

- b. **Odin xomjak** sbežal iz kletki.

one hamster ran.away from cage

‘One hamster ran away from the cage.’

Similar to Experiment 1, the context scenarios were presented as text, but the response sentences were presented as audio recordings with neutral prosody. The audio recording

⁹Hereinafter, I use the English *the*-noun in the translation of the stimuli in the unique condition because that would be an expected form in English here, and the English *a*-noun in the translation of the non-unique condition stimuli. I use *one* in all translations of Russian stimuli with *odin* for simplicity, even though their meaning can be close to that of *some*-indefinites, especially in postverbal configurations. For the experimental stimuli, I do not assume that the Russian sentences operate the same exact way their English translations do.

and post-processing procedure was the same as in Experiment 1. In total, 16 experimental recordings were prepared: 8 sentences with bare singulars and 8 with *odin*-marked singulars.

Figure 2.4 shows a screenshot of one experimental trial (in Russian, corresponding to example (40) above). Each trial displayed two unlabeled slider bars, both initialized at the midpoint. The left and right ends of each slider were labeled “Doesn’t fit at all” and “Perfectly fits as an answer to this question,” respectively.

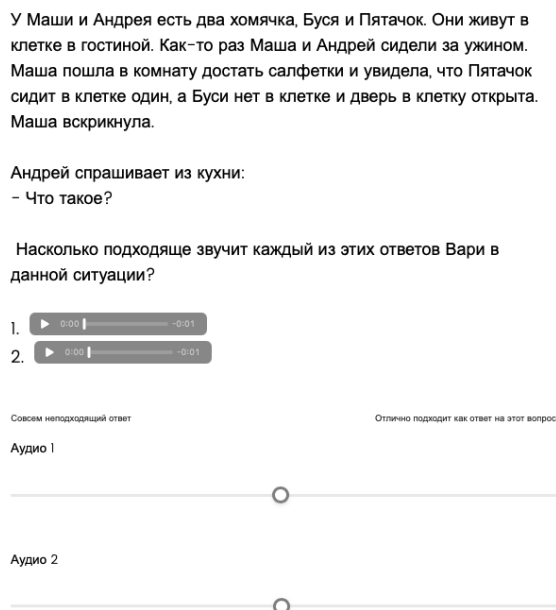


Figure 2.4: Screenshot of an experimental trial from Experiment 2A (corresponds to (40)).

2.3.2.2 Experiment 2B: uniqueness when familiarity is suppressed

Experiment 2B was designed as a parallel task to Experiment 2A. As before, participants read short scenarios describing a conversation between two interlocutors. In this experiment, however, one interlocutor reported an event involving an unfamiliar individual. Two types of scenarios were included; the type of the scenario was not considered an experimental factor but was balanced. The *news* contexts described a person watching a TV news segment about an event that happened to either a unique or a non-unique non-familiar individual.

The second interlocutor then asked, “What happened in the news?”. An example pair is shown below:

(41) **Unique condition**

Mom is watching the news. Today’s news is about a zoo in a small town. The report says that this morning, a staff member came to feed a raccoon — the only one in this zoo — and saw that the cage was open and the raccoon was gone.

Dad asks: “What did they say on the news?”

How appropriate is each of these possible replies from Mom in this situation?

- a. Iz odnogo zooparka sbežal enot.
from one zoo ran.away raccoon
‘A raccoon ran away from a zoo.’
- b. Iz odnogo zooparka sbežal odin enot.
from one zoo ran.away one raccoon
‘One raccoon ran away from a zoo.’

(42) **Non-Unique condition**

Mom is watching the news. Today’s news is about a zoo in a small town. The report says that this morning, a staff member came to feed two raccoons and saw that the cage was open and one of the raccoons was gone. The other was still sleeping in its little house.

Dad asks: “What did they say on the news?”

How appropriate is each of these possible replies from Mom in this situation?

- a. Iz odnogo zooparka sbežal enot.
from one zoo ran.away raccoon
‘A raccoon ran away from a zoo.’
- b. Iz odnogo zooparka sbežal odin enot.
from one zoo ran.away one raccoon
‘One raccoon ran away from a zoo.’

The second type of context, labeled *unfamiliar sound*, involved two interlocutors hearing

an unfamiliar noise. One of them looked to identify the source and then responded to the other's question, "What's going on?". An example pair is given below:

(43) **Unique condition**

Ira and Lena arrived at an animal shelter. While they're talking with the director, they hear barking from the yard. Ira looks out the window. In the yard, there's only one dog. It's sitting by the gate and barking at the cars.

Lena asks: "What's going on?"

How appropriate is each of these possible replies from Ira in this situation?

- a. U vorot laet sobaka.
by gate barks dog
'A dog is barking by the gate.'
- b. U vorot laet odna sobaka.
by gate barks one dog
'One dog is barking by the gate.'

(44) **Non-Unique condition**

Ira and Lena arrived at an animal shelter. While they're talking with the director, barking can be heard from the yard. Ira looks out the window. In the yard, there are three dogs: two are running around, and one is sitting by the gate barking at the cars.

Lena asks: "What's going on?"

How appropriate is each of these possible replies from Ira in this situation?

- a. U vorot laet sobaka.
by gate barks dog
'A dog is barking by the gate.'
- b. U vorot laet odna sobaka.
by gate barks one dog
'One dog is barking by the gate.'

Two properties of the response sentences differed in Experiment 2B from Experiment 2A.

First, word order was reversed: the locative phrase preceded the verb and subject NP. Second, in the *news* contexts, the locative phrase itself contained a *one*-phrase to reinforce non-familiarity.

In total, 16 experimental contexts were prepared (8 pairs of unique vs. non-unique contexts). Eight were of the *news* type and eight of the *unfamiliar sound* type. Similar to Experiment 2A, the context scenarios were presented as text, while the response sentences were presented as audio. Recordings were produced and post-processed in the same way as in Experiments 1 and 2A. The experimental trial screen and the slider bar design were also identical to those in Experiment 2A.

2.3.2.3 Experimental lists

For Experiment 2A, 8 pairs of context scenarios were constructed, each pair comprising one unique and one non-unique version of the same situation (differing only in the number of introduced potential referents). Thus, every experimental item had two matched variants. Experiment 2B contained 8 pairs of context scenarios as well, 4 of the “news” type and 4 of the “unfamiliar sound” type. The contexts between Experiments 2A and 2B were not connected to each other and were not treated as paired.

To ensure participant engagement and data quality, both Experiments 2A and 2 B included filler tasks. These fillers were unrelated to the experimental manipulation and maintained a 3:2 ratio of filler to experimental items. The filler items were identical across both experiments.

Each experiment consisted of two experimental lists, and participants were randomly assigned to one of them. In both experiments, each list contained 8 experimental items and 12 filler tasks. Lists were counterbalanced so that no participants saw both the unique and the non-unique versions of the same scenario. With each trial, the order of the two response options (bare vs. marked with ‘one’) was counterbalanced across participants.

2.3.2.4 Participants and outlier removal

A total of 61 self-identified native speakers of Russian participated in Experiment 2A, and 60 in Experiment 2B. Participants were recruited through Prolific and word of mouth. The two participant pools did not intersect with each other. The median completion time was approximately 12 minutes for Experiment A and 13 minutes for Experiment B.

Filler tasks were used to filter for inadequate answers. Filler responses were standardized within each participant to account for individual differences in slider use. Participants were evaluated on several filler types: “directional” fillers (where one option should clearly be rated higher than the other one) and “good”/“bad” fillers (where both options should receive similar ratings, either high or low). For “directional” fillers, a response was considered correct if the difference between the two ratings was in the expected direction and exceeded a small participant-specific threshold, calculated as $D_{threshold} = 0.3 \times SD \times \sqrt{2}$, where SD is that participant’s variability in slider use (standard deviation). “Good” or “bad” fillers were considered consistent if both options received ratings on the expected side of the scale (above or below average) and differed by less than 0.4 standard deviations. Response patterns that were flagged for exclusion included overall accuracy below 0.5, poor performance on directional fillers (below 0.75), or minimal slider movement. Participants meeting any of these exclusion criteria were removed from the dataset. This procedure led to the exclusion of 3 of the 61 participants from Experiment A and 3 of the 60 participants from Experiment B.

2.3.3 Results

2.3.3.1 Experiment 2A: clear effect of uniqueness manipulation in preverbal familiar nouns

Within the unique condition, bare singulars were rated overwhelmingly higher than singulars with ‘one’: the mean rate for bare singulars in that context was 95.0, 95% CI[94.22, 95.75],

and the mean rate for singulars with ‘one’ was 29.5, 95% CI[27.37, 31.64]. Within the non-unique condition, the preference is the opposite, but the effect size is much smaller. The mean rate for singulars with ‘one’ was 92.4, 95% CI[91.47, 93.43], and the mean rate for bare singulars was 79.6, 95% CI[78.24, 80.89]. Average raw ratings are visualized in Figure 2.5. Figure 2.6 visualizes ratings that were standardized within participants as a boxplot, and it shows that the variability around the center is low (as the interquartile range boxes are relatively narrow). This indicates that the distribution of the ratings is compact, symmetric, and homogeneous. On a higher interpretation level, it means that participants largely agreed with each other, and the task was not confusing or ambiguous.

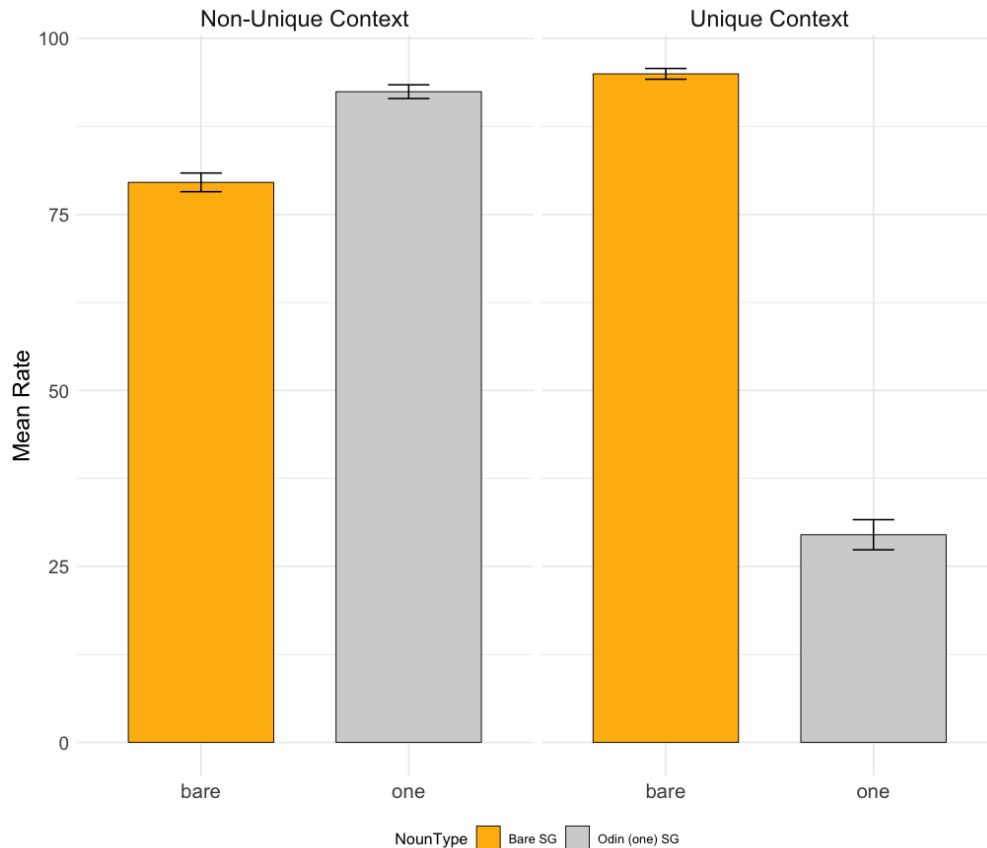


Figure 2.5: Mean ratings with standard error bars for each condition in Experiment 2A (familiar preverbal contexts).

For statistical analysis, a linear mixed-effects model¹⁰ was fit to z-scored ratings with

¹⁰For modeling, the package `lme4` was used (Bates et al., 2015). In these kind of models, p-values are not

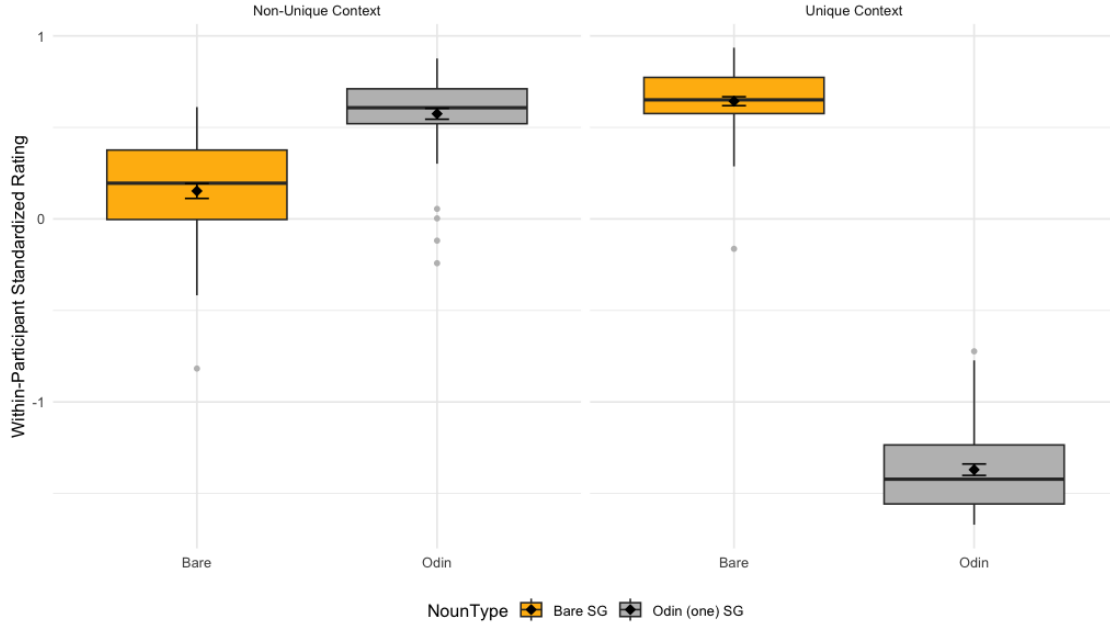


Figure 2.6: Within-participants standardized ratings for each condition in Experiment 2A (familiar preverbal contexts). The diamonds are means with standard errors, and the lines are medians.

Uniqueness (unique vs. non-unique) and **Noun Type** (bare vs. marked by *odin* ‘one’) as fixed effects and random intercepts for participant and item: $\text{zRate} \sim \text{Uniqueness} * \text{NounType} + (1|\text{ResponseId}) + (1|\text{Item})$. The model revealed a strong interaction between **Uniqueness** and **NounType** ($\beta = -2.31$, $\text{SE} = 0.07$, $t(870) = -31.16$, $p < 0.001$). This means that how natural the sentence was judged depended on both noun type (bare vs. marked by ‘one’) and uniqueness, resulting in a strong crossover pattern.

Pairwise comparisons of estimated marginal means were conducted using the Kenward-Roger degrees-of-freedom method with Tukey adjustment for multiple comparisons (using **emmeans**, Lenth and Piaskowski (2025)). All pairwise comparisons reached statistical significance. When the context implied non-uniqueness, bare singulars were rated on average 0.38 standard deviations lower than singulars marked with *odin* ‘one’, $t(860) = -7.24$, $p < 0.001$. This effect can be considered medium to large (Cohen’s $D = 0.67$). In opposite, when the context

provided by default, and effects are evaluated based on t-values (absolute t-values above 2 are considered reliable). To obtain Satterthwaite-approximated degrees of freedom and p-values, I used the **lmerTest** package (Kuznetsova et al., 2017).

introduced a clearly unique individual, bare nouns were rated on average 1.93 standard deviations higher, $t(860) = 36.82$, $p < 0.001$, which corresponds to a very large effect size (Cohen’s $D = 4.09$).

The residuals of the initial linear model showed a roughly normal distribution, but somewhat unequal variance (heteroscedasticity). To account for this, and to assess the influence of outliers observed in the initial visualization, a robust mixed-effects linear model of the same architecture was used, using the `robustlmm` package (Koller, 2016). The robust model yielded the same qualitative pattern of results, with a similarly strong interaction effect ($\beta = -2.59$, $SE = 0.05$, $t(870) = -48.97$, $p < 0.001$) and slightly smaller estimated for both random effects. The robust estimation successfully down-weighted a small subset of extreme responses, as residual variance decreased substantially (from 0.32 to 0.15). Together, non-robust and robust modeling demonstrated that the observed interaction is highly reliable and not driven by outliers.

In simpler terms, these results show that in contexts supporting familiarity, Russian speakers strongly prefer bare singulars when there is only one relevant individual in the situation, and they prefer singulars with *odin* ‘one’ when there are several equally possible referents. The crossover pattern is clear and robust across analyses: bare singulars sound natural when the referent is unique, while *odin*-marked forms sound natural when it is not. Even after accounting for individual differences and a few unusual responses, the overall conclusion remains the same: bare singulars in Russian are sensitive to uniqueness when familiar.

2.3.3.2 Experiment 2B: no visible uniqueness requirement in postverbal non-familiar nouns

The pattern we observed in Experiment 2B was completely different from what we saw in Experiment 2A. First, bare singulars were rated equally high in both the unique and the non-unique condition. In contexts supporting uniqueness, bare singulars received an average

rating of 90.4, 95% CI[87.9, 92.9]; in contexts supporting non-uniqueness, the average rating of a sentence containing a bare singular subject was 92.8, 95% CI[91.0, 94.6]. The control condition, *odin*-marked singulars, was rated lower than bare singulars in both unique and non-unique contexts, with mean ratings of 66.0, 95% CI[61.9, 70.1], and 70.3, 95% CI[66.5, 74.1], respectively. The raw mean ratings for all conditions in Experiment 2B are visualized in Figure 2.7.

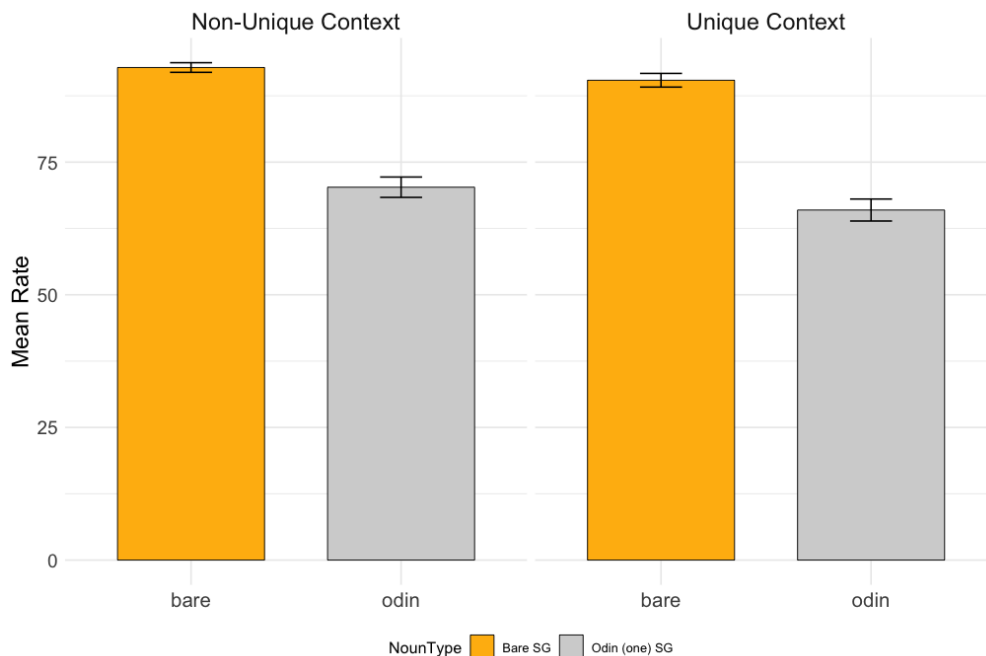


Figure 2.7: Mean ratings with standard error bars for each condition in Experiment 2B (non-familiar postverbal contexts).

The distribution of within-participant standardized ratings is visualized in Figure 2.8. The variability of ratings for *odin*-marked nouns was also significantly higher, which can be seen from larger interquartile ranges. Together, Figures 2.7 and 2.8 show no interaction effect: bare singulars and singulars marked with ‘one’ demonstrate roughly the same level of acceptability across different levels of the UNIQUENESS condition.

Again, a mixed-effects linear model was used to predict z-scored ratings from UNIQUENESS, NOUNTYPE, and their interaction, with random intercepts for participants and items: $\text{zRate} \sim \text{Uniqueness} * \text{NounType} + (1|\text{ResponseId}) + (1|\text{Item})$. The analysis showed that

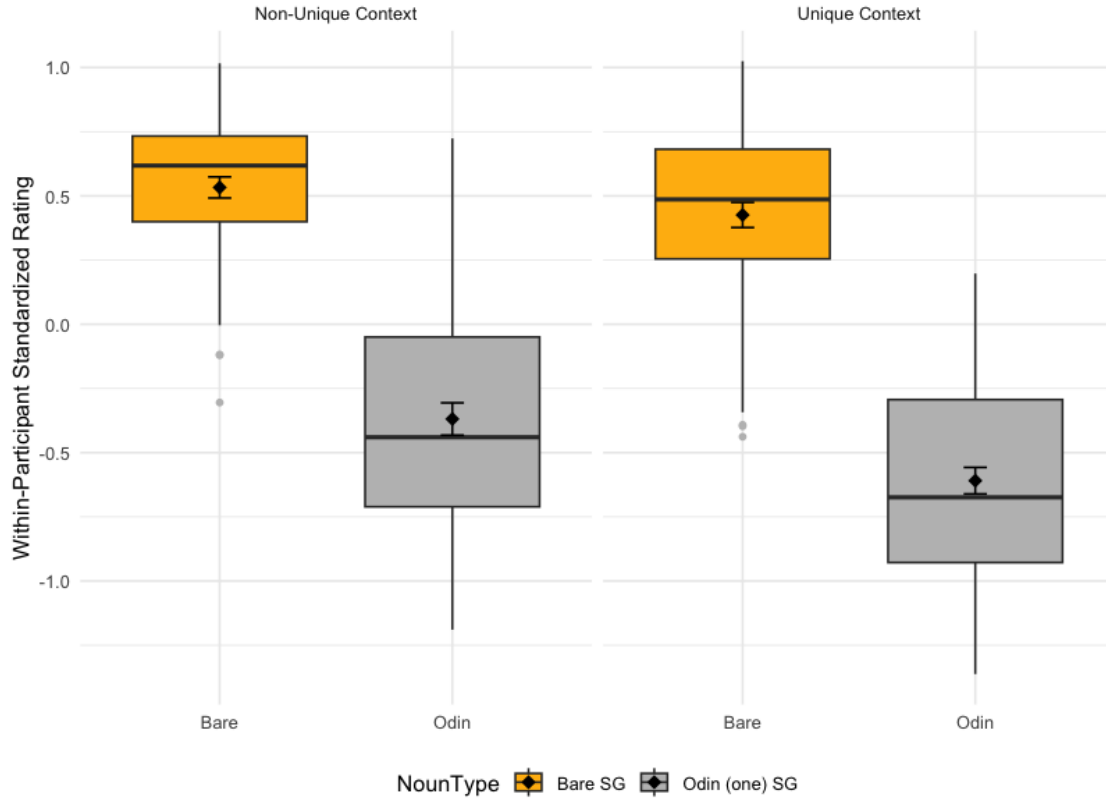


Figure 2.8: Within-participants standardized ratings for each condition in Experiment 2B (non-familiar postverbal contexts). The diamonds are means with standard errors, and the lines are medians.

nouns marked with ODIN were rated on average 0.85 standard deviations lower than bare singulars across different types of contexts, and this effect is statistically significant and strong ($\beta = -0.85$, $SE = 0.07$, $t = -11.65$, $p < 0.001$). In contrast, the difference between bare singulars in unique and non-unique conditions did not reach significance ($p = 0.24$), and neither did the interaction between UNIQUENESS and NOUNTYPE ($p = 0.5$). This indicates that the difference between bare singulars and *odin*-marked nouns was consistent across unique and non-unique conditions. Random intercepts for participants accounted for most of the random variation, while the variation between experimental items contributed minimally. This suggests that the main source of variability beyond the fixed effects came from the differences between participants, not the differences between items.

Overall, the results of Experiment 2B did not reveal any effect of uniqueness on bare

singular acceptability in postverbal contexts. There was no significant differences between the acceptability of postverbal bare singulars in context supporting uniqueness and contexts incompatible with uniqueness. In both types of contexts, bare singulars receive relatively good, above average ratings.

2.3.4 Discussion and further steps

We observed two different behavioral patterns in Experiment 2A vs. Experiment 2B. When the bare singular subject NP was preverbal and its referent was familiar, there was a strong effect of uniqueness: sentences were rated as more felicitous when the referent of the subject NP was unique. By contrast, when the bare singular subject was postverbal and its referent was non-familiar, no uniqueness effect was observed: the manipulation of uniqueness did not influence acceptability ratings. Regardless of whether the referent of the subject NP was unique or not, sentences with postverbal non-familiar bare singular subjects were rated relatively high. The contrast between familiar preverbal and non-familiar postverbal bare singulars is robust: in Experiment 2A, the effect size is large, and the null effect in Experiment 2B is consistent across statistical analyses. Since the filler tasks and the procedure for low-quality answers exclusion did not differ across the two experiments, the null effect in Experiment 2B does not come from noise responses. In this way, our interim result from Experiment 2A and 2B is the following: the sensitivity of bare singulars to uniqueness is not uniform across configurations.

If bare singulars were pure existentials, uniqueness manipulation should never matter. Therefore, the No Uniqueness Hypothesis (37) appears empirically incorrect: we do find strong sensitivity to uniqueness in preverbal familiar bare singulars, and an even stronger interaction effect of uniqueness manipulation vs. noun type (bare nouns vs. *odin*-nouns).

The Strong Uniqueness Hypothesis (that bare singulars require uniqueness independently of familiarity or syntactic position; (35)) is supported only partially. Our results show that bare singulars can show strong sensitivity to uniqueness, but this effect does not shows up

uniformly across familiar vs. non-familiar contexts.

Finally, the Flexible Uniqueness Hypothesis (36) seems to be the closest descriptively: the uniqueness effect appears in preverbal and familiar bare singulars only. The most straightforward explanation of this asymmetry would be through lexical ambiguity: bare singulars are realized through *iota* when preverbal and familiar and through some sort of existential operation when postverbal and non-familiar. This view resembles Geist (2010)’s approach and would capture the facts descriptively.

However, such an analysis makes semantic predictions that are perhaps too strong: it predicts true existentiality in postverbal non-familiar bare singulars. If such nouns are bona fide existential quantifiers, they should freely allow reference to one out of several equally salient individuals of the same kind. Under a lexical ambiguity account, the postverbal bare singular would simply introduce an existentially bound variable, and its felicity should not depend on whether other individuals satisfying the nominal predicate are present in the context. This is why English indefinite singulars are felicitous in sentences with incompatible predicates:

- (45) a. A dog is sleeping and a dog is running around.
 b. $\exists x[\text{dog}(x) \wedge \text{is-sleeping}(x)] \wedge \exists y[\text{dog}(x) \wedge \text{is-running-around}(x)]$

Another reason to raise doubt about the existential nature of postverbal bare singulars comes from the elicitation data presented in (Soloveva et al., 2026), who claim that Russian bare nouns do not show a typical profile of indefinites. First, their scopal behavior is not that of a full-fledged existential quantifier; second, their data suggest that bare singulars in Russian generally have a uniqueness requirement even in their existential-like readings, discussed in more detail in Dayal (2026b).

Let us take a closer look at the contexts used in Experiment 2B. Initially, they were designed to be presentational. They included two types of situations: one describing someone talking about what happened on the news, and another one describing someone’s reaction

to an unfamiliar sound. Our contexts did not introduce a topic explicitly, and the question asked was *What did they say on the news?* for the ‘news’ stimuli and *What’s going on?* for the ‘unfamiliar sound’ stimuli. Effectively, the question targeted either the event reported in the news, or the source of a perceptually salient sound. This means that in both cases, the domain relevant for the interpretation of the bare nominal may have been narrowed to individuals that were event-relevant, and that domain did not necessarily include other individuals present in the broader situation. For example, in the context like (42) about one of the two raccoons escaping from the zoo, the raccoons that were present but did not escape may simply be outside the domain relevant for evaluating the felicity of the answer; similarly, in contexts like (44), non-barking dogs are irrelevant for evaluating an answer to the question that essentially means *What is this sound?*

This means that the results of Experiment 2B with the postverbal noun being interpreted as situationally unique:

- (46) a. Na ulice la-et sobaka.
 in street bark-PRS.3SG dog
 b. $\exists x \in D. \text{dog}(x) \wedge \text{is-barking}(x) \wedge \forall y[\text{dog}(y) \rightarrow y = x]$,
 where $D = \{\text{sound-making creatures}\}$

If so, then even in contexts that were globally non-unique, postverbal bare singulars may have been evaluated relative to a smaller, potentially singleton domain of contextually relevant individuals¹¹.

If this is true, and our contexts in Experiment 2B in fact had domains restricted to a singleton, accounting for the different results in Experiment 2A vs. Experiment 2B does not necessarily require bare singulars to be lexically ambiguous between definite and existential depending on syntactic position. Instead, it would reflect differences in how

¹¹Similar effects have been documented for overt definites in article languages. Reed (2016, 2020) argues that informational sufficiency and conversational relevance can license definite descriptions even in globally non-unique situations. Likewise, Srinivas et al. (2026) demonstrates experimentally that sufficiency of information can license a definite NP even when the referent is not unique, especially when the set of potential referents is relatively small.

the contextual domain is constructed in different syntactic configurations. In postverbal non-familiar contexts, the domain may be implicitly restricted to the event-relevant individual, masking potential uniqueness violations.

At this stage, both explanations are viable: the ambiguity explanation and the domain-restriction explanation. To distinguish between these possibilities, we must consider contexts in which domain restriction cannot happen. If postverbal bare singulars are truly existential, they should remain fully acceptable in such contexts, even if uniqueness is violated. If, however, they still have a uniqueness requirement that is masked by domain restriction in certain settings, then forcing reference to multiple individuals of the same kind should result in degradation in the acceptability of such sentences.

The next experiment is designed precisely to test this prediction.

2.4 Blocking domain restriction: incompatible predicates

In previous tasks, which involved simple predications with bare nouns vs. *odin* ‘one’-marked nouns, we found that Russian bare singulars are rated higher than Russian bare singulars marked with *odin* ‘one’ only when their referent is familiar and the word order is direct. In contrast, when their referent is non-familiar and the word order is inverted, uniqueness had no effect, and bare singulars were judged the same (and relatively high) no matter whether their referent was unique or not.

At face value, these results might suggest a lexical ambiguity: bare singulars are definite when preverbal and indefinite when postverbal. However, in elicited sentences with incompatible predicates, bare singulars are degraded both pre- and postverbally¹²:

- (47) Masha and Lena are playing with a kaleidoscope that shows different pictures.
Masha looks into the kaleidoscope, and sees two squirrels. Lena asks her: “What do you see there?”

¹²Also see examples provided in Soloveva et al. (2026).

-
- a. #**Belka** prygaet i **belka** spit.
 squirrel jumps and squirrel sleeps
- b. #Prygaet **belka** i spit **belka**.
 jumps squirrel and sleeps squirrel
 Intended reading: ‘A squirrel is jumping and a squirrel is sleeping.’

However, as discussed in the previous section, it is possible that in our previous task the domain was pragmatically restricted to singleton, making the uniqueness condition trivially satisfied in postverbal bare nouns. This is what motivates the final experiment in this chapter: if the reference to more than one individual is forced, like in contexts with incompatible predicates, but the bare noun is degraded, we can argue that under conditions where it is not possible to shrink the domain to singleton, the uniqueness requirement becomes visible.

2.4.1 Experiment 3: Design and materials

Similarly to Experiments 2A and 2B, Experiment 3 was a 2×2 acceptability-judgment task manipulating UNIQUENESS (+unique vs. -unique) and NOUN TYPE (bare noun vs. one-marked noun). Participants read short context scenarios and rated the two options presented simultaneously on a scale. The crucial difference was in the way the context scenario was constructed. To ensure that the context is presentational (thetic), the task was set up as a game with a View-Master – a toy that shows pictures one-by-one. The participants were presented the following instructions, supported by the visual material in :

- (48) Imagine that two girlfriends, Masha and Liza, are playing with a View-Master – a toy that looks like binoculars. When you look into this toy, you see a picture. There are different slides inside the toy that you can click through. Here’s what it looks like – you might remember them!

So, Masha and Liza are playing with a new View-Master. Masha looks into the toy and sees pictures, and Liza asks her what she sees in there. Masha clicks through

the pictures one by one and answers Liza.

You will be given a very brief description of what object or objects Masha sees in the toy. You will need to use a scale to rate two possible answers that Masha could give to Liza’s question. Pay close attention to how many objects, animals, or people Masha sees in the picture according to the description!

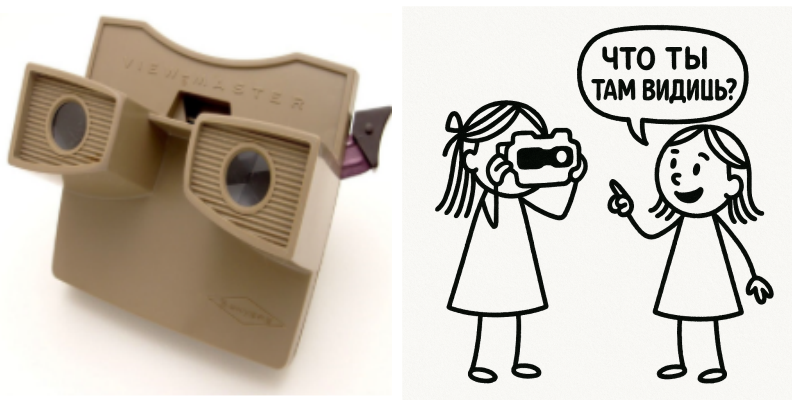


Figure 2.9: The visual material presented together with the instructions (48) in Experiment 3. The translation of the text on the right: “What do you see there?”

After the instruction block, participants were presented short contexts mentioning what objects Masha sees, and two sentences – potential Masha’s answers to Liza’s questions about what she sees. In this way, Experiment 3 is an acceptability-judgment experiment with joint presentation of both levels of NOUNTYPE, which is parallel to the design employed in Experiments 2A and 2B. However, the way different noun types were presented slightly differed from the previous tasks. In Experiment 2B, we saw that the ‘one’-marked nouns, our control condition, were rated low no matter the uniqueness manipulation. Pilot testing revealed that NPs with *odin* ‘one’ are independently degraded in postverbal position, especially under repetition. Since the priority was to have a well-formed and acceptable comparison for the treatment condition (bare singulars), we used preverbal ‘one’-marked NPs as the control.

The contexts manipulated uniqueness in the following way:

(49) -Uniqueness

Masha looks into the toy and sees two dogs. Liza asks: "What do you see there?"

- a. Na kartinke spit **sobaka** i begaet po krugu **sobaka**.
in picture sleeps dog and runs in circles dog
- b. Na kartinke **odna sobaka** spit i **odna sobaka** begaet po krugu .
in picture one dog sleeps and one dog runs in circles
Intended: 'In the picture, a/one dog is sleeping and a/one dog is running in circles'.

(50) +Uniqueness

Masha looks into the toy and sees a dog and a bunny. Liza asks: "What do you see there?"

- a. Na kartinke spit **sobaka** i begaet po krugu **zajčik**.
in picture sleeps dog and runs in circles bunny
- b. Na kartinke **odna sobaka** spit i **odin zajčik** begaet po krugu.
in picture one dog sleeps and one bunny runs in circles
Intended: 'In the picture, a/one dog is sleeping and a/one bunny is running in circles'.¹³

In this way, in -Uniqueness contexts, Masha saw two equally salient individuals of the same kind (e.g. two dogs). In +Uniqueness contexts, Masha saw two different individuals, e.g. a dog and a bunny, which made the dog in the context unique. The action that the two individuals were performing in the picture was not introduced in the written context, only in the response options.

Participants were asked to rate each of the possible answers on a scale from 1 to 100 with the labels hidden. The options (sentences with bare nouns vs. sentences with *odin*-nouns) were randomized. The experimental lists were organized in a way that if a participant saw the -Uniqueness version of a context, they would not be presented the +Uniqueness version of the same context. The screenshot of an experimental trial, corresponding to (50), is shown

¹³The tentative translations use *a*, because that would be the expected form in the non-familiar context. I do not assume that the translations of the stimuli reflect the felicity of the Russian stimuli.

Маша смотрит в игрушку и видит в ней собачку и зайчика.

Лиза: – Что ты там видишь?

Оцените, насколько подходяще звучит каждый из следующих возможных ответов Маши:

На картинке спит одна собачка и бегают по кругу один зайчик.



На картинке спит собачка и бегают по кругу зайчик.



Figure 2.10: Screenshot of an experimental trial in Experiment 3, corresponding to (50).

in Figure 2.10.

We prepared 8 pairs of contexts (resulting in 16 experimental items) paired by Uniqueness and 16 filler tasks. Each experimental list consisted of 8 experimental items, balanced by Uniqueness, and 16 filler tasks.

60 self-reported native speakers of Russian were recruited for this task. Some of the filler tasks were used to filter for inadequate responses, following the procedure similar to the first two experiments of the series. I used the same technique as in Experiments 2A and 2B to clean the data from unreliable responses. As a result, the data from 4 participants were removed from the final analysis.

2.4.2 Results

In contexts that force a non-unique interpretation of the referent, bare singulars were rated relatively low (mean raw rating $M=44.3$, 95% CI[42.2, 46.3]). In the same contexts, singular NPs modified by *odin* ‘one’ were rated much higher at 92.4, 95% CI[91.5, 93.3]. In contrast, when the context that supported the uniqueness of the referent, both alternatives received similarly high ratings. Bare singulars were rated on average at 81.5, 95% CI[79.9, 83.1], while *odin*-marked nouns received a mean raw rating of 79.0, 95% CI[77.5, 80.6]. Figure 2.11 represents raw mean ratings with standard errors by condition.

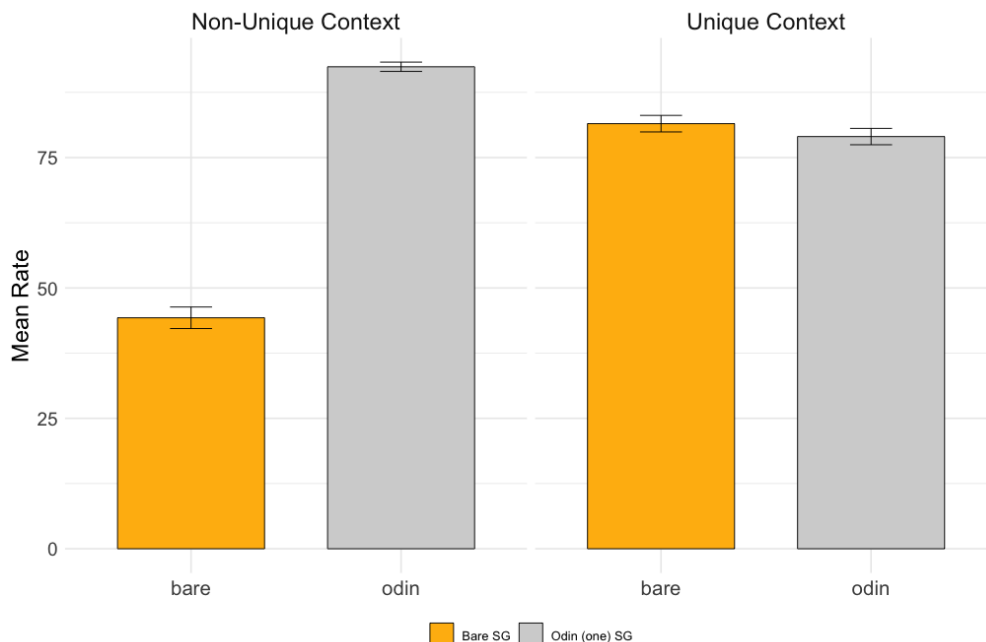


Figure 2.11: Mean ratings with standard error bars for each condition in Experiment 3.

Figure 2.12 visualizes the results using ratings standardized within each participant, which makes differences between conditions easier to compare while controlling for individual differences in scale use. This figure is informative in a way that it shows that the interquartile ranges of the answers are relatively narrow, and definitely much narrower than in Experiment 2B. This means that there was less variation in participants’ responses. Similarly, we see that the mean ratings for each condition lay on the median line, or are extremely close to it. This suggests that across condition, the distribution of ratings is symmetric, and the variability on either side of the average rating is similar. All in all, this plot shows that the results are robust: the participants largely agreed with each other and were not confused by the task.

To formally evaluate these patterns, we fit a linear mixed-effects model to z-scored (within the sample) ratings with **Uniqueness** and **NounType** as fixed effects, including their interaction, and random intercepts for **ResponseID** and **Item**: $\text{zscore} \sim \text{Uniqueness} \times \text{NounType} + (1|\text{ResponseID}) + (1|\text{Item})$. Statistical modeling revealed a significant interaction effect between **Uniqueness** and **NounType** ($\beta = -1.7, t = -18.34$). This means that the way

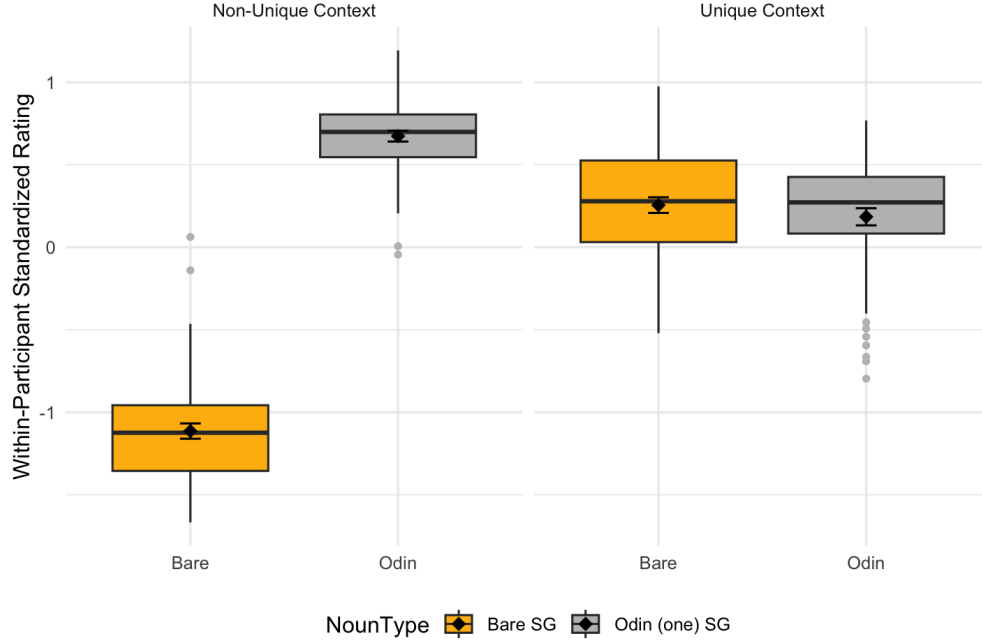


Figure 2.12: Within-participant standardized ratings for each condition in Experiment 3. The diamonds are means with standard errors, and the lines are medians.

context manipulation influenced acceptability was substantially different for bare singulars and *odin*-marked nouns. The strong positive effect of **+Uniqueness** observed in bare singulars is reversed in *odin*-marked nouns.

In order to clarify this interaction, we analyzed pairwise comparisons of estimated marginal means with Tukey adjustments. For bare singulars, **Uniqueness** manipulation had a strong and robust effect ($\beta = 1.25, t(834) = -19, p < .001$): on average, bare singulars were rated 1.25 standard deviations higher when their referent was unique, compared to contexts where the non-uniqueness of their referent was forced. This corresponds to a very large effect size (Cohen’s $D = 1.8$). For ‘one’-marked nouns, the effect went in the opposite direction and was smaller in magnitude ($\beta = 0.45, t(834) = 6.87, p < .001$). On average, *odin*-marked nouns were rated 0.45 standard deviations lower when their uniqueness was presupposed, compared to contexts where non-uniqueness was forced. This corresponds to a moderate effect size. The difference between the average ratings of bare singulars and *odin*-marked nouns in contexts supporting uniqueness did not reach statistical significance ($t(830) = 1.27, p = 0.58$).

The fixed effects (**Uniqueness** and **NounType**) explained about 36.6% of the variance in the model. An additional 15.5% of variance can be attributed to random intercepts, primarily reflecting difference between participants rather than experimental items. Residual analysis showed no systematic deviations from model assumptions: linearity was met, there were no extreme outliers, and residuals remained centered around zero across the range of fitted values. All of this suggests a relatively good fit of the model to the data.

Taken together, these results show that when non-uniqueness is enforced by incompatible predicates, the uniqueness requirement of bare singulars becomes clearly visible. We see that in such contexts bare singulars are significantly degraded, not only compared to a better alternative (a ‘one’-marked singular), but also relative to their own acceptability in similar configurations where uniqueness is presupposed. We also see that the observed effect of uniqueness in bare singulars is much stronger than in Experiment 2A, which involved familiar and preverbal bare singulars. The comparison the results of all the three tasks is presented in Figure 2.13.

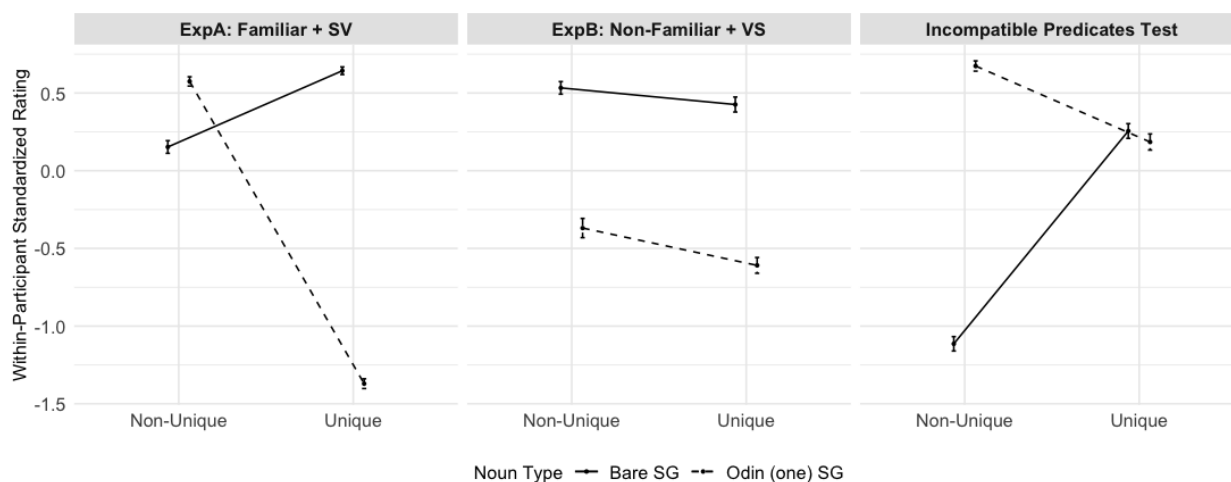


Figure 2.13: Interaction plots for Experiments 2A (Familiar + SV), 2B (Non-Familiar + VS), and 3 (Incompatible Predicates Test).

2.4.3 Consequences

The results of the Experiment 3 clarifies the results of Experiments 2A and 2B. Experiment 2B, which manipulated uniqueness in postverbal and non-familiar bare singulars, showed no visible uniqueness effect. Taken in isolation, this pattern is compatible with a lexical ambiguity account, on which preverbal and familiar bare singulars are interpreted like *iota*-expressions, and postverbal non-familiar bare singulars allow an existential interpretation.

Experiment 3, however, showed that the design of the stimuli in Experiment 2B contained a critical confound. By forcing simultaneous reference to two equally salient individuals within a single presentational sentence, Experiment 3 blocked the possibility of contextual domain restriction. Under these conditions, postverbal and non-familiar bare singulars became clearly degraded under forced non-uniqueness.

This result is problematic for a fully existential analysis of postverbal non-familiar bare singulars. If they were introduced by existential quantification, they would remain fully acceptable in contexts where two individuals of the same predicate are simultaneously described with incompatible predicates. English indefinite singulars behave exactly this way. This is not what we observed in Russian postverbal bare singulars.

Taken together, the three uniqueness-related experiments reported in this chapter suggest that the presence or absence of the uniqueness effect in our tasks did not track a lexical ambiguity between “definite” and “indefinite” bare singulars. Instead, the visibility of the effect depended on whether the context allowed domain restriction (resulting in the uniqueness requirement being trivially satisfied) or not. The syntactic position of the bare singular, along with its information-structural status, influences how that domain is constructed.

2.5 Interim summary: Uniqueness, domain restriction, and the status of bare singulars

This chapter set out to test whether Russian bare singulars impose a uniqueness requirement when familiar and when non-familiar. Our experiments converged on a single conclusion: Russian bare singulars are sensitive to uniqueness, but the visibility of this requirement depends on contextual factors.

Experiment 2A showed that when a bare singular subject is preverbal and its referent is familiar, violations of uniqueness lead to a sharp degradation. In these configurations, bare singulars behave like *iota*-expressions. By contrast, Experiment 2B demonstrated no visible effect of uniqueness in postverbal and non-familiar bare singular subjects.

As argued in section 2.3.4, the null result in Experiment 2B is better explained by domain restriction than by lexical ambiguity. Experiment 3 confirmed this suspicion. Its design forced simultaneous reference to two equally salient individuals in presentational contexts, thus blocking any possible domain restriction. In this configuration, postverbal and non-familiar bare singulars became clearly degraded when non-unique. This shows that postverbal non-familiar bare singulars are not purely existential. They are subject to a uniqueness constraint, too. The problem is that this constraint is easy to satisfy trivially, when the relevant domain is pragmatically narrowed.

Our results are incompatible with a fully existential analysis of bare singulars: if bare singulars were simply existential quantifiers, forced non-uniqueness should never reduce acceptability. However, this is exactly what we observed in Experiments 2A and 3. This rules out fully existential approaches to bare singulars like (Seres and Borik, 2021).

The implications for various ambiguity theories are more nuanced. First of all, our results are compatible with bare singulars being *iota* expressions, as the very definition of *iota* is tied to the relevant domain. However, our results are also compatible with the information-structure restricted ambiguity account proposed by Geist (2010). The absence

of a uniqueness effect in Experiment 2B, combined with its reappearance in Experiment 3, in principle aligns with a view on which non-topical bare singulars are ambiguous between *iota* and existential construal. However, the results suggest that the existential construal available to comment bare singulars must be more restricted than Geist assumes. Within the set of comment bare singulars, the choice between *iota* and existential construal does not appear to be unweighted. Instead, comment bare singulars seem to default to a uniqueness-based construal, unless independent factors license an existential reading.

Another important caveat is that we have not ruled out the possibility that the acceptable cases in Experiment 2B involve existential quantification over a restricted domain. In such settings, restricted existentials may be empirically indistinguishable from *iota*-expressions. This possibility would require additional machinery, but it cannot be ruled out. I do not address this issue directly; however, in Chapter 5 I will discuss the relevant diagnostics that may shed light on this question, which come from downward-entailing environments.

In the following chapter, we will switch to the plural counterpart of uniqueness – maximality.

3 Bare plurals and maximality in episodic contexts

While singular definite descriptions are associated with uniqueness, plural definite descriptions are typically taken to refer to the maximal plural entity in the relevant domain that satisfies the description of the predicate. In this sense, maximality can be considered the “plural” counterpart of uniqueness, often formalized as the σ operator (the generalized version of *iota* introduced by Link (1983) (1-a); a simplified version suggested by Ogihara (1987) is in):

$$(1) \quad \text{a.} \quad \sigma x Px := \iota x (*Px \wedge \forall y (*Py \rightarrow y \sqsubseteq x)) \quad (\text{Link, 1983, D.14})$$

$$\text{b.} \quad \sigma x.P(x) = \cap x [P(x) \wedge \forall y [P(y) \rightarrow y \oplus x = x]] \quad (\text{Ogihara, 1987})$$

a maximal plural individual consisting of all individuals satisfying P¹

Across languages with and without articles, bare plurals exhibit both existential-like and kind readings. In episodic contexts, they are often analyzed as introducing existential quantification, either directly or via Derived Kind Predication (DKP; see section 1.3.2). At the same time, bare plurals in many languages display properties similar to definite plurals, including sensitivity to maximality.

¹Here $*$ is the pluralization closure, extending a predicate to sums and pluralities, and $\sqsubseteq$ denotes the parthood relation on the mereological lattice. The maximality condition $\forall y (*Py \rightarrow y \sqsubseteq x)$ ensures that x is not merely a sum of P -individuals but the greatest such sum, which should be unique when the lattice has a supremum for P .

In this chapter, I take maximality to be diagnostic for (in)definiteness of bare plurals in episodic contexts. The main empirical question is whether episodic bare plurals are freely compatible with a non-maximal interpretation, as expected under an existential analysis, or they show preference for maximal interpretations similar to definite descriptions. Another question relevant in this chapter is whether bare plurals in a system without articles, like Russian, differ in this respect from bare plurals in a language with articles, like English.

To address these questions, I analyze the data from acceptability judgments that contrast maximal and non-maximal interpretations in a presentational context. The primary evidence comes from a controlled experiment on Russian bare plurals in the subject position, with comparison to English bare and definite plurals. Additionally, I consider a smaller dataset on the same types of plurals in the object position.

If episodic bare plurals are fundamentally existential (even if their existentiality is derivative from DKP), they should be compatible with non-maximal readings. If, by contrast, they show definite-like behavior, maximal interpretations should be preferred. The results presented in this chapter bear on this distinction.

3.1 Theoretical background

3.1.1 The existential component in the semantics of bare plurals

As overviewed in Chapter 1, the neo-Carlsonian approach, as proposed by Carlson (1977a) and developed in Chierchia (1998), treats bare plurals as kinds, not just in combination with kind-level predicates but also in the case of object-level predication. This means that bare plurals are arguments of type e , derived from the lexical noun predicate of type $\langle e, t \rangle$ via the kind-forming operator *nom* (Chierchia, 1998).

In order to get to the predictions of this approach about maximality, let us go back to how episodic bare plurals are derived within the neo-Carlsonian approach. Since bare plurals are treated as kinds, the use of bare plurals with kind-selecting predicates is straightforward.

Such predicates accept a kind-denoting argument directly:

$$(2) \quad \llbracket \mathbf{Dogs} \text{ are widespread} \rrbracket = \text{widespread}(\cap \text{dogs}) \quad (\text{Chierchia, 1998, p. 363})$$

Combining kinds with object-level predicates is more challenging. Chierchia proposes the rule of Derived Kind Predication (DKP), which mediates between kind-level arguments and object-level predicates. DKP maps a kind k to existential quantification over its instantiations at a given index. The derivation in (3) asserts that there exist instantiations of the lion-kind that are ruining the garden in the relevant situation:

$$(3) \quad \begin{aligned} \llbracket \mathbf{Lions} \text{ are ruining my garden} \rrbracket &= \\ &= \text{ruining-my-garden}(\cap \text{lions}) \iff (\text{via DKP}) \exists x[\cup \cap \text{lions}(x) \wedge \text{ruining-my-garden}(x)] \\ &(\text{Chierchia, 1998, p. 364}) \end{aligned}$$

A direct consequence of DKP is that the existential component associated with episodic bare plurals is structurally local. Existential quantification is introduced only at the point where the predicate applies to instantiations of the kind. This means that unlike ordinary indefinites, bare plurals are not flexible in scope. The obligatory narrow scope of bare plurals has been taken as a diagnostic of kind-based analyses, since it clearly distinguishes them from ordinary existentials.

In the earlier versions of the neo-Carlsonian approach, maximality is not imposed to bare plurals. While the existential quantification ranges over the instantiations of the kind, nothing in the DKP proposed by Chierchia (1998) suggests that these instantiations should exhaust the relevant domain, or that they should form the maximal plural individual satisfying the nominal predicate. In this respect, Chierchia's approach does not predict episodic bare plurals to have a maximality interpretation; if anything, they should pattern with indefinites rather than with definites.

Krifka (2003) proposes an alternative account that is different conceptually but gives the

same predictions about maximality. On that account, the existential interpretation of bare plurals comes from a local existential closure directly from the property type $\langle e, t \rangle$, without going through *nom* and DKP first. This yields an existential reading with obligatorily narrow scope. For example, (4-a) is analyzed as asserting the existence of a plurality of dogs that satisfy the predicate:

- (4) a. **Dogs** are barking.
 b. $\lambda w \exists x. [\cup \text{dogs}(w)(x) \wedge \text{barking}(w)(x)]$ the result of Chierchia's DKP
 c. $\lambda w \exists x [\text{dogs}(w)(x) \wedge \text{barking}(w)]$ Krifka's simplified account

Importantly, nothing in either Chierchia's derivation (4-b) or Krifka's derivation (4-c) requires that x correspond to the maximal sum of all contextually relevant dogs. Suppose there are five dogs in the yard, and only two of them are barking. Under the DKP analysis, (4-a) is true if there exists a plurality of dog-instantiations such that if they are barking. This means that in the non-maximal situation with two out of five dogs barking, (4-a) is predicted to be true. Under the local existential analysis, (4-a) is predicted to be true in that situation as well: there is no requirement that the sum of all five dogs satisfy the predicate *barking*.

At the same time, Krifka does observe some uses of bare plurals that have effectively maximal interpretations. He analyzes them as kinds, claiming that certain predicates can apply to either kinds or regular objects; but these uses cannot be considered fully episodic:

- (5) $\llbracket \mathbf{Rats} \text{ reached Australia in 1770} \rrbracket =$ Krifka (2003, p.199)
 $= \lambda w. [\text{reach-Australia-in-1770}(w)(\cap \llbracket \text{rats} \rrbracket(w))]$ $=$
 $= \lambda w. [\text{reach-Australia-in-1770}(w)(\iota \lambda x \exists n [\text{rat}(w)(n)(x)])]$

In this way, the empirical predictions of both Chierchia (1998) and Krifka (2003) with respect to maximality in episodic bare plurals are the same, and they are the same for Russian

and English bare plurals. Under either of these approaches, bare plurals should be compatible with non-maximal interpretation. If episodic bare plurals systematically resist non-maximal interpretations, this would suggest that something beyond existential quantification and cumulativity is involved in their interpretation. In the following sections, I introduce two approaches that consider this possibility.

3.1.2 Dayal’s revision of Derived Kind Predication

Dayal (2013) revisits the neo-Carlsonian analysis of episodic bare plurals and argues that DKP overgenerates existential readings. Specifically, she demonstrates that the interpretation of bare plurals is more constrained than the DKP analysis would predict. One reason to reconsider existential quantification as part of lexical meaning of bare plurals comes from their *iota*-like uses, first noted by Condoravdi (1994, 1997):

- (6) There was a ghost on campus. **Students/The students** were afraid. cited from
Dayal (2013, p. 57)

Essentially, Dayal (2013) proposes to eliminate existential quantification from DKP altogether. On the original version of DKP, episodic bare plurals introduce existential quantification over instantiations of a kind:

- (7) Derived Kind Predication (Chierchia, 1998)
If P applies to objects and k denotes a kind, then $P(k) = \exists x[\cup k(x) \wedge P(x)]$.

Dayal proposes two modifications to DKP. First, she suggests eliminating existential quantification from it, and having the sort mismatch between the object-level predicate and the kind term repaired by taking the extension of the kind, namely, the maximal entity that instantiates the kind at the relevant index:

- (8) DKP-Modification 1 (Maximality) (Dayal, 2013, p. 68)

If $P(s)$ applies to objects and k is a kind, then $P(s)(k) = P(s)(k_s)$,
 where k_s is the extension of the kind at s .

The suggestion to remove existential quantification from the denotation of episodic bare plurals comes from the observation that the referents of a bare plural cannot be listed (9-a). In this, bare plurals differ from indefinite plurals (9-b). This contrast is expected if bare plurals refer to a single entity rather than quantify over instantiations of a kind.

- (9) a. **Dogs**, (#namely Spotty and Rover), are barking.
 b. **Some dogs**, namely Spotty and Rover, are barking.

The second modification, Proper Widening, states that the instantiations of the kind in the widened domain must properly include the instantiations in the base situation:

- (10) DKP-Modification 2 (Proper Widening) (Dayal, 2013, p. 68)
 If $P(s)$ applies to objects and k is a kind,
 then $\llbracket P(s)(k) \rrbracket = 1/0$ if $\llbracket \exists s'[s < s' \wedge P(s)(k_{s'})] \rrbracket = 1/0$,
 where $\exists x[x \leq k_{s'} \wedge \neg \text{in-}s(x)]$ and is undefined otherwise.

This modification was proposed based on the fact that English bare plurals tend to be maximal within a contextually relevant domain. On the standard test for maximality, the incompatible predicates diagnostic, bare plurals are typically degraded as much as English definite plurals:

- (11) Dayal (2013)
 a. #**Dogs** are barking and dogs are sleeping.
 b. #**The dogs** are barking and **the dogs** are sleeping.

However, if the context is such that the scene can be broken down into sub-domains over

instances of a kind denoted by the bare plural, repeated bare plurals in contexts with incompatible predicates are felicitous:

- (12) *Context: live report from a disaster scene* (Dayal, 2013, p. 76)
 People are screaming, **people** are jumping out of the windows, **people** are trying
 to force doors open...

On the newly proposed version of DKP, bare plurals denote a unique maximal plural individual, defined over a contextually widened domain. According to Dayal, examples like (12) involve multiple sub-domains, and within each sub-domain, the plural individual denoted by the bare plural is maximal. Apparent existential interpretations of bare plurals arise not because DKP introduces an existential quantifier, but because the maximal plural individual can support weaker construals in context. A predicate may hold of a salient representative subgroup of the maximal entity without entailing that it holds distributively of every atomic member. In this way, episodic bare plurals can be true in contexts where not all the individuals satisfy the predicate and preserve maximal reference at the level of denotation.

What is important about Dayal’s revision is that maximality is treated as a part of the semantic core of episodic bare plurals. Non-maximal interpretations are derived pragmatically or via plural predication mechanisms. This is the main difference with both the classic DKP, which assumes that existential closure over instantiations of a kind is the primary source of episodic interpretation, and the pure local existential closure approach of Krifka. Unlike those accounts, Dayal predicts episodic bare plurals to pattern with definite plurals in showing a preference for maximality.

A more recent refinement of the account to English bare plurals, developed in Chierchia (2022b), retains the kind-based derivation but derives maximality-like effects through exhaustification of subdomain alternatives; this account is discussed in more detail section 3.5.2 in light of the experimental results. For now what is relevant is that it also builds maximality on top of Derived Kind Predication.

3.1.3 Bare plurals as weak definites

Dayal’s proposal reanalyzed bare plurals within the neo-Carlsonian approach, maintaining the idea that episodic bare plurals are derived from kinds. There are also approaches that go further and align bare plurals directly with definite descriptions. In particular, Berulava and Mayr (2024) argue that in languages without articles, bare plurals are best analyzed not as existential indefinites and not as kind-denoting expressions, but as weak definites. On this view, bare plurals contribute maximality and existential presupposition. Apparent non-maximal interpretations arise from independently motivated mechanisms of homogeneity and domain restriction.

Berulava and Mayr start from the observation that in six languages without articles, including Russian, bare plurals do not behave like existential indefinites. They resist non-maximal readings in contexts where plural definites resist them. Bare plurals in systems like that also pattern with definites with respect to domain restriction. In a scenario like (13), if at least one door is enough to reach the safe, both English definite plurals and Russian bare plurals are licensed.

- (13) A bank robbery is to take place. There are four doors, each of which leads to the safe. Last night, only one door was locked. John asks Bill — the burglar — if he could reach the safe. Bill says:
- a. Yes, **the doors** were open.
 - b. Yes, **some (of the) doors** were open.
 - c. Da, dver-i by-l-i otrkyt-y.
yes door-PL be-PST-PL open-PL

In a scenario like (14), where one needs to open all doors in order to reach the safe, neither the English definite plural nor the Russian bare plural is licensed.

- (14) A bank robbery is to take place. There are four doors that must be passed to reach

the safe. Last night, one door was locked. John asks Bill — the burglar — if he could reach the safe. Bill says:

- a. #No, but **the doors** were open.
- b. No, but **some (of the) doors** were open.
- c. #Net, no **dver-i** by-l-i otrkyt-y
no but door-PL be-PST-PL open-PL

Based on these observations, Berulava and Mayr propose that bare plurals denote maximal plural individuals, structurally similar to σ -denotations. However, unlike strong definites, bare plurals do not encode maximality in the same way. Instead, they presuppose the existence of a relevant domain and refer to the maximal plurality within that domain.

Similar to Dayal, Berulava and Mayr suggest that apparent existential readings of bare plurals do not come from existential quantification. Instead, they attribute such readings to the interaction of two independent mechanisms. The first is the weak definite operator THE, which introduces a contextually supplied restriction $g(C)$ that narrows the domain of quantification to a relevant subset of individuals, while its assertive component contributes maximality over that restricted domain. In this way, the maximal plurality may be defined over a contextually restricted set of individuals. The second mechanism is related to homogeneity: when a distributive predicate applies to a plural individual, speakers often tolerate situations in which the predicate holds of most, but not all atomic parts, which weakens the maximality requirement. In such cases, the sentence may not be strictly true in a bivalent sense, but can still be judged as acceptable if it is informative enough for the discourse at hand. The interaction of these effects is what we observe in a situation like (13): there occurs an impression of an existential interpretation, even though the underlying denotation remains maximal.

Berulava and Mayr claim that the weak definite analysis applies to bare plurals, at least in systems without articles. They do not derive any explicit predictions for bare plurals in languages with definite plurals, like English. For bare plurals in languages without articles,

like Russian, the predictions are clear. Since bare plurals denote maximal plural individuals, they are expected to display a systematic preference for maximal interpretations. Apparent existential interpretations should be restricted to contexts where the discourse goal can be satisfied by a proper subset of the relevant individuals.

In this way, the weak definite analysis of bare plurals has the following predictions: bare plurals should be judged as highly acceptable in maximal contexts. In non-maximal contexts, judgments should be lower, but not necessarily categorically rejected.

3.1.4 The indefiniteness hypothesis

A different line of work argues that bare plurals, and bare nouns in general, are fundamentally indefinite. On this view, they introduce a variable over individuals, and their interpretation is determined by the interaction of semantic underspecification with discourse and information structure. For Russian, this approach has been developed in Seres et al. (2019) and Seres (2020). The main claim is that bare nominals are interpreted existentially by default, while definite and generic readings arise through pragmatic strengthening.

The existential approach does not treat maximality as part of the core semantic of episodic bare plurals. Seres observes that generic bare plurals exhibit maximality, but attributes it to the semantics of genericity rather than to the bare plural itself. According to Seres, generic sentences involve quantification over a contextually restricted domain of instances, and the generic operator GEN evaluates the predicate with respect to all relevant cases. This can give rise to domain-exhaustive interpretations, which superficially resemble maximality as introduced by the σ -operator, but is essentially not inherent maximality.

In this way, the presence of maximality effects in generic bare plurals does not extend to episodic contexts, and episodic bare plurals are expected to be compatible with non-maximal situations. In this respect, the indefiniteness hypothesis is compatible with approaches such as Krifka (2003), where episodic bare plurals undergo local type-shifting to an existential. It just goes further in treating the existential interpretation as the default and in deriving

definite readings through pragmatic strengthening rather than through type-shifting.

3.2 Maximality as a diagnostic

The theories that I presented in 3.1 differ in how exactly they derive episodic bare plurals. However, there is a single dimension that divides them into two groups: whether they predict bare plurals to be compatible with non-maximal interpretations or not. Both the classic neo-Carlsonian approach (Chierchia, 1998; Dayal, 2004) and the type-shifting approach of Krifka (2003) treat bare plurals as existential quantifiers at the end of the day (be that over instantiations of a kind or over individuals). This means that a sentence with a bare plural subject is predicted to be true whenever there exists a plurality satisfying the predicate of the noun and of the VP. There is no requirement for all contextually relevant individuals to be included in that plurality, which means that non-maximal interpretations should be acceptable.

A similar prediction follows under the indefiniteness hypothesis (Seres et al., 2019; Seres, 2020). If bare plurals are semantically indefinite and never realized through *iota*/ σ , then maximality is not part of their core semantics. Non-maximal interpretations should remain generally available in episodic sentences.

Approaches that assign definite-like semantics to bare plurals make a different prediction. On Dayal (2013)’s revision of DKP, and Chierchia (2022b)’s dynamic account, bare plurals denote maximal plural individuals, defined as the extension of a kind at a given index. The weak definite analysis of Berulava and Mayr (2024) suggests that bare plurals denote maximal pluralities within a contextually relevant domain. All these analyses suggest that maximality is part of the semantic contribution of the bare plural per se. This means that non-maximal interpretations should be dispreferred, and bare plurals are expected to pattern with definite plurals in showing sensitivity to maximality manipulation.

It is important to mention that even under maximality-based accounts, non-maximal

bare plurals are not necessarily predicted to be completely infelicitous. Domain restriction and homogeneity may license intermediate judgments. So the empirical evidence we will be looking for is relative acceptability rather than categorical acceptability.

The present study tests the predictions of these two groups of theories about maximality in bare plurals. The task involves felicity judgments in presentational contexts, where (non)-maximality is inferred through visual stimuli. Our main focus will be on episodic bare plurals in Russian, with comparison to English bare and definite plurals. The comparison with English definite plurals serves as a point of reference rather than an independent object of investigation: overt definites are standardly analyzed as maximal. The behavior of English bare plurals, though, is not completely clear and still debated in the literature. However, there are contexts in which Russian and English bare plurals do not align. For example, in certain bridging configurations the English bare plural is infelicitous (15-a), but the Russian bare plural is perfectly fine (15-b):

- (15) a. I bought a car. The wheels/ *Wheels need to be replaced. Dayal (2013)
 b. Ja kupi-l-a mašin-u. Kolēs-a nado zamenit'.
 I buy-PST-F.SG car-ACC wheel-PL needed.to change.INF
 ‘I bought a car. The wheels need to be replaced.’

This suggests that the two systems may differ in how reference is established in the absence of articles. Therefore, the comparison of Russian and English bare plurals is partly exploratory. It compares bare plurals in a system with no articles whatsoever (Russian) to bare plurals in a system with a dedicated definite plural (English).

The main experiment (Experiment 4) investigates preverbal plural subject NPs. In addition, I also consider a smaller dataset involving object plural NPs, which provides a preliminary extension of our study to a different syntactic environment.

3.3 Experiment 4: Subject position

So far we have identified that maximality is the parameter that distinguishes between competing analyses of bare plurals. Existential approaches and indefinite-based approaches predict that bare plurals should remain compatible with non-maximal interpretations, whereas definite-like approaches predict a preference for maximal interpretation. Experiment 4 tests these predictions under controlled settings for plural subject NP, which are evaluated in contexts that contrast maximal and non-maximal interpretations. Russian bare plurals are compared to English bare plurals and English definite plurals.

The contexts are set up as presentational and essentially topicless, since our goal is to provide a conservative ground for potential maximality effect. If bare plurals show sensitivity to maximality in this environment, this would suggest that such effects are not limited to marked information structure configurations, such as topic or narrow focus.

3.3.1 Methods, materials, and participants

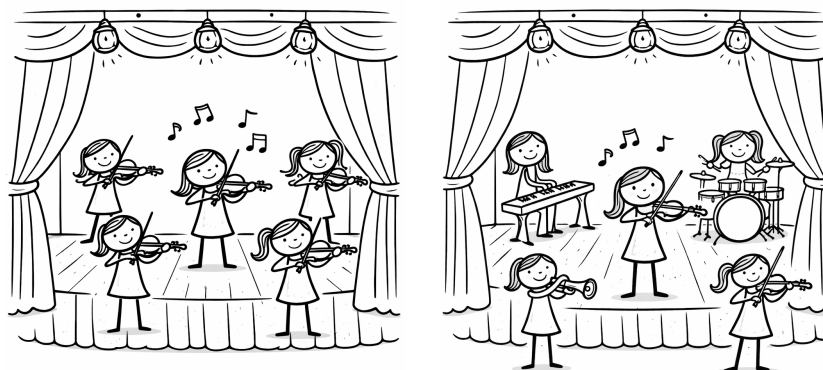
The goal of Experiment 4 was to test whether Russian bare plurals and English bare plurals prefer a maximal interpretation, and if yes, whether the magnitude of this preference differs from English definite plurals. To do this, we compared the acceptability of sentences with plural subjects in two types of visual contexts: maximal and non-maximal.

The Russian version of the experiment did not manipulate the form of the subject NP: it was always a preverbal bare plural. The English version of the experiment included definiteness as an experimental condition, and tested parallel sentences with definite vs. bare plural subjects. In this way, the Russian task only compared ratings in 2 conditions, and the English task extended the design to a 2×2 manipulation crossing Maximality (maximal vs. non-maximal) with Definiteness (bare vs. definite). An example of the set of tested sentences is given below ((25) for Russian and (26) for English):

- (16) Devočki igraju-ut na skripk-e.
 girl-PL play-PRS.3SG on violin-INSTR
 ‘The girls/Girls are playing the violin.’

- (17) a. The girls are playing the violin.
 b. Girls are playing the violin.

Maximality was manipulated visually. Experimental sentences were presented with two different pictures. In the +MAXIMAL condition, all the individuals present in the picture satisfied the description of both the noun predicate and the VP; in the case of (25)-(26), the picture depicted five girls, all of whom were playing the violin. The -MAXIMAL condition had all of the individuals satisfy the predicate of the noun, but only some of them satisfying the VP. In the case of the examples above, the -MAXIMAL picture would present five girls playing different instruments, and two of them would be playing the violin. These pictures are presented in figure 3.1.



(a) [+MAXIMAL]

(b) [-MAXIMAL]

Figure 3.1: Experiment 4 (subject position): Examples of the maximal (a) and non-maximal (b) picture stimuli.

The procedure was an acceptability task framed as a language game (Braine and Romain,

1981; Papafragou and Musolino, 2003; Guasti et al., 2005). At the beginning of the task, participants were presented the following prompt:

- (18) Ann and Diana have a friend Ellie, who is not an English/Russian speaker. They help Ellie to learn English/Russian by playing a language game with her. Ellie takes a picture, looks at it, and says a sentence that she thinks is good to describe what's happening in the picture. The girls correct her, saying how good Ellie's sentences are.

Then participants were presented a picture, a target sentence (an alleged Ellie's response describing the picture), and a slider bar, displayed simultaneously. The picture displayed either a maximal or a non-maximal scenario. Using the slider bar, participants rated how natural Ellie's response sounded to them. The slider bar was not labeled in intermediate points, and the start of the slider was fixed in the middle to avoid positional bias. A screenshot of an experimental screen is given in Figure 3.2.

A total of 12 experimental pictures was prepared for the task (6 pairs of $\pm$ MAXIMAL). An experimental item consisted of a picture and a sentence describing the scene. This resulted in 12 experimental items in the Russian version of the task (6 sentences, each tested in a maximal and non-maximal condition), and 24 experimental items in the English version of the task (6 sentences with bare plurals + 6 sentences with definite plurals, each tested in the maximal and non-maximal condition). The items were organized in experimental lists (2 lists for the Russian task, and 4 lists for the English task). The lists used a Latin square design: each participant saw every experimental sentences only one picture from the $\pm$ MAXIMAL, but across the two lists each sentence appeared once in each maximality condition. Each participant saw 6 experimental trials in total, interspersed with 12 filler items. This ensured that the maximal and non-maximal picture conditions were represented equally across the experiment while avoiding repetition of the same item within a participant.

30 self-reported monolingual Russian speakers and 48 self-reported monolingual English



Figure 3.2: Experiment 4 (subject position): Example experimental trial screen, featuring a sentence with an English bare plural.

speakers were recruited through Prolific. The median completion time was 5 minutes for both Russian and English versions. Filler items also served as catch trials to screen participants for attentiveness to the task. Catch trials came in three groups: “good” fillers that were expected to be rated high, “medium” fillers that were partially acceptable, and “bad” fillers that were expected to be rated very low. Participants were excluded if their average ratings for each filler group did not reflect the expected ordering, or if the difference between the highest and the lowest filler group ratings was smaller than 1 SD. This procedure excluded the data from 6 participants from the Russian dataset, and the data from 6 participants from the English dataset.

3.3.2 Results

All three plurals (Russian bare plurals, English bare plurals, and English definite plurals) demonstrated higher ratings in the maximal condition. Russian bare plurals were rated on

average as 8.34, 95% CI[7.83, 8.85] in the maximal condition and as 4.11, 95% CI[3.39, 4.83]. English bare plurals and English definite plurals demonstrated similar ratings as well: in the maximal condition, English bare plurals were rated as 8.83, 95% CI[8.37, 9.29], and English definite plurals were rated as 8.9, 95% CI[8.37, 9.45]. In the non-maximal condition, the average ratings of both English bare and English definite plurals dropped to 5.87, 95% CI[4.95, 6.79] and 5.42, 95% CI[4.65, 6.19], respectively. The raw means with standard error bars are visualized in Figure 3.3.

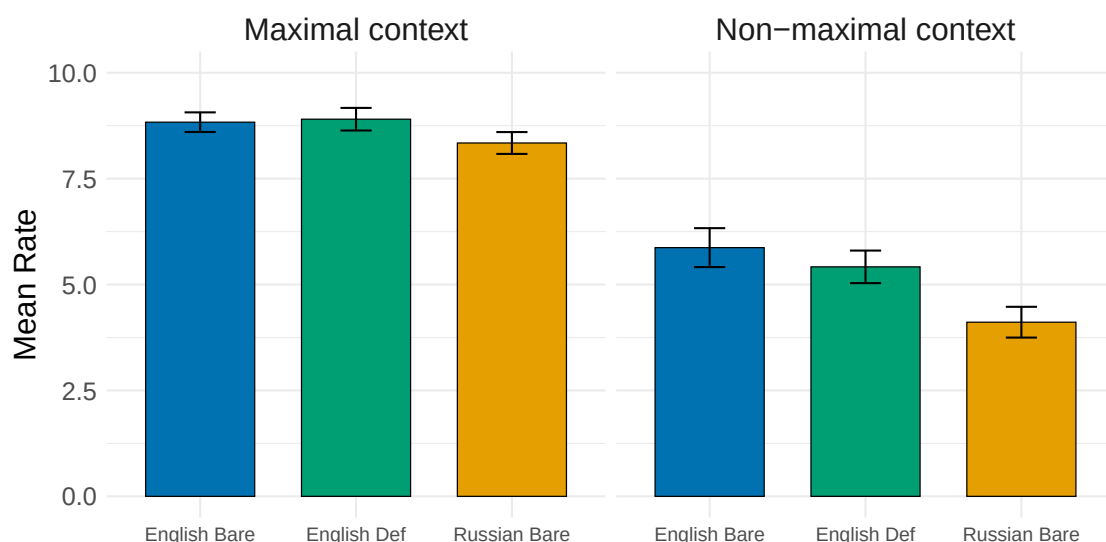


Figure 3.3: Experiment 4 (subject position): Mean rates with standard error bars of English bare plurals, English definite plurals, and Russian bare plurals in maximal and non-maximal contexts.

Figure 3.4 shows the same results standardized within participants. We can see that in the maximal condition, the means are lower than the medians, which generally means that most people rated all plurals in maximal contexts very high, but there are several extreme low ratings (visualized as grey dots in the plot) that influenced the means. In the non-maximal contexts, the situation is the opposite: means are generally higher than medians, which means that all plurals were rated generally low, but sporadic outliers influenced the means.

For statistical analysis, the Russian and the English datasets were combined into one and the rates were standardized. A linear mixed-effects model was fit to the data to predict

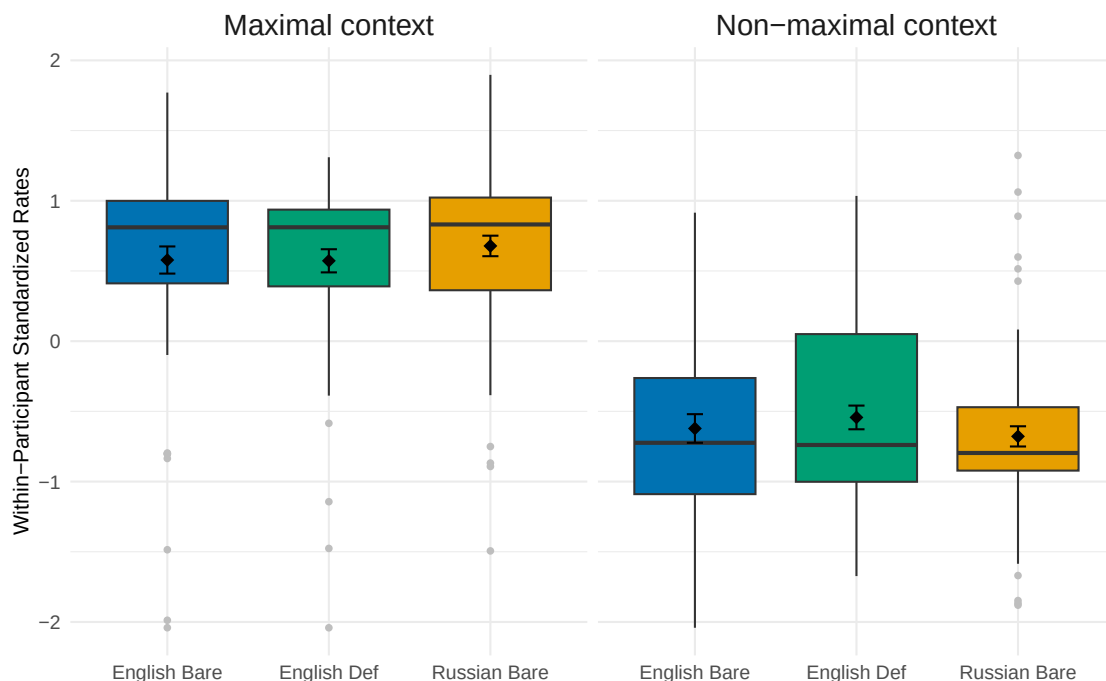


Figure 3.4: Experiment 4 (subject position): Distributions of ratings, standardized within each participants. Diamonds are means with standard error bars; middle lines are medians.

standardized scores from the maximality parameter, the plural type (Russian bare plural vs. English bare plural vs. English definite plural), and their interaction. The model included random intercepts for experimental items and participants, as well as by-participant random slopes for maximality: $\text{zRate} \sim \text{Maximality} * \text{PluralType} + (1 + \text{Maximality} | \text{ResponseId}) + (1 | \text{Item})$. This structure of the model allowed participants to differ in both their overall rating level and in how strongly they responded to the maximality manipulation. The structure also reflects the structure of the combined dataset, in which maximality was manipulated within participants, but a subset of participants only rated Russian bare plurals, while the rest of participants rated both English definite and English bare plurals.

The model showed a strong main effect of maximality manipulation: across different plural types, on average, the ratings in non-maximal condition were 1.03 SD lower than in the maximal condition ($t(105) = -6.84$, $\text{SE} = 0.15$, $p < 0.001$). However, there was no significant difference in ratings across Russian bare plurals, English bare plurals, and English definite plurals, and no significant interaction effect between **Maximality** and **PluralType**.

Predicted z-scored ratings by this model are visualized in Figure 3.5.

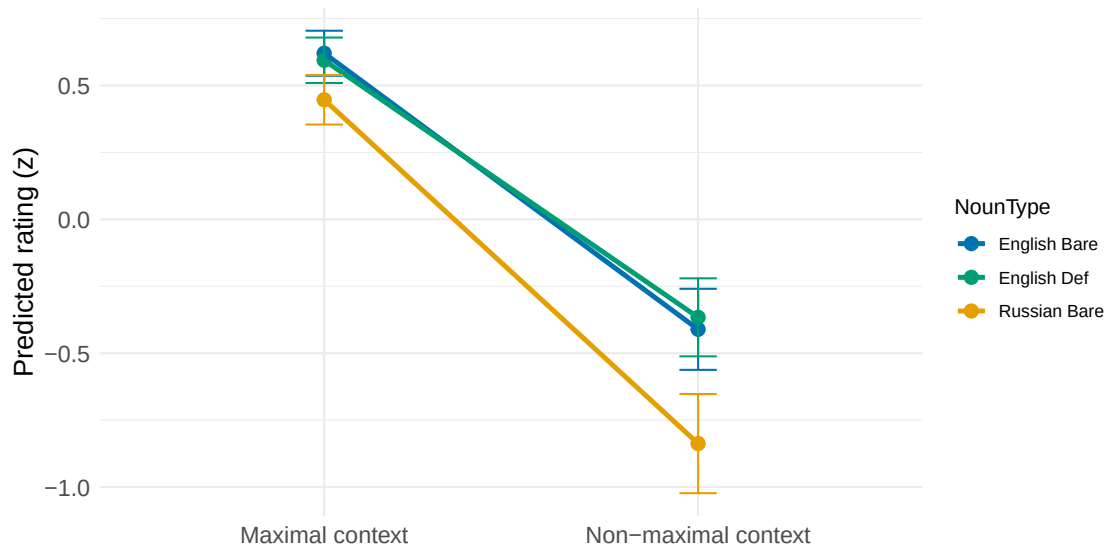


Figure 3.5: Experiment 4 (subject position): Mean predicted ratings (z-scored) for sentences describing maximal and non-maximal contexts across the three noun types tested.

Fixed effects explained about 32% of the variance, and random effects explained additional 42% of the variance, which shows that the model was a good fit to the data. The analysis of variance components showed that most random variation came from the by-participant slope for **Maximality**, which means that participants differed significantly in how strongly they responded to the maximality manipulation. Participant intercept variance was smaller, which means that the baseline ratings of the participants differed less, and item-level variance was negligible.

3.3.3 Interim discussion

Our goal in Experiment 4 was to test whether bare plural subjects are sensitive to maximality in episodic presentational contexts. The design contrasted maximal and non-maximal interpretations, evaluating the same sentences with each of them. The comparison with English definite plurals provided a reference point for interpreting the magnitude of the effect.

Overall, we see a clear effect of maximality. Across different types of plurals we tested (Russian bare plurals, English bare plurals, and English definite plurals), sentences were

rated higher in maximal than in non-maximal conditions. This indicates that non-maximal interpretations were indeed systematically dispreferred in our experimental design. At the same time, non-maximal contexts were not categorically rejected, and received intermediate ratings rather than uniformly low ones; to compare, some of our intentionally infelicitous catch trials received average ratings as low as 1–2, which is significantly lower than the ratings in the non-maximal conditions we observed. This suggests that maximality is not a strict truth-conditional requirement, but rather a preference reflected in relative judgments.

The observed pattern is not straightforwardly predicted by the classic DKP analysis of bare plurals that involves existential quantification, or the indefinite-based analysis of bare plurals along the lines of Seres et al. (2019). If bare plurals involved existential quantification, sentences should be true as long as some relevant plurality of individuals satisfies the predicate.

On the other hand, our results are consistent with approaches that treat bare plurals as definite-like (Dayal, 2013; Chierchia, 2022b; Berulava and Mayr, 2024). Bare plurals, both Russian and English, behaved nearly identical to English definite plurals with respect to our experimental manipulation. At this point, our data is compatible with either the Maximality amendment to DKP proposed by Dayal (2013), or the weak definite analysis proposed in Berulava and Mayr (2024); we will come back to a more detailed discussion of the pros and cons of each approach with regard to our data in general discussion.

The comparison between bare plurals and definite plurals does not show a strong difference in how sensitive they are to the maximality manipulation. This null effect should be interpreted with caution. First, Russian bare plurals were evaluated by a different population. Second, although the participants of the English task were shown both bare and definite plurals across the experiment, these conditions were not directly compared within the same item. Each sentence-picture pair was displayed in only one condition for a given participant, which means that Definiteness was evaluated across different items. This can potentially reduce sensitivity to differences between definite and bare plurals.

Finally, it is important to keep in mind that Experiment 4 is limited to preverbal subject NPs. As we have shown for bare singulars in Chapter 2, and as demonstrated in the literature for both bare singulars (Geist, 2010) and bare plurals (Seres et al., 2019), preverbal position in Russian is associated with familiarity and topic-like interpretations. This can potentially favor stronger readings. Therefore, the results leave open the question of whether we’d observe the same consistent maximality effect in other syntactic positions, specifically, in postverbal plurals. In the next section, I present exploratory data from object NPs, which is an initial step in this direction.

3.4 Preliminary evidence from object NPs

3.4.1 Why look at object NPs?

Experiment 4 focused on preverbal subject NPs in canonical SVO sentences. In Russian, preverbal position can favor definite-like readings for independent reasons. This raises the question: is the robust maximality effect observed in Russian bare plurals in Experiment 4 driven by their semantic properties, or is it partly a reflection of certain information structure related to the preverbal position? To address this concern, we should look at syntactic positions that are less strongly associated with topicality.

Object NPs in canonical SVO sentences provide a natural testing ground for this question. Postverbal elements can be either topical or not (Geist, 2010); this means that indefinite-like readings are potentially more accessible. If bare plurals show the same strong maximality preference in the object position as in the subject position, this would suggest that the effect is not tied to the preverbal position (and, consequently, information structure), but reflects a more general semantic property. Examining objects thus allows us to probe whether the syntactic position creates conditions under which non-maximal interpretations become more accessible, and whether any such effect is uniform across Russian, a language with flexible word order, and English, a language with a more fixed object position.

3.4.2 Methods, materials, and participants

The data reported in this section come from a broader acceptability-judgment experiment originally designed to investigate the interpretation of singular and plural nouns in Russian and English. The original experiment manipulated both number and maximality/uniqueness. In the present chapter, only the plural conditions are considered, as our focus is on maximality effects with bare plurals.

The task was designed to capture the interpretation of direct objects in sentences with neutral word order. Working with the object position, it was important to exclude atelicity or imperfectivity, as they often come along with non-definiteness and plurality; it has been demonstrated for Russian specifically that imperfectivity can endorse the weak referential reading of bare singular objects (Grønn, 2004; Mueller-Reichau, 2015). Therefore, this task used sentences with perfective verbs in the past tense, in combination with video stimuli that pictured an action that was clearly completed.

Two limitations of the dataset should be noted. First, the maximality manipulation relied on a single maximal/non-maximal stimulus pair in the plural object condition. Second, the plural items were embedded in a broader design that also tested singular noun phrases. The present analysis therefore constitutes a reanalysis of previously collected data, and the results should be interpreted as exploratory rather than as a fully balanced test of the maximality manipulation.

The following English and Russian sentences were tested:

- (19) a. The woman put **the bags** on the table.
 b. The woman put **bags** on the table.
- (20) Devuška postavi-l-a **paket-y** na stol.
 young.woman put-PST-F bag-PL on table
 ‘The young woman put the bags/bags on the table.’

The sentences were tested each with a +MAXIMAL and -MAXIMAL video. In the +MAXIMAL

video, the woman enters the room with four bags, places them all on the table, and leaves. In the -MAXIMAL video, the woman enters the room with five bags, places some of them on the table, some of them on the floor, and leaves with the rest. Figure 3.6 shows the screenshots of the two experimental videos.

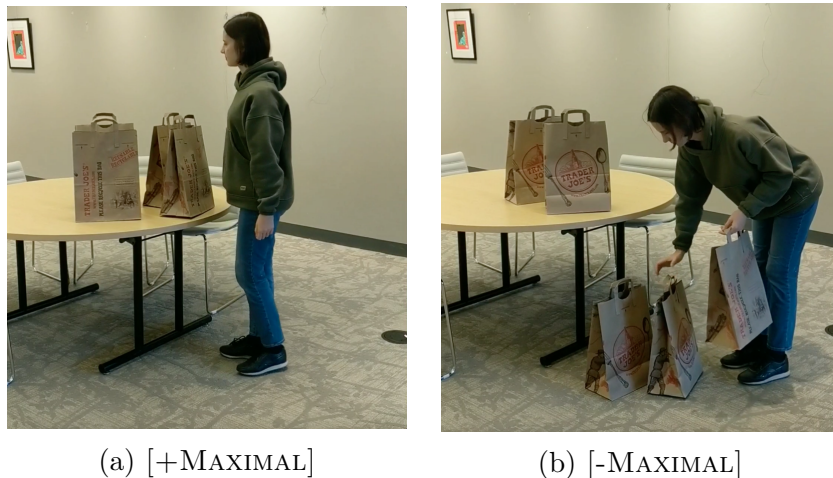


Figure 3.6: Object dataset: Screenshots of the maximal (a) and non-maximal (b) video stimuli used in the task.

Another important feature of this survey is that the context we tested focused the object NP. Each experimental screen included the question *What did the woman put on the table?* on the top, then the experimental video, then the sentence to rate, and a scale on which the participants rated the acceptability of the sentence as the answer to the question. The scale had no numerical labels, but had three text labels: “Totally disagree”, “Neither agree nor disagree”, and “Totally agree”. The slider was set up to start in the middle of the scale. A screenshot of an experimental screen is provided in 3.7.

60 self-identified monolingual speakers of English, recruited through Prolific, evaluated the English sentences, and 27 self-identified monolingual speakers of Russian, recruited through Yandex.Toloka, evaluated the Russian stimuli. Since the data on plurals was collected as a part of a larger dataset, there were 8 non-plural experimental stimuli, and 8 catch trials in each task. Within the Russian task, the maximal and the non-maximal condition was rated within-subjects. In the English task, definite plurals were tested within-subjects,

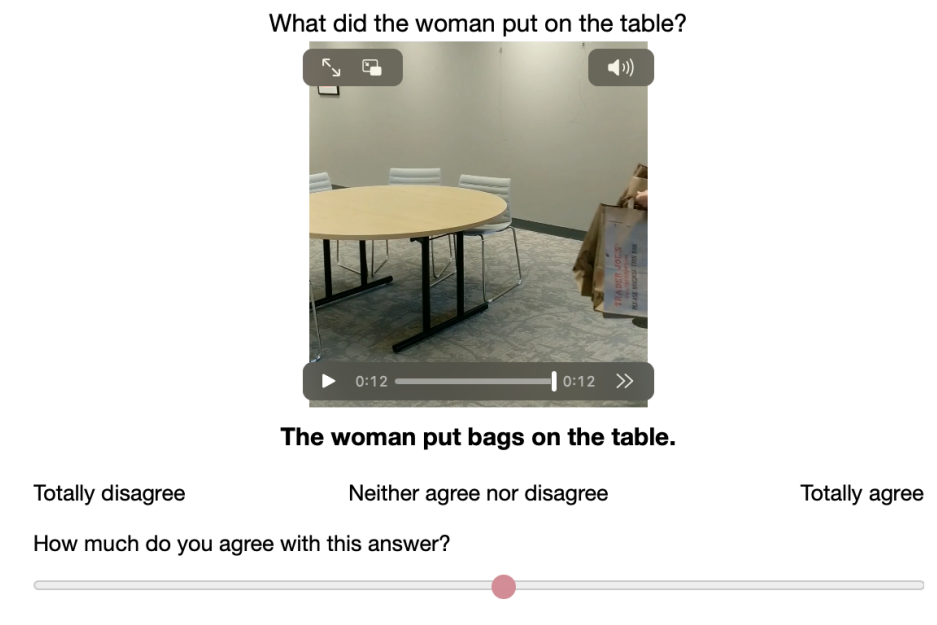


Figure 3.7: Object dataset: Example experimental trial screen, featuring a sentence with an English bare plural.

while bare plurals were tested between-subjects.

3.4.3 Results

The tested sentence with the Russian bare plural received the average rating of 9.92, 95%CI[9.81, 10.00] in the +MAXIMAL condition, and the average rating of 6.62, 95% CI[5.49, 7.75] in the -MAXIMAL condition. This difference is highly significant, as supported by the Wilcoxon Signed Rank test ($V = 171, p < 0.001$; Cohen's $D = 1.66$, large effect size)².

English definite plurals showed a similar tendency, as they were uniformly rated as 10.00 in the maximal condition. In the non-maximal condition, they were rated on average as 6.47, 95%CI[5.66, 7.28]. The difference between the ratings in the maximal and non-maximal conditions is highly significant, as shown by a paired sample t-test, $t(29) = -8.88, p < 0.001$; Cohen's $D = 2.3$, large effect size.

English bare plurals were tested between-subjects, and received perfect ratings of uniformly

²The non-parametric version was used since the sample size is relatively small ($n = 26$), and the difference between the ratings in the two conditions was not distributed normally (Shapiro-Wilk normality test, $W=0.88, p < 0.01$).

10.00 in the maximal condition as well. In the non-maximal condition, they were rated lower at 8.50, 95% CI[7.45, 9.55] on average. This difference is significant, as shown by the Mann-Whitney U-test, $W = 615, p < 0.001$, but the effect size is moderate rather than large (Cohen's $D = 0.78$).

The difference between the average ratings of English bare plurals and English definite plurals within the non-maximal condition can be considered significant ($W=649, p < 0.001$, with a large effect size, Cohen's $D = 0.83$), as well as the difference between the average ratings of English bare plurals vs. Russian bare plurals ($W=203, p < 0.01$; the effect size can be considered moderate, Cohen's $D = 0.68$).

There is no significant difference between the average ratings of Russian bare plurals and English definite plurals within the non-maximal condition, $W=400, p=0.87$.

Figure 3.8 visualizes the distributions of ratings for different types of plurals taken from this dataset³. From this plot, we see that all tested types of plurals are rated near-perfectly in the maximal condition. We see that the interquartile range is the largest for Russian bare plurals, and the tails of the distribution extend to the whole scale, which points at a larger variability within that condition. For English bare plurals, we see that the median is much higher than the mean, and the mean is outside of the interquartile range. This means that the distribution of ratings for them was negatively skewed: most people rated English bare plurals as perfect in the non-maximal context, and the asymmetry comes from a small number of extreme judgments, visible in the plot as outliers. As for English definite plurals, we see that the mean is higher than the median, which means that the distribution was positively skewed: most people gave lower ratings to a definite plural in the non-maximal context, but a fraction of participants rated them very high. Such pattern signals uncertain or marginal acceptability.

All in all, all tested plurals – Russian bare plurals, English bare plurals, and English definite plurals – demonstrated better ratings when the context is maximal compared to

³Unlike in Figure 3.4, ratings are not standardized within participants, as each participant contributed a single rating per condition.

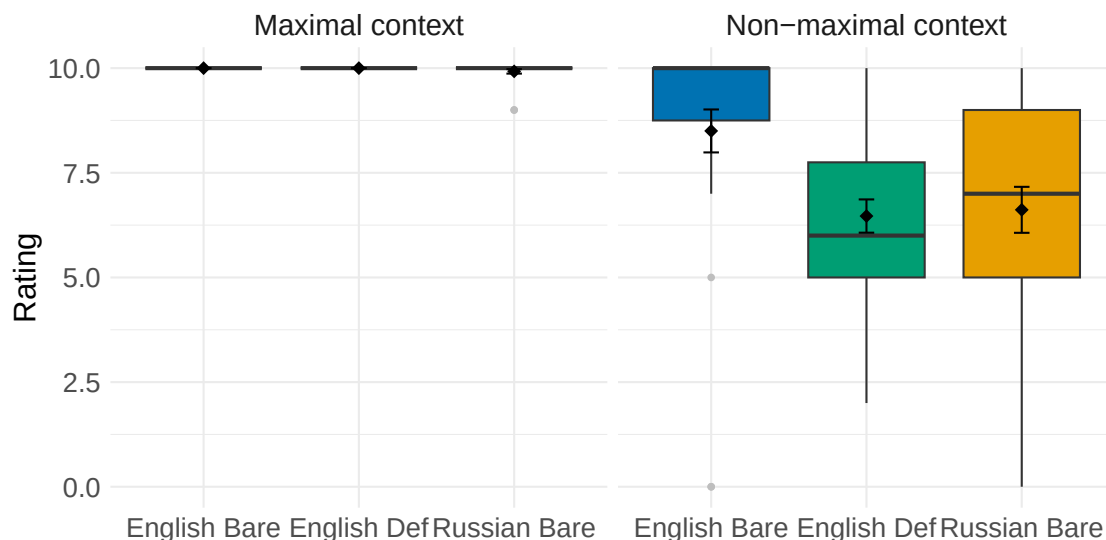


Figure 3.8: Object dataset: Distributions of raw ratings of English bare plurals, English definite plurals, and Russian bare plurals in maximal and non-maximal contexts. Diamonds are means with standard error bars; middle lines are medians.

when the context is non-maximal. In the case of Russian bare plurals and English definite plurals, this tendency has a large effect size. In the case of English bare plurals, the preference for a maximal interpretation is of moderate size. Russian bare plurals and English definite plurals showed nearly identical behavior on this set of stimuli.

3.4.4 Summary of the object data

The observations from the object dataset extend the general pattern that we observed in Experiment 4. The results can be summarized as follows. Similar to the subject experiment, Russian bare plurals demonstrated a strong preference for maximal interpretations. On that, they aligned with English definite plural object NPs, which were also rated as much more acceptable in maximal contexts compared to non-maximal contexts. English bare plural objects preferred a maximal interpretation on average, but were more compatible with non-maximal interpretation than definite plurals or Russian bare plurals.

So what we see is that unlike in the subject position, in the object position the behavior of Russian bare plurals and English bare plurals diverge. On this set of items, Russian

bare plurals show a strong preference for a maximal interpretation, while English bare plurals don't. This difference is not categorical: maximal readings of English bare plurals remain acceptable and often preferred, but their non-maximal readings are significantly more acceptable than in the subject position. At the same time, this contrast should be interpreted with caution. The object dataset is more limited than the subject dataset, and the observed effect concerns the degree to which non-maximal interpretations are penalized. The relevant effect may be sensitive to the properties of the specific stimuli we used. Another general result we see from comparing the subject and the object data sets is that non-maximal interpretations seem to be penalized less in the object position than in the subject position (based on our limited dataset).

In what follows, I consider how the results from both Experiment 4 and the object dataset fit into a broader accounts of bare plural semantics and apparent exceptions from maximality.

3.5 General discussion

The experiments reported in this chapter tested one dimension on which competing analyses of episodic bare plurals make different predictions: whether bare plurals are compatible with non-maximal interpretations in presentational contexts. The results are consistent across both datasets. In the subject position, Russian bare plurals, English bare plurals, and English definite plurals all showed a robust preference for maximal interpretations: participants ratings dropped substantially when the predicate held of only a subset of the contextually salient individuals. In the object position, we observed the same directional preference for Russian bare plurals and English definite plurals, but English bare plurals showed a weaker effect. These findings allow us to evaluate the theoretical accounts introduced in section 3.1 along two dimensions: whether the general maximality preference is captured, and whether the asymmetry between Russian vs. English bare plurals in the object position is accommodated.

3.5.1 Against purely existential approaches

First of all, our findings are clearly not compatible with accounts that ultimately involve existential quantification in the denotation of episodic bare plurals without any additional maximization mechanisms. Under the classic DKP analysis of Chierchia (1998), episodic bare plurals introduce existential quantification over instantiations of a kind. Under Krifka (2003)’s analysis, the same result is achieved via local existential closure over individuals. The hypotheses suggested in Seres et al. (2019); Seres (2020) take it further, and treat existential quantification as the default across all bare nominal uses, with definite or generic readings arising through pragmatic strengthening. All three of these positions share the following prediction: a sentence with a bare plural subject should be true as long as at least one relevant plurality satisfies the predicate, and there is no constraint that would force the predication to cover the maximal set of individuals. Therefore, non-maximal interpretations are predicted to be freely available.

However, across both languages and both syntactic positions tested, participants systematically penalized non-maximal readings in bare plurals relative to maximal ones. This was not a ceiling effect or an artifact of the design, as non-maximal contexts received intermediate ratings, compared to extremely low ratings that we saw in intentionally infelicitous catch trials. The pattern that we observe is not one of categorical ungrammaticality but one of consistent and statistically robust dispreference. Neither a pure existential approach nor the DKP version of Chierchia (1998) have a mechanism for deriving this contrast in a consistent way. One could potentially suggest a mechanism of pragmatic preference for strong interpretations, but it would need to be strong enough to produce effects of the magnitude we observed, and we observed an effect comparable in size with English definite plurals, which carry maximality as part of their lexical semantics. For a purely pragmatic account, this would be hard to model, particularly in presentational contexts that we used, as they do not impose a strong question under discussion that would favor exhaustive interpretations.

3.5.2 Evaluating maximality-based approaches and their limits

The results are more naturally accommodated by approaches that include maximality explicitly in the semantics of bare plurals. In this section, I evaluate three such approaches against our findings in Experiment 4: that Russian bare plurals, English bare plurals, and English definite plurals all show a comparable preference for maximal interpretations in presentational contexts. The accounts differ in how they derive this preference and, consequently, in how well they handle the fact that the ratings for non-maximal interpretations observed in Experiment 4 were intermediate rather than uniformly low.

Under Dayal (2013) proposal, the DKP rule is modified such that predication applies to the maximal instantiation of a kind rather than to existentially quantified instances of a kind. This does predict that bare plurals should prefer maximal readings. However, taken in isolation, this amendment would predict that non-maximal readings are simply unavailable, since the denotation of the bare plural is tied to the maximal instantiation of a kind and existential quantification is eliminated. But our results are not that categorical: the ratings in the non-maximal condition are lower but not at floor. Dayal does allow for apparent non-maximality through representative group readings (when the predicate is evaluated relative to a contextually salient subgroup), but these amount to a pragmatic weakening mechanism operating on top of a maximality-denoting element. Our data is generally compatible with this, but again, the intermediate ratings we observe suggest that whatever mechanism produces apparent non-maximality works in the environments we tested, not just in specially constructed representative-group contexts.

Another account that retains an existential based while deriving maximality is Chierchia (2022b). It is especially relevant since it makes a clear distinction between plural definites and bare plurals. Chierchia starts from an observation that both English definite plurals and English bare plurals allow universal and often existential readings in the positive context, but are strictly existential in the negative context⁴:

⁴Examples are cited from (Chierchia, 2022b).

(21) Definite plurals

- a. I interviewed **the incoming students**.

$\rightsquigarrow$ I interviewed all incoming students. $\checkmark \forall$

possible: $\rightsquigarrow$ I interviewed some but not all incoming students. $\checkmark \exists$

- b. I didn't interview **the incoming students**.

$\rightsquigarrow$ I did not interview any of the incoming students. $\checkmark \neg \exists$

$\times$ I did not interview all of the incoming students. $\times \neg \forall$

(22) Bare plurals

- a. Bears are now hibernating around here.

$\rightsquigarrow$ All bears are now hibernating. $\checkmark \forall$

possible: $\rightsquigarrow$ Some bears are now hibernating. $\checkmark \exists$

- b. Bears are not hibernating yet.

$\rightsquigarrow$ No bears are hibernating yet. $\checkmark \neg \exists$

$\times$ Not all bears are hibernating yet. $\times \neg \forall$

To account for this, Chierchia suggests that both plural definites and bare plurals involve existential quantification, which is introduced not by the nominal itself, but by the theta-role applicative head. The applicative head introduces a discourse referent existentially over the argument it saturates. For plural definites, the argument is the maximal individual picked out within a contextually relevant domain by the *iota* operator (23-a), and the existential ranges over atomic parts of that individual (23-b).

- (23) a. $\llbracket \text{the students} \rrbracket = \iota x \in D. \text{students}(x)$

- b. $\llbracket \text{I interviewed the students} \rrbracket =$
 $= \exists e \exists z_1 \in D(\iota x. \text{students}(x)) [\text{AG}(e)(\text{I}) \wedge \text{TH}(e)(z_1) \wedge \text{interview}(e)]$

In upward-entailing contexts, these alternatives are exhaustified, requiring the proposition

to hold for every subdomain including every singleton, which results in a universal reading. Under negation, the negated existential already entails the negation of all subdomain alternatives, so the interpretation remains existential.

For bare plurals, the derivation runs parallel but starts differently. The bare plural denotes a kind and undergoes DKP when fed to an object-level predicate, as in the classic neo-Carlsonian approach. But then again, the applicative head introduces an existential discourse referent. Unlike in definite plurals, in bare plurals the existential ranges over instances of a kind in the relevant world and domain, not over atomic parts of the maximal individual. Subdomains are activated in the same way as for the plural definites, and the universal reading in upward-entailing contexts arises similarly through exhaustification.

The surface readings of the two sentences are therefore identical, but the derivations differ in the source of the domain D . For definite plurals, it is fixed by the presuppositional *iota* operator as the set of atoms of a contextually maximal individual whose existence must be already established in the discourse (24-a). For bare plurals, the domain must be supplied externally by the applicative head, without any presuppositional anchoring: the kind exists as an abstract entity regardless of whether any contextually relevant instances have been established.

- (24) a. For plural definites: $D(\iota x.P(x))$
 b. For bare plurals⁵: $D(\cap P)$

This single difference produces three empirical consequences. First, definites but not bare plurals carry an existence presupposition. Second, because the domain for bare plurals is not presuppositionally fixed, subdomain alternatives can be easier to prune, which weakens exhaustification and makes existential readings more accessible. On the empirical level,

⁵This is a simplified notation. In the original approach, kind-level and object-level proceed simultaneously: the kind is the theme of the main event as a whole, and instances of the kind are introduced as themes of sub-events via a separate existential. The domain $D(\cap P)$ here corresponds to the set of contextually relevant instances of the kind. I abstract away from the two-event structure, which is needed for Chierchia's account of differentiated scope but is not directly relevant to the maximality question addressed here.

it means that bare plurals tolerate non-maximal readings better than definite plurals in upward-entailing contexts. Third, the two derivations introduce discourse referents of different types. For definites, the discourse referent ranges over parts of a contextually bounded individual and is anchored to that individual in the discourse, which makes them anaphoric. For bare plurals, the discourse referent ranges over instances of a kind without presuppositional anchoring, which is why English bare plurals resist being anaphoric.

Chierchia’s account is compatible with our results from the experiment on the subject position. The visual stimuli we used in Experiment 4 showed equally salient individuals, which made the domain fully explicit. This means that subdomain alternatives must have been fully determined and exhaustification must have applied without pruning, which predicts a strong preference for maximal readings. The account also predicts the gradient nature of the ratings in the non-maximal condition: if non-maximal readings arise from a pragmatic mechanism such as the pruning of subdomain alternatives, we’d expect intermediate rather than floor-level ratings. In this way, Chierchia’s account might handle the gradient of our results better.

Finally, one should evaluate our results with the respect to the theory presented in Berulava and Mayr (2024). Rather than deriving maximality pragmatically, they encode it directly in the denotation of bare plurals via a covert definite operator. This means that on their account, bare plurals in articleless languages are structurally parallel to overt definites. The source of non-maximal readings is homogeneity: a sentence with a bare plural is neither strictly true nor strictly false when the predicate holds of some but not all atomic members of the maximal plurality, and it can be judged acceptable in case it adequately resolves the question under discussion. This predicts intermediate ratings precisely where we observe them.

In this way, with respect to the data from subject plurals, all three maximality-based accounts are broadly compatible with our findings. Our experimental results alone cannot distinguish between the three accounts definitively. As we will see in the next section, the

preliminary data from object plurals is more intriguing.

3.5.3 The object-subject asymmetry and its theoretical implications

The preliminary data from object NPs introduce a complication that deserves a separate discussion, even though the scope of that dataset was limited. In the object position, Russian bare plurals and English definite plurals patterned alike: both showed a strong preference towards maximal interpretations, with large effect sizes and comparably lower ratings in the non-maximal condition. However, English bare plurals behaved differently, and showed a moderate and more variable effect: they preferred maximal readings on average, but their ratings in the non-maximal condition were significantly higher than those of Russian bare plurals and English definite plurals. This divergence between English and Russian bare plurals was not observed in the subject position, where all the three plural types behaved nearly identically.

The accounts that apply the same derivational mechanism to bare plurals in a system with a definite plural vs. in an articleless system do not predict this asymmetry. If Chierchia’s exhaustification-based strengthening works uniformly across both languages (which it should, since both Russian and English bare plurals would undergo DKP and the subsequent activation of subdomain alternatives), there is no obvious reason to expect Russian bare plurals in object position to be more resistant to non-maximal readings than English bare plurals. The contrast is explained more naturally by the account of Berulava and Mayr (2024), who connect the definite-like behavior of bare plurals to the absence of an overt definite article in the system. On their account, bare plurals in languages like Russian encode maximality through a covert definite operator, while languages with a competing definite plural would block that covert definite operator. That would predict exactly the asymmetry we observed: English bare plurals would have to derive any maximality effects through independent mechanisms, and Russian bare plurals would behave like overt definites.

However, we did not observe the same divergence in the subject position. If Russian

bare plurals involve a covert definite operator, and English bare plurals do not, we would the asymmetry to hold across different syntactic positions. Without giving a definitive solution, let us discuss several factors that may have suppressed it in our subject data. As discussed in the literature (Slioussar, 2007; Geist, 2010; Bailyn, 2012, among others), and confirmed through our own data in Experiment 1 (Chapter 2), preverbal position in Russian is associated with familiarity and topic-like interpretations. This may have independently driven maximal construal for both English and Russian bare plurals in the subject position, masking any underlying semantic differences. The object position, being less strongly associated with topicality, may have allowed the semantic difference to surface. The interaction between syntactic position and maximality source would be worth investigating in future work.

3.5.4 What the data leave open

Our data from Experiment 4 and from the object dataset together suggest that episodic bare plurals are not straightforward indefinites, and that Russian bare plurals pattern more closely with definite descriptions than with English bare plurals, at least in the contexts we tested. What the present data cannot decide is whether Russian bare plurals involve a covert *iota*-like operator and are introduced broadly the same way English definites are, or whether Russian bare plurals and English bare plurals are introduced the same way, but there is some sort of language-specific calibration of exhaustification that reflected in our results. The two accounts make identical predictions for the upward-entailing contexts.

What kind of data would be able to tell lexical maximality from derivative maximality apart? One direction would be the behavior of bare plurals under negation. Both Chierchia (2022b) and Berulava and Mayr (2024) predict that non-maximal readings are less available under negation than in positive sentences, but for different reasons, and these reasons make somewhat different predictions. Chierchia claims that the difference between upward- and downward-entailing environments follows from the logical structure of exhaustification, which

appears vacuous under negation, because the negated existential already entails the negation of every subdomain alternative. Exhaustification computation alone is responsible for the effect. By contrast, Berulava and Mayr claim that the effect follows from how negation interacts with the trivalent truth-value structure of weak definites. In positive sentences, a gap-valued sentence (where the predicate holds of some but not all atomic parts of the maximal plurality) can be pragmatically true enough if the gap condition and the truth condition resolve the QUD in the same way. Under negation, the gap is mapped to falsity by the negation operator, so there is no hap to be rescued by the QUD, which blocks non-maximal readings. They provide the data from Georgian showing that bare plurals resist non-maximal readings under negation in articleless systems, too, and this observation holds in Russian (25); however, this preliminary observation is consistent with both accounts and does not distinguish between them.

(25) **Sobak-i** ne laj-ut.

dog-PL NEG bark-PRS.3SG

✓ ‘There are no dogs that are barking.’ $\neg\exists$

× ‘Some dogs are not barking.’ $\exists\neg$

A more promising diagnostic concerns the presuppositional behavior of bare plurals in negative contexts. According to Berulava and Mayr (2024), the Russian bare plural should introduce an existential presupposition via the weak definite operator. This presupposition can be locally accommodated under negation where no salient individuals are established, which results in an intermediate reading and contrasts with an overt definite, as they demonstrate for Georgian bare plural (26-b) vs. English definite plural (26-a):

(26) Scenario: Checking into a hotel, Ann is relieved to find that there is no barking and says:

a. #The dogs aren’t barking.

-
- b. ?zaɣl-eb-i ar kep'en
 dog-PL-NOM NEG bark
 'Dogs are not barking.'

(Berulava and Mayr, 2024, p. 15)

In Chierchia's account, no such presupposition is involved: the kind exists as an abstract entity regardless of whether any contextually relevant instances have been established. This means that using a bare plural in a context like (26), where no individuals were established, should have no presupposition accommodation cost. This yields the following diagnostic prediction: if Russian bare plurals are clearly degraded in out-of-the-blue negative contexts like (26), the weak definite analysis is more plausible. If Russian bare plurals involve kind-based DKP, they should pattern with English bare plurals in being felicitous. At this point, we don't have systematic evidence for Russian about this, but this is an intriguing direction of future work⁶.

A second diagnostic concerns anaphoricity. As discussed in section 3.5.2, under Chierchia's account, the derivations for the English definite plural vs. English bare plural introduce discourse referents that are structurally different. In the definite derivation, the discourse referent is anchored to a contextually bound individual fixed by the *iota* operator, which is anaphoric. In the bare plural derivation, the discourse referent ranges over instances of a kind without presuppositional anchoring, which predicts non-anaphoricity. Russian bare plurals, however, can readily serve as antecedents for anaphora, unlike what was reported for English bare plurals:

- (27) a. Deti_i zaš-l-i v komnat-u. Deti_i se-l-i za stol.
 children enter-PST-PL in room-ACC children sit-PST-PL at table
 'Children/The children entered the room. The children sat at the table.'

elicitation data

- b. Some children came in. *Children sat down. (Dayal, 2013)

⁶My informal judgments generally align with Mayr and Berulava's findings; however, since the argument operates on relative acceptability rather than sharp contrasts, a larger-scale investigation would be preferable.

Chapter 4 investigates this question by examining the anaphoric behavior of Russian and English bare plurals under systematically varied discourse conditions. Whether the anaphoric potential of Russian bare plurals reflects a definiteness-based architecture or a different route to discourse reference is the central question we will be addressing.

4 Anaphoric potential of bare plurals

In the previous chapters, we discussed uniqueness and maximality. In this chapter, I turn to a different diagnostic for (in)definiteness: anaphoricity. While uniqueness/maximality are properties of what a noun phrase denotes, anaphoricity is about how the noun phrase updates the discourse: a noun phrase is anaphoric if it picks up a referent already established in the discourse, and non-anaphoric if it introduces a novel referent. In that sense, by switching to the discussion of anaphoricity, we move on to discourse-level representation.

In languages with articles, the difference between anaphoric and non-anaphoric nominals is encoded in the grammar. An indefinite typically introduces a new referent (1-a). A definite refers to something that has already been introduced (1-b).

- (1) a. **A girl** entered the house.
- b. **The girl** sat down.

In this ability to be linked to something in the prior discourse, the definite noun aligns with pronouns:

- (2) a. **A girl** entered the house.
- b. **She** sat down.

Based on this similarity, Heim (1982) proposes a dynamic account to (in)definiteness in English. Essentially, she claims that definite noun phrases, like pronouns, come with an index. That index has to be already established (or familiar) in the discourse. In this way,

definite noun phrases are anaphoric, as in, they have to be linked to a previously mentioned discourse referent. Indefinite noun phrases, by contrast, are non-anaphoric: they introduce new indices. In English, the most natural interpretation of (3-b) is that the girl who sat down is a different girl.

- (3) a. **A girl_i** and a boy entered the house.
 b. **A girl_{*i/j}** sat down.

The situation becomes more intriguing in languages without articles. Bare arguments in languages without articles have been shown to have both anaphoric potential (which aligns them with definites) and the ability to introduce new entities into the discourse. This has been reported for Hindi (Dayal, 2015), Kannada (Srinivas, 2021), Shan (Moroney, 2021), Kaingang (Navarro, 2025), Khanty (Tiutiunnikova et al., 2025), as well as the sample of articleless languages considered in Dayal (2026b), and various Slavic languages including Russian.

- (4) a. Devočka_i i mal'čik zaš-l-i v dom.
 girl and boy enter-PST-PL in house
 'A girl and a boy entered the house.'
 b. Devočka_{i/*j} se-l-a.
 girl sit-PST-F
 'The girl sat down.'

Moreover, it seems like the Russian bare singular has to be anaphoric if there is a suitable referent: the noun *devočka* 'girl' in (4-b) cannot refer to a different girl, contrasting with the English indefinite in (3-b). In order to switch reference, an explicit marker like *drugaja* 'other.F' is required (5-b):

- (5) a. Devočka_i i mal'čik zaš-l-i v dom.
 girl and boy enter-PST-PL in house
 'A girl and a boy entered the house.'

-
- b. *(Drug-aja) devočka_j se-l-a.
 other-F girl sit-PST-F
 ‘Another girl sat down.’

What is especially intriguing is that a bare argument coexisting with an overtly marked definite does not have the same anaphoric potential as a bare noun in a system without articles. Let us look at the bare plural in English, which is opposed to the definite plural, and in Russian, where there is no definiteness marker. It has been claimed in the literature that English bare plurals are essentially non-anaphoric (Dayal, 2013):

- (6) Some children_i came in. *Children_i sat down. (Dayal, 2013)

Russian bare plurals, in contrast, have no issue referring to pluralities that have already been introduced:

- (7) Kakie-to deti_i zaš-l-i v komnat-u. Deti_{i/??j} seli za stol.
 some children come-PST-PL in room-ACC.SG children sit-PST-PL at table
 ‘Some children entered the room. The children sat down.’

In this way, what we observe empirically is that bare nouns in languages without articles can clearly be both novel and anaphoric, but not in a completely unrestricted way. Moreover, we see that the absence of morphological marking alone does not give rise to this kind of ambiguity automatically: the specifics are dependent on whether the language has competing marked forms or not. This is challenging both theoretically and empirically. Empirically, the full picture for a language without articles, to my knowledge, has not been described yet. Theoretically, one would have to model both relative freedom and restriction into an account of bare noun reference, and such a model should account for whether the whole system has competing forms or not.

The final series of experimental studies in this dissertation is designed to approach this problem. I report two parallel forced-choice experiments on Russian and English, and I

concentrate on plural nouns. Experiment 5A uses contexts designed to favor anaphoric interpretations and asks whether Russian bare plurals achieve the same level of anaphoricity as English definite plurals under such conditions. Experiment 5B uses contexts designed to disfavor anaphoric interpretations and asks whether Russian bare plurals resist non-anaphoric construals even when the context pulls towards them (which would be expected from an overtly marked definite). Combined, the two experiments allow us to assess both the overall level of the anaphoric potential of bare plurals and their resilience to non-anaphoric interpretations.

4.1 Anaphoricity as a diagnostic for (in)definiteness

4.1.1 Novelty and familiarity

Heim (1982) provides a formalization for the underlying principle distinguishing between definites and indefinites: the Novelty-Familiarity condition. This principle has been developed for systems with overt (in)definiteness markers. Essentially, it claims that the referent of a noun phrase marked as indefinite cannot belong to the domain that maps from discourse context C to discourse referents NP_i ; that is, it must be novel. By contrast, a noun marked as definite must have a referent that is entailed in the domain of the context in question.

(8) The Extended Novelty-Familiarity Condition (Heim, 1982, p. 369-370)

For a δ to be felicitous w.r.t. a context C , it is required for every NP_i that:

- a. if NP_i is $[-\text{definite}]$, then $i \notin \text{Dom}(C)$;
- b. if NP_i is $[\text{+definite}]$, then
 - (i) $i \in \text{Dom}(C)$, and
 - (ii) if NP_i is a formula, C entails NP_i .

Importantly, indefinite noun phrases are analyzed not as quantifiers but as free variables

further bound by existential closure. In this way, indefinite and definite NPs are treated similarly in every respect except for the value of their (in)definiteness feature.

4.1.2 Application to systems without articles

While Heim does not analyze systems without articles in any detail, this parallelism between indefinites and definites allows her to hypothesize what should happen in languages without articles. In Heim’s system, the Novelty-Familiarity distinction is tied to the morphological marking: definite determiners are [+definite], indefinite determiners are [−definite]. In the absence of morphological marking, bare nouns are ambiguous, but only with respect to the feature [$\pm$ definite], not ambiguous between variables and quantifiers.

Dayal (2026b) shows how the Novelty-Familiarity condition can be extended onto systems without articles. According to Dayal, the Novelty-Familiarity condition is triggered by morphological marking contained in the D-layer of the syntactic structure, and bare nouns are NPs rather than DPs¹. Therefore, in the absence of the (in)definiteness marking they are neither definite nor indefinite, and can go either way. For bare plurals, the anaphoric reading is achieved through *iota*, and the non-anaphoric reading is derived through DKP, as originally proposed by Chierchia (1998). For bare singulars, Dayal proposes an approach within neo-Davidsonian event semantics, where verbs are predicates of events and arguments attach via theta heads; she builds the ambiguity not within nominal semantics itself, but within the theta head, which allows her to derive narrow scope in non-anaphoric bare singulars as well.

But there are two intriguing asymmetries in the anaphoric behavior of bare nouns, which need to be addressed. The first one is the difference in the anaphoric potential of bare nouns

¹This claim is not uncontroversial for Russian. Diagnostics from nominal modifier configurations and semantic vs. syntactic agreement argue for the presence of a D layer in Russian argument nouns, responsible for various referential properties of the nominal, such as semantic gender agreement (Pereltsvaig, 2006, 2013; Lyutikova, 2018). If bare nouns do project up to DP, the contrast between bare noun interpretation and overtly marked (in)definite nouns does not stem from the absence of a necessary syntactic projection, but from the absence of surface overt marking of (in)definiteness. However, this does not affect the predictions tested in this chapter.

in different systems. As shown in the beginning of this chapter, bare nouns in languages without articles in general can be both novel and anaphoric, while bare nouns in systems having definite determiners have been claimed to be strictly non-anaphoric. Specifically, English bare plurals are non-anaphoric, see (6) from (Dayal, 2013); Russian bare plurals do not exhibit the same restriction (7).

The second asymmetry is that Russian bare nouns also strongly prefer to be anaphoric on second mention, rather than just permit it, as demonstrated in (4)-(5). If the Novelty-Familiarity condition is triggered by morphological marking, bare nouns in systems with no (in)definiteness marking are exempt from it. However, this would predict availability of both novel and anaphoric readings on the second mention. Yet, at least our preliminary elicitation material shows that the anaphoric reading is strongly preferred.

The anaphoric behavior of Russian bare nouns is further complicated by the fact that they compete with other anaphoric expressions, such as demonstratives or pronouns. The relative felicity of competing anaphoric expressions is dependent on discourse structure. For instance, pronouns are generally preferred over bare nouns when the referent is maximally salient and unambiguous, while bare nouns are more natural when the predicate content is informative or when the referent needs to be reidentified. Example (9) only introduces the girl; in a context like that, the pronoun (9-a) is considered a better alternative than the bare noun (9-b)².

- (9) Devočka zaš-l-a v komnat-u.
 girl entered in room
 ‘A girl entered the room.’
- a. **Ona** sela na stul.
 she sit-PST-F on chair
 ‘She sat on a chair.’

²My respondents were only able to notice the contrast in (9) when presented with both alternatives at the same time. Elicited in isolation, (9-b) was unanimously judged as an acceptable continuation. The same holds for (10).

-
- b. ?**Devočka** se-l-a na stul.
girl sit-PST-F on chair
'The girl sat on a chair.'

By contrast, example (10) introduces a coordinated structure, and the continuation with a bare noun (10-b) becomes more acceptable than the continuation with a pronoun (10-a):

- (10) Devočka i mal'čik zaš-l-i v komnat-u.
girl and boy entered in room
'A girl and a boy entered the room.'

- a. ?**Ona** sela na stul.
she sit-PST-F on chair
'She sat on a chair.'
- b. **Devočka** se-l-a na stul.
girl sit-PST-F on chair
'The girl sat on a chair.'

This pattern suggests that the competition between bare noun and pronoun is sensitive to how many discourse referents are simultaneously active. When only one referent is available, the pronoun, being the more economical form, is preferred; when two referents are active, the bare noun's descriptive content becomes an asset rather than redundancy, because it specifies which of the two is being picked up (Saha, 2026). The anaphoric status of the bare noun in Russian is therefore not uniform across different contexts.

In this way, there are several questions one may ask. First, does the empirical observation that the presence or absence of a competing definite form in the system influences the anaphoric potential of the bare noun hold on larger-scale evidence? Second, how does the architecture of the (in)definiteness system shape the anaphoric potential of bare plurals specifically? Third, is the anaphoric preference of Russian bare nouns grammatically encoded or pragmatically derived?

In the remainder of this chapter, I concentrate on bare plurals, where the cross-linguistic contrast between Russian and English is most revealing. In the next section, I lay out specific

predictions from the neo-Carlsonian account about how Russian and English bare plurals should differ in their anaphoric behavior, and I outline the experimental design testing these predictions.

4.2 Theoretical background and predictions

4.2.1 Ranking, blocking, and the cross-linguistic asymmetry

The neo-Carlsonian approach, introduced in Chapter 1, maintains that bare nouns in languages like Russian always have access to an *iota* interpretation. Bare singulars are either definite or taxonomic kinds, which at the end of the day still shift through *iota*. Therefore, the bare singular is predicted to be strictly anaphoric. The account of bare plurals is more nuanced. On the neo-Carlsonian account, bare plurals are kind-denoting expressions. In episodic contexts, they receive existential force through DKP: when an object-level predicate combines with a kind-denoting argument, DKP repairs the sort mismatch by introducing a narrow-scope existential over instances of a kind. This is the source of non-anaphoric readings of bare plurals. To receive an anaphoric reading, the bare plural would have to shift through *iota*.

In fact, the neo-Carlsonian approach does predict the asymmetry in the anaphoric potential of English vs. Russian bare plurals in (6)-(7). The Blocking Principle (Chierchia, 1998) states that if a language has an overt morphological determiner responsible for a certain type-shifting operation, covert application of the same operation is blocked; Dayal (2004, 2026b) adopts this principle as well. In English, *iota* is realized through the overt definite article. Therefore, the English bare plural cannot access it covertly, and anaphoric readings are predicted to be unavailable. In Russian, no overt definite article exists, so nothing blocks *iota* from happening covertly. This predicts free availability of anaphoric readings to Russian bare nouns.

As discussed in Chapter 3, Dayal (2013) proposes a more radical revision, which eliminates

the existential force from DKP entirely. On that account, bare plurals are essentially definites over widened domains, and any indefinite-like reading represents reference to a representative group of the maximal entity. In the previous chapter, I discussed the consequences of this revision for maximality. For anaphoricity, the new theory makes an even stronger prediction: since bare plurals never have genuine existential force, they are expected to be either strongly anaphoric, if the language has no articles, or strongly non-anaphoric, if there is a definite plural available.

So in principle, the standard neo-Carlsonian approach predicts free availability of anaphoric readings to Russian bare plurals, and unavailability of such readings for English bare plurals. It does not predict unavailability of non-anaphoric readings to Russian bare plurals, as they have access to DKP. The revised account (Dayal, 2013) makes a stronger claim, predicting that bare plurals in a language without articles have to be anaphoric. At this point, we don't have decisive empirical data on whether Russian bare plurals prefer anaphoricity as strongly as Russian bare singulars do; one of the goals of the experimental studies will be to investigate that.

4.2.2 Predictions

All in all, the neo-Carlsonian approach generates the following predictions. First, English bare plurals should show low rates of anaphoric interpretations, because the Blocking Principle prevents covert application of *iota* in the presence of a competing overt *iota*-alternative. Second, Russian bare plurals should show high rates of anaphoric interpretations, if the context provides a suitable antecedent, because *iota* is freely available in the absence of a definite article.

Importantly, the neo-Carlsonian account does not predict that the non-anaphoric readings should be unavailable or even dispreferred in Russian bare plurals: the DKP remains available as an alternative route. However, the preliminary data from elicitation suggest that bare nouns in Russian strongly prefer anaphoric construals if there is a suitable antecedent. The

way this preference operates is still an open question. If the preference is purely pragmatic, then it should be defeasible: manipulating the context to disfavor anaphoricity should weaken the anaphoric preference. If it reflects a grammatical specification, it should be resilient even under contextual pressure.

This theoretical situation motivated our experimental design. The final series of experiments reported in this dissertation focuses on the anaphoric potential of Russian vs. English bare plurals in contexts biased towards and against anaphoric interpretation. If we only tested bare plurals in contexts that are easily compatible with anaphoricity, we could not distinguish between two very different scenarios: we would not be able to tell apart grammatically privileged anaphoricity from pragmatically derived anaphoricity. This is why we needed to test contexts where the anaphoric interpretation is unlikely as well.

Therefore, two parallel experiments were carried out. Experiment 5A focused on the anaphoric potential of bare plurals in contexts that are easily compatible with anaphoricity. Experiment 5B tests whether that anaphoric potential holds up when the context is biased towards a non-anaphoric interpretation. As control condition, the experimental studies employ English definite plurals, which are expected to oppose non-anaphoric interpretations no matter the context, and indefinite plurals with *some* in English. In Russian, the control condition is the indefinite plural marked with *kakie-to* ‘some.PL’, one of the series of Russian indefinite pronouns formed with the suffix *-to*. Such constructions constitute a valid contrast for bare plurals: they introduce a new referent and have been analyzed as choice-functional indefinites (Yanovich, 2005; Geist, 2010)³. Therefore, plurals with *kakie-to* ‘some’ are expected to favor non-anaphoric construals.

Within each experiment, different plural types were tested pairwise: one sub-task tested English bare plurals vs. English definite plurals, and another sub-task tested English bare plurals vs. English indefinite plurals; the Russian task tested bare plurals vs. indefinite plurals (see Table 4.1).

³While these works claim obligatory wide scope for *to*-pronouns, Eremina (2012) shows that they can have apparent narrow scope readings as well.

Language	English		Russian
Task	English Task 1	English Task 2	Russian Task
Comparison	Definite plurals vs. Bare plurals	Indefinite plurals vs. Bare plurals	Indefinite plurals vs. Bare plurals

Table 4.1: Pairwise comparisons in Experiments 5A and 5B.

Now, we can formulate the exact predictions for each experiment. In Experiment 5A, Russian bare plurals should demonstrate strong tendency to be anaphoric, approaching the level set by overt definite plurals in English. English bare plurals should show substantially lower preference to be interpreted anaphorically, because *iota* is blocked by the definite plural. The overt indefinites (both the Russian *kakie-to* ‘some.PL’ and the English *some*-plurals) are predicted to favor non-anaphoric construals in both languages. A more intriguing question is how much the preference for anaphoric readings will drop in Experiment 5B compared to Experiment 5A. For Russian bare plurals, DKP should remain available, so when the contextual support for anaphoric reading is weakened, Russian bare plurals are expected to be interpreted anaphorically less. The main prediction is that Russian bare plurals will show lower preference for anaphoric interpretations, but this drop will be smaller than for English bare plurals. If Russian and English bare plurals drop by the same amount and converge at the same level in Experiment 5B, that would suggest the anaphoric preference in Russian is purely pragmatic and not related to the privileged availability of *iota*. If Russian bare plurals show greater resilience, that is consistent with *iota* being prioritized for bare plurals in Russian in a way that has no parallel in English.

4.3 Experiment 5A: anaphoric-biased contexts

Experiment 5A was designed to answer the following question: are Russian bare plurals interpreted anaphorically at rates compatible to English definite plurals in contexts that support anaphoric construal? It is set up as a forced binary choice interpretation paradigm. Participants read short contexts scenario and a target sentence containing a plural noun

phrase. They were asked to choose between two possible interpretations of this target sentence: the one rephrasing an anaphoric interpretation vs. the one rephrasing a non-anaphoric interpretation of the target plural. This design was chosen to minimize the ambiguity in response patterns, as it gives the participants an explicit choice between an anaphoric and a non-anaphoric interpretation of a given noun phrase. In this way, we map the binary distinction between familiarity and novelty directly onto the response options.

As mentioned in the previous section, two parallel versions of the experiments were conducted, one in Russian and one in English. Across these two versions, different plurals were presented pairwise. For English, bare plurals were compared to definite plurals and separately to indefinite plurals (modified by *some*). For Russian, bare plurals were compared to plurals modified by the indefinite pronoun *kakie-to* (which.PL-INDEF) ‘some’.

4.3.1 Materials and participants

The stimuli consisted of short context scenarios followed by a target sentence containing either a bare plural, a definite plural, or an indefinite plural, depending on the condition. Each context introduced a set of entities in the first sentence, and the target sentence used one of the plural types in a continuation. Participants chose between two interpretations: an anaphoric one, where the plural noun phrase referred back to the set introduced in the context, vs. a non-anaphoric one, where the noun phrase introduced a distinct set of entities. An example of a target item in the English version of the experiment is provided in (11); a screenshot of the same experimental trial is presented in 4.1.

- (11) Ethan ordered plates and cups online. When the delivery arrived, he put the stuff in the kitchen. Later, he washed **plates** / **the plates** / **some plates**.

Choose which best matches the context:

- | | |
|---|---------------|
| a. Ethan washed the plates he ordered | anaphoric |
| b. Ethan washed other plates he had in the kitchen. | non-anaphoric |

Ethan ordered plates and cups online. When the delivery arrived, he put the stuff in the kitchen. Later, he washed plates.

Choose which best matches the context:

- ☐ Ethan washed the plates he ordered.
- ☐ Ethan washed other plates he had in the kitchen.

Figure 4.1: Experiment 5A: Example of experimental trial in one of the English tasks, testing the bare plural.

Importantly, all the scenarios introduces the set of entities as a part of a coordinated structure, like *plates and cups* in (11) above. The reason for this was briefly discussed in section 4.1.2: the presence of a contrasting noun in the context sentence affects the acceptability of bare nouns in the continuation in a positive way, making them a more plausible anaphoric continuation than pronouns or demonstratives. This allows us to look at the anaphoric potential of target plural types (bare, definite, and indefinite) without any confound effect from competing anaphoric expressions.

The Russian version used the same context scenarios, translated and adapted to control for word order, as it is known to interact with (in)definiteness in Russian. All target plural nouns were placed in the direct object position immediately following the verb, and all verbs were in the past tense and perfective aspect, to avoid the influence of pluractionality on interpretation. (12) illustrates the experimental item from the Russian task, corresponding to the English item in (11):

- (12) Ivan zakazal tarelk-i i bokal-y onlajn. Kogda dostavka priexala, on prinës
Ivan ordered plate-PL and glass-PL online when delivery arrived he brought
pokupki na kuxnju. Pozže on pomyl **tarelki / kakie-to tarelki**.
shopping on kitchen later he washed plate-PL which-INDEF plate-PL
'Ivan ordered plates and glasses online. When the delivery arrived, he brought his
shopping to the kitchen. Later, he washed plates / some plates.'

-
- a. Ivan pomyl tarelk-i, kotorye on zakazal.
 Ivan washed plate-PL which.PL he ordered
 ‘Ivan washed the plates he had ordered.’ anaphoric
- b. Ivam pomyl drugie tarelk-i, kotorye uže byli u nego na kuxne.
 Ivan washed other.PL plate-PL which.PL already were by him on kitchen
 ‘Ivan washed other plates that were already in his kitchen.’ non-anaphoric

In Experiment 5A, all stimuli were designed in a way that favored the anaphoric interpretation. By this I mean that the actor of the situation was the same, and the location of the situation did not change. 16 scenarios were constructed in English and then translated into Russian.

As mentioned at the beginning of this section, the plural types were tested pairwise: English bare plurals vs. English definite plurals; English bare plurals vs. English *some*-plurals; Russian bare plurals vs. Russian *kakie-to* plurals. This resulted in 3 sub-experiments, which I will be referring to as English Task 1 (bare vs. definite), English Task 2 (bare vs. *some*), and Russian Task. Within each task, participants were assigned to one of the two experimental lists, counterbalanced so that participants only saw each scenario with only one type of plural. Each experimental list contained 16 experimental types (8 of each plural type tested). Experimental items were interspersed with 24 filler questions of a similar structure, half of which controlled for participant attention by having a specific expected correct answer, and the other half introduced some interpretive uncertainty. The two choices (anaphoric vs. non-anaphoric) were presented in randomized order. Participants completed a training block before the main task, and also had a minimum 60 second required break half way through the task.

For the English experiment, participants were recruited via Prolific and restricted to self-reported monolingual English speakers located in the US. For the Russian experiment, participants were recruited via Prolific and word of mouth and restricted to self-reported monolingual Russian speakers. After exclusions, the final sample consisted of 30 participants for each English experiment and 40 for the Russian experiment. The median completion time was 16 minutes for English task 1 (bare vs. definite plurals), 17 minutes for English task 2

(bare vs. *some*-plurals), and 20 minutes for Russian.

4.3.2 Results

The results of Experiment 5A are visualized in Figure 4.2; below, I discuss the pairwise comparisons within sub-experiments.

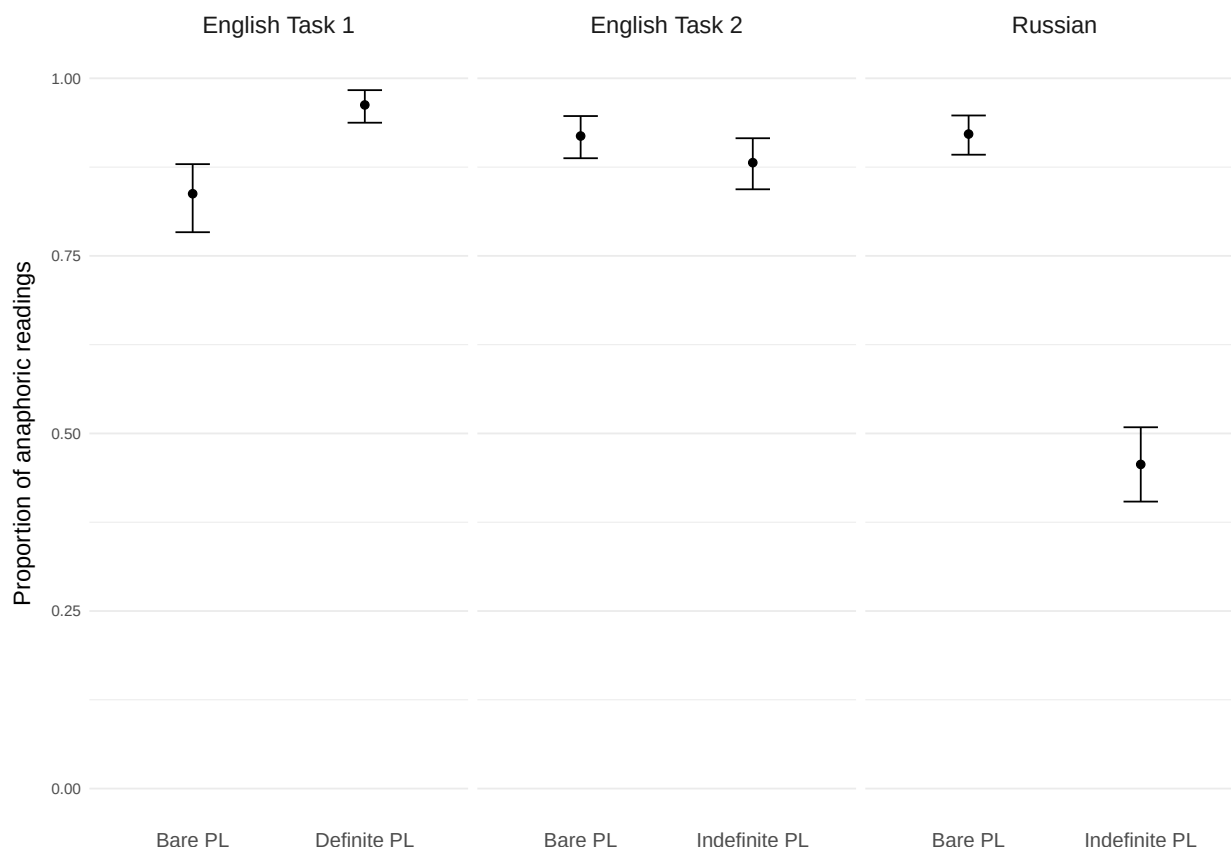


Figure 4.2: Experiment 5A (anaphoric bias): Proportion of anaphoric interpretations by plural type across the three pairwise comparisons, with standard error bars.

English Task 1: bare plurals vs. definite plurals. In the English task 1, experimental trials contained bare plurals vs. definite plurals. In our experimental trials with bare plurals, participants chose the anaphoric interpretation in 83.8% of trials, 95%CI [79.1, 88.4]. For definite plurals, participants chose the anaphoric interpretation in 96.2% of trials, 95% CI[93.8, 98.7]. To assess whether the difference between these proportions was significantly different, a linear mixed-effects logistic model was fitted to the data, predicting

the interpretation choice from the plural type as a fixed effect and random intercepts for participant and item⁴. The model showed that in contexts with definite plurals, participants were significantly more likely to choose an anaphoric interpretation, compared to contexts with bare plurals ($\beta = 1.92$, $SE = 0.43$, $p < .001$). The odds of selecting an anaphoric interpretation for a definite plural were 6.8 times greater than for a bare plural. While the proportions of anaphoric interpretations for both plural types are near-ceiling and the effect is modest on the probability scale, it is large on the log-odds scale.

To better assess this pattern, subject-level responses were analyzed. As a result, we can conclude that the effect is driven primarily by the reduction in variability: responses in the bare plural condition showed much greater dispersion, with some participants choosing the non-anaphoric interpretation, but the responses in the definite condition were clustered near the top. Since the contexts we tested were biased towards an anaphoric interpretation, we can conclude that bare plurals do not resist anaphoric readings strongly, but the presence of the definite marking strengthens and stabilizes an already strong preference for an anaphoric reading.

English Task 2: bare plurals vs. indefinite (*some*) plurals. In this subexperiment, participants chose between anaphoric and non-anaphoric interpretations of bare plurals vs. plurals modified by *some*. In this task, for bare plurals, participants chose the anaphoric interpretation in 91.9% of trials, 95% CI[88.9, 94.9]. For indefinite plurals, the anaphoric interpretation was preferred in 88.1% of trials, 95% CI[84.6, 91.7].

To evaluate this pattern statistically, a linear mixed-effects logistic regression model with the same structure was fit to the data (with the plural type as a fixed effect and participant and item as random effects). While *some*-marked plurals received anaphoric readings in a smaller percentage of trials descriptively, this effect is marginal ($p = 0.08$), which means that the evidence is suggestive of a trend but not strong. The random effects in the model

⁴All logistic models reported in this chapter had the same formula, applied to different datasets: `choice ~ pluralType + (1|item) + (1|ResponseId)`, `family = binomial`. The models were fitted using the `lme4` package (Bates et al., 2015). Explained variance was estimated by R^2 values, obtained using the `MuMIn` package (Bartoń, 2024).

showed substantial variability, especially across items with the variance of 2.5, compared to participants with the variance of 0.83. This suggests that differences between items contributed more to response variability than differences between participants. Most of the explained variance in the model comes from the random effects (about 50%), while the fixed effect only explains about 1% of the variance. This indicates that plural type manipulation had little to do with which interpretation (anaphoric or non-anaphoric) the participants preferred.

One can see that bare plurals showed a greater proportion of preferred anaphoric interpretations when the contrasting condition was the indefinite plural, rather than the definite plural. I will discuss the theoretical conclusions from this asymmetry later. To evaluate whether this difference is enough to be statistically significant, the datasets from both English task were combined and subsetting to the bare plural responses. Another logistic regression of the similar structure was fitted to this subset of the data, predicting the likelihood of choosing an anaphoric interpretation for the bare plural depending on what they were contrasted with, with random intercepts for participants and items. The analysis showed that bare plurals were significantly more likely to be interpreted anaphorically when the contrasting condition was the indefinite plural (in the English task 2), $\beta = 1.00$, $SE = 0.47$, $p < 0.05$. In other words, participants of the Task 1, which contained stimuli with bare plurals and definite plurals, were 2.7 times less likely to choose the anaphoric interpretation for the bare plural than participants of the Task 2, which included stimuli with bare plurals and indefinite plurals. This corresponds to a moderate effect size. Again, most variability came from the random effects of items and participants (46% of the explained variance) than from the fixed effect (4% of the explained variance).

Russian bare plurals vs. Russian indefinite (*kakie-to* ‘some’) plurals. In the Russian task, we see a much sharper contrast. For bare plurals, the anaphoric interpretation was preferred in 92.2% of trials, 95% CI[89.3, 95.0]. By contrast, for indefinite plurals, the anaphoric interpretation was chosen in 45.6% of trials, 95% CI[40.4, 50.9]. A mixed-effects

logistic regression model of the same architecture was fit to the data, and it showed a strong effect of plural type on the likelihood of choosing the anaphoric interpretation ($\beta = 3.8$, $SE = 0.34$, $p < 0.001$). Participants were 45 times more likely to choose an anaphoric interpretation for the bare plurals than for the indefinite plural. The random effects showed substantial variability across participants, with the variance of 2.3, than across items, with the variance of 0.82. This means that individual differences between participants contribute more to overall variability than item differences. Unlike in the previous models, the plural type alone explains a substantial portion of the variance (36%), and random effects add additional explanatory power (about 31% of explained variance) on top of this already strong fixed effect.

4.3.3 Interim summary

We see that Russian bare plurals are interpreted anaphorically at high rates in contexts that support anaphoric construal, comparable to the rates observed for English definite plurals. The sharpest contrast we observe is between Russian bare plurals and Russian indefinite plurals: bare plurals were interpreted as anaphoric in 92.2% of trials, compared to 45.6% for *kakie-to* indefinites. This result is consistent with the prediction that Russian bare plurals have privileged access to *iota*. The *kakie-to* indefinites showed near-chance rates of being interpreted anaphorically. This suggests confirming that they resist anaphoric construals even when the discourse supports them.

The English results were more complex. While the definite plurals showed high levels of anaphoric interpretations, as expected from an overtly marked definite expression, English bare plurals showed relatively high anaphoric rates, too (83.8% in task 1 and 91.9% in Task 2). This was not predicted by the blocking account. This suggests that in strongly anaphoric-biased contexts, even English bare plurals can be accommodated as anaphoric through pragmatic mechanisms.

The near-ceiling rates for both English bare plurals and English definites in anaphoric-biased

contexts mean that Experiment 5A cannot on its own distinguish grammatically driven anaphoricity from pragmatically derived anaphoricity in English. Experiment 5B addresses this directly. By using contexts designed to disfavor anaphoric interpretations, it tests whether the anaphoric preferences observed in 5A are resilient under contextual pressure, as would be expected from a purely pragmatic preference.

4.4 Experiment 5B: non-anaphoric-biased contexts

4.4.1 Materials, participants, and design

The design of the Experiment 5B was identical to that of Experiment 5A with one exception: the scenarios were built in a way that consistently biased the participants towards a non-anaphoric interpretation. As in the previous task, participants read short context scenarios where a set of entities was introduced and chose between an anaphoric and a non-anaphoric interpretation of a plural noun phrase. The way the bias towards non-anaphoric interpretation was operationalized was the following. The context introduced a change of situation between the first and the second mention of the target plural NP: the actor of the situation changed and the location of the plural referent changed. This made it plausible that the noun phrase in the target sentence referred to a new set of entities rather than the one introduced in the context. For example, a context parallel to (11) from Experiment 5A but biased towards non-anaphoric interpretation was the following:

- (13) Ethan ordered plates and cups online. When the delivery arrived, he opened the packages in the kitchen. Later that evening, his grandma was cleaning up after dinner and noticed some dishes in the sink. She washed **plates / the plates / some plates**.

Choose which best matches the context:

- a. Ethan's grandma washed the same plates Ethan had ordered. anaphoric

b. Ethan’s grandma washed other plates.

non-anaphoric

Experiment 5B included 8 out of the original 16 scenarios of Experiment 5A adapted for this purpose, and 8 newly built scenarios that introduced the change of the actor and the location as well. The same pairwise comparisons were used (see Table 4.1).

All other aspects of the design were identical to Experiment 5A. Participants were assigned to one of the two counterbalanced experimental lists, each list consisting of 16 experimental trials and 24 fillers tasks. The filler tasks were reused from Experiment 5A. Participant recruitment and exclusion criteria were also identical. After exclusions, the final sample consisted of 42 participants for each English experiment and 45 for the Russian experiment. The median completion time was 23 minutes for the Russian task and the English task 1 (bare vs. definite plurals), and 24 minutes for the English task 2 (bare vs. *some*-plurals).

4.4.2 Results

Figure 4.3 visualizes the results of Experiment 5B. Below I interpret the pairwise comparisons within each sub-task.

English bare plurals vs. English definite plurals. Unlike in Experiment 5A, we observe a significant contrast between bare plurals and definite plurals. Bare plurals received an anaphoric interpretation in 34.3% of trials, 95%CI[29.0, 39.6], while definite plurals received an anaphoric interpretation in 67.6% of trials, 95% CI[62.4, 72.8]. A mixed-effects logistic regression confirmed that this difference is significant and of a large effect size, $\beta = 1.92$, $SE = 0.22$, $p < 0.001$. Definite plurals have about 6.8 times higher odds of an anaphoric interpretation than bare plurals. 15% of the variance in the model was explained by the fixed effect, and the full model including random effects explains about 48% of the variance. Random effects indicate substantial variability across both participants and items, with somewhat greater variance attributable to items.

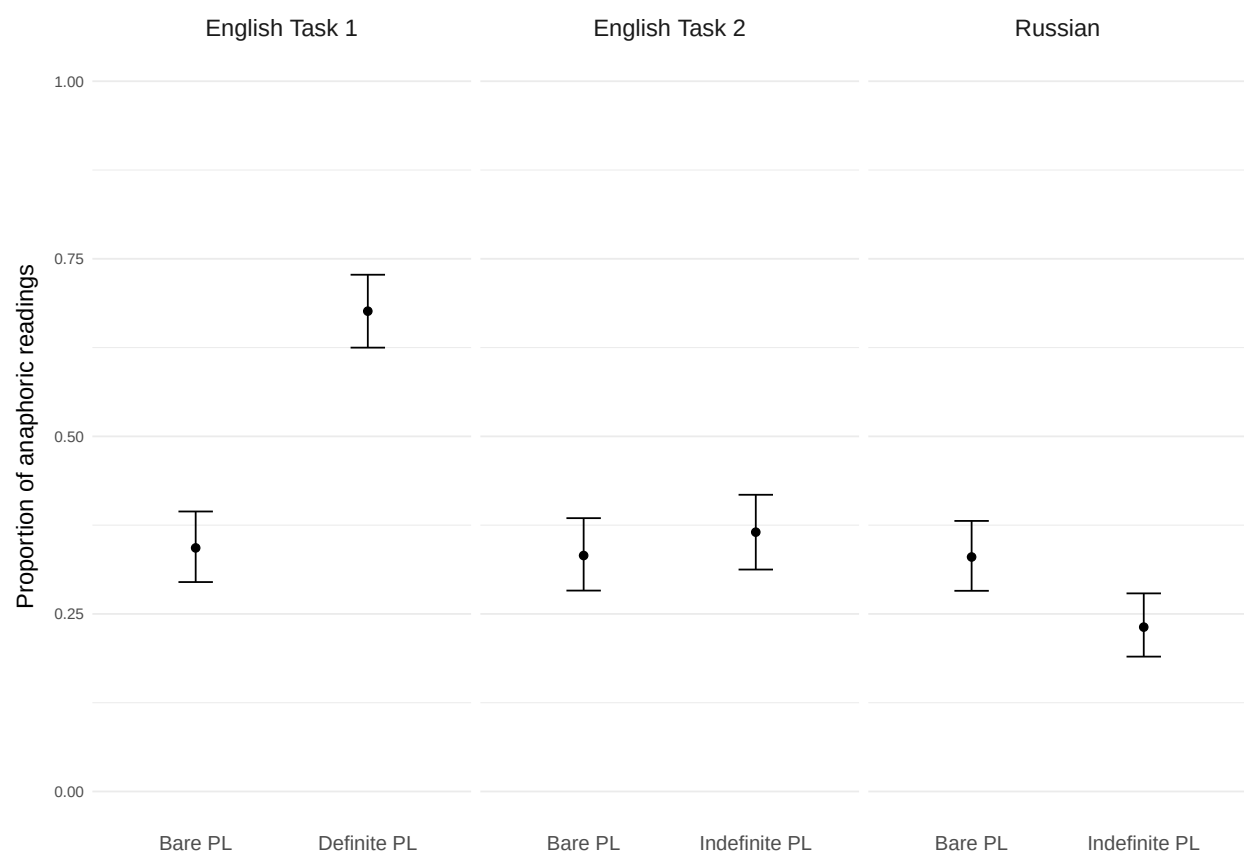


Figure 4.3: Experiment 5B (non-anaphoric bias): Proportion of anaphoric interpretations by plural type across the three pairwise comparisons, with standard error bars.

English bare plurals vs. English indefinite plurals. Like in Experiment 5A, bare plurals and indefinite plurals showed similar likelihood of being interpreted anaphorically. Bare plurals received an anaphoric interpretation in 33.2% of trials, 95% CI[27.9, 38.5], while indefinite plurals received an anaphoric interpretation in 36.5% of trials, 95% CI[31.1, 41.9]. This difference did not appear significant, as confirmed by mixed-effect logistic regression modeling ($p = 0.28$).

Russian bare plurals vs. Russian indefinite plurals. The proportion of anaphoric interpretations dropped significantly for Russian bare plurals in Experiment 5B compared to Experiment 5A. When the context was biased towards non-anaphoricity, Russian bare plurals received an anaphoric interpretation in 33% of trials, 95% CI[27.8, 38.2]. Russian indefinite plurals received an anaphoric interpretation in 23.1% of trials, 95% CI[18.6, 27.6]. This difference is nevertheless significant, as mixed-effect logistic modeling showed ($\beta = -0.69$, $SE = 0.24$ $p < 0.01$). Participants were approximately twice more likely to chose an anaphoric interpretation in a context biased towards non-anaphoricity for a bare plural, than for an indefinite plural. This result is statistically reliable but moderate in size. The random effects showed substantial variability, especially between items, but between participants as well.

4.5 General discussion

Let us summarize the results of Experiments 5A and 5B. Russian bare plurals appeared to be very sensitive to context manipulation. They showed high probability of being interpreted as anaphoric in Experiment 5A, but significantly lower probability of anaphoric interpretations when the context biased against anaphoricity. However, in each task they remained above Russian indefinite plurals with *kakie-to*, which resisted anaphoric interpretations across the board.

English bare plurals showed very high rates of anaphoric interpretations when the context was biased towards anaphoricity, but that rate was nevertheless lower than those of overt

definite plurals. However, when presented along with overt indefinites marked with *some*, English bare plurals showed higher tendency to be anaphoric. This suggests that they are highly flexible and the competition account is plausible. In Experiment 5B, bare plurals showed very low probability of being interpreted as anaphoric. Overt definites resisted non-anaphoric interpretations (although we observe a drop there, too); bare plurals did not.

The most important results come from 5B, the resilience task. I will proceed interpreting these results in the following order. First, I discuss our results in the light of the Blocking Principle, central to the neo-Carlsonian approach 4.5.1. I then proceed to a more detailed discussion of Russian bare plurals in the resilience task 4.5.2. Third, I will discuss the asymmetry between Task 1 and Task 2 in English 4.5.3. Finally, I will discuss what the behavior of Russian indefinite plurals observed in the experimental results contributes to their treatment in the existing literature 4.5.4.

4.5.1 The results and the Blocking Principle

First, let's evaluate our results for English bare plurals in the light of the Blocking Principle. The original version of this principle (Chierchia, 1998) implies that English bare plurals should not be able to access *iota* at all. This predicts that they should show low rates of anaphoric preferences even in contexts that support anaphoricity. However, the observed picture is informative but less radical. The contexts with English bare plurals showed anaphoric readings in 83.8% of trials when the other stimuli contained English definite plurals, and in 91.9% of trials when the other stimuli contained English indefinite plurals with *some*. While we do see a contrast between English bare plurals and English definite plurals, we need to acknowledge that the categorical version of the Blocking Principle is not supported by our results.

Another important result is the *similarity* of English bare plurals with English *some*-plurals. This suggests that whatever is responsible for high rates of anaphoric readings in Experiment 5A is not specific to bare plurals, but applies to any plural form that does not actively resist

anaphoric construal. Since *some* is known to have a partitive reading, this may be what explains its high anaphoric rates. What about the high anaphoric rates of the bare plural, then?

We can evaluate whether the more radical version of the neo-Carlsonian account, proposed in (Dayal, 2013), can explain our findings about English bare plurals. On that view, existential force is eliminated from DKP completely, and bare plurals are analyzed as definites over a widened domain; Dayal claims unavailability of anaphoric readings for bare plurals in English, since there is a definite plural available. However, as Chierchia (2022b) notices, nothing in this account blocks anaphoric readings of English bare plurals when Proper Widening is satisfied.

Let us entertain this possible explanation in more detail. According to Dayal (2013), the bare plural in a situation s must be interpretable with respect to a wider situation s' properly containing more instances of a kind; the truth conditions are evaluated with respect to the maximal entity instantiating the kind in that wider situation s' . Proper Widening also comes with a presupposition, which requires that the wider situation s' contains at least one instance of the kind not present in the base situation s .

Now, let us look back at the experimental scenarios that were biased towards the anaphoric interpretation (Experiment 5A), such as (11). The context introduces a set of plates P_0 that Ethan ordered, and he places them in the kitchen. The target sentence uses a bare plural *plates* in a continuation where Ethan washes plates in the same kitchen. Let s_0 be the situation introduced by the context sentence (which contains P_0), and let s_1 be the situation of the target sentence. One can stipulate that since the actor and the location are the same, s_1 is essentially an extension of s_0 , and since s_1 and s_0 coincide, the maximal entity instantiating the kind is exactly P_0 , the set introduced in the prior discourse. This results in an anaphoric reading. The bare plural picks up P_0 not because of *iota*, but because the contextually most salient widening of the situation is one that contains exactly the plates already introduced, making them the maximal instantiation of a kind in s' .

Let us look at the parallel scenarios from Experiment 5B, which was biased towards non-anaphoricity. In all of them, the actor and the location of the situation were changed. It is possible that this modification was sufficient to make s_2 a distinct situation in which P_0 is no longer present, rather than an extension of the base situation. The maximal instantiation of the kind in the most salient s' is no longer P_0 .

A potential objection to this would be that English *some*-plurals showed anaphoric rates nearly identical to bare plurals across both 5A and 5B. Proper Widening, though, is specific to kind-denoting bare plurals and does not apply to ordinary indefinites. However, *some*-plurals have an independent route to anaphoric-like readings as they can have partitive readings. Therefore, high anaphoric readings of bare plurals and *some*-plurals in Experiment 5B may arise from different mechanisms: *some*-plurals may be interpreted partitively, while bare plurals may rely on Proper Widening or some other pragmatic process. What is important for our purposes is that neither involve grammatically encoded anaphoricity, as they do not resist non-anaphoric interpretations in a way definite plurals do.

While this is a mere stipulation rather than a full-fledged formalized account, it is a potential direction of explanation of the difference between the anaphoric potential of English bare plurals in Experiment 5A vs. Experiment 5B. In essence, it would go along the following lines: in Experiment 5A, the convergence of s_0 and s_1 results in the anaphoric reading being the most natural output of Proper Widening; in Experiment 5B, the base situation and the situation of the target sentence diverge, which does not recover the set introduced in the base situation as the natural output of Proper Widening. I therefore treat the account proposed in Dayal (2013) as a possible but not fully formalized explanation of the overall high baseline of English bare plural anaphoricity in anaphoric-biased contexts.⁵ Why that baseline differs depending on the comparison condition is a separate concern, not explainable by Proper Widening; I will address it in 4.5.3. But first, let us discuss the results for Russian.

⁵It is worth mentioning, though, that the formal proposal from Dayal (2013) faces independent empirical challenges in English, discussed in Chierchia (2022b).

4.5.2 Russian bare plurals and context manipulation

We see that the percentage of anaphoric interpretations for Russian bare plurals dropped significantly between Experiment 5A and 5B, from nearly 92% to 33%. This is a very large effect. The magnitude of the observed effect suggests that anaphoric interpretation is highly context-dependent. This result is compatible with the theoretical picture described in section 4.2, namely, the idea that both *iota* and DKP route are freely available to Russian bare plurals. When the context disfavors anaphoricity, the DKP becomes more competitive, and the bare plural is realized through DKP and receives a non-anaphoric interpretation. The neo-Carlsonian framework predicts that Russian bare plurals should have access to anaphoric interpretations, not that they should resist non-anaphoric interpretations categorically.

The most meaningful result for Russian bare plurals is not their absolute level of anaphoricity in Experiment 5B, but the persistent significant difference between Russian bare plurals and Russian indefinite plurals with *kakie-to* across both tasks. In Experiment 5A, bare plurals were judged as anaphoric in 92.2% of trials, while indefinite plurals were anaphoric in 45.6% of trials; in Experiment 5B, Russian bare plurals received an anaphoric interpretation in 33% of trials, while the indefinite plurals did so in 23.1% of trials. While the gap between Russian bare plurals and indefinite plurals is smaller in Experiment 5B, it remains significant.

How can one explain this persistent effect? There are multiple ways. One would be to say that Russian plurals with *kakie-to* are morphologically marked as [−definite], and therefore the Novelty-Familiarity condition blocks them from being anaphoric. Russian bare plurals have no morphological (in)definiteness marking, and therefore are exempt from the Novelty-Familiarity condition. However, this explanation would be inconsistent with the Blocking Principle: if indefinite pronouns like *kakie-to* count as the morphological marking of [−definite], then the presence of overt indefiniteness marking in the system would block non-anaphoric readings of Russian bare plurals⁶. This is not what we observe in the results:

⁶This line of explanation only works if familiarity is derived from *iota*, as the Blocking Principle applies to argument-forming operators and does not concern Novelty-Familiarity directly (Dayal, p.c.).

in many trials, the non-anaphoric interpretation was preferred for bare plurals as well.

A more promising explanation would be that *kakie-to* blocks or at least disfavors anaphoric readings on its own, defaulting to a novel reading. But Russian bare plurals, by contrast, do not carry such a default in either direction. They are genuinely underspecified with respect to Novelty-Familiarity, and therefore track the discourse context almost entirely, being strongly anaphoric when the context supports it and rather non-anaphoric when the context disfavors anaphoricity.

To sum up, the residual gap between bare plurals and indefinite plurals in Russian likely does not show that bare plurals are grammatically anaphoric, but rather that indefinite plurals are grammatically anti-anaphoric. In section 4.5.4, I will discuss potential implementations of this requirement into the semantic profile of *to*-pronouns in more detail. The anaphoric potential of the Russian bare plural is derived from the absence of a competing default, not from a grammatical specification towards anaphoricity.

4.5.3 The contrast effect in English bare plurals

One result that cannot be overlooked is the asymmetry between the anaphoricity rates of English bare plurals across different pairwise comparisons, noticeable in Experiment 5A. English bare plurals showed significantly higher anaphoricity rates when contrasted with English *some*-indefinites, compared to when contrasted with definite plurals. This is best interpreted as a contrast effect within the forced-choice paradigm. Even though the task contained as many fillers as experimental items, and different types of plurals were presented across different experimental items, the nature of the comparison condition influenced how participants interpreted the bare plural. When the alternative was a definite plural, participants were nudged toward the non-anaphoric interpretation of the bare plural; when the alternative was an indefinite plural, participants were nudged towards the definite-like interpretation. This behavior could be reinforced by the nature of the task, as forced-choice paradigms allow for less flexibility.

However, if it were a purely methodological artifact, we would expect to observe the contrast effect no matter whether the context was biased towards anaphoricity or not. Yet, we only see it in Experiment 5A, where all the contexts were biased towards anaphoricity, but not in Experiment 5B, where all the scenarios favored a non-anaphoric continuation. In Experiment 5B, the English bare plural converges with the indefinite plural and diverges sharply from the definite plural.

I suggest that the contrast effect visible in contexts biased towards anaphoricity reflects the Blocking Principle. In particular, it reflects that participants had some implicit knowledge that the definite is the grammatically appropriate anaphoric form in English. When the definite was present as an alternative, participants treated the bare plural as less optimal. When the alternative is the plural with *some*, the bare plural acts as the more anaphoric-like form by default. The asymmetry disappears in Experiment 5B, because the anaphoric potential of the bare plural is entirely context-dependent and not encoded grammatically (unlike the definite plural, which lexicalizes *iota*). As a result, when the context strongly disfavors anaphoricity, the bare plural is treated the same as the genuine indefinite.

4.5.4 Notes on Russian *to*-pronouns

An interesting side result is that we see a much sharper resistance to anaphoric reading in the Russian indefinite plurals with *kakie-to* compared to English indefinite plurals with *some*. English indefinite plurals showed only a marginal and statistically non-significant tendency away from English bare plurals. In the Russian task, participants were 45 times less likely to choose an anaphoric interpretation for *kakie-to* plurals compared to bare plurals. In essence, the probability of an anaphoric interpretation for *kakie-to* plurals was near-chance. This shows that the anaphoric reading is not categorically impossible, but is strongly and systematically dispreferred, and suggests that *to*-pronouns impose some (potentially defeasible but nevertheless strong) constraint against picking up discourse-familiar referents.

There exist other data that distinguishes English *some* and Russian *kakie-to*. For example, Russian *kakie-to*, resists felicitous continuation listing the members of the set introduced by it, especially if the speaker is in a position to identify the referent. In (14), the continuation is judged infelicitous because using *kakie-to* signals ignorance, and in a first-person report of one’s own actions, demonstrating the full knowledge of what has been bought is pragmatically anomalous. This contrast weakens if the subject is third-person (15): it is compatible with genuine speaker ignorance about what was purchased, and the continuation resolves that ignorance rather than contradicts it.

- (14) Ja kupi-l-a **kak-ie-to** **knig-i**, #a imenno Bulgakov-a i
 I buy-PST-F which-PL-INDEF book-PL and namely Bulgakov-GEN and
 Čexov-a.
 Chekhov-GEN
 ‘I bought some books, #namely Bulgakov and Chekhov.’
- (15) Papa kupi-l **kak-ie-to** **knig-i**, ?a imenno Bulgakov-a i
 dad buy-PST which-PL-INDEF book-PL and namely Bulgakov-GEN and
 Čexov-a.
 Chekhov-GEN
 ‘Dad bought some books, namely Bulgakov and Chekhov.’

This is not something we observe with English *some* at all (noted also in Dayal (2013)):

- (16) I bought **some books**, namely Bulgakov and Chekhov.

While our focus is on bare nouns, these results are also meaningful for the existing research on indefinite pronouns in Russian. The account of *to*-pronouns suggested by Yanovich (2005) treats them as generalized choice functions known to the speaker. This correctly predicts their specificity, but fails to capture the resistance to backward anaphora or the ignorance component of meaning observed in (14). Speaker knowledge of the choice function is compatible, in principle, with the referent being discourse-familiar and accessible.

Instead, these patterns are consistent with the framework developed by Degano and

Aloni (2025), who classify Russian indefinites with *-to* as epistemic indefinites. In their account, such indefinites are formally associated with a variation condition $var(\emptyset, x)$, which requires the value of the variable introduced by the indefinite to vary across the speaker’s epistemic possibilities. Essentially, Degano and Aloni place Russian *to*-indefinites alongside other indefinites marking ignorance, such as the German *irgend-* or Italian *un qualche*. This account of *-to* indefinites is compatible with our findings, namely, with the fact that *to*-indefinites resisted anaphoric interpretation even in contexts biased towards it. Familiar referents that have already been introduced and identified in the discourse have by definition a constant value across the speaker’s epistemic possibilities. Using an epistemic indefinite to refer back to an already established referent therefore creates a contradiction: the indefinite signals that the speaker’s epistemic possibilities vary, but the prior discourse has already collapsed those possibilities to a single identified individual⁷. English *some*, which carries no analogous ignorance condition when it modifies a plural noun phrase⁸, has no restriction on picking up familiar referents, and this is what we observe in our experimental results.

In this way, while this is not the focus of the current research, our experimental results also shed some light on the nature of Russian indefinites with *-to*. We observed their resilience to anaphoric interpretations even when the context was biased towards anaphoricity, and we didn’t observe the same in English indefinites with *some*. Our experimental results lend support to accounts that include an ignorance component into the semantics of Russian *to*-indefinites, and are not straightforwardly derivable from the choice-functional account alone.

⁷The same logic extends to the contrast observed in sentences with appositive identification (14)-(15). When the speaker is genuinely ignorant (15), $var(\emptyset, x)$ is satisfied at the point of the utterance and the continuation simply resolves it. When the speaker cannot be ignorant, as in the first-person case (14), $var(\emptyset, x)$ was never satisfiable to begin with, and the continuation makes this contradiction explicit.

⁸Singular nouns modified by *some*, however, do impose ignorance and have been accounted in terms of epistemic indefinites (Alonso-Ovalle and Menéndez-Benito, 2003).

4.6 Sharper predictions for the singular

Before concluding, let me make a final remark about the predictions for the bare singulars. The predictions of the neo-Carlsonian approach about bare singulars in terms of anaphoricity are more straightforward. In episodic contexts, bare singulars are expected to shift only through *iota*. This means that if we were to test them in the contexts of Experiment 5B, which was biased towards non-anaphoric interpretations, we would expect them to resist non-anaphoricity and display a significantly higher rate of anaphoric interpretations than the bare plural. The data from elicitation, presented at the beginning of this chapter, showed a particularly sharp bias towards an anaphoric interpretation (see (4)-(5)), supporting this prediction. A controlled experiment measuring how much the bare singular resists non-anaphoric interpretations remains a direction for future research.

4.7 Conclusion

This chapter investigated the anaphoric potential of bare plurals in English vs. Russian. Specifically, we were interested whether the following contrast, predicted by the neo-Carlsonian approach, holds: Russian bare plurals have access to *iota* and therefore can be anaphoric; English bare plurals do not have access to *iota* (because of the presence of the competing definite plural) and therefore cannot be anaphoric.

The results of our experimental investigation yielded a complex picture. Russian bare plurals do have access to anaphoric interpretations in a way that is different from overt indefinites, which are anti-anaphoric. However, this access is modulated by the context. This is compatible with the idea that Russian bare plurals always have access to some non-anaphoric construal, be that through some version of DKP or through a pure existential construal; this option is readily adopted when the discourse favors it.

The conclusions for the English bare plural are more complicated. Any account that

blocks any anaphoric reading from English bare plurals would not be supported by our results. However, our results are compatible with English bare plurals being able to have anaphoric readings through some pragmatic mechanism, such as Proper Widening. We also see that English bare plurals always show lower anaphoric rates compared to English definite plurals, no matter the context manipulation. This suggests that anaphoric readings are somewhat restricted for English bare plurals, but not categorically unavailable. The Blocking Principle captures this asymmetry, but only if it is understood as a preference rather than an absolute constraint.

To conclude, the neo-Carlsonian framework correctly predicts the broad contrast in the anaphoric potential between Russian and English bare plurals, but our results show that more attention needs to be paid to how pragmatic context interacts with the available argument-forming operations. Refining this complexity and modeling it formally remains a direction for future work.

5 Conclusions and open questions

5.1 Summary of the findings

In this dissertation, I investigated the phenomenon of (in)definiteness in Russian, a language that lacks any sort of articles. The main goal was to evaluate the leading approaches to bare noun semantics in such systems through controlled experimentation. In doing so, I analyzed the following dimensions of (in)definiteness: uniqueness, maximality, and anaphoricity. Below, I briefly summarize my findings and their theoretical contributions.

5.1.1 Uniqueness

Chapter 2 investigated whether bare singulars in Russian impose a uniqueness requirement, typically associated with definite-like semantics achieved through *iota*. I demonstrated that bare singulars in Russian are systematically sensitive to uniqueness manipulation. Across three experimental studies, I demonstrated that sentences with bare singular subjects are judged as more felicitous when the referent of the bare singular is unique within a given domain. While this does not allow us to conclusively decide whether Russian bare singulars are always definite or not, we can safely conclude that the semantics of Russian bare singulars should involve some sort of uniqueness constraint. Importantly, since uniqueness is evaluated relative to a relevant domain, the constraint is most visible in contexts that block domain restriction.

These findings challenge a classic existential analysis, whether implemented via standard

existential closure or choice functions. The uniqueness requirement is robust, and an adequate account must include it into the semantic profile of bare singulars. However, our results do not show definitively what the nature of that requirement is and whether it is at-issue or not. They do not definitively decide if bare singulars shift via a genuine *iota* operation, or can be existential with a custom uniqueness requirement.

5.1.2 Maximality

In Chapter 3, I examined whether Russian vs. English bare plurals in episodic contexts display maximality, which would align them with definite descriptions rather than existentials. Through a controlled acceptability-judgment study using visual stimuli, I provided definitive evidence that in preverbal subject position, Russian bare plurals strongly prefer a maximal interpretation in presentational contexts, and the same holds for English bare plurals. Preliminary data from the object position replicated the preference for maximal interpretation for Russian bare plurals but not for English bare plurals; the latter tolerated non-maximal interpretations easier in the object position.

Yet again, I consider these findings an argument against purely existential approaches to bare plurals. These findings also demonstrate that a kind-based approach to episodic bare plurals such as Derived Kind Predication does require an additional maximization clause in order to capture the facts correctly. The results are broadly consistent with accounts that do encode maximality into the semantics of bare plurals. However, the data from this chapter alone cannot adjudicate between competing implementations of the maximality component, specifically, whether it is lexically encoded or pragmatically derived through exhaustification. The subject-object asymmetry observed in English but not in Russian points towards a higher-level difference between the two languages, most likely, the presence of a competing overt definite in English.

5.1.3 Anaphoricity

Finally, in Chapter 4, I turn from truth-conditional conditions like uniqueness/maximality to the discourse-level behavior of bare plurals, and focus on the anaphoric potential of bare plurals in English vs. Russian. I reported two parallel forced-choice experiments that investigate whether Russian vs. English bare plurals can function anaphorically in a way predicted by the neo-Carlsonian approach.

I provided novel data about anaphora preference in contexts that were biased towards or against anaphoricity. I demonstrated that while their preference for anaphoric interpretation is highly modulated by the context, Russian bare plurals consistently have access to anaphoric interpretations in a way that differentiates them from overtly indefinite constructions that resist anaphoric interpretations, such as indefinite pronouns. This is broadly compatible with what the neo-Carlsonian approach predicts: Russian bare plurals have consistent access to covert *iota*, but also have access to non-anaphoric interpretations through Derived Kind Predication. English bare plurals, while also showing high rates of anaphoricity when the context favored that and low rates of anaphoricity when the context disfavored that, were consistently less likely to be interpreted as anaphoric when compared to English definite plurals. This may reflect the Blocking Principle, one of the pillars of the neo-Carlsonian approach. However, the extreme context-dependent flexibility of English bare plurals suggests that the elicitation data reported in previous works is not as categorical and straightforward, and it might be the time to reinterpret the Blocking Principle as a preference rather than a categorical restriction.

5.2 Future directions

Below, I discuss some issues that fell beyond the scope of this thesis, but would be great directions for future research, as they would complement the findings presented in this study and the discussion of (in)definiteness in languages without articles in general.

5.2.1 The nature of the D layer in Russian

In this dissertation, I do not really discuss what the syntactic structure of Russian bare nouns is, and how it interfaces with the semantic profile I’ve developed. Whether bare nouns in languages without articles are DPs or NPs is a subject of intense debate, especially in Slavic linguistics. While some authors claim that Slavic bare nouns are NPs (Bošković, 2005, 2008, 2009), others claim that in any language without articles, at least some bare nouns project up to the DP (Progovac, 1998; Pereltsvaig, 2006; Lyutikova, 2018). I adopt the latter view, in the form that Russian bare nouns are structurally ambiguous. The same surface string can potentially correspond to different functional structure depending on the syntactic context. I further assume that bare arguments in subject and object positions, analyzed in this dissertation, are full-fledged DPs.

I follow Lyutikova (2018)’s analysis of the Russian noun phrase, which is along the following lines. While Russian has no specialized lexical items whose inherent category is D, the D projection is nevertheless available. The presence of this projection is visible from the interaction of adjective-like elements, such as demonstratives or possessives, which yield different readings depending on their position. When they are located higher up in the structure (above NumP), they receive an exhaustive interpretation. When the same elements are located lower (below NumP), they remain within the lexical layer of the functional structure and do not yield a definite-like interpretation. What determines the interpretation is the position rather than the lexical category.

Let us look at a couple of examples illustrating that the position above the numeral contributes to a definite interpretation. One of the elements that can “activate” the D-layer is demonstratives. When the demonstrative is placed above the numeral, it results in an exhaustive interpretation. When the demonstrative is placed below the numeral, it is NP-internal and results in a partitive interpretation¹. According to Lyutikova, this

¹Examples (1)-(2) were inspired by a corpus example from Lyutikova (2018, p. 39) but elicited independently.

demonstrates that the higher syntactic position contributes definite semantics independently of the lexical item that fills it.

- (1) Éti pjat' aksiom nado vyučit' k èkzamen-u.
 these five axiom.GEN.PL is.needed learn.INF by exam-DAT
 'One needs to learn these five axioms for the exam.'

↪ there are only five axioms on the list

- (2) Pjat' étix aksiom nado vyučit' k èkzamen-u.
 five these.GEN.PL axiom.GEN.PL is.needed learn.INF by exam-DAT
 'One needs to learn five of these axioms for the exam.'

↪ there more than five axioms on the list, but you only need to learn any five of them

A similar contrast occurs with possessives. A possessive preceding the numeral is interpreted exhaustively (3); a possessive following the numeral does not entail that the possessor only has two sons (4).

- (3) Moi dva syn-a uže okonči-l-i škol-u.
 my.PL two son-GEN already graduate-PST-PL school-ACC
 'My two sons have already graduated high school.' (Lyutikova, 2018, p. 39)

↪ I only have two sons

- (4) Dva moix syn-a uže okonči-l-i škol-u.in these
 two my.GEN.PL son-GEN already graduate-PST-PL school-ACC
 'Two of my sons have already graduated high school.' (Lyutikova, 2018, p. 39)

↪ I possibly have more than two sons

Lyutikova is non-committal on whether demonstratives and possessives in Russian examples like (1) or (3) are located in Spec,D or D itself. However, it is clear that the exhaustive, definite-like semantics comes from the syntactic position rather than from the demonstrative or possessive itself. While I am not developing a full-fledged syntactic account here, I suppose that what the null D actually hosts is a null argument-forming operator. In its

definite-flavored realization, like in the examples above, it is the *iota*. The moved elements, whether they are in Spec,D or in lower dedicated projections dominated by D, enter into a configuration with this operator, and the composition yields a definite interpretation for the whole nominal. In this way, the demonstrative itself does not contribute the *iota*, it just co-occurs with an *iota*-bearing null D.

A natural question occurs: if the null D can host *iota* for definite readings, can it host other argument-forming operators, too? What about the kind-forming operator? I believe there is clear evidence that the possibility of a kind reading may also be associated with that higher syntactic position. This evidence comes from so-called *sortal inversion* (Trugman, 2007). Generally, the adjective precedes the noun in Russian in episodic contexts:

- (5) U menja v stakane krasnoe vino / *vino krasnoe.
 by me in glass red-NEUT red wine wine red
 ‘In my glass, there is red wine.’

However, when a noun phrase names a species, variety, or product type within a larger class, the adjective may be placed after the noun. Such construction is referred to in the literature as *sortal inversion*. It is especially common in classification contexts (like listing (6)) or contexts with kind-level predicates (7), but unavailable in episodic predicated like in (5) above.

- (6) Vino krasnoe 1/2 stakana.
 wine red 1/2 glass-GEN
 (From a list of ingredients in a recipe) ‘Red wine, 1/2 glass.’
- (7) Gagarka beskrylaja isčez-l-a s lica zemli v 1844 godu
 auk wingless disappear-PST-F from face.GEN earth.GEN in 1844 year-PREP
 ‘The great auk (*lit.* the wingless auk) became extinct in 1844.’
 (Lyutikova, 2018, p. 249)

Trugman (2007) proposes that *sortal inversion* involves a special null D head, which

she calls sortal D. That sortal D turns predicates of the type $\langle e, t \rangle$ into kind-denoting expressions. Constructions with inversion therefore are analyzed as N-to-D movement, even though Russian has no overt D lexical items.

However, it seems like posting a separate sortal D separate from the ordinary D may be redundant. A simpler move is to say that there is one null D head in Russian, which hosts a variable operator². This operator embodies the Meaning Preservation Hierarchy (Dayal, 2004), and can be realized as either *iota* or *nom* (8). Whether this variable operator sits in the D head itself, in Spec,D, or in some other position within the extended DP projection is a question I leave open.

(8)

$$ARG = \begin{cases} \iota x. P(x) \\ \lambda P. \cap P \end{cases}$$

I should emphasize that this is a preliminary suggestion rather a fully worked-out syntactic proposal. I remain agnostic whether this operator sits in D or Spec,D, nor do I specify the exact mechanics that determine its resolution in a given derivation.

All in all, the syntactic analysis of the Russian noun phrase structure proposed by Lyutikova is highly compatible with our findings from the semantic part. My preliminary proposal is that Russian bare arguments that project up to the DP level contain a null operator variable that can be realized as either *iota* or *nom* (but not $\exists$). Supposedly, cross-linguistically these realizations would be subject to blocking. Its implementation remains an intriguing direction for future research.

From this follows, though, that smaller nominals must not shift through *ARG*. In the next section, I briefly discuss one such case.

²The idea of a variable operator ranging over argument forming operations comes from Chierchia (2022a).

5.2.2 Genitive of negation

Another phenomenon that fell outside the scope of the main thesis is the genitive of negation. It has been widely observed that negative sentences with nominative vs. genitive subjects have different interpretation with respect to (in)definiteness, as exemplified by the canonical examples from Borschev and Partee (2008):

- (9) Otvét iz polk-a ne prišē-l.
 answer[M].NOM from regiment-GEN NEG come-PST.M.SG
 ‘The answer from the regiment has not arrived.’
- (10) Otvét-a iz polk-a ne priš-l-o.
 answer[M]-GEN from regiment-GEN NEG come-PST-N.SG
 ‘There was no answer from the regiment.’

In (9), the nominative subject presupposes the existence of a particular answer and asserts that it failed to arrive; a parallel sentence with a genitive subject denies the existence of any answer. This interpretive contrast has been treated as evidence that the genitive subject is structurally a smaller nominal, namely, a NP of property type $\langle e, t \rangle$, while the nominative is a full DP with referential semantics.

The bare arguments examined in the main chapters do not include genitive-marked subjects or objects under negation. My preliminary proposal about the variable argument-forming operator should be understood as targeting only bare arguments that project a full DP. While much of the literature discusses whether genitive arguments under negation remain $\langle e, t \rangle$ and compose in a way akin to semantic incorporation or in some other property-specific way (Borschev and Partee, 2008; Kagan, 2005, 2012), or they eventually undergo existential closure at the VP level (Diesing, 1992), I do not take a stance on it here. This thesis looks at (in)definiteness through the lens of the dimensions of uniqueness/maximality and anaphoricity; whether these properties emerge in the context of genitive of negation is an open empirical question. My hypothesis for now is that negated genitives should not exhibit

uniqueness, while negated nominatives should. I discuss the latter prediction in the next section.

5.2.3 Downward-entailing environments and uniqueness

In Chapter 2, I concluded that the data did not decide between *iota* and domain-restricted existential for postverbal bare singulars, mentioning that the most revealing diagnostics would come from downward-entailing environments. This idea comes from Dayal (2026b), who shows that postverbal bare singulars in downward-entailing contexts have no existential reading at all – neither a narrow-scope nor a wide-scope reading is available. This is demonstrated based on examples from Hindi and Russian. Consider the following example:

- (11) ??Na trav-e ne sid-it **sobaka**.
 on grass-PREP NEG sit-PRS.3SG dog
 Intended reading: ‘There isn’t a dog sitting on the grass.’ (Dayal, 2026b)

First of all, the sentence in (11) is only marginally acceptable. More importantly, if there is any meaning that works for it, it is the definite-like reading, where the dog is familiar or particularly salient. For example, such construction is perfectly felicitous in contrastive contexts, if the speaker expected to see a dog on the grass but instead saw a cat:

- (12) Na trave ne sid-it sobaka, na trave sidit koška.
 on grass-PREP NEG sit-PRS.3SG dog on grass-PREP sit-PRS.3SG cat
 ‘It is not a/the dog that is sitting on the grass, it is a/the cat that is sitting on the grass.’

But there is no genuine narrow-scope existential reading to (11). By contrast, bare plurals do not exhibit this behavior, and yield narrow-scope readings without any problems:

- (13) Na trav-e ne sid-jat **sobak-i**.
 on grass-PREP NEG sit-PRS.3PL dog-PL
 ‘There aren’t dogs sitting on the grass.’ Dayal (2026b)

The fact that there are contexts where the bare singular does not have any existential reading (neither narrow- nor wide-scope) suggests that there is little support for the existential semantics of bare singulars in postverbal position. Dayal proposes that bare singulars are semantically restricted to denote singleton sets. When a bare noun introduces a new discourse referent, it carries an additional condition of number sensitivity. This sets bare singulars apart from bare plurals, which impose no such restriction. This uniqueness component can be described as backgrounded assertion: it is not the primary focus of the sentence but rather part of the underlying context. The restriction projects across operators. In downward-entailing contexts, the uniqueness condition projects beyond the scope of the operator, making its effects more prominent.

This additional condition imposed on bare singulars is important since it does not include the presupposition of existence. If existence were presupposed, we would not be able to introduce a new discourse referent. Instead, bare singulars are acceptable when there is an expectation: if something exists, this something is unique.

However, before we can accept this as support for the *iota* analysis of bare singulars, there is a methodological concern. Negative speech acts are typically uttered in contexts that are presuppositionally richer than the contexts typical for affirmative utterances (Givón, 1975). This means that negation is not quite felicitous in out-of-the-blue contexts. The complex not-at-issue structure of negative sentences creates a challenge for linguists testing them. As Borschev and Partee (2008) note, not every negative sentence has a corresponding affirmative sentence of the same information structure. This means that when testing negation, one may inadvertently pronounce a negative sentence as a denial of an implicit preceding affirmative sentence, which does not operate as an ordinary sentential negation. As a result, it is not easy to find a natural context for a negative sentence that does not involve such a denial.

For a postverbal bare singular, which tends to be non-familiar, this means the infelicity of (11) might come from pragmatic requirements of negation rather than from the uniqueness of the bare singular itself. Borschev and Partee (2008) classify sentences like (11) as negative

declarative sentences, and claim that such sentences require a presupposed identifiable subject; the negation then denies something about that presupposed individual. But a postverbal bare singular is unlikely to be familiar and cannot be presupposed. It could be the case that negative sentences with postverbal bare singulars like (13) are infelicitous because of the general difficulty of using negation without adequate presuppositional support. At the same time, the plural/singular contrast still survives, so the confound of presuppositional requirements of negation does not fully explain the data.

In order to test this contrast further, one would have to build a context that cleanly tests the uniqueness hypothesis. A context like that would be such that the uniqueness of the referent is presupposed, but the referent itself is novel; at the same time, it also would have to manipulate cardinality of the subject, in a way similar to our Experiments 2A, 2B, and 3. The following pair of contexts seems to be on the right track³. Imagine the following situation:

(14) Context: There is a rule in the local hospital that every doctor must be assisted by a nurse. It is always a different nurse. But today, I went to my appointment, and I thought it was weird, since...

- a. Doktor-u ne pomoga-l-a medsestra.
doctor-DAT NEG help.IPFV-PST-F nurse
'A nurse was not assisting a/the doctor.'

In a context like above, the uniqueness of the unique nurse is presupposed, and my informants judged the sentence as felicitous. By contrast, if we construct a situation where more than one nurse is expected, the sentence sounds odd:

(15) Context: There is a rule in the local hospital that every doctor must be assisted by at least two nurses. They are always different. But today, I went to my appointment, and I thought it was weird, since...

³These diagnostic contexts were suggested by Gennaro Chierchia, p.c.

-
- a. #Doktor-u ne pomoga-l-a medsestra.
doctor-DAT NEG help.IPFV-PST-F nurse
Intended: ‘A nurse was not assisting a/the doctor.’

This suggests that bare singulars remain unique even when both postverbal and under negation, and they do not yield any existential reading in this context. Again, whether this is best analyzed as *iota* or as a tightly constrained domain-restricted existential is not quite decided from this data. However, *iota* captures uniqueness naturally, while the existential route involves additional explanatory burden. A simpler conclusion would be that bare singulars are *iota* expressions, possibly underspecified for novelty/familiarity, as suggested by Dayal (2026b). A larger-scale investigation across a range of downward-entailing environments would be needed to confirm this.

5.3 Final remarks

In this dissertation, I hope to have contributed to our empirical and theoretical understanding of systems without articles. The findings point towards a constraint-based system that prioritizes some argument-forming operations over others. Though the question of the exact implementation of these constraints remains open (especially at the syntactic level), I believe that the theoretical moral is that one should be looking for a simpler account, rather than for the most flexible one.

Appendix A Experimental stimuli for Experiment 1 (Familiarity vs. Word Order)

This appendix contains the context scenarios (in Russian, glossed and translated) and transcripts of the audio response options tested in Experiment 1. Context scenarios are presented in pairs, corresponding to the two levels of the Familiarity condition.

As mentioned in Chapter 2, post-hoc diagnostics revealed that one of the items behaved so differently from the others that it had to be removed from analysis. This was Item 11. I hypothesize that this is because this item used a past tense unaccusative verb with an inanimate subject; no other item uses this configuration. While several other items do feature inanimate subjects or unaccusative verbs, in all such cases the verb appears in the present tense. I leave a more precise explanation of this phenomenon to future research.

Here and in other appendices, the glossing of the contexts is simplified.

- Item 1

- (1) U Maši i Andreja est' mašina, kotoraja inogda samoproizvol'no vključaet at Maša and Andrej is car which sometimes spontaneously turns-on signalizaciju. Andrej zvonit Maše po telefonu i slyšit signalizaciju. Èto ix alarm. Andrej calls Maša on phone and hears alarm. This their mašina snova vključilas' sama. Andrej sprašivaet: Čto proisxodit? car again turned-on itself. Andrej asks: What happens? 'Masha and Andrey have a car that sometimes sets off its alarm on its own. Andrey calls Masha on the phone and hears the alarm. It is their car that has gone off again by itself. Andrey asks: What's going on?' *+Familiar*
- (2) Andrej zvonit Maše po telefonu i slyšit avtomobil'nye gudki. Maša vygljadyvaet Andrej calls Maša on phone and hears car honks. Maša looks v okno i vidit neskol'ko mašin na perekrestke. Andrej sprašivaet: in window and sees several cars at intersection. Andrej asks: Čto proisxodit? What happens? 'Andrey calls Masha on the phone and hears car horns. Masha looks out the window and sees several cars at the intersection. Andrey asks: What's going on?' *-Familiar*

Response options:

- (3) a. Mašina signalit na ulice.
car signals in street
- b. Na ulice signalit mašina.
in street signals car

- Item 2

- (4) U Kati i Vani est' syn Egor, kotoryj každyj den' igraet na ulice. Vanja at Katja and Vanja is son Egor who every day plays on street. Vanja razgovarivaet s Katej po telefonu i slyšit gromkij smex. Katja talks with Katja on phone and hears loud laughter. Katja vygljadyvaet iz okna i vidit: èto snova Egor. Vanja sprašivaet: Čto looks-out from window and sees: this again Egor. Vanja asks: What proisxodit? happens?
'Katja and Vanja have a son Egor who plays outside every day. Vanja is talking to Katja on the phone and hears loud laughter. Katja looks out the window and sees: it's Egor again. Vanja asks: What's going on?' *+Familiar*
- (5) Katja i Vanja razgovarivajut po telefonu. Vanja slyšit detskij smex. Katja Katja and Vanja talk on phone. Vanja hears child's laughter. Katja vygljadyvaet na ulicu i vidit, čto kakie-to deti igrajut na uglu. Vanja looks-out onto street and sees, that some children play on corner. Vanja sprašivaet: Čto proisxodit? asks: What happens?
'Katya and Vanya are talking on the phone. Vanya hears children's laughter. Katya looks out onto the street and sees that some children are playing on the corner. Vanya asks: What's going on?' *-Familiar*

Response options:

- (6) a. Rebënok smeëtsja na ulice.
child laughs in street
- b. Na ulice smeëtsja rebënok.
in street laughs child

- Item 3

- (7) Marina i Anton znajut, čto každuju subbotu sosedskij mal'čik Vitja treniruet Marina and Anton know that every Saturday neighbor boy Vitja practices basketbol'nye podači na ploščadke vo dvore. Segodnja subbota, i Anton slyšit basketball shots on court in yard. Today Saturday, and Anton hears stuk mjačika. Marina vygljadyvaet i vidit Vitju. Anton sprašivaet: thud of-ball. Marina looks-out and sees Vitja. Anton asks: Čto proisxodit? What happens? 'Marina and Anton know that every Saturday the neighbor boy Vitya practices basketball shots on the court in the yard. Today is Saturday, and Anton hears the thud of a ball. Marina looks out and sees Vitya. Anton asks: What's going on?'
+Familiar
- (8) Marina i Anton razgovarivajut. Anton slyšit stuk mjačika. Marina vygljadyvaet Marina and Anton talk. Anton hears thud of-ball. Marina looks-out i vidit trenera s neskol'kimi rebjatami – oni trenirujut basketbol'nye and sees coach with several kids – they practice basketball priemy. Anton sprašivaet: Čto proisxodit? techniques. Anton asks: What happens? 'Marina and Anton are talking. Anton hears the thud of a ball. Marina looks out and sees a coach with several kids – they are practicing basketball techniques. Anton asks: What's going on?'
-Familiar

Response options:

- (9) a. Mal'čik treniruetsja na ploščadke.
boy practices in court
- b. Na ploščadke treniruetsja mal'čik.
in court practices boy

• Item 4

- (10) Sosed i Ani i Lëvy peredelyvajut kryšu na ètoj nedele. Anja i Lëva neighbors of-Anja and Lëva redo roof on this week. Anja and Lëva pomogli im najti rabočego – Nikolaja, kotoryj ran’še pomog im samim helped them find worker – Nikolaj, who earlier helped them themselves s remontom doma. Segodnja Lëva snova slyšit stuk. Anja vygljadyvaet with repair of-house. Today Lëva again hears banging. Anja looks v okno i vidit Nikolaja za rabotoj. Lëva sprašivaet: Čto proisxodit? in window and sees Nikolaj at work. Lëva asks: What happens? ‘Anya and Lëva’s neighbors are redoing their roof this week. Anya and Lëva helped them find a worker – Nikolay, who previously helped them with their own house repairs. Today Lëva hears banging again. Anya looks out the window and sees Nikolay at work. Lëva asks: What’s going on?’ *+Familiar*
- (11) Anja i Lëva razgovarivajut po telefonu. Lëva slyšit zvuki molotka. Anja Anja and Lëva talk on phone. Lëva hears sounds of-hammer. Anja vygljadyvaet i vidit, čto srazu v neskol’kix sosedskix domax idët looks-out and sees that at-once in several neighboring houses goes remont. Lëva sprašivaet: Čto proisxodit? repair. Lëva asks: What happens? ‘Anya and Lëva are talking on the phone. Lëva hears the sounds of a hammer. Anya looks out and sees that several neighboring houses are being repaired at the same time. Lëva asks: What’s going on?’ *-Familiar*

Response options:

- (12) a. Rabočij šumit u sosedej.
worker makes.noise by neighbours
- b. U sosedej šumit rabočij.
by neighbours makes.noise worker

• Item 5

- (13) U Leny i Timofeja est' sobaka po kličke Maks, kotoryj laet na vsech
at Lena and Timofej is dog by name Maks, who barks at all
proxožix. Timofej zvonit Lene i snova slyšit laj. Lena vidit, čto èto
passers-by. Timofej calls Lena and again hears barking. Lena sees that this
Maks, kotoryj begaet vo dvore. Timofej sprašivaet: Čto proisxodit?
Maks who runs in yard. Timofej asks: What happens?
'Lena and Timofey have a dog named Max who barks at all passers-by. Timofey
calls Lena and hears barking again. Lena sees that it is Max, who is running around
in the yard. Timofej asks: What's going on?' *+Familiar*
- (14) U Leny i Timofeja net domašnix životnyx. Timofej zvonit Lene i slyšit
at Lena and Timofej no domestic animals. Timofej calls Lena and hears
laj na fone. Lena vygljadyvaet i vidit, čto neskol'ko bezdomnyx
barking in background. Lena looks-out and sees that several homeless
sobak zabežali vo dvor. Timofej sprašivaet: Čto proisxodit?
dogs ran into yard. Timofej asks: What happens?
'Lena and Timofey have no pets. Timofey calls Lena and hears barking in the
background. Lena looks out and sees that several stray dogs have run into the yard.
Timofey asks: What's going on?' *-Familiar*

Response options:

- (15) a. Sobaka laet vo dvore.
dog barks in yard
- b. Vo dvore laet sobaka.
in yard barks dog

• Item 6

- (16) Nataša i Slava znajut syna sosedej, Kolju, kotoryj každyj den' gonjaet na Nataša and Slava know son of-neighbors, Kolja, who every day races on èlektrosamokate tuda-sjuda po ulice. Slava slyšit gudenie motora. e-scooter back-and-forth along street. Slava hears humming of-motor. Nataša vygljadyvaet v okno i vidit Kolju. Slava sprašivaet: Čto Nataša looks in window and sees Kolja. Slava asks: What proisxodit? happens? 'Natasha and Slava know the neighbors' son, Kolja, who races back and forth on his e-scooter along the street every day. Slava hears the humming of a motor. Natasha looks out the window and sees Kolja. Slava asks: What's going on?' *+Familiar*
- (17) Nataša i Slava razgovarivajut po telefonu. Slava slyšit gudenie. Nataša Natasha and Slava talk on phone. Slava hears humming. Natasha vidit gruppu podrostkov na velosipedax i èlektrosamokatax vnizu ulicy. Slava sees group of-teenagers on bicycles and e-scooters down street. Slava sprašivaet: Čto proisxodit? asks: What happens? 'Natasha and Slava are talking on the phone. Slava hears humming. Natasha sees a group of teenagers on bicycles and e-scooters down the street. Slava asks: What's going on?' *-Familiar*
- (18) a. Rebënok kataetsja na ulitse. child rides on street
b. Na ulitse kataetsja rebënok. in street rides child

• Item 7

- (19) Každyj god naprotiv doma Julja i Olega ustraivaetsja muzykal'nyj
 every year across-from house of-Julja and Oleg is-held music
 festival'. Segodnja ètot festival' dolžen načat'sja. Oleg zamečæet, čto na ploščadi
 festival. Today this festival should begin. Oleg notices that on square
 sobirajutsja ljudi s muzykal'nymi instrumentami. Oleg sprašivaet: Čto
 gather people with musical instruments. Oleg asks: What
 proisxodit?
 happens?
 'Every year a music festival is held across from Julja and Oleg's house. Today this
 festival is supposed to begin. Oleg notices that people with musical instruments are
 gathering on the square. Oleg asks: What's going on?' *+Familiar*
- (20) Julja i Oleg razgovarivajut. Oleg slyšit zvuki barabanov na fone. Julja
 Julja and Oleg talk. Oleg hears sounds of-drums in background. Julja
 smotrit v okno i zamečæet, čto na ploščadi sobirajutsja ljudi s
 looks in window and notices that on square gather people with
 muzykal'nymi instrumentami. Stranno – takogo obyčno nikogda ne
 musical instruments. Strange – such-thing usually never not
 proisxodilo. Oleg sprašivaet: Čto proisxodit?
 happened. Oleg asks: What happens?
 'Julja and Oleg are talking. Oleg hears the sounds of drums in the background.
 Julja looks out the window and notices that people with musical instruments are
 gathering on the square. Strange – this has never usually happened. Oleg asks:
 What's going on?' *-Familiar*

Response options:

- (21) a. Festival' načinaetsja na ulice.
 festival starts on street
- b. Na ulice načinaetsja festival'.
 on street starts festival

- Item 8

- (22) Vadim i Polina často vidjat parnja na velosipede, kotoryj proezžaet mimo
Vadim and Polina often see guy on bicycle who rides by
každyj den'. Èto ix xorošij znakomyj. Vadim slyšit velosipednyj zvonok.
every day. This their good acquaintance. Vadim hears bicycle bell.
Polina vygljadyvaet i vidit znakomogo velosipedista. Vadim sprašivaet: Čto
Polina looks-out and sees familiar cyclist. Vadim asks: What
proisxodit?
happens?
'Vadim and Polina often see a guy on a bicycle who rides by every day. He is a good
acquaintance of theirs. Vadim hears a bicycle bell. Polina looks out and sees the
familiar cyclist. Vadim asks: What's going on?' *+Familiar*
- (23) Vadim i Polina razgovarivajut. Vadim slyšit velosipednyj zvonok. Polina
Vadim and Polina talk. Vadim hears bicycle bell. Polina
vygljadyvaet i vidit neskol'ko velosipedistov. Vadim sprašivaet: Čto
looks-out and sees several cyclists. Vadim asks: What
proisxodit?
happens?
'Vadim and Polina are talking. Vadim hears a bicycle bell. Polina looks out and
sees several cyclists. Vadim asks: What's going on?' *-Familiar*

Response options:

- (24) a. Velosipedist proezžaet mimo doma.
cyclist passes by house
- b. Mimo doma proezžaet velosipedist.
by house passes cyclist

• Item 9

- (25) U Zoi i L'va est' popugaj po imeni Kivi, kotoryj ljubiti svistet'. Lev slyšit
at Zoya and Lev is parrot by name Kivi who likes to-whistle. Lev hears
svist. Èto Kivi u okna. Lev sprašivaet: Čto proisxodit?
whistle. This Kivi at window. Lev asks: What happens?
'Zoya and Lev have a parrot named Kivi who likes to whistle. Lev hears a whistle.

It's Kivi by the window. Lev asks: What's going on?' *+Familiar*

- (26) Zoya i Lev razgovarivajut. Lev slyšit xlopan'e kryl'ev i pronzitel'nyj
Zoya and Lev talk. Lev hears flapping of-wings and piercing
svist. Zoya vidit ptic, proletajuščix mimo okna. Lev sprašivaet: Čto
whistle. Zoya sees birds flying past window. Lev asks: What
proisxodit?
happens?
'Zoya and Lev are talking. Lev hears the flapping of wings and a piercing whistle.

Zoya sees birds flying past the window. Lev asks: What's going on?' *-Familiar*

Response options:

- (27) a. Ptica svistit za oknom.
bird whistles behind window
b. Za oknom svistit ptica.
behind window whistles bird

- Item 10

- (28) U Fainy i Marka est' sumasšedšij sosed Miša, kotoryj každoe utro at Faina and Mark is crazy neighbor Miša who every morning pinaet musornyj bak, tak čto on katitsja po doroge i skrežeščet. Mark kicks trash can, so that it rolls along road and grates. Mark snova slyšit skrežet. On sprašivaet: Čto proisxodit? again hears grating. He asks: What happens? 'Faina and Mark have a crazy neighbor Misha who every morning kicks the trash can, so that it rolls along the road and grates. Mark hears the grating again. He asks: What's going on?' *+Familiar*
- (29) Faina i Mark razgovarivajut. Mark slyšit skrežet metalla. Faina vygljadyvaet Faina and Mark talk. Mark hears grating of-metal. Faina looks-out i vidit, kak veter oprokinul musornyj bak i on katitsja po doroge. and sees how wind knocked-over trash can and it rolls along road. Mark sprašivaet: Čto proisxodit? Mark asks: What happens? 'Faina and Mark are talking. Mark hears the grating of metal. Faina looks out and sees how the wind has knocked over the trash can and it is rolling along the road. Mark asks: What's going on?' *-Familiar*

Response options:

- (30) a. Musornyj bak katitsja po doroge.
trash can rolls along road
- b. Po doroge katitsja musornyj bak.
along road rolls trash can

• Item 11

- (31) Ženja i Il'ja znajut, čto gorod ustraivaet saljuty každuju pjatnicu večerom.
Ženja and Il'ja know that city holds fireworks every Friday evening.
Il'ja slyšit xlopki. Il'ja sprašivaet: Čto proisxodit?
Il'ja hears pops. Il'ja asks: What happens?
'Zhenya and Ilya know that the city holds fireworks every Friday evening. Il'ja hears
pops. Ilya asks: What's going on?' *+Familiar*
- (32) Ženja i Il'ja razgovarivajut po telefonu. Il'ja slyšit xlopki. Ženja vidit
Ženja and Il'ja talk on phone. Il'ja hears pops. Ženja sees
vdali saljut. Stranno – segodnja vtornik i ne prazdničnyj den'. Il'ja
in-distance fireworks. Strange – today Tuesday and not holiday day. Il'ja
sprašivaet: Čto proisxodit?
asks: What happens?
'Zhenya and Ilya are talking on the phone. Ilya hears pops. Zhenya sees fireworks
in the distance. Strange – today is Tuesday and not a holiday. Ilya asks: What's
going on?' *-Familiar*

Response options:

- (33) a. Saljut načalsja vdali.
firework started in.distance
- b. Vdali načalsja saljut.
in.distance started firework

• Item 12

- (34) U Tani i Bori est' domašnjaja myška Smoki, kotoroj možno begat' po
 at Tanja and Borja is domestic mouse Smoki whom allowed to-run around
 domu. Borja slyšit pisk i šorox'. Èto Smoki kopošitsja u musorki. Borja
 house. Borja hears squeak and rustle. This Smoki fidgets by trash. Borja
 sprašivaet: Čto proisxodit?
 asks: What happens?
 'Tanya and Borya have a pet mouse Smokey who is allowed to run around the house.
 Borja hears a squeak and a rustle. It's Smokey rummaging by the trash. Borya asks:
 What's going on?' *+Familiar*
- (35) Tanja i Borja razgovarivajut. Borja slyšit pisk i šorox'. Tanja
 Tanja and Borja talk. Borja hears squeak and rustle. Tanja
 vygljadyvaet i vidit, čto čto-to dvižetsja vozle musornyx bakov. Kažetsja,
 looks-out and sees that something moves near trash cans. seems
 èto myš'! Borja sprašivaet: Čto proisxodit?
 this mouse! Borja asks: What happens?
 'Tanya and Borya are talking. Borya hears a squeak and a rustle. Tanya looks out
 and sees that something is moving near the trash cans. It seems to be a mouse!
 Borya asks: What's going on?' *-Familiar*

Response options:

- (36) a. Myš kopošitsja u musorki.
 mouse fidgets by trash.can
- b. U musorki kopošitsja myš.
 by trash.can fidgets mouse

• Item 13

- (37) U Ali i Viktora est' enot Saša, ktorogo oni nedavno prijutuli. Viktor at Alja and Viktor is raccoon Saša whom they recently took-in. Viktor slyšit skrežet. Alja zagljadyvaet na kuxnju i vidit Sašu, kotoryj pytaetsja hears scratching. Alja peeks into kitchen and sees Saša who tries otkryt' musorku. Viktor sprašivaet: Čto proisxodit? to-open trash. Viktor asks: What happens? 'Alya and Viktor have a raccoon Sasha whom they recently took in. Viktor hears scratching. Alya peeks into the kitchen and sees Sasha trying to open the trash can. Viktor asks: What's going on?' *+Familiar*
- (38) Alja i Viktor razgovarivajut. Viktor slyšit skrežet i pisk. Alja Alja and Viktor talk. Viktor hears scratching and squeak. Alja zamečae kakix-to životnyx u musornyx bakov, no ne možet ponjat', kto notices some animals at trash cans, but not can understand who èto. Vnezapno ona ponimaet, čto èto enot – vot stranno, nikogda ne this. Suddenly she realizes that this raccoon – here strange, never not videla zdes' enotov! Viktor sprašivaet: Čto proisxodit? saw here raccoons! Viktor asks: What happens? 'Alya and Viktor are talking. Viktor hears scratching and a squeak. Alya notices some animals by the trash cans, but can't figure out what they are. Suddenly she realizes it's a raccoon – how strange, she has never seen raccoons here! Viktor asks: What's going on?' *-Familiar*

Response options:

- (39) a. Enot šumit u musorki.
raccoon makes.noise by trash
- b. U musorki šumit enot.
by trash makes.noise raccoon

- Item 14

- (40) Majja i Artëm život rjedom s ljud'mi, u kotoryx postojanno zvenit telefon. Majja and Artëm live next-to with people at whom constantly rings phone. Oni daže mogu predugadat', kogda on zazvonit – rovno v 10 utra každyj den' razdačsja pervyj zvonok. Sejčas 10 utra, i Artëm slyšit zvonok. They even can predict when it will-ring – exactly at 10 morning every day sounds first ring. Now 10 morning, and Artëm hears ring. Artëm sprašivaet: Čto proisxodit? Artëm asks: What happens? 'Majja and Artëm live next to people whose phone rings constantly. They can even predict when it will ring – exactly at 10 a.m. every day the first ring sounds. It is now 10 a.m., and Artëm hears a ring. Artëm asks: What's going on?' *+Familiar*
- (41) Majja i Artëm razgovarivajut. Artëm slyšit slabyj zvon. Majja vygljadyvaet, Majja and Artëm talk. Artëm hears faint ringing. Majja looks-out, no nikogo ne vidit; èto možet byt' čej-to telefon. Artëm sprašivaet: Čto but no-one not sees; this can be someone's phone. Artëm asks: What proisxodit? happens? 'Majja and Artëm are talking. Artëm hears a faint ringing. Majja looks out, but does not see anyone; it may be someone's phone. Artëm asks: What's going on?' *-Familiar*

Response options:

- (42) a. Telefon zvenit u sosedej.
phone rings at neighbours
- b. U sosedej zvenit telefon.
at neighbours rings phone

- Item 15

- (43) U Tani i Nikity est' sosedka Irina, professional'naja opernaja pevica, at Tanja and Nikita is neighbor Irina, professional opera singer, kotoraja každyj vyxodnoj repetiruet na lužajke. Nikita zvonit Tane po telefonu who every weekend rehearses on lawn. Nikita calls Tanja on phone i slyšit penie. Èto Irina raspevaetsja na ulice. Nikita sprašivaet: Čto and hears singing. This Irina warms-up on street. Nikita asks: What proisxodit? happens?
'Tanya and Nikita have a neighbor Irina, a professional opera singer, who rehearses on the lawn every weekend. Nikita calls Tanya on the phone and hears singing. It is Irina warming up her voice outside. Nikita asks: What's going on?' *+Familiar*
- (44) Tanja i Nikita razgovarivajut po telefonu. Nikita slyšit v trubke penie. Tanja and Nikita talk on phone. Nikita hears in receiver singing. Tanja vidit gruppu muzykantov, kotorye repetirujut koncert. Neskol'Ko devušek Tanja sees group of-musicians who rehearse concert. several girls po očeredi raspevajutsja. Nikita sprašivaet: Čto proisxodit? in turn warm-up. Nikita asks: What happens?
'Tanya and Nikita are talking on the phone. Nikita hears singing on the receiver. Tanya sees a group of musicians who are rehearsing a concert. Several young women take turns warming up. Nikita asks: What's going on?' *-Familiar*

Response options:

- (45) a. Pevica raspevaetsja na ulice.
singer warms.up in street
- b. Na ulice raspevaetsja pevica.
in street warms.up singer

• Item 16

- (46) U Èmmy i Il'i est' sobaka Maks, kotoryj vseгда ryčit vo vremja igry.
at Èmma and Il'ja is dog Maks who always growls during play. Il'ja
Il'ja zvonit Èmme po telefonu i slyšit kakoj-to šum. Èmma
calls Èmma on phone and hears some noise. Èmma looks
vygljadyvaet v okno i vidit, čto Maks igraet na ulice. Il'ja
in window and sees that Maks plays on street. Il'ja asks:
sprašivaet: Čto proisxodit?
What happens?
'Èmma and Il'ja have a dog Max who always growls during play. Ilya calls Èmma
on the phone and hears some noise. Èmma looks out the window and sees that Max
is playing outside. Ilya asks: What's going on?' *+Familiar*
- (47) Èmma i Il'ja razgovarivajut po telefonu. Il'ja slyšit ryčanie. Èmma vidit
Èmma and Il'ja talk on phone. Il'ja hears growling. Èmma sees
kakix-to sobak, kotorye borjutsja meždu soboj, odna iz nix gromko
some dogs who wrestle between themselves, one of them loudly
ryčit. Ona nikogda ran'se ne videla sobak rjedom s ix domom. Il'ja
growls. She never before not saw dogs near with their house. Il'ja
sprašivaet: Čto proisxodit?
asks: What happens?
'Èmma and Ilya are talking on the phone. Il'ja hears growling. Èmma sees some
dogs wrestling with each other, one of them is growling loudly. She has never seen
dogs near their house before. Ilya asks: What's going on?' *-Familiar*

Response options:

- (48) a. Sobaka ryčit na ulice.
dog growls in street
- b. Na ulice ryčit sobaka.
in street growls dog

Appendix B Experimental stimuli for Experiments 2A, 2B, and 3 (Uniqueness studies)

This appendix contains the context scenarios (in Russian, glossed and translated) and transcripts of the audio response options tested in Experiments 2A, 2B, and 3. Context scenarios are presented in pairs, corresponding to the two levels of the Uniqueness condition.

B.1 Experiment 2A: familiar preverbal nouns

- Item 1

+Unique:

- (1) U Maši i Andreja est' xomjak po kličke Pjatačok. On živet v kletke
at Masha's and Andrey's is hamster by nickname Pyatachok. he lives in cage
v gostinoj. Kak-to raz Maša i Andrej sideli za užinom. Maša pošla v
in living-room. once time Masha and Andrey sat at dinner. Masha went into
komnatu dostat' salfetki i uvidela, čto Pjatačka net v kletke i dver'
room to-get napkins and saw that Pyatachok not-there in cage and door
v kletku otkryta. Maša vskriknula. Andrej sprašivaet iz kuxni: - Čto
to cage is-open. Masha screamed. Andrey asks from kitchen: - What
takoe?
happened?
'Masha and Andrey have a hamster named Pyatachok. He lives in a cage in the living
room. One time, Masha and Andrey were sitting at dinner. Masha went into the
room to get some napkins and saw that Pyatachok was not in his cage and the cage
door was open. Masha screamed. Andrey calls from the kitchen: What happened?'

-Unique:

-
- (2) U Maši i Andreja est' dva xomjačka, Busja i Pjatačok. Oni živut v at Masha's and Andrey's are two hamsters, Busya and Pyatachok. they live in kletke v gostinoj. Kak-to raz Maša i Andrej sideli za užinom. Maša cage in living-room. once time Masha and Andrey sat at dinner. Masha pošla v komnatu dostat' salfetki i uvidela, što Pjatačok sidit v kletke went into room to-get napkins and saw that Pyatachok sits in cage odin, a Busi net v kletke i dver' v kletku otkryta. Maša alone, and Busya not-there in cage and door to cage is-open. Masha vskriknula. Andrej sprašivaet iz kuxni: - Čto takoe? screamed. Andrey asks from kitchen: - What happened? 'Masha and Andrey have two hamsters, Busya and Pyatachok. They live in a cage in the living room. One time, Masha and Andrey were sitting at dinner. Masha went into the room to get some napkins and saw that Pyatachok was sitting alone in the cage, and Busya was not in the cage and the door was open. Masha screamed. Andrey calls from the kitchen: What happened?'

Response options:

- (3) a. Xomjak sbežal iz kletki.
hamster ran.away from cage
- b. Odin xomjak sbežal iz kletki.
one hamster ran.away from cage

- Item 2

+Unique:

- (4) U Valentiny Ivanovny i Nikolaja Petroviča živet domašnjaja kurica Katja.
 at Valentina Ivanovna's and Nikolay Petrovich's lives pet hen Katya.
 Kak-to raz Valentina Ivanovna pošla proverit' kurjatnik i uvidela, što Katja
 once time Valentina Ivanovna went to-check henhouse and saw that Katja
 upala s žerdočki i povredila lapku. Ona pobežala domoj za aptečkoj.
 fell from perch and hurt her-paw. she ran home for first-aid-kit.
 Nikolaj Ivanovič sprašivaet: - Čto slučilos'?
 Nikolay Ivanovich asks: – What happened?
 'Valentina Ivanovna and Nikolay Petrovich have a pet hen named Katya. One time,
 Valentina Ivanovna went to check the henhouse and saw that Katya had fallen off the
 perch and hurt her leg. She ran home to get the first-aid kit. Nikolay Ivanovich asks:
 What happened?'

-Unique:

- (5) U Valentiny Ivanovny i Nikolaja Petroviča živut dve kuricy, belaja i
 at Valentina Ivanovna's and Nikolay Petrovich's live two hens, white and
 seraja. Kak-to raz Valentina Ivanovna pošla proverit' kurjatnik i uvidela, što
 grey. once time Valentina Ivanovna went to-check henhouse and saw that
 belaja kurica upala s žerdočki i povredila lapku. Ona pobežala domoj za
 white hen fell from perch and hurt her-paw. she ran home for
 aptečkoj. Nikolaj Ivanovič sprašivaet: - Čto slučilos'?
 first-aid-kit. Nikolay Ivanovich asks: – What happened?
 'Valentina Ivanovna and Nikolay Petrovich have two hens, one white and one grey.
 One time, Valentina Ivanovna went to check the henhouse and saw that the white
 hen had fallen off the perch and hurt her leg. She ran home to get the first-aid kit.
 Nikolay Ivanovich asks: What happened?'

Response options:

- (6) a. Kurica povredila lapku.
 chicken injured leg
 b. Odná kurica povredila lapku.
 one chicken injured leg

- Item 3

+Unique:

- (7) V mestnom zooparke živet tol'ko odna makaka, i ee zovut Anfisa. Segodnja in local zoo lives only one macaque, and her name-is Anfisa. today volonter Alena prišla čistit' ee kletku i uvidela, što Anfisa sidit v uglu volunteer Alena came to-clean her cage and saw that Anfisa sits in corner i kašljaet. Alena zvonit svoemu koordinatoru i govorit, što nužen and coughs. Alena calls her coordinator and says that needed veterinar. Koordinator sprašivaet: - Čto slučilos'? vet. coordinator asks: – What happened?
'The local zoo has only one macaque, and her name is Anfisa. Today, volunteer Alena came to clean her cage and saw that Anfisa was sitting in the corner and coughing. Alena calls her coordinator and says that they need a vet. The coordinator asks: What happened?'

-Unique:

- (8) V moskovskom zooparke v odnoj kletke život dve makaki, mal'čik i devočka. in Moscow zoo in one cage live two macaques, male and female. Segodnja volonter Alena prišla čistit' ix kletku i uvidela, što today volunteer Alena came to-clean their cage and saw that makaka-mal'čik sidit v uglu i kašljaet. Makaka-devočka vygljadit zdorovoj. macaque-male sits in corner and coughs. macaque-female looks healthy. Alena zvonit svoemu koordinatoru i govorit, što nužen veterinar. Koordinator Alena calls her coordinator and says that needed vet. coordinator sprašivaet: - Čto slučilos'? asks: – What happened?
'At the Moscow Zoo, two macaques—a male and a female—live in one cage. Today, volunteer Alena came to clean their cage and saw that the male macaque was sitting in the corner and coughing. The female macaque looks healthy. Alena calls her coordinator and says that they need a vet. The coordinator asks: What happened?'

Response options:

- (9) a. Makaka zbolela prostudoj.
macaque got.sick with.cold
- b. Odna makaka zbolela prostudoj.
one macaque got.sick with.cold

-
- Item 4

+Unique:

- (10) U Frosi est' zolotaja rybka po imeni Nemo. Frosja ukrasila akvarium at Frosya's is golden fish by name Nemo. Frosya decorated aquarium kamnjami i posadila vodorosli. Segodnja ona uvidela, čto Nemo zastrjal with-stones and planted seaweed. today she saw that Nemo got-stuck meždu kamnjami. Frosja pobežala k mame rasskazat', čto slučilos'. between stones. Frosya ran to mom to-tell what happened. 'Frosya has a goldfish named Nemo. Frosya decorated the aquarium with stones and planted some seaweed. Today she saw that Nemo had gotten stuck between the stones. Frosya ran to her mom to tell her what happened.'

-Unique:

- (11) U Ali est' dve rybki, belaja i zolotaja. Ona ukrasila ix akvarium at Alya's are two fish, white and golden. she decorated their aquarium kamnjami i posadila vodorosli. Segodnja ona zametila, čto zolotaja rybka with-stones and planted seaweed. today she noticed that golden fish plavaet, kak obyčno, a belaja zastrjala meždu kamnjami. Alja pobežala k swims as usual, but white got-stuck between stones. Alya ran to mame rasskazat', čto slučilos'. mom to-tell what happened. 'Alya has two fish, one white and one golden. She decorated their aquarium with stones and planted some seaweed. Today she noticed that the golden fish was swimming around as usual, but the white one had gotten stuck between the stones. Alya ran to her mom to tell her what happened.'

Response options:

- (12) a. Rybka zastrjala meždu kamnej.
fish got.stuck in.between stones
- b. Odna rybka zastrjala meždu kamnej.
one fish got.stuck in.between stones

-
- Item 5

+Unique:

- (13) Katja pojmla v stakan muxu i radostno pokazala mame. Mama poxvalila Katya caught in glass fly and joyfully showed to-mom. mom praised ee i pošla na kuxnju. Čerez minutu Katja zakričala. Mama pribežala her and went to kitchen. after minute Katya screamed. mom came-running i uvidela, što muxa nepodvižna. Mama sprašivaet: - Čto takoe? and saw that fly is-motionless. mom asks: – What happened? ‘Katya caught a fly in a glass and happily showed it to her mom. Her mom praised her and went to the kitchen. A minute later, Katya screamed. Mom came running and saw that the fly was motionless. Mom asks: What happened?’

-Unique:

- (14) Katja pojmla v stakan dvux mux i radostno pokazala mame. Mama Katya caught in glass two flies and joyfully showed to-mom. mom poxvalila ee i pošla na kuxnju. Čerez minutu Katja zakričala. Mama praised her and went to kitchen. after minute Katya screamed. mom pribežala i uvidela, što v stakane odna muxa nepodvižna, a drugaja came-running and saw that in glass one fly is-motionless, and other vsë eščë polzaet po stenke. Mama sprašivaet: - Čto takoe? still yet crawls along wall. mom asks: – What happened? ‘Katya caught two flies in a glass and happily showed them to her mom. Her mom praised her and went to the kitchen. A minute later, Katya screamed. Mom came running and saw that one fly in the glass was motionless, while the other was still crawling along the side. Mom asks: What happened?’

Response options:

- (15) a. Muxa perestala dvigat’sja.
fly stopped moving
- b. Odna muxa perestala dvigat’sja.
one fly stopped moving

- Item 6

+Unique:

- (16) V škol'nom terrariume živet čerepaxa Valentina. Učitel'nica poprosila
 in school terrarium lives tortoise Valentina. teacher asked
 dežurnuju Alenu proverit', vsě li s nej v porjadke. Alena
 duty-student Alena to-check whether everything with her in order. Alena
 podošla i uvidela, čto Valentina ležit na spine. Učitel'nica sprašivaet: -
 came-up and saw that Valentina lies on back. teacher asks: -
 Čto tam?
 What's there?
 'The school terrarium is home to a tortoise named Valentina. The teacher asked the
 student on duty, Alena, to go check whether everything was alright with her. Alena
 went over and saw that Valentina was lying on her back. The teacher asks: What's
 going on?'

-Unique:

- (17) V škol'nom terrariume živut dve čerepaxi. Učitel'nica poprosila dežurnuju
 in school terrarium live two tortoises. teacher asked duty-student
 Alenu proverit', vsě li s nimi v porjadke. Alena podošla i
 Alena to-check whether everything with them in order. Alena came-up and
 uvidela, čto odna čerepaxa ležit na spine. Vtoraja spokojno spit v uglu.
 saw that one tortoise lies on back. second calmly sleeps in corner.
 Učitel'nica sprašivaet: - Čto tam?
 teacher asks: - What's there?
 'The school terrarium is home to two tortoises. The teacher asked the student on
 duty, Alena, to go check whether everything was alright with them. Alena went
 over and saw that one tortoise was lying on its back. The second one was sleeping
 peacefully in the corner. The teacher asks: What's going on?'

Response options:

- (18) a. Čerepaxa perevernulas' na spinu.
 turtle turned into back
- b. Odna čerepaxa perevernulas' na spinu.
 one turtle turned into back

-
- Item 7

+Unique:

- (19) U Lizy est' domašnjaja ptica Asja. Ona živet v bol'soj kletke na podokonnike.
 at Liza's is pet bird Asya. she lives in big cage on windowsill.
 Odnazdy Liza podošla k kletke i uvidela, što dverca otкрыta, a Asi
 once Liza came-up to cage and saw that door is-open, and Asya
 nigde net. Mama sprašivaet: - Čto slučilos'?
 nowhere not-there. mom asks: - What happened?
 'Liza has a pet bird named Asya. She lives in a big cage on the windowsill. One day,
 Liza came up to the cage and saw that the door was open and Asya was nowhere to
 be found. Mom asks: What happened?'

-Unique:

- (20) U Lizy est' dve malen'kie domašnie pticy, Asja i Frosja. Oni živut v
 at Liza's are two small pet birds, Asya and Frosya. they live in
 bol'soj kletke na podokonnike. Odnazdy Liza podošla k kletke i uvidela, što
 big cage on windowsill. once Liza came-up to cage and saw that
 dverca otкрыta, a Frosi nigde net. Asja vse tak že sidit na žerdočke
 door is-open, and Frosya nowhere not-there. Asya still just as sits on perch
 v kletke. Mama sprašivaet: - Čto slučilos'?
 in cage. mom asks: - What happened?
 'Liza has two small pet birds, Asya and Frosya. They live in a big cage on the
 windowsill. One day, Liza came up to the cage and saw that the door was open and
 Frosya was nowhere to be found. Asya was still sitting on her perch in the cage as
 usual. Mom asks: What happened?'

Response options:

- (21) a. Ptica vyletela iz kletki.
 bird flew.out from cage
- b. Odn ptica vyletela iz kletki.
 one bird flew.out from cage

- Item 8

+Unique:

- (22) U Oli živet domašnjaja jaščerica Ekaterina. Odnazdy Olja pošla posmotret',
 at Olya's lives pet lizard Ekaterina. once Olya went to-see
 kak Ekaterina poživaet v svoej kletke i ne srazu ee uvidela.
 how Ekaterina is-doing in her cage and not immediately her saw.
 Okazyvaetsja, Ekaterina zalezla pod kamen'. Olja pokazyvaet mame
 it-turns-out, Ekaterina had-climbed under stone. Olya shows to-mom
 kletku.
 cage.
 'Olya has a pet lizard named Ekaterina. One day, Olya went to check on Ekaterina
 in her cage but couldn't spot her right away. It turned out that Ekaterina had
 climbed under a stone. Olya shows her mom the cage.'

-Unique:

- (23) U Oli est' dve jaščericy, zelënaja i koričnevaja. Ona zametila, čto zelënaja
 at Olya's are two lizards, green and brown. she noticed that green
 jaščerica sidit na vetke, a drugoj nigde net. Okazyvaetsja,
 lizard sits on branch, and other nowhere not-there. it-turns-out,
 koričnevaja jaščerica zalezla pod kamen'. Olja pokazyvaet mame kletku.
 brown lizard had-climbed under stone. Olya shows to-mom cage.
 'Olya has two lizards, one green and one brown. She noticed that the green lizard
 was sitting on a branch, but the other was nowhere to be found. It turned out that
 the brown lizard had climbed under a stone. Olya shows her mom the cage.'

Response options:

- (24) a. Jaščerica sprjatalas' pod kamen'.
 lizard hid under stone
- b. Odn jaščerica sprjatalas' pod kamen'.
 one lizard hid under stone

B.2 Experiment 2B: non-familiar postverbal nouns

- Item 1

+Unique:

- (25) Mama smotrit novosti. Segodnja novosti pro zoopark v odnom nebol'som mom watches news. today news about zoo in one small gorode. Sjužet rasskazyvaet, čto segodnja utrom sotrudnica prišla kormit' city. story tells that today morning employee came to-feed enota, edinstvennogo v ètom zooparke, i uvidela, čto kletka otkryta, i raccoon, only-one in this zoo, and saw that cage is-open, and enot isčez. Papa sprašivaet: - Čto rasskazali v novostjax? raccoon disappeared. dad asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about a zoo in a small town. The story reports that this morning an employee came to feed the raccoon (the only one in this zoo) and found that the cage was open and the raccoon had disappeared. Dad asks: What did they say on the news?'

-Unique:

- (26) Mama smotrit novosti. Segodnja novosti pro zoopark v odnom nebol'som mom watches news. today news about zoo in one small gorode. Sjužet rasskazyvaet, čto segodnja utrom sotrudnica prišla kormit' city. story tells that today morning employee came to-feed dvux enotov i uvidela, čto kletka otkryta, i odin iz enotov two raccoons and saw that cage is-open, and one of raccoons isčez. Vtoroj tak i spal v svoem domike. Papa sprašivaet: - Čto disappeared. second just so slept in its house. dad asks: - What rasskazali v novostjax? did-they-say in news? 'Mom is watching the news. Today's segment is about a zoo in a small town. The story reports that this morning an employee came to feed two raccoons and found that the cage was open and one of the raccoons had disappeared. The other one was still asleep in its little house. Dad asks: What did they say on the news?'

Response options:

- (27) a. V odnom zooparke sbežal enot.
in one zoo ran.away raccoon

-
- b. V odnom zooparke sbežal odin enot.
in one zoo ran.away one raccoon

-
- Item 2

+Unique:

- (28) Mama smotrit novosti. Segodnja novosti pro pervyj v istorii odnogo mom watches news. today news about first in history of-one mestnogo muzeja vizit inostrannogo turista. Vpročem, vizit okazalsja local museum visit of-foreign tourist. however, visit turned-out neudačnym. Delo v tom, čto ètot inostranec kakim-to obrazom okazalsja unsuccessful. thing in that, that this foreigner in-some way turned-out zapertym v muzee posle zakrytija, i emu prišlos' perenočevat' sredi locked in museum after closing, and he had-to spend-night among èksponatov. Poterjavšegosja turista našli tol'ko na sledujuščij den'. Papa exhibits. lost tourist found only on next day. dad sprašivaet: - Čto rasskazali v novostjax? asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about the first-ever visit by a foreign tourist to a local museum. However, the visit did not go well. It turned out that the tourist somehow ended up locked inside the museum after closing time and had to spend the night among the exhibits. The lost tourist was only found the next day. Dad asks: What did they say on the news?'

-Unique:

- (29) Mama smotrit novosti. Segodnja novosti pro to, kak dvoe inostrancev mom watches news. today news about that, how two foreigners rešili posetit' odin iz mestnyx muzeev. Oni proveli tam stol'ko vremeni, decided to-visit one of local museums. they spent there so-much time, čto odin iz inostrancev poterjalsja, da nastol'ko, čto emu prišlos' that one of foreigners got-lost, even so-much-so, that he had-to perenočevat' sredi èksponatov. Ego tovarišč iskal ego v koridorax spend-night among exhibits. his companion searched-for him in corridors ves' den', ne smog ètogo sdelat' i pošel pisat' zajavlenie v policiju. all day, not managed this do and went to-write statement to police. Poterjavšegosja turista našli tol'ko na sledujuščij den'. Papa sprašivaet: - Čto lost tourist found only on next day. dad asks: - What rasskazali v novostjax? did-they-say in news? 'Mom is watching the news. Today's segment is about two foreigners who decided to visit one of the local museums. They spent so much time there that one of them got lost; so lost that he had to spend the night among the exhibits. His companion

searched for him in the corridors all day, couldn't find him, and went to file a report with the police. The lost tourist was only found the next day. Dad asks: What did they say on the news?'

Response options:

- (30) a. V odnom muzee poterjalsja inostranec.
 in one museum got.lost foreigner
- b. V odnom muzee poterjalsja odin inostranec.
 in one museum got.lost one foreigner

-
- Item 3

+Unique:

- (31) Mama smotrit novosti. Segodnja novosti pro neudačnyj koncert v odnom iz mom watches news. today news about unsuccessful concert in one of bol'six koncertnyx zalov. Tam vystupal znamenityj skripač, i on tak big concert halls. there performed famous violinist, and he so uvleksja igroj, čto upal so sceny v orkestruvuju jamu. got-carried-away with-playing that fell from stage into orchestra pit. Koncert prišlos' prervat'. Papa sprašivaet: - Čto rasskazali v novostjax? concert had-to interrupt. dad asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about an ill-fated concert at one of the big concert halls. A famous violinist was performing, and he got so carried away playing that he fell off the stage into the orchestra pit. The concert had to be stopped. Dad asks: What did they say on the news?'

-Unique:

- (32) Mama smotrit novosti. Segodnja novosti pro neudačnyj koncert v odnom iz mom watches news. today news about unsuccessful concert in one of bol'six koncertnyx zalov. Tam vystupal znamenityj duet skripačej, i odin big concert halls. there performed famous duo of-violinists, and one iz ispolnitelej tak uvleksja igroj, čto upal so sceny v of performers so got-carried-away with-playing that fell from stage into orkestruvuju jamu. Koncert prišlos' prervat', vtoroj skripač daže ne doigral orchestra pit. concert had-to interrupt, second violinist even not finished p'esu do konca. Papa sprašivaet: - Čto rasskazali v novostjax? piece to end. dad asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about an ill-fated concert at one of the big concert halls. A famous violin duo was performing, and one of the musicians got so carried away playing that he fell off the stage into the orchestra pit. The concert had to be stopped, and the second violinist didn't even get to finish the piece. Dad asks: What did they say on the news?'

Response options:

- (33) a. V odnom zale upal so sceny skripač.
in one hall fell from stage violinist

-
- b. V odnom zale upal so sceny odin skripač.
in one hall fell from stage one violinist

- Item 4

+Unique:

- (34) Mama smotrit novosti. Segodnja novosti pro odnogo monaxa, kotoryj žil v mom watches news. today news about one monk, who lived in polnom odinočestve v odnoj derevne. Ètot monax sošel s uma i načal complete solitude in one village. this monk went off mind and began vyt', kak volk, i myčat', kak korova. Tol'ko čerez kakoe-to vremja to-howl, like wolf, and to-moo, like cow. only after some time žiteli derevni smogli obratit'sja za pomošč'ju v inhabitants of-village managed to-contact for help psychiatric psixiatričeskuju bol'nicu, čtoby derevenskogo monaxa osmotreli hospital, so-that village monk would-be-examined by vrači. Papa sprašivaet: - Čto rasskazali v novostjax? doctors. dad asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about a monk who lived in complete solitude in a village. The monk went mad and started howling like a wolf and mooing like a cow. It was only after some time that the villagers managed to contact a psychiatric hospital to have the monk examined by doctors. Dad asks: What did they say on the news?'

-Unique:

- (35) Mama smotrit novosti. Segodnja novosti pro odin nebol'šoj skit v odnoj mom watches news. today news about one small hermitage in one derevne, gde živut vsego dva monaxa. Odin iz ètix monaxov sošel s uma village, where live just two monks. one of these monks went off mind i načal vyt', kak volk, i myčat', kak korova. Vtoromu monaxu prišlos' and began to-howl, like wolf, and to-moo, like cow. second monk had-to prervat' obet molčaniya i obratit'sja za pomošč'ju v psixiatričeskuju bol'nicu. break vow of-silence and to-contact for help in psychiatric hospital. Papa sprašivaet: - Čto rasskazali v novostjax? dad asks: - What did-they-say in news? 'Mom is watching the news. Today's segment is about a small hermitage in a village where only two monks live. One of the monks went mad and started howling like a wolf and mooing like a cow. The second monk had to break his vow of silence and contact a psychiatric hospital for help. Dad asks: What did they say on the news?'

Response options:

-
- (36) a. V odnoj derevne sošel s uma monax.
in one village went from spirit monk
- b. V odnoj derevne sošel s uma odin monax.
in one village went from spirit one monk

- Item 5

+Unique:

- (37) Katja i Vanja — social'nye rabotniki. Na ètoj nedele oni poseščajut školy.
 Katya and Vanya — social workers. on this week they visit schools.
 Segodnja, sidja v odnoj iz rajonnyx škol, Katja i Vanja zapolnjali
 today, sitting in one of district schools, Katya and Vanya were-filling-out
 dokumenty i vdrug uslyšali gromkij šlepok. Katja vygljanula v okno.
 documents and suddenly heard loud thud. Katya looked-out into window.
 Vo dvore odin rebenok, i on, sudja po vsemu, upal s dereva.
 in yard one child, and he, judging by all-appearances, fell from tree.
 Vanja sprašivaet: - Čto slučilos'?'
 Vanya asks: - What happened?
 'Katya and Vanya are social workers. This week they are visiting schools. Today,
 while sitting in one of the district schools filling out paperwork, they suddenly heard
 a loud thud. Katya looked out the window. There was one child in the yard, and
 he had apparently fallen out of a tree. Vanya asks: What happened?'

-Unique:

- (38) Katja i Vanja — social'nye rabotniki. Na ètoj nedele oni poseščajut školy.
 Katya and Vanya — social workers. on this week they visit schools.
 Segodnja, sidja v odnoj iz rajonnyx škol, Katja i Vanja zapolnjali
 today, sitting in one of district schools, Katya and Vanya were-filling-out
 dokumenty i vdrug uslyšali gromkij šlepok. Katja vygljanula v okno.
 documents and suddenly heard loud thud. Katya looked-out into window.
 Vo dvore školy tri rebenka, i odin iz nix upal s dereva. Dvoe
 in yard of-school three children, and one of them fell from tree. two
 drugix pytajutsja emu pomoč'. Vanja sprašivaet: - Čto slučilos'?'
 others try to-him to-help. Vanya asks: - What happened?
 'Katya and Vanya are social workers. This week they are visiting schools. Today,
 while sitting in one of the district schools filling out paperwork, they suddenly
 heard a loud thud. Katya looked out the window. There were three children in
 the schoolyard, and one of them had fallen out of a tree. The other two were trying
 to help him. Vanya asks: What happened?'

Response options:

- (39) a. S dereva upal rebënok.
 from tree fell child

-
- b. S dereva upal odin rebënok.
from tree fell one child

- Item 6

+Unique:

- (40) Ira i Lena priexali v prijut dlja životnyx. Poka oni besedujut s
Ira and Lena arrived at shelter for animals. while they talk with
direktorom, iz dvora slyšitsja laj. Ira vygljadyvaet v okno. Na
director, from yard is-heard barking. Ira peeks out window. in
ploščadke vsego odna sobaka. Ona sidit u vorot i laet na mašiny. Lena
yard just one dog. she sits by gate and barks at cars. Lena
sprašivaet: - Čto takoe?
asks: – What’s going-on?
‘Ira and Lena have come to an animal shelter. While they are talking with the
director, barking can be heard from the yard. Ira peeks out the window. There is
only one dog in the yard. It is sitting by the gate and barking at passing cars. Lena
asks: What’s going on?’

-Unique:

- (41) Ira i Lena priexali v prijut dlja životnyx. Poka oni besedujut s
Ira and Lena arrived at shelter for animals. while they talk with
direktorom, iz dvora slyšitsja laj. Ira vygljadyvaet v okno. Na
director, from yard is-heard barking. Ira peeks out window. in
ploščadke tri sobaki: dve sobaki begajut, a odna sidit u vorot i laet
yard three dogs: two dogs run-around, and one sits by gate and barks
na mašiny. Lena sprašivaet: - Čto takoe?
at cars. Lena asks: – What’s going-on?
‘Ira and Lena have come to an animal shelter. While they are talking with the
director, barking can be heard from the yard. Ira peeks out the window. There
are three dogs in the yard: two are running around, and one is sitting by the gate
barking at passing cars. Lena asks: What’s going on?’

Response options:

- (42) a. U vorot laet sobaka.
by gate barks dog
b. U vorot laet odna sobaka.
by gate barks one dog

-
- Item 7

+Unique:

- (43) Anna i Andrej prišli v bol'nicu zabrat' rezul'taty analizov. Oni sidjat v Anna and Andrej came to hospital to-pick-up results of-tests. they sit in kabinete vrača i ždut, poka tot zapolnit dokumenty. Vdrug oni slyšat office of-doctor and wait while he fills-out documents. suddenly they hear gromkuju rukan'. Anna vygljadyvaet iz kabineta i vidit, čto u kabineta loud cursing. Anna peeks out-of office and sees that by office sidit vsego odin pacient, i on gromko rugaetsja — kak budto s samim sits just one patient, and he loudly curses — as if with his soboj, ved' tam bol'se nikogo net. Andrej sprašivaet: - Čto own-self, after-all there no-one else not-there. Andrej asks: - What's takoe? going-on?
'Anna and Andrej have come to the hospital to pick up test results. They are sitting in the doctor's office waiting while he fills out some paperwork. Suddenly they hear someone cursing loudly. Anna peeks out of the office and sees that there is only one patient sitting outside, and he is cursing loudly—as if talking to himself, since there is no one else there. Andrej asks: What's going on?'

-Unique:

- (44) Anna i Andrej prišli v bol'nicu zabrat' rezul'taty analizov. Oni sidjat v Anna and Andrej came to hospital to-pick-up results of-tests. they sit in kabinete vrača i ždut, poka tot zapolnit dokumenty. Vdrug oni slyšat office of-doctor and wait while he fills-out documents. suddenly they hear gromkuju rukan'. Anna vygljadyvaet iz kabineta i vidit očered' iz trex loud cursing. Anna peeks out-of office and sees queue of three pacientov, i odin iz nix gromko rugaetsja. Dvoe ostal'nyx smotrjat s patients, and one of them loudly curses. two others watch with nedoumeniem. Andrej sprašivaet: - Čto takoe? bewilderment. Andrej asks: - What's going-on?
'Anna and Andrej have come to the hospital to pick up test results. They are sitting in the doctor's office waiting while he fills out some paperwork. Suddenly they hear someone cursing loudly. Anna peeks out of the office and sees a queue of three patients, and one of them is cursing loudly. The other two are watching in bewilderment. Andrej asks: What's going on?'

Response options:

- (45)
- a. V očeredi razozlilsja pacient.
in queue got.angry patient
 - b. V očeredi razozlilsja odin pacient.
in queue got.angry one patient

-
- Item 8

+Unique:

- (46) Polina i Vasja rabotajut v odnom ofise. Vdrug oni slyšat šum
 Polina and Vasya work in one office. suddenly they hear noise
 posredi rabočego dnja. Polina vygljadyvaet iz okna i vidit odnogo
 in-middle-of work day. Polina peeks out-of window and sees one
 čeloveka. On, sudja po vsemu, upal i ne mozet vstat'. Vasja
 person. he, judging by all-appearances, fell and not can get-up. Vasya
 sprašivaet: - Čto takoe?
 asks: – What happened?
 'Polina and Vasya work in the same office. Suddenly, in the middle of the workday,
 they hear a noise. Polina peeks out the window and sees one person. He apparently
 fell and cannot get up. Vasya asks: What's going on?'

-Unique:

- (47) Polina i Vasja rabotajut v odnom ofise. Vdrug oni slyšat šum
 Polina and Vasya work in one office. suddenly they hear noise
 posredi rabočego dnja. Polina vygljadyvaet iz okna i vidit trex
 in-middle-of work day. Polina peeks out-of window and sees three
 proxožix. Odin iz nix, sudja po vsemu, upal, a dva drugix
 passers-by. one of them, judging by all-appearances, fell, and two others
 pytajutsja emu pomoč'. Vasja sprašivaet: - Čto takoe?
 try to-him to-help. Vasya asks: – What happened?
 'Polina and Vasya work in the same office. Suddenly, in the middle of the workday,
 they hear a noise. Polina peeks out the window and sees three passersby. One of
 them apparently fell, and the other two are trying to help him. Vasya asks: What's
 going on?'

Response options:

- (48) a. Na ulice upal proxožij.
 in street fell passerby
- b. Na ulice upal odin proxožij.
 in street fell one passerby

B.3 Experiment 3: incompatible predicates

In Experiment 3, the response options were presented as text, not audio.

- Item 1

+Unique:

- (49) Maša smotrit v igrušku i vidit v nej sobačku i zajčika. Liza: - Čto ty Masha looks in toy and sees in it puppy and bunny. Liza: – What you tam vidiš'? there see? 'Masha is looking in the toy and sees a puppy and a bunny in it. Liza: What do you see in there?'

Response options:

- (50) a. Na kartinke spit sobačka i begaet po krugu zajčik.
in picture sleeps puppy and runs in circle bunny
- b. Na kartinke odna sobačka spit i odin zajčik begaet po krugu.
in picture one puppy sleeps and one bunny runs in circle

-Unique:

- (51) Maša smotrit v igrušku i vidit v nej dvux sobaček. Liza: - Čto ty tam Masha looks in toy and sees in it two puppies. Liza: – What you there vidiš'? see? 'Masha is looking in the toy and sees two puppies in it. Liza: What do you see in there?'

Response options:

- (52) a. Na kartinke spit sobačka i begaet po krugu sobačka.
in picture sleeps puppy and runs in circle puppy
- b. Na kartinke odna sobačka spit i odna sobačka begaet po krugu
in picture one puppy sleeps and one puppy runs in circle

-
- Item 2

+Unique:

- (53) Maša smotrit v igrušku i vidit v nej devočku i mal'čika. Liza: - Čto
Masha looks in toy and sees in it girl and boy. Liza: – What
ty tam vidiš'?
you there see?
'Masha is looking in the toy and sees a girl and a boy in it. Liza: What do you see
in there?'

Response options:

- (54) a. Na kartinke čitaet devočlka i prygaet na skakalke mal'čik.
in picture reads girl and jumps on rope boy
b. Na kartinke odna devočlka čitaet i odin mal'čik prygaet na skakalke.
in picture one girl reads and one boy jumps on rope

-Unique:

- (55) Maša smotrit v igrušku i vidit v nej dvux devoček v škol'nom dvore. Liza:
Masha looks in toy and sees in it two girls in school yard. Liza:
- Čto ty tam vidiš'?
– What you there see?
'Masha is looking in the toy and sees two girls in a schoolyard in it. Liza: What do
you see in there?'

Response options:

- (56) a. Na kartinke čitaet devočka i prygaet na skakalke devočlka.
in picture reads girl and jumps on rope girl
b. Na kartinke odna devočka čitaet i odna devočka prygaet na skakalke.
in picture one girl reads and one girl jumps on rope

-
- Item 3

+Unique:

- (57) Maša smotrit v igrušku i vidit v nej gimnastku i klouna. Liza: - Čto
Masha looks in toy and sees in it gymnast and clown. Liza: – What
ty tam vidiš’?
you there see?
‘Masha is looking in the toy and sees a gymnast and a clown in it. Liza: What do
you see in there?’

Response options:

- (58) a. Na kartinke sidit v špagate gimnastka i delaet koleso kloun.
in picture sits in split gymnast and makes wheel clown
b. Na kartinke odna gimnastka sidit v špagate i odin kloun delaet koleso .
in picture one gymnast sits in split and one clown makes wheel

-Unique:

- (59) Maša smotrit v igrušku i vidit v nej dvux gimnastok. Liza: - Čto ty
Masha looks in toy and sees in it two gymnasts. Liza: – What you
tam vidiš’?
there see?
‘Masha is looking in the toy and sees two gymnasts in it. Liza: What do you see in
there?’

Response options:

- (60) a. Na kartinke sidit v špagate gimnastka i delaet koleso gimnastka.
in picture sits in split gymnast and makes wheel gymnast
b. Na kartinke odna gimnastka sidit v špagate i odna gimnastka delaet
in picture one gymnast sits in split and one gymnast makes
koleso.
wheel

-
- Item 4

+Unique:

- (61) Maša smotrit v igrušku i vidit v nej voronu i sovu. Liza: - Čto ty
Masha looks in toy and sees in it crow and owl. Liza: – What you
tam vidiš’?
there see?
‘Masha is looking in the toy and sees a crow and an owl in it. Liza: What do you
see in there?’

Response options:

- (62) a. Na kartinke letaet vorona i spit sova.
in picture flies crow and sleeps owl

b. Na kartinke odna vorona letaet i odna sova spit.
in picture one crow flies and one owl sleeps

-Unique:

- (63) Maša smotrit v igrušku i vidit v nej dvux voron. Liza: - Čto ty tam
Masha looks in toy and sees in it two crows. Liza: – What you there
vidiš’?
see?
‘Masha is looking in the toy and sees two crows in it. Liza: What do you see in
there?’

Response options:

- (64) a. Na kartinke letaet vorona i spit vorona.
in picture flies crow and sleeps crow

b. Na kartinke odna vorona letaet i odna vorona spit.
in picture one crow flies and one crow sleeps

-
- Item 5

+Unique:

- (65) Maša smotrit v igrušku i vidit v nej popugaja i vorob'ja. Liza: - Čto
Masha looks in toy and sees in it parrot and sparrow. Liza: – What
ty tam vidiš'?
you there see?
'Masha is looking in the toy and sees a parrot and a sparrow in it. Liza: What do
you see in there?'

Response options:

- (66) a. Na kartinke poët pesnju popugaj i p'ët vodu vorobej.
in picture sings song parrot and drinks water sparrow
b. Na kartinke odin popugaj poët pesnju i odin vorobej p'ët vodu.
in picture one parrot sings song and one sparrow drinks water

-Unique:

- (67) Maša smotrit v igrušku i vidit v nej dvux popugaev. Liza: - Čto ty tam
Masha looks in toy and sees in it two parrots. Liza: – What you there
vidiš'?
see?
'Masha is looking in the toy and sees two parrots in it. Liza: What do you see in
there?'

Response options:

- (68) a. Na kartinke poët pesnju popugaj i p'ët vodu popugaj.
in picture sings song parrot and drinks water parrot
b. Na kartinke odin popugaj poët pesnju i odin popugaj p'ët vodu.
in picture one parrot sings song and one parrot drinks water

-
- Item 6

+Unique:

- (69) Maša smotrit v igrušku i vidit v nej červjaka i ulitku. Liza: - Čto ty
Masha looks in toy and sees in it worm and snail. Liza: – What you
tam vidiš’?
there see?
‘Masha is looking in the toy and sees a worm and a snail in it. Liza: What do you
see in there?’

Response options:

- (70) a. Na kartinke igraet na dudočke červjak i est listik ulitka.
in picture plays on reedpipe worm and eats leaf snail
b. Na kartinke odin červjak igraet na dudočke i odna ulitka est listik.
in picture one worm plays on reedpipe and one snail eats leaf

-Unique:

- (71) Maša smotrit v igrušku i vidit v nej dvux červjakov. Liza: - Čto ty tam
Masha looks in toy and sees in it two worms. Liza: – What you there
vidiš’?
see?
‘Masha is looking in the toy and sees two worms in it. Liza: What do you see in
there?’

Response options:

- (72) a. Na kartinke igraet na dudočke červjak i est listik červjak.
in picture plays on reedpipe worm and eats leaf worm
b. Na kartinke odin červjak igraet na dudočke i odin červjak est listik.
in picture one worm plays on reedpipe and one worm eats leaf

-
- Item 7

+Unique:

- (73) Maša smotrit v igrušku i vidit v nej čerepaxu i ljagušku. Liza: - Čto
Masha looks in toy and sees in it tortoise and frog. Liza: – What
ty tam vidiš’?
you there see?
‘Masha is looking in the toy and sees a tortoise and a frog in it. Liza: What do you
see in there?’

Response options:

- (74) a. Na kartinke čitaet knižku čerepaxa i spit ljaguška.
in picture reads book turtle and sleeps frog
b. Na kartinke odna čerepaxa čitaet knižku i odna ljaguška spit.
in picture one turtle reads book and one frog sleeps

-Unique:

- (75) Maša smotrit v igrušku i vidit v nej dvux čerepax. Liza: - Čto ty tam
Masha looks in toy and sees in it two tortoises. Liza: – What you there
vidiš’?
see?
‘Masha is looking in the toy and sees two tortoises in it. Liza: What do you see in
there?’

Response options:

- (76) a. Na kartinke čitaet knižku čerepaxa i spit čerepaxa.
in picture reads book turtle and sleeps turtle
b. Na kartinke odna čerepaxa čitaet knižku i odna čerepaxa spit.
in picture one turtle reads book and one turtle sleeps

-
- Item 8

+Unique:

- (77) Maša smotrit v igrušku i vidit v nej obez'janku i tukana. Liza: - Čto ty tam vidiš'?
Masha looks in toy and sees in it monkey and toucan. Liza: – What you there see?
'Masha is looking in the toy and sees a monkey and a toucan in it. Liza: What do you see in there?'
- (78) a. Na kartinke tancuet obez'janka i sidit na dereve tukan.
in picutre dances monkey and sits on tree toucan
- b. Na kartinke odna obez'janka tancuet i odin tukan sidit na dereve.
in picture one monkey dances and one toucan sits on tree

-Unique:

- (79) Maša smotrit v igrušku i vidit v nej dvux obez'janok. Liza: - Čto ty Masha looks in toy and sees in it two monkeys. Liza: – What you tam vidiš'?
there see?
'Masha is looking in the toy and sees two monkeys in it. Liza: What do you see in there?'

Response options:

- (80) a. Na kartinke tancuet obez'janka i sidit na dereve obez'janka.
in picutre dances monkey and sits on tree monkey
- b. Na kartinke odna obez'janka tancuet i odna obez'janka sidit na dereve.
in picture one monkey dances and one monkey sits on tree

Appendix C Experimental stimuli for Experiment 4 (Maximality)

This appendix contains textual and visual stimuli for Experiment 4. The pictures were generated using ChatGPT 5.2.

- Item 1

- (1) a. Boys are holding soccer balls. *English, bare PL*
b. The boys are holding soccer balls. *English, definite PL*
- (2) Mal'čik-i derž-at mjač-i.
boy-PL hold-PRS.3PL ball-PL
'The boys/Boys are holding balls.' *Russian, bare PL*



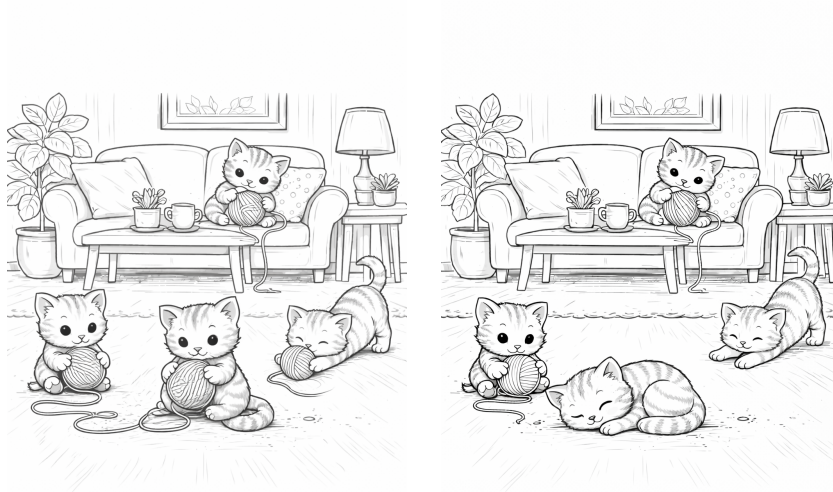
(a) [+MAXIMAL]



(b) [-MAXIMAL]

- Item 2

- (3) a. Cats are playing with yarn. *English, bare PL*
 b. The cats are playing with yarn. *English, definite PL*
- (4) Kotj-ata igraj-ut s nitk-ami.
 kitten.PL play-PRS.3PL with thread-INSTR.PL
 ‘Kittens/The kittens are playing with threads.’ *Russian, bare PL*

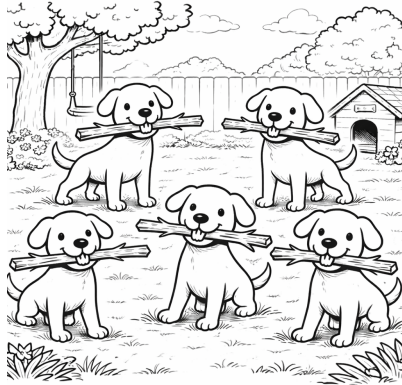


(a) [+MAXIMAL]

(b) [-MAXIMAL]

- Item 3

- (5) a. Dogs are holding sticks. *English, bare PL*
 b. The dogs are holding sticks. *English, definite PL*
- (6) Собак-и дер̣ж-ат в зуб-ах палк-и.
 dog-PL hold-PRS.3PL in tooth-PREP.PL stick-PL
 ‘The dogs/Dogs are holding sticks in the mouth.’ *Russian, bare PL*



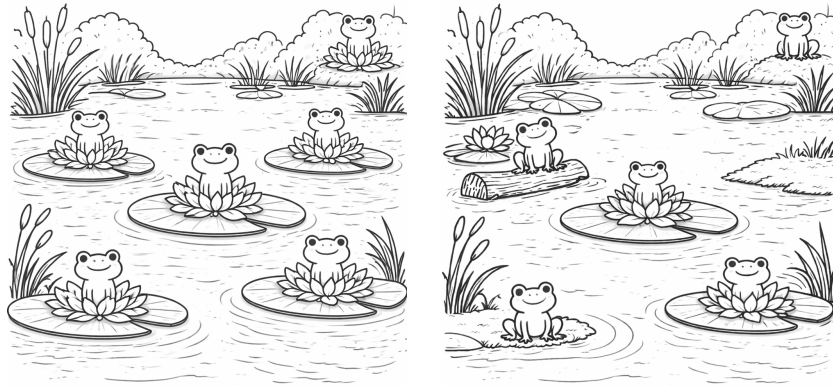
(a) [+MAXIMAL]



(b) [-MAXIMAL]

- Item 4

- (7) a. Frogs are sitting on lily pads. *English, bare PL*
 b. The frogs are sitting on lily pads. *English, definite PL*
- (8) Ljagušk-i sidj-at na kuvšink-ax.
 frog-PL sit-PRS.3PL on lily.pad-PREP.PL
 ‘Frogs/The frogs are sitting on lily pads.’ *Russian, bare PL*



(a) [+MAXIMAL]

(b) [-MAXIMAL]

• Item 5

- (9) a. Girls are playing the violin. *English, bare PL*
 b. The girls are playing the violin. *English, definite PL*
- (10) Devočk-i igra-j-ut na skripk-e.
 girl-PL play-PRS.3PL on violin-PREP
 ‘Girls/The girls are playing the violin.’ *Russian, bare PL*



(a) [+MAXIMAL]



(b) [-MAXIMAL]

- Item 6

- (11) a. Grannies have their umbrellas open. *English, bare PL*
 b. The grannies have their umbrellas open. *English, definite PL*
- (12) Starušk-i otkry-l-i zont-ik-i.
 granny-PL open-PST-PL umbrella-DIM-PL
 ‘Grannies/The grannies opened their umbrellas.’ *Russian, bare PL*



(a) [+MAXIMAL]



(b) [-MAXIMAL]

Appendix D Experimental stimuli for Experiments 5A and 5B (Anaphoricity studies)

This appendix contains contexts tested in the experiments 5A and 5B. They are grouped by experiment and language, as some of the translations are not exactly the same (due to syntactic and stylistic differences between the two languages). Here I do not provide the Russian versions of the response options, but they were presented in Russian.

D.1 Experiment 5A (biased towards anaphoricity)

D.1.1 English

(1) Jacob harvested carrots and potatoes from his farm. He stored the vegetables in the pantry. When he got hungry that evening, he cooked potatoes / the potatoes / some potatoes for dinner.

- a. Jacob cooked the potatoes he harvested earlier that day. *anaphoric*
- b. Jacob cooked other potatoes he had. *non-anaphoric*

(2) Olivia bought candles and small paintings for her room. She added them to her room decor and really liked the result. Later that night, she wanted to have a cozy night and lit candles / the candles / some candles.

- a. Olivia lit the candles she bought earlier that day. *anaphoric*
- b. Olivia lit other candles she had. *non-anaphoric*

-
- (3) Alex went grocery shopping and bought apples and candies. He brought the groceries home. He realized that he was hungry, so he ate apples / the apples / some apples.
- a. Alex ate the apples he bought. *anaphoric*
- b. Alex ate other apples. *non-anaphoric*
- (4) In the morning, Sarah picked roses and tulips in the garden for her flower shop. She brought the flowers to the shop. Before opening, she arranged roses / the roses / some roses in a vase.
- a. Sarah arranged the roses she picked earlier. *anaphoric*
- b. Sarah arranged other roses. *non-anaphoric*
- (5) Ethan ordered plates and cups online. When the delivery arrived, he put the stuff in the kitchen. Later, he washed plates / the plates / some plates.
- a. Ethan washed the plates he ordered. *anaphoric*
- b. Ethan washed other plates he had in the kitchen. *non-anaphoric*
- (6) Emma picked tomatoes and cucumbers from her garden. She placed them on the kitchen counter. Later, she was making a salad and put tomatoes / the tomatoes / some tomatoes in it.
- a. Emma put the tomatoes she picked in the salad. *anaphoric*
- b. Emma put other tomatoes she had in the salad. *non-anaphoric*
- (7) Alex gathered notebooks and folders from the classroom. He left them on his desk. Later, he organized folders / the folders / some folders for the final review.
- a. Alex organized the folders he gathered. *anaphoric*
- b. Alex organized other folders on the desk. *non-anaphoric*
- (8) Yesterday, Clara washed scarves and gloves. She put them in her drawer. The next day, she wore gloves / the gloves / some gloves because it was cold.

-
- a. Clara wore the gloves she washed the day before. *anaphoric*
- b. Clara wore other gloves. *non-anaphoric*
- (9) Ashley borrowed books and DVDs from the library. Next day, she went to the library again to return books / the books / some books.
- a. Ashley returned the books she borrowed the day before. *anaphoric*
- b. Ashley returned other books that were due. *non-anaphoric*
- (10) Chloe picked leaves and flowers for her art project. Later that evening, she made a watercolor sketch of leaves / the leaves / some leaves.
- a. Chloe sketched the leaves she earlier picked. *anaphoric*
- b. Chloe sketched other leaves. *non-anaphoric*
- (11) Lily purchased blueberries and peaches at the farmer's market. She brought the fruit to her kitchen. After dinner, she ate blueberries / the blueberries / some blueberries.
- a. Lily ate the blueberries she purchased. *anaphoric*
- b. Lily ate other blueberries she had. *non-anaphoric*
- (12) Jack borrowed some tools from his neighbor: a hammer and nails. He left them on his workbench. The next day, he used nails / the nails / some nails for his project.
- a. Jack used the nails he borrowed. *anaphoric*
- b. Jack used other nails. *non-anaphoric*
- (13) Alex ran a workshop on coding and robotics. Participants built robots and wrote basic programs. At the end of the workshop, Alex demonstrated robots / the robots / some robots to the class.
- a. Alex demonstrated the robots from the workshop. *anaphoric*
- b. Alex demonstrated other robots. *non-anaphoric*

-
- (14) Emma taught a pottery class. During the class, her students made bowls and vases. Emma also demonstrated different techniques for painting them. Towards the end, the students painted bowls / the bowls / some bowls.
- a. The students painted the bowls they made during the class. *anaphoric*
- b. The students painted other bowls. *non-anaphoric*
- (15) Last weekend, Oliver bought blankets and pillows. His roommate Daniel was packing for a camping trip. He borrowed blankets / the blankets / some blankets from Oliver.
- a. Daniel borrowed the blankets Oliver bought last weekend. *anaphoric*
- b. Daniel borrowed other blankets Oliver had. *non-anaphoric*
- (16) Max baked cupcakes and cookies for the party. He also packed up cupcakes / the cupcakes / some cupcakes in a nice box for his friend Dana, who had her birthday.
- a. Max packed up the cupcakes he baked. *anaphoric*
- b. Max packed up other cupcakes. *non-anaphoric*

D.1.2 Russian

- (17) Vanja pošel v magazin i kupil **jabloki** i konfety. On prinjos pokupki
 Vanja went in store and bought apples and candies He brought purchases
 domoj. Potom on progolodalsja i s'jel jabloki / kakie-to jabloki.
 home then he got.hungry and eat apples some apples
 'Vanya went to the store and bought apples and candy. He brought the shopping
 home. Then he got hungry and ate **apples** / **some apples**.'
- a. Vanya ate the apples he had bought. *anaphoric*
- b. Vanya ate different apples. *non-anaphoric*
- (18) Rano utrom Sonja pošla v sad i sobrala svežie **rozy** i tjul'pany dlja
 early morning Sonja went in garden and gathered fresh roses and tulips for
 svoego cvetočnogo magazina. Ona prinesla cvety v magazin. Pered otkrytiem
 self flower store she brought flowers in store before opening

ona postavila rozy / kakie-to rozy v vazu.
 she put roses some roses in vase
 ‘Early in the morning Sonya went to the garden and gathered fresh roses and tulips
 for her flower shop. She brought the flowers to the shop. Before opening, she put
roses / some roses in a vase.’

- a. Sonya put the roses she had gathered early that morning in the vase. *anaphoric*
- b. Sonya put other roses in the vase, some that were already in her shop. *non-anaphoric*

- (19) Ivan zakazal **tarelki** i bokaly onlajn. Kogda dostavka priekhala, on prinjos
 Ivan ordered plates and glasses online when delivery arrived he brought
 pokupki na kukhnju. Pozže on pomyl tarelki / kakie-to tarelki.
 purchases to kitchen later he washed plates some plates
 ‘Ivan ordered plates and glasses online. When the delivery arrived, he brought the
 purchases to the kitchen. Later he washed **plates / some plates**.’

- a. Ivan washed the plates he had ordered. *anaphoric*
- b. Ivan washed other plates that were already in his kitchen. *non-anaphoric*

- (20) Natal’ja sobrala **omidory** i ogurcy so svoego ogoroda. Ona položila
 Natal’ja gathered tomatoes and cucumbers from self garden she put
 ikh na kukhonnyj stol. Pozže ona gotovila salat i položila pomidory /
 them on kitchen table later she prepared salad and put tomatoes
 kakie-to pomidory v nego.
 some tomatoes in it
 ‘Natalya gathered tomatoes and cucumbers from her garden. She put them on the
 kitchen table. Later she was making a salad and put **tomatoes / some tomatoes**
 in it.’

- a. Natalya put in the salad the tomatoes she had just gathered from the garden. *anaphoric*
- b. Natalya put in the salad other tomatoes she had had before. *non-anaphoric*

- (21) Učitel’nica sobrala **tetradi** i papki u klassa. Ona ostavila ikh na
 teacher collected notebooks and folders from class she left them on
 svoëm stole. Pozže ona razložila papki / kakie-to papki po jaščikam.
 self desk later she sorted folders some folders into drawers

-
- ‘The teacher collected notebooks and folders from the class. She left them on her desk. Later she sorted **folders** / **some folders** into drawers.’
- a. The teacher sorted the folders she had collected from the class. *anaphoric*
- b. The teacher sorted other folders that had been lying on her desk. *non-anaphoric*
- (22) Karina postirala šarfy i varežki. Ona položila ikh v komod. Na sledujuščij den’ pokholodalo, i Karina nadela varežki / kakie-to varežki.
 Karina washed scarves and mittens she put them in dresser on next day it.got.cold and Karina put.on mittens some mittens
 ‘Karina washed scarves and mittens. She put them in the dresser. The next day it got cold, and Karina put on **mittens** / **some mittens**.’
- a. Karina put on the mittens she had washed the day before. *anaphoric*
- b. Karina put on other mittens she had. *non-anaphoric*
- (23) Asja vzjala v biblioteke knigi i DVD-diski. Na sledujuščij den’ ona snova Asja took in library books and DVD-disks on next day she again prišla v biblioteku i vernula knigi / kakie-to knigi.
 came in library and returned books some books
 ‘Asya checked out books and DVDs from the library. The next day she went back to the library and returned **books** / **some books**.’
- a. Asya returned the books she had checked out the day before. *anaphoric*
- b. Asya returned other books that were due. *non-anaphoric*
- (24) Katja sobrala osennie list’ja i cvety, čtoby ispol’zovat’ ikh v Katja gathered autumn leaves and flowers in-order-to use them in kačestve natury dlja risovanija. Večerom ona narisovala list’ja / kakie-to quality.of model for drawing in.the.evening she drew leaves some list’ja akvarelju.
 leaves with.watercolor
 ‘Katya gathered autumn leaves and flowers to use as models for drawing. In the evening she painted **leaves** / **some leaves** in watercolor.’
- a. Katya painted the leaves she had gathered during the day. *anaphoric*
- b. Katya painted other leaves. *non-anaphoric*

-
- (25) Lena kupila **klubniku** i persiki na rynke. Ona postavila pokupki v
 Lena bought strawberry and peaches at market she put purchases in
 holodil'nik. Na užin Lena prigotovila laz'anju. Posle užina, Lena s'ela persiki /
 fridge for dinner Lena prepared lasagna after dinner Lena ate peaches
 kakie-to persiki.
 some peaches
 'Lena bought strawberries and peaches at the market. She put the shopping in the
 fridge. For dinner she made lasagna. After dinner, Lena ate **peaches** / **some peaches**.'
- a. Lena ate the peaches she had bought at the market. *anaphoric*
- b. Lena ate other peaches she had. *non-anaphoric*
- (26) Oleg zajal u sosedu instrumenty: **gvozdi** i molotok. On položil ikh
 Oleg borrowed from neighbor tools nails and hammer he put them
 na skamejku vozle doma. Na sledujuščij den', on ispol'zoval gvozdi / kakie-to
 on bench near house on next day he used nails some
 gvozdi v rabote.
 nails in work
 'Oleg borrowed tools from his neighbor: nails and a hammer. He put them on a
 bench near the house. The next day, he used **nails** / **some nails** for work.'
- a. Oleg used the nails he had borrowed from his neighbor. *anaphoric*
- b. Oleg used other nails. *non-anaphoric*
- (27) Aleksej provël master-klass dlja škol'nikov po robotostrojeniju i
 Aleksej conducted workshop for students on robotics and
 programirovaniju. Rebjata sdělali malen'kih **robotov** i napisali neskol'ko
 programming kids made small robots and wrote several
 prostyh programm. V konce uroka Aleksej prodemonstiroval robotov / kakix-to
 simple programs at end lesson Aleksej demonstrated robots some
 robotov klassu.
 robots to.class
 'Alexei ran a workshop for students on robotics and programming. The kids made
 small robots and wrote several simple programs. At the end of the lesson, Alexei
 demonstrated **robots** / **some robots** to the class.'
- a. Alexei demonstrated the robots the kids had made in the workshop. *anaphoric*
- b. Alexei demonstrated other robots. *non-anaphoric*

- (28) Alina provela master-klass po lepke. Vo vremena uroka, rebjata sdělali Alina conducted workshop on sculpting in time.of lesson kids made **podsvěčniki** i vazy. Alina takže pokazala, kak raskrašivat' izdelija. V konce candlesticks and vases Alina also showed how to.paint objects at end uroka rebjata raskrasili podsvěčniki / kakie-to podsvěčniki. lesson kids painted candlesticks some candlesticks 'Alina ran a sculpting workshop. During the lesson, the kids made candlesticks and vases. Alina also showed how to paint the pieces. At the end of the lesson the kids painted **candlesticks** / **some candlesticks**.'
- a. The kids painted the candlesticks they had made in the workshop. *anaphoric*
- b. The kids painted other candlesticks that were in the workshop. *non-anaphoric*
- (29) Na prošloj nedele Oleg kupil **poduški** i odejala. Ego žena provodila bol'suju on past week Oleg bought pillows and blankets his wife did big stirku i postirala odejala / kakie-to odejala. laundry and washed blankets some blankets 'Last week Oleg bought pillows and blankets. His wife was doing a big laundry and washed **blankets** / **some blankets**.'
- a. The wife washed the blankets Oleg had bought. *anaphoric*
- b. The wife washed other blankets. *non-anaphoric*
- (30) Marina ispekla **pečen'ja** i keksy dlja večerinki. Eščë ona upakovala Marina baked cookies and cupcakes for party also she packed pečen'ja / kakie-to pečen'ja v krasivuju korobočku dlja svoej podrugi Niny, u cookies some cookies in pretty box for self friend Nina by kotoroj byl den' roždenija. which was day of.birth 'Marina baked cookies and cupcakes for a party. She also packed **cookies** / **some cookies** in a pretty box for her friend Nina, whose birthday it was.'
- a. Marina packed the cookies she had baked. *anaphoric*
- b. Marina packed other cookies. *non-anaphoric*
- (31) Jana sobrala **kabački** i pomidory s ogoroda. Ona položila ovošči v Jana gathered zucchinis and tomatoes from garden she put vegetables in

kukhonnyj škaf. Kogda ona progolodalas', ona prigotovila kabački / kakie-to
 kitchen cupboard when she got.hungry she cooked zucchinis some
 kabački.
 zucchinis

'Yana gathered zucchini and tomatoes from the garden. She put the vegetables in the
 kitchen cupboard. When she got hungry, she cooked **zucchinis** / **some zucchinis**.'

- a. Yana cooked the zucchini she had gathered from the garden. *anaphoric*
- b. Yana cooked other zucchini. *non-anaphoric*

- (32) Olja kupila **sveči** i kartiny dlja svoej komnaty. Ona ukrasila imi
 Olja bought candles and paintings for self room she decorated witht.them
 komnatu i ostalas' dovol'na rezul'tatom. Večerom ej zahotelos'
 room and remained pleased by.result in.the.evening she wanted
 romantičnoj atmosfery, i Olja zažgla sveči / kakie-to sveči.
 romantic atmosphere and Olja lit candles some candles
 'Olya bought candles and paintings for her room. She decorated her room with them
 and was pleased with the result. In the evening she wanted a romantic atmosphere,
 and Olya lit **candles** / **some candles**.'

- a. Olya lit the candles she had bought that day. *anaphoric*
- b. Olya lit other candles she already had. *non-anaphoric*

D.2 Experiment 5B (biased against anaphoricity)

D.2.1 English

- (33) Alex went grocery shopping and bought oranges, apples, and peaches. He brought
 the groceries home and sat on the couch to watch TV. His wife, Kayla, came home
 from work with groceries as well. She unpacked them, and afterward, she realized
 she was hungry. Wanting a healthy snack, she ate apples / the apples / some apples.

- a. Kayla ate the same apples Alex had brought. *anaphoric*
- b. Kayla ate other apples. *non-anaphoric*

-
- (34) Ethan ordered plates and cups online. When the delivery arrived, he opened the packages in the kitchen. Later that evening, his grandma was cleaning up after dinner and noticed some dishes in the sink. She washed plates / the plates / some plates.
- a. Ethan's grandma washed the same plates Ethan had ordered. *anaphoric*
 - b. Ethan's grandma washed other plates. *non-anaphoric*
- (35) Ashley borrowed thrillers and crime novels from the library. She placed them on a shelf with other borrowed library materials at home. The next day, her daughter, Anna, went to a different library location. Anna returned books / the books / some books.
- a. Anna returned the same books Ashley had borrowed the previous day. *anaphoric*
 - b. Anna returned other books. *non-anaphoric*
- (36) Chloe picked leaves and flowers for her art project. Later that evening, she attended an art class with her friend, Audrey. Chloe painted a cityscape, while Audrey sketched leaves / the leaves / some leaves.
- a. Audrey sketched the same leaves Chloe had picked earlier. *anaphoric*
 - b. Audrey sketched other leaves. *non-anaphoric*
- (37) Katie bought candles and small paintings. She used them to decorate her room. In the evening, her roommate Valerie decided to make the living room cozier and lit candles / the candles / some candles.
- a. Valerie lit the same candles Katie had bought earlier. *anaphoric*
 - b. Valerie lit other candles. *non-anaphoric*
- (38) Harris works as a librarian. Today, he sorted books and magazines. He placed the latest magazine issues on the shelf in the periodicals section. In the evening,

-
- his colleague Tony was tidying up the reading room and found magazines / the magazines / some magazines.
- a. Tony found the same magazines Harris had earlier placed in the periodicals section.
anaphoric
 - b. Tony found other magazines. *non-anaphoric*
- (39) Ashley borrowed books and DVDs from the library. Next day, she went to the library again to return books / the books / some books.
- a. Ashley returned the books she borrowed the day before. *anaphoric*
 - b. Ashley returned other books that were due. *non-anaphoric*
- (40) Clara works as a costume designer in the theater. She prepared jackets and dresses for the next day's performance, placed them on a designated shelf, and went home. The next day, her colleague Alina arrived at the theater. She entered the laundry room and found dresses / the dresses / some dresses.
- a. Alina found the same dresses Clara had earlier placed on the shelf. *anaphoric*
 - b. Alina found other dresses. *non-anaphoric*
- (41) Finn brought balls and field markers to the sports center. He unloaded the equipment in the coach's room. After lunch, Coach Grant was preparing the field for training, went into the men's locker room, and found balls / the balls / some balls.
- a. Coach Grant found the same balls Finn had earlier brought. *anaphoric*
 - b. Coach Grant found other balls. *non-anaphoric*
- (42) The students of the music school took violins and violas to the instrument storage room. In the evening, the janitor was mopping the floors in the orchestra classroom and found violins / the violins / some violins.
- a. The janitor found the same violins the students had earlier taken to the storage room.
anaphoric

-
- b. The janitor found other violins. *non-anaphoric*
- (43) The swimming coach, Courtney, packed towels and swimsuits for her team into a bag, placed it in the coach's room, and went to talk to the administrator. Another coach, Karen, went into the restroom and found towels / the towels / some towels.
- a. Coach Karen found the same towels Coach Courtney had earlier packed into a bag. *anaphoric*
- b. Coach Karen found other towels. *non-anaphoric*
- (44) Colin will represent his company at the job fair. Before the event, he laid out brochures and printed job descriptions at his company's booth. Later, the fair organizer was checking the venue's readiness. She went into the women's restroom and found brochures / the brochures / some brochures.
- a. The organizer found the same brochures Colin had earlier laid out at his company's booth. *anaphoric*
- b. The organizer found other brochures. *non-anaphoric*
- (45) Sonia sorted the jackets and scarves that people had donated to the charity organization into baskets. The next morning, another volunteer, Hunter, was arranging the items before the event, stepped outside for a smoke, and found scarves / the scarves / some scarves.
- a. Hunter found the same scarves Sonia had earlier sorted. *anaphoric*
- b. Hunter found other scarves. *non-anaphoric*
- (46) Beverly works as a barista in a café. She took out fresh croissants and buns, carefully arranged them in the display case, and went on a break. An hour later, the waiter, Max, was checking the stock of cups at the counter, looked into the fridge, and found croissants / the croissants / some croissants.

-
- a. Max found the same croissants Beverly had earlier arranged in the display case.

anaphoric

- b. Max found other croissants.

non-anaphoric

- (47) Hayley works in a flower shop. She received a new delivery of roses and tulips, placed them in the flower refrigerator, and went to arrange the display. In the evening, Lindy came to pick up the orders for delivery, looked into the storage room, and found tulips / the tulips / some tulips.

- a. Lindy found the same tulips Hayley had earlier placed in the refrigerator. *anaphoric*

- b. Lindy found other tulips.

non-anaphoric

- (48) Jen works at a school. She prepared printouts and new notebooks for the students, put them in the cabinet, and left for a teachers' meeting. Another teacher, Tess, went into Jen's classroom to get some tea, approached the desk, and saw notebooks / the notebooks / some notebooks.

- a. Tess found the same notebooks Jen had earlier put in the cabinet.

anaphoric

- b. Tess found other notebooks.

non-anaphoric

D.2.2 Russian

- (49) Leša pošel za produktami i kupil apel'siny, **jabloki** i persiki. On prinjos
Leša went for groceries and bought oranges apples and peaches he brought
pokupki domoj i sjel na divan smotret' televizor. Ego žena Karina prišla
purchases home and sat on sofa to.watch TV his wife Karina came
domoj s raboty i tože prinesla produkty. Ona razbrala produkty. Posle
home from work and also brought groceries she unpacked groceries after
etogo Karina rešila, što ona progolodalas' i hočet poleznyj perekas i
this Karina decided that she got.hungry and wanted healthy snack and
s'ela jabloki / kakie-to jabloki.
ate apples some apples
'Lesha went grocery shopping and bought oranges, apples, and peaches. He brought
the shopping home and sat on the sofa to watch TV. His wife Karina came home from

work and also brought groceries. She unpacked the groceries. After that, Karina decided she was hungry and wanted a healthy snack, and ate **apples** / **some apples**.’

- a. Karina ate the apples Lesha had bought. *anaphoric*
b. Karina ate different apples. *non-anaphoric*

- (50) Timofej zakazal v onlajn-magazine **tarelki** i stakany. Kogda prišla dostavka, Timofej ordered in online-store plates and glasses when arrived delivery on otkryl pokupki na kuhne. Pozže večerom, ego babuška Anna he opened purchases in kitchen later that-evening his grandmother Anna Sergeevna ubirala kukhnju posle užina i zametila, čto v rakavine est’ Sergeevna cleaned kitchen after dinner and noticed that in sink were posuda. Anna Sergeevna pomyla tarelki / kakie-to tarelki. dishes Anna Sergeevna washed plates some plates
‘Timofei ordered plates and glasses from an online store. When the delivery arrived, he unpacked them in the kitchen. Later that evening, his grandmother Anna Sergeyevna was cleaning the kitchen after dinner and noticed there were dishes in the sink. Anna Sergeyevna washed **plates** / **some plates**.’

- a. Anna Sergeyevna washed the plates Timofei had unpacked in the kitchen. *anaphoric*
b. Anna Sergeyevna washed other plates. *non-anaphoric*

- (51) Ol’ga vzjala v biblioteke trillery i detektivy. Doma ona postavila ikh Ol’ga took in library thrillers and crime.novels at-home she put them na polku vmeste s drugimi bibliotečnymi materialami. Na sledujuščij den’ on shelf together with other library materials on next day ee dočka Arina tože pošla v biblioteku, no už v drugoe otdelenie. her daughter Arina also went in library but already to different branch Arina vernula knigi / kakie-to knigi. Arina returned books some books
‘Olga checked out thrillers and detective novels from the library. At home she put them on a shelf together with other library materials. The next day her daughter Arina also went to the library, but to a different branch. Arina returned **books** / **some books**.’

- a. Arina returned the books Olga had checked out the day before. *anaphoric*

b. Arina returned other books.

non-anaphoric

- (52) Katja sobrala list'ja i cvety dlja togo, čtoby risovat' s natury.
Katja gathered leaves and flowers for that what draw from life
Večerom ona pošla na master-klass po risovaniju so svoej podrugoj Tanej.
in/evening she went to workshop on drawing with self friend Tanja
Katja narisovala frukty na podnoce, a Tanja narisovala cvety / kakie-to cvety.
Katja drew fruits on tray and Tanja drew flowers some flowers
'Katya gathered leaves and flowers to draw from life. In the evening she went to a
drawing workshop with her friend Tanya. Katya drew fruit on a tray, and Tanya
drew **flowers / some flowers.**'

a. Tanya drew the flowers Katya had gathered earlier.

anaphoric

b. Tanya drew other flowers.

non-anaphoric

- (53) Olja kupila sveči i nebol'sie kartiny. Ona ukasila imi svoju
Olja bought candles and small paintings she decorated with.them self
komnatu. Večerom ee sosedka po kvartire Alëna rešila sdelat' gostinuju
room evening her neighbour by flat Alëna decided make living-room
pouijutnee i zažgla sveči / kakie-to sveči.
cozier and lit candles some candles
'Olya bought candles and small paintings. She decorated her room with them. In
the evening her flatmate Alyona decided to make the living room cozier and lit
candles / some candles.'

a. Alyona lit the candles Olya had bought earlier.

anaphoric

b. Alyona lit other candles.

non-anaphoric

- (54) Lëva rabotaet bibliotekarem. Segodnja on razobral **knigi** i žurnaly. On
Lëva works as.librarian today he sorted books and magazines he
raspoložil svežie vypuski žurnalov na polke v otdele periodiki.
placed fresh issues of.magazines on shelf in section of.periodicals.
Večerom ego kollega Anton ubiralsja v čital'nom zale i našël
in.evening his colleague Anton was.cleaning in reading hall and found
žurnaly / kakie-to žurnaly.
magazines some magazines

‘Lyova works as a librarian. Today he sorted books and magazines. He placed fresh issues of the magazines on the shelf in the periodicals section. That evening his colleague Anton was tidying the reading hall and found **magazines** / **some magazines**.’

- a. Anton found the magazines Lyova had placed in the periodicals section earlier. *anaphoric*
- b. Anton found other magazines. *non-anaphoric*

(55) Boris rabotaet v pračecnoj. On privëz postirannye **prostyni** i polotenca v Boris works in laundromat he brought washed sheets and towels to gostinicu, kotoraja zakazala stirku. Boris vygruzil zakaz v komnate dlja hotel which ordered laundry Boris unloaded order in room for khraneniya postel'nogo bel'ja. Tem že večerom gorničnaja gotovila storing bed linen that same evening maid was preparing komnaty k zaseleniyu, zašla v nomer na pervom etaže i našla polotenca / rooms for check-in entered in room on first floor and found towels kakie-to polotenca.
some towels
‘Boris works at a laundromat. He brought freshly washed sheets and towels to a hotel that had placed an order. Boris unloaded the order in the linen storage room. That same evening a housekeeper was preparing rooms for check-in, entered a room on the first floor, and found **towels** / **some towels**.’

- a. The housekeeper found the towels Boris had unloaded earlier. *anaphoric*
- b. The housekeeper found other towels. *non-anaphoric*

(56) Klara rabotaet kostjumerom v teatre. Ona podgotovila **pidžaki** Klara works as costumer in theater she prepared i plat'ja dlja zavtrašnego spektaklja, položila ikh na special'nuju jackets and dresses for tomorrow's performance put them on polku i ušla domoj. Na sledujuščij den' ee kollega Alina prišla v special shelf and went home on next day her colleague Alina came teatr. Ona zašla v komnatu dlja stirki i našla plat'ja / kakie-to to theater she entered in room for laundry and found dresses plat'ja.
some dresses
‘Klara works as a costume manager at a theatre. She prepared jackets and dresses for tomorrow's performance, put them on a special shelf, and went home. The next

day her colleague Alina came to the theatre, went into the laundry room, and found
dresses / some dresses.

- a. Alina found the dresses Klara had prepared earlier. *anaphoric*
- b. Alina found other dresses. *non-anaphoric*

- (57) Roman privëz v sportivnyj centr **mači** i fiški dlja razmetki polja. On
Roman brought to sports center balls and cones for marking field he
razgruzil oborudovanie v trenerskuju komnatu. Posle obeda trener Aleksej
unloaded equipment in coaches' room after lunch coach Aleksej
Petrovič gotovil pole k trenirovke, zašël v mužskuju razdevalku
Petrovič was preparing field for practice entered in men's changing-room
i našël mači / kakie-to mači.
and found balls some balls
'Roman brought balls and cones for marking the field to the sports centre. He
unloaded the equipment in the coaches' room. After lunch, coach Alexei Petrovich
was preparing the field for practice, entered the men's changing room, and found
balls / some balls.

- a. Alexei Petrovich found the balls Roman had brought earlier. *anaphoric*
- b. Alexei Petrovich found other balls. *non-anaphoric*

- (58) Učeniiki muzykal'noj školy otnesli **skripki** i al'ty v komnatu dlja
students music school took violins and violas to room for
khraneniija instrumentov. Večerom uborščica Nina Ivanovna myla poly
storing instruments in evening cleaner Nina Ivanovna was washing floors
v orkeistrovom klasse i našla skripki / kakie-to skripki.
in orchestra classroom and found violins some violins
'The music school students took violins and violas to the instrument storage room.
That evening the cleaner Nina Ivanovna was washing the floors in the orchestra
classroom and found **violins / some violins.**

- a. Nina Ivanovna found the violins the students had been using. *anaphoric*
- b. Nina Ivanovna found other violins. *non-anaphoric*

(59) Trener po plavaniju Ol'ga Aleksandrovna upakovala **polotenca** i kupal'niki coach in swimming Ol'ga Aleksandrovna packed towels and swimsuits dlja svoej komandy v sumku, položila eë v trenerskuju i ušla pogovorit' for self team into bag put it in coaches' room and went talk s administratorom. Drugoj trener, Elena Gennad'evna, zašla v tualet with administrator another coach Elena Gennad'evna entered in restroom i našla polotenca / kakie-to polotenca. and found towels some towels 'Swimming coach Olga Alexandrovna packed towels and swimsuits for her team into a bag, left it in the coaches' room, and went to talk with the administrator. Another coach, Elena Gennadyevna, went into the restroom and found **towels** / **some towels**.'

a. Elena Gennadyevna found the very same towels.

anaphoric

b. Elena Gennadyevna found other towels.

non-anaphoric

(60) Oleg budet predstavlat' svoju kompaniju na jarmarke vakansij. Pered Oleg will represent self company at fair of jobs before meroprijatijem on razložil **brošjury** i raspečatki s opisaniem vakansij event he laid out brochures and printouts with description of vacancies na stojke svoej kompanii. Pozže organizator jarmarki provorjala gotovnost' at stand self company later organizer of fair was checking readiness pomeščeniya k meroprijatiju, zašla v ženskiju ubornuju i našla brošjury / venue for event entered in women's restroom and found brochures kakie-to brošjury. some brochures 'Oleg was going to represent his company at a job fair. Before the event he laid out brochures and job description printouts at his company's stand. Later the job fair organizer was checking the venue's readiness, went into the women's restroom, and found **brochures** / **some brochures**.'

a. The organizer found Oleg's brochures.

anaphoric

b. The organizer found other brochures.

non-anaphoric

(61) Sonja rassortirovala **kurtki** i šarfy, kotorye ljudi požertvovali Sonja sorted jackets and scarves which people donated blagotvoritel'noj organizacii, po korzinam. Na sledujuščee utro drugoj charitable organization into baskets on next morning another

volontër, Ivan, rakladyval vešči pered meroprijatiem, vyšel pokurit' i
 volunteer Ivan was.arranging things before event went.out to.smoke and
 našel šarfy / kakie-to šarfy.
 found scarves some scarves
 'Sonya sorted jackets and scarves donated to a charity into baskets. The next
 morning, another volunteer, Ivan, was arranging items before the event, stepped
 outside for a cigarette, and found **scarves / some scarves.**'

- a. Ivan found the scarves Sonya had sorted earlier. *anaphoric*
- b. Ivan found other scarves. *non-anaphoric*

- (62) Katja rabotaet baristoj v kafe. Ona dostala svežie **kruassany** i buločki,
 Katja works as.barista in cafe she took.out fresh croissants and rolls
 akkuratno razložila ikh v vitrine i ušla na perervy. Čerez čas oficiant
 neatly arranged them in display-case and went on break in hour waiter
 Artëm provorjal zapas stakanov na stojke, zagljanul v holodil'nik i našel
 Artëm checked supply cups at counter looked in fridge and found
 kruassany / kakie-to kruassany.
 croissants some croissants
 'Katya works as a barista at a café. She took out fresh croissants and rolls, neatly
 arranged them in the display case, and went on her break. An hour later, waiter
 Artyom was checking the cup supply at the counter, looked in the fridge, and found
croissants / some croissants.'

- a. Artyom found the croissants Katya had been arranging in the display case earlier.
anaphoric
- b. Artyom found other croissants. *non-anaphoric*

- (63) Marina rabotaet v cvetočnom magazine. Ona prinjala novuju postavku **roz**
 Marina works in flower store she received new delivery of.roses
 i tjul'panov, postavila ikh v holodil'nik dlja cvetov i pošla oformljat'
 and of.tulips put them in fridge for flowers and went to.decorate
 vitrinu. Večerom kur'er Pëtr zašël za zakazami dlja dostavki,
 display-window in/evening courier Pëtr came-in for orders for delivery
 zagljanul v podsobku i našel tjul'pany / kakie-to tjul'pany.
 looked in storeroom and found tulips some tulips

‘Marina works at a flower shop. She received a new delivery of roses and tulips, put them in the flower fridge, and went to arrange the display window. That evening courier Pyotr came in to pick up delivery orders, looked in the storeroom, and found **tulips / some tulips.**’

- a. Pyotr found the tulips Marina had received earlier. *anaphoric*
- b. Pyotr found other tulips. *non-anaphoric*

(64) Nataša rabotaet v škole. Ona podgotovila raspečatki i novye **tetradi** dlja
 Nataša works in school she prepared printouts and new notebooks for
 učnikov, položila ikh v škaf i ušla na pedsovet. Drugaja
 students put them in cupboard and went to staff.meeting another
 učitel'nica, Marina Pavlovna, zašla v kabinet Nataši za čaem,
 teacher Marina Pavlovna entered in classroom of.Natasha for tea
 podošla k stolu i uvidela tetradi / kakie-to tetradi.
 approached to desk and saw notebooks some notebooks
 ‘Natasha works at a school. She prepared printouts and new notebooks for the
 students, put them in a cupboard, and went to a staff meeting. Another teacher,
 Marina Pavlovna, came into Natasha’s classroom for tea, walked over to the desk,
 and saw **notebooks / some notebooks.**’

- a. Marina Pavlovna saw the notebooks Natasha had put in the cupboard earlier. *anaphoric*
- b. Marina Pavlovna saw other notebooks. *non-anaphoric*

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